A CONTINUUM ACTION-DENSITY THEORY FOR FOUR-DIMENSIONAL SU(2) YANG–MILLS: A PROOF OUTLINE

Abstract. We give a detailed proof outline for constructing the scalar action density of four-dimensional SU(2) Yang–Mills theory from Wilson lattice measures. The construction takes the infinite-volume limit before removing the lattice spacing. Its target is a translation component whose moments are Euclidean invariant, reflection positive and clustering, with factorial bounds on Schwartz tests flat at coincidences. The connected two-point function should have size $d^{-8}(\log(1/d))^{-2}$ for fixed scaled bumps. We develop the renormalisation, large-field and source arguments, the passage from cubic symmetry to rotations, and the selection of a clustering phase. Uniform construction estimates that still need complete verification are identified at their point of use.

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1. The construction and its target

For an oriented nearest-neighbour bond b of a four-dimensional cubic lattice, let $U_b \in \mathrm{SU}(2)$, with $U_{-b} = U_b^{-1}$. For an elementary plaquette p, let U_p be the ordered product around its boundary and set

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} U_p, \qquad \mathcal{O}_a(f) = \sum_p f(x(p))a_p(U). \tag{1}$$

The sum takes each unoriented plaquette once, and $x(p)$ is its centre. On a periodic lattice of spacing a and side ℓ, Wilson's measure is

$$d\mu_{a,\ell,\beta}(U) = Z_{a,\ell,\beta}^{-1} \exp\left(-\beta \sum_p a_p(U)\right) \prod_b dU_b, \tag{2}$$

where the product is over one orientation of each bond, dU_b is normalised Haar measure, and $\beta = g_0^{-2}$ defines the bare coupling g_0. The continuum observable sought here is the scalar action density, formally proportional to $\operatorname{tr}(F_{\mu\nu}F_{\mu\nu})$. Its normalisation is fixed by (1); there is no additional power of a in that sum.

We first take $\ell \to \infty$ at fixed (a, β), and only afterwards let $a \to 0$ and $g_0 \to 0$. A small reference parameter g determines the terminal inverse-coupling window in (72); exact tuning to g is not required. Subsequence selection is allowed. If ν_a is the fixed-spacing infinite-volume reference state, we form moments of $\mathcal{O}_a(f) - \nu_a(\mathcal{O}_a(f))$, without requiring a finite uncentred one-point limit. A continuum limit can mix phases. An ergodic component means one phase in its decomposition

under translations. It retains this reference-state subtraction and can therefore have a nonzero mean. Clustering is sought in that component.

The Euclidean test space in the theorem must be specified. Let

$$\mathcal{S}_{\neq}(\mathbb{R}^{4n}) = \{\varphi \in \mathcal{S}(\mathbb{R}^{4n}) : \partial^{\alpha}\varphi(x) = 0 \text{ whenever } x_i = x_j \text{ for some } i \neq j, \text{ for every } \alpha\}. \tag{3}$$

Thus the test function and every derivative vanish at coincidences. For $n = 1$ this is the usual Schwartz space, and $\mathcal{S}_{\neq}(\mathbb{R}^0) = \mathbb{C}$. Write

$$|\varphi|_k = \sup_{x,\,|\alpha|\le k} (1+|x|^2)^{k/2}|\partial^{\alpha}\varphi(x)|.$$

For a separated tuple $X = (f_1, \dots, f_n)$, the functions are smooth, compactly supported, and have pairwise disjoint supports at positive distance. We write $S(X) = S_n(f_1 \otimes \cdots \otimes f_n)$. Let $\theta(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, -x_4)$ and $(\Theta\varphi)(x) = \overline{\varphi(\theta x_1, \dots, \theta x_n)}$.

The following target organizes the proof outline. The sections below develop the construction and the estimates leading to it; the outline does not claim that every analytic and geometric verification is complete.

Target theorem A (Euclidean action-density construction)**.** For every sufficiently small reference parameter $g > 0$, there are bare couplings $g_{0,j} \to 0$ and spacings $a_j \to 0$ for which the centred, volume-first action-density correlations have a common subsequential limit on all separated tuples. This limit has an ergodic component with moments S_n on $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$, $S_0 = 1$, with the following properties.

(i) The moments are invariant under translations and O(4), symmetric under permutation of the arguments, and real on real tests. Here Euclidean motions act on every argument by the same pullback. The first moment is $S_1(f) = b\int f$ for a finite real constant b.

(ii) They are reflection positive:

$$\sum_{q,q'\ge 0} S_{q+q'}(\Theta\varphi_q \otimes \varphi_{q'}) \ge 0 \tag{4}$$

for every finite sequence $\varphi_q \in \mathcal{S}_{\neq}(\mathbb{R}^{4q})$ supported where all time coordinates are positive, with $\varphi_0 \in \mathbb{C}$.

(iii) There are $A, B > 0$, an integer $k \ge 1$, and $p < \infty$, independent of n, such that

$$|S_n(\varphi)| \le AB^n(n!)^p|\varphi|_{kn} \qquad (n \ge 1). \tag{5}$$

(iv) For separated compactly supported tuples X, Y, there are $C_{X,Y}, r_{X,Y} < \infty$ such that

$$|S(X, \tau_v Y) - S(X)S(Y)| \le C_{X,Y}|v|^{-2} \qquad (|v| \ge r_{X,Y}), \tag{6}$$

where $(\tau_v f)(x) = f(x - v)$.

(v) Fix a nonnegative smooth function χ of integral one, supported in $B(0, 1/8)$, and put

$$f_d(x) = d^{-4}\chi(x/d), \qquad h_d(x) = d^{-4}\chi((x - 3de_1)/d).$$

There are $c, C, d_0 > 0$, depending only on g and χ, such that

$$\frac{c\,d^{-8}}{(\log(1/d))^2} \le S_2(f_d \otimes h_d) - S_1(f_d)S_1(h_d) \le \frac{C\,d^{-8}}{(\log(1/d))^2} \qquad (0 < d \le d_0). \tag{7}$$

The smallness threshold for g may depend on the fixed choice of χ; all other constants may depend on g and, in this clause, on χ.

■□

We use the diagonal-flat test-space convention of [OS73, §§2–3, pp. 85–89] for the Osterwalder–Schrader properties listed above. The quantitative growth and reconstruction questions are distinguished in [OS75, §IV.1, pp. 287–288]; here only the Euclidean properties listed in the target are sought. The lower bound in (7) says that the scalar field is not a constant. The statement concerns

this observable sector; it asserts neither a mass gap nor uniqueness of the continuum limit. It does not specify contact terms, that is, the values of distributions supported at coincident points. We do not include a reconstruction theorem in the target statement.

The construction begins with one common sequence of effective action coefficients. In four dimensions the quadratic curvature term is marginal: its small changes accumulate over arbitrarily many scales. It must therefore be extracted at each step and put into the running inverse coupling. The other local terms are controlled on analytic domains with independent field and current variables. Agreement on nonwrapping regions makes this one coefficient sequence serve all volumes. Stopping at the first passage into a fixed small-coupling window then gives the lattice spacings used in the limit.

Large fields cannot be removed by imposing a global small-field restriction. The integral is decomposed by its actual failed tests, and every retained region carries the fields and operations stored there. A component that has stayed small and quiet for sufficiently many steps is prepared, integrated and replaced by local terms. This operation prevents a rare old event from charging every subsequent scale. Its proof has to control whole histories with an exterior supremum inside the component integral; a bound for each individual history would leave the history count unpaid. The effective density and its induction are developed in sections 2 and 3.

Sources are inserted by freezing those real operations and changing the weight to which they are applied. Local scalar source energies are subtracted before taking absolute values. The same subtraction can then be used in different volumes and Gibbs states, even when the uncentred one-point functions diverge. This gives bounded centred moments, estimates at each fixed order, and a graded estimate that keeps one factor for each active insertion. Two observation scales separate the leading free-curvature covariance from the local and live-component errors; the latter must be small relative to the leading $g(d)^4$ term. These are the source and two-point parts of sections 4 and 5.

The limiting construction initially retains only cubic symmetry. Coordinate and diagonal reflection forms supply Hilbert spaces and bounded transfer inequalities. Passing those bounded inequalities to the continuum gives the sharp coordinate cones. A local insertion bound independent of the other fields makes planar point insertions into bounded forms. Their ordered products are holomorphic, with an exponent in the tube margin growing only linearly in their number. Once the ordered angular charts have been glued, the margin bound gives only finitely many Fourier coefficients in the rotation angle. Reflection positivity gives convexity of the highest frequency, and a polygon zero-row argument turns it into linearity. A short-distance upper bound and a second positivity sign at a diagonal mirror settle the two-point base case. Tracking the coupling dependence of source stability supplies a sufficient upper bound without the relative Gaussian comparison. Ordered tubes, an angular differential equation and slab translations give the continuation through different orderings in section 7.

The reference continuum correlation hierarchy may be a mixture of phases. At each fixed lattice spacing its translation-ergodic decomposition is again Gibbs for the same Wilson specification. A uniform estimate of the approach to the invariant plateau is then needed to pass this decomposition to the continuum and retain clustering. The proof uses the transfer spectrum in four coordinate pictures and is developed in section 8. Finally, the graded correlation bounds extend the separated distributions to tests flat at coincidences. The extension, restoration of the finite component mean, and order of the common construction are described in section 9.

2. Renormalisation and the choice of bare couplings

A renormalisation step replaces a field by its averages on blocks of side L. Its purpose is to express the same integral in terms of fewer variables, while retaining quantitative information about the fluctuations that have been integrated out. In four dimensions this requires a running coefficient

of the action. A bound on all the other terms, without separating that coefficient, does not remain uniform in the number of steps.

Choose the remainder exponent κ_0 sufficiently large for the scale sums below; in particular, $\kappa_0 \geq 7$. Fix L odd and sufficiently large and put $a = a_K = L^{-K}$. The auxiliary cube sizes are chosen as powers of L. The torus estimates use sides $\ell = 2L^m$, with m an integer large enough to satisfy the volume floor in eq. (71). Below, "tori" refers to this family, except for the adapted period lattices used for diagonal mirrors in section 6. After k steps the lattice is $\mathbb{T}^{(k)} = (L^k a\mathbb{Z}/\ell\mathbb{Z})^4$, with spacing $L^k a$. When lengths are measured in units of this lattice, the bare spacing is $\eta_k = L^{-k}$. We write $A(U) = \sum_p a_p(U)$ and $x_k = g_k^{-2}$. For a scalar weight v, also write $A(v, U) = \sum_p v(x(p))a_p(U)$. Supremum norms on matrix-valued fields use a fixed matrix operator norm; on Lie-algebra fields they use a fixed Euclidean norm. The symbol ∂U denotes a plaquette product. Covariant differences ∇_U and the infinitesimal gauge action D_U use the spacing of the lattice on which they act; distances in a level-k estimate use that level's units unless physical units are specified. Thus D_U maps site gauge parameters to bond potentials, and D_U^* is its adjoint. Write d_U for the covariant exterior derivative from bond potentials to plaquette two-forms, with the same spacing convention. Write $\Delta_U = D_U^* D_U$ for the covariant Laplacian on site functions. The symbol Δ denotes the positive flat lattice Laplacian, acting componentwise on forms. The construction uses two auxiliary cube sizes: M_1 for local propagators and M for local interactions. The constants are chosen in the order L, M_1, the decay rate κ, then M large, the interaction bound E_0, and finally a small coupling ceiling γ. In particular, the small coupling is chosen after the constants that multiply it. Write W_4 for the group of signed permutations of the four coordinate axes, acting on the lattice and its fields.

Complex fields take values in $\mathrm{SL}(2, \mathbb{C})$. For them, A and its weighted version use the holomorphic plaquette term $a_p(\mathbf{U}) = 1 - \frac{1}{2}\operatorname{tr}\mathbf{U}_p$. This agrees with (1) on $\mathrm{SU}(2)$, whose elements have real trace. All actions evaluated on complex fields below use this continuation. Adjoints are taken with respect to the fixed real-field inner products. At complex fields the starred operator formulas denote their holomorphic continuations, rather than Hermitian adjoints taken at the complex argument.

The small-field construction follows Bałaban's [B87]. The work is to retain its locality and analytic dependence uniformly in the cutoff and torus, and then to include the large-field terms. We describe the small-field integral first. The operation on the complementary regions and the exact summation over their histories are treated in section 3. The two parts share the same running coupling and must close one induction.

The argument separates exact integral identities from the estimates used to iterate them. Technical verifications still needed for the construction are identified where they enter. Exact preservation of an integral does not itself give a uniform bound on its local expansion.

2.1. **Averaging and gauge fixing.** Group-valued fields require an average that respects multiplication by gauge transformations. For a family of nearby group elements V_i, one first forms a mean V determined locally by $\sum_i \log(V_i V^{-1}) = 0$. This mean is analytic near a constant family, commutes with left and right multiplication, and has ordinary arithmetic averaging as its linear part in the Lie algebra. For a site y at the centre of a block $B(y)$, let $U(y, x)$ be the mean of the transports from y to $x \in B(y)$ along shortest paths that traverse the coordinate directions in all possible orders. Write $U(x, y) = U(y, x)^{-1}$.

Let c be a coarse bond, let $[x, x']$ be its parallel copy starting at $x \in B(c_-)$, and let $U[c]$ be straight transport along c. The block average is

$$\mathcal{M}(U)(c) = \exp\left\{ L^{-4} \sum_{x \in B(c_-)} \log\Big[U(c_-, x)U([x, x'])U(x', c_+)U[c]^{-1}\Big] \right\} U[c]. \tag{8}$$

The logarithms in this formula concern closed transports based at c_-. On real fields with plaquettes within cL^{-2} of the identity in the two adjacent blocks, the contours are close enough to the identity for this formula to be analytic. A complex field requires the corresponding contour conditions as

well: a small bound on its plaquettes alone is insufficient, since conjugation by a complex gauge transformation need not preserve the matrix norm. Outside the analytic domain the construction uses covariant Borel extensions of the average. Such an extension is needed because the exact integral is over all configurations.

The centred blocks and the mean over all coordinate orders make $\mathcal{M}$ covariant under the hypercubic group. It is gauge covariant and local in the two blocks meeting c. The transformation $\mathbf{T}$ is the image measure under this map:

$$\int F\, d(\mathbf{T}\rho) = \int F(\mathcal{M}(U))\rho(U)\, dU, \qquad \int \mathbf{T}\rho = \int \rho.$$

This equality is exact. It concerns the averaging transformation $\mathbf{T}$; the observables accompanying the subsequent large-field operation require the insertion identities of section 3.

Uniform control of repeated averages uses a gauge correction at every step. For a regular real U and a small complex potential A, the expansion has the form

$$\mathcal{M}^k(e^{i\eta_k A}U) = \left(e^{i[Q_k^\sharp(U)A + C_k(U,A)]}\mathcal{M}^k U\right)^{v_k}, \qquad |C_k(U,A)| \le C|A|^2, \quad |v_k - 1| \le C|A|. \tag{9}$$

Here the block gauge factor produced by one average is removed before the next is taken. The linear operator $Q_k^\sharp$ is the derivative of this corrected average, rather than the plain derivative of $\mathcal{M}^k$. Its leading term is the straight covariant average; the difference has relative size $O(\varepsilon)$ at curvature ε. The two agree at the trivial field.

This distinction accounts for uniformity in k. If $A^{(n)}$ is the potential after n corrected averages, set $t_n = \sup|A^{(n)}|/(L^n\eta_k)$, and let ϖ_n bound the curvature after those averages. The corrected-average estimate is

$$t_{n+1} \le t_n(1 + C\varpi_n + CL^n\eta_k t_n), \qquad \sum_n \varpi_n \le C\varepsilon, \qquad \sum_{n<k} L^n\eta_k \le 2L^{-1}.$$

The error sums are geometric. By comparison, multiplying estimates for the uncorrected linear averages produces a factor larger than $2L$ per step instead of the spacing ratio L. It does not give the required uniform estimate. The corrected operator also satisfies the gauge identity $Q_k^\sharp(U)D_U\lambda = D_{\mathcal{M}^k U}Q'\lambda$. This is the identity used in the gauge projection below. The site operator Q' is the linearisation of the iterated gauge-transformation average. At one step it averages a site parameter after adjoint transport to the block centre: $\lambda \mapsto L^{-4}\sum_{x\in B(y)} \mathrm{Ad}(U(y,x))\lambda(x)$. Its iterate uses the corresponding fields at successive levels.

The symmetric average here also has to be compared with the corner-block average in Bałaban's background-field propagator construction [B85]. The comparison is between their gauge-corrected linear parts. After aligning the blocks by a half-block translation, it has the form $Q_k^\sharp = \mathrm{Ad}(t)(Q_k^{\mathrm{B}} + \mathsf{D}_k)$, where t transports the block base point and D_k has relative size $C\varepsilon$ in row and column norms, uniformly in k. At the trivial field these linear parts coincide. Thus the transfer of propagator estimates uses a controlled comparison of the actual averages.

Gauge fixing removes zero directions of the Hessian. On a block there are $3(L^4-1)$ such directions for SU(2), corresponding to gauge transformations equal to the identity at the coarse site. Insert a factor $\mathfrak{G}_k$ whose average over these gauge transformations is one. In terms of the mean contour variables $V(y,x)$, its local factors have the form

$$\mathfrak{z}(x_k,\varepsilon_k)^{-1}\mathbf{1}_{\{|V(y,x)-1|<\varepsilon_k\}}\exp\Big[-x_k\left(1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr}V(y,x)\right)\Big].$$

The exponential term is a soft gauge fixing. With the fluctuation scaled as $g_k B$, its quadratic part is of order one and is strictly positive on the residual gauge directions. Its indicators equal one on the smaller fluctuation domain used in the integral. The ratio of these two domains is fixed after L. The normalising factors $\mathfrak{z}$ enter the vacuum energy, and hence must be included in the exact bookkeeping of constants.

Axial gauges choose representatives of gauge orbits. The estimates use a Landau slice, with gauge transformations constrained by their averages. For a regular reference field U_0, nearby regular fields can be written, after such a gauge transformation, as $e^{i\eta_k A}U_0$ with $R_{\mathrm{comb}}D^*_{U_0}A = 0$ and $|A| + |\nabla_{U_0}A| \le C(\varepsilon + \alpha_1)$. Here R_{comb} is the slice projection for the corner-comb site averages described below, and α_1 bounds the change in the prescribed averages. On a cube of side $O(M)$, curvature and current bounds of size ε give a local gauge with $|A| + |\nabla A| \le CM\varepsilon$ when $M\varepsilon$ is small. These local charts are the ones used to compare the background operators with their values at the trivial field.

The plain derivative will also be needed. Put

$$\mathcal{Q}_k(U;A) = \frac{1}{i}\log\left[\mathcal{M}^k(e^{i\eta_k A}U)(\mathcal{M}^k U)^{-1}\right], \qquad Q_k(U) = \partial_A \mathcal{Q}_k(U;0).$$

If $\dot v_k$ is the infinitesimal node factor in (9), then, with the gauge-action convention used there,

$$Q_k = Q_k^\sharp - D\dot v_k, \qquad Q_k D_U\lambda = D_{\mathcal{M}^k U}(\lambda|_{\mathbb{T}^{(k)}}). \tag{10}$$

The corresponding right-hand side for $Q_k^\sharp$ contains the average $Q'\lambda$. A gauge parameter vanishing at the coarse sites therefore preserves the plain constraint, whereas a gauge parameter of zero average preserves the corrected constraint. These are different subspaces, already at the trivial field. The propagators use the second one. The fibre coordinates below solve the plain constraint and include its compensating node factor.

The local mean and the continuation of the average have different purposes. Near a constant family, writing $V_i = e^{X_i}V_0$ and $V = e^X V_0$ reduces the mean equation to

$$\sum_i \log(e^{X_i}e^{-X}) = 0.$$

At $X_i = X = 0$ its derivative in X is minus the number of members times the identity. The local implicit solution thus has linear part $X = \frac{1}{\#\{i\}}\sum_i X_i$. Conjugation commutes with the equation, and uniqueness gives local covariance. This argument supplies no logarithm at every group element or unique mean for an arbitrary family. In particular, the principal matrix logarithm at -1 is not an $\mathfrak{su}(2)$ logarithm. Borel rules away from the analytic neighbourhood are fixed before the expansion.

The recurrence for corrected averages gives an elementary uniform bound once its one-step estimate is available. Fix a small upper bound $t_\star$ for the initial potential. Up to a first index with $t_n > 2t_\star$, the product of the recurrence factors is at most

$$\exp\left(C\sum_n \varpi_n + 2Ct_\star\sum_{n<k} L^n\eta_k\right) \le \exp(C\varepsilon + 4Ct_\star/L).$$

Choose $\varepsilon, t_\star$ so this is less than two. There can then be no such first index. The uniform one-step estimate, its quadratic error, and its row and column bounds remain the averaging input; the displayed product explains how that input prevents an extra factor depending on k.

For later use, a level-j activity is holomorphic in an independent pair $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ represented by a splitting $\mathbf{U} = e^{i\eta_j A'}U$, with U real. On each relevant cube the analytic domain requires

$$\begin{gathered}|\partial U - 1|, |\partial\mathbf{U} - 1| < \alpha_{0,j}\eta_j^2, \qquad \max\{|A'|, |\nabla_U A'|\} < \alpha_{1,j}, \qquad |\mathbf{J}| < \alpha_{0,j},\\ CM\alpha_{0,j} < \alpha_{1,j}, \qquad \alpha_{1,j}^2 \le C\alpha_{0,j}.\end{gathered} \tag{11}$$

It also requires the local real gauges above and regularity of the backgrounds formed from averages on the subdomains used by the construction. These latter conditions are part of the domain. Two allowed splittings of one pair must give the same value to every activity. No saturation under arbitrary complex gauge transformations is asserted.

Expanding the plaquette of $e^{i\eta_j A'}U$, the two plaquette conditions give

$$|\mathrm{d}_U A'| \le C(\alpha_{0,j} + \alpha_{1,j}^2).$$

The quadratic term has the order of the allowed curvature and scales accordingly when the background is restricted to a later level. Response and interpolation estimates must preserve the entire domain, including the conditions on subdomain backgrounds.

The block factor removed at a single step is explicit. If $W = \mathcal{M}^n U$ and $W' = e^{iA^{(n)}} W$ is the already straightened field, the factor at y is

$$v^{(n)}(y) = \exp\left(L^{-4} \sum_{x \in B(y)} \log[W'(y,x) W(y,x)^{-1}] \right). \tag{12}$$

It is formed from W', not from the plain iterate of the perturbed field. Removing it before the next average is what leaves the spacing factor L in the row norm. The product of the transported block factors is v_k in (9). If the perturbed field has the same contour variables as U at each preceding level, it is axial relative to U, and these corrections are trivial.

For a gauge function u, the corresponding block average at the reference field W is

$$(\overline{R}_W u)(y) = u(y) \exp\left(L^{-4} \sum_{x \in B(y)} \log[u(y)^{-1} \operatorname{Ad}(W(y,x)) u(x)] \right). \tag{13}$$

At first order this gives Q'. The nonlinear condition $\overline{R}_U^{(k)} u = 1$ describes the gauge transformations that leave the straightened constraint unchanged at a relatively axial field. Away from that field the condition depends on the field itself. It should not be treated as a fixed subgroup. The transformations equal to one at the coarse sites do form a subgroup; the node factor relates these two descriptions of an orbit.

This distinction also explains the choice of soft gauge fixing. Delta functions on single tree transports can be solved by ordering the tree. Delta functions on the contour means in (8) instead couple all the paths and have no such triangular solution. The soft factor retains hypercubic covariance and its normalisation is still exact: at fixed V, $V^u(y,x) = V(y,x) u(x)^{-1}$, and Haar invariance reduces the integral of each local factor to $\mathfrak{z}/\mathfrak{z} = 1$.

Axial representatives on a sequence of layers are pinned outside its nodes. At a seam a finer node belongs to a block of the coarser layer, so its transport is pinned as well. An orbit of the group equal to one at all nodes then meets the axial gauge once. The subsequent Landau estimate uses transformations with trivial averages and therefore need not keep those node values fixed; their coarse effect is compensated in the datum. The site averages defining R_{comb} are formed by transport along combs starting at block corners. The comparison of this slice with the Landau slice for the original averages must be checked uniformly in the level. The two projections cannot be identified merely because both remove infinitesimal gauge directions.

2.2. **The constrained background and its response.** The background is a regular critical configuration with prescribed averages. Only uniqueness among regular critical configurations is needed. No assertion about all stationary points, or all absolute minima in the fibre, enters the construction. We first specify the problem on a sequence of regions; the problem on one coarse lattice is its special case in which every region is the torus.

Determining regions and the variational problem. Fix the last level k and use its units. The bare spacing is $\eta_k = L^{-k}$, and the spacing at level $j \le k$ is $\xi_j = L^{j-k}$. Let

$$\Omega_0 \supset \Omega_1 \supset \cdots \supset \Omega_k, \qquad \Omega_{k+1} = \varnothing,$$

where Ω_j is a union of cubes of side $M_1 \xi_j$ and

$$\operatorname{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > \rho_{\mathrm{geom}} \xi_j, \qquad \rho_{\mathrm{geom}} = R_1 M_1.$$

The parameters M_1, R_1 are large powers of the fixed odd integer L. The regions may be disconnected and their shapes are unrestricted. Their separation, rather than regularity of their boundaries, is the geometric input.

The layer of level j is $\Omega_j \setminus \Omega_{j+1}$. Its tops are the sites of $\mathbb{T}^{(j)}$ in that layer. A top y has scale $\xi_y = \xi_j$ and pillar $B^j(y)$. The pillars partition the determining region. At a seam, a site may belong to a finer lattice while lying in a pillar of a coarser level; following its successive block centres leads to its top. This ascent is what fixes the gauge convention at seams. For a bare bond or plaquette e, write $\mathsf{L}(e)$ for the largest scale attached to its corners.

Let Γ_j consist of the bonds of $\mathbb{T}^{(j)}$ with both endpoints in Ω_0, at least one endpoint in Ω_j, and neither endpoint in Ω_{j+1}. Put

$$\mathbf{B} = \bigcup_{j=0}^{k} \Gamma_j, \qquad M_{\mathbf{B}}(U) = (\mathcal{M}^j U|_{\Gamma_j})_j.$$

An interior row joins two tops in one layer. An outward row has an endpoint in the adjacent finer layer; the bonds inside that endpoint's block are retained at the finer level. There is no row on an inward bond entering Ω_{j+1}. This last convention is important when the next step imposes its block axial gauge: an extra inward constraint can prevent that gauge from being reached. In the strip construction, where a different set of nodes is retained, axial gauge is imposed outside those nodes instead.

For datum $\mathbf{V} = (V_j|_{\Gamma_j})_j$, the fibre and action are

$$C(\mathbf{B}, \mathbf{V}) = \{U : M_{\mathbf{B}}(U) = \mathbf{V}\}, \qquad A_{\Omega_0}(U) = \sum_{p \subset \Omega_0} a_p(U).$$

Criticality refers to curves in this fibre. The action contains the inner plaquettes only. Its current includes their first variation at the boundary bonds; it must not be replaced there by the current of an arbitrarily continued whole-torus field. The residual gauge group is

$$\mathcal{G}^{\mathrm{res}}(\mathbf{B}) = \{u : u = 1 \text{ at every node of } \mathbf{B}\}.$$

The datum is regular when its test-loop products are within ε of one. Inside a layer the loops are its plaquettes; across a seam, the finer values are averaged before forming the loop. A regular configuration satisfies

$$|\partial U(p) - 1| < \varepsilon_{\mathrm{reg}} \eta_k^2 \mathsf{L}(p)^{-2}, \qquad |J(U)(b)| < \varepsilon_{\mathrm{reg}} \mathsf{L}(b)^{-3}.$$

Thus curvature has dimension two and current dimension three in the units of the layer where they are read.

The background construction gives one orbit of regular critical configurations when the datum is sufficiently regular. We write it $U_{\mathbf{B}}(\mathbf{V})$ or $U_k(V)$ on a single lattice. The regular radius is fixed above $C\varepsilon$ and below a small absolute ceiling. The torus must have room for the double blocks and the enlarged local cubes used below; in particular these cubes are shorter than half its side. Increasing the volume introduces no new constant.

On a cube of side $O(M)$ the local regularity statement is

$$\begin{aligned} |\partial U_k - 1| \le C\varepsilon\eta_k^2, \qquad |J(U_k)| \le C\varepsilon, \qquad U_k = e^{i\eta_k A}, \\ |A| + |\nabla A| + \|\nabla A\|_{C^{\alpha_{\mathrm{H}}}} \le CM\varepsilon, \qquad 0 < \alpha_{\mathrm{H}} < 1. \end{aligned} \tag{14}$$

Here $M\varepsilon$ is small, and the gauge is chosen on a slightly larger cube. On a layer of scale ξ_j the quantities bounded by $CM\varepsilon$ are $\xi_j|A|$, $\xi_j^2|\nabla A|$ and $\xi_j^{2+\alpha_{\mathrm{H}}}\|\nabla A\|_{C^{\alpha_{\mathrm{H}}}}$. Thus their unweighted bounds contain the corresponding inverse powers of ξ_j. The Hölder exponent is fixed; constants need not remain bounded as it tends to one. The linear and nonlinear constructions that supply these estimates are developed below.

The distance used by the inverse estimates. Measure length in the units of the layer it traverses:

$$d_{\mathbf{B}}(x,x') = \inf_{\Gamma \ni x,x'} \left[\sum_{j\geq 1} \xi_j^{-1} \operatorname{length}(\Gamma \cap (\Omega_j \setminus \Omega_{j+1})) + \eta_k^{-1} \operatorname{length}(\Gamma \setminus \Omega_1) \right].$$

The infimum is over connected paths in the torus. In particular the distance outside Ω_0 uses the bare unit; leaving the determining region does not provide a cheaper route. The first member Ω_0 does not otherwise enter the definition.

If y, y' have levels j, j', each intervening layer costs the gap:

$$d_{\mathbf{B}}(y,y') \geq \rho_{\mathrm{geom}}(|j-j'|-1)_+.$$

This controls the entropy of summing over layers. Within a layer, sums of an exponential are bounded by the corresponding lattice integral. On moving one level outward, at most a factor L^4 is lost in the count of tops. Choose the gap, for every fixed rate $\tau > 0$ and scale power q that will be used, so that $\tau\rho_{\mathrm{geom}} \geq C(1+q)\log L$. Then

$$\sup_y \sum_{y'} e^{-\tau d_{\mathbf{B}}(y,y')} \leq C_{\tau,L}, \qquad \left(\frac{\xi_{y'}}{\xi_y}\right)^q \leq L^q e^{\tau d_{\mathbf{B}}(y,y')}.$$

A fixed fraction of a decay rate can therefore absorb scale ratios without producing a factor depending on the number of regions. This is the summation used in the Green functions, the response, and the later replacement of global backgrounds by local ones.

The raw distance is insufficient as a weight when second differences are taken. Use a regularized distance $f_{y'}$ with

$$|f_{y'} - d_{\mathbf{B}}(\cdot,y')| \leq C, \quad |\nabla f_{y'}| \leq C\xi^{-1}, \quad |\nabla^2 f_{y'}| \leq C\xi^{-2}$$

in the local scale. Pairing the opposite directions in the conjugated Laplacian gives a perturbation $O(\theta)$ for the weight $e^{\theta f_{y'}}$. Estimating its directed bonds separately would instead introduce the inadmissible ratio ξ/η_k. The mass and the weight must be scaled together.

The chart and the row datum. Fix a real regular reference U_0. Put $U = e^{i\eta_k A'} U_0$, and let $F(A') = A_{\Omega_0}(U)$. The straightened average is the nonlinear function $\mathcal{Q}_k^\sharp(A')$ whose site factors were removed in (9). Its derivative will be denoted $K_{A'}$; at zero it consists of one local row for each bond of $\mathbf{B}$. Each row sees only the double block beneath that bond. The plain logarithm of the average has an additional coarse gauge term, which is present even at the trivial field. It cannot be discarded as a curvature error.

At an interior row the reduced datum equals B. At an outward row whose outer endpoint is z, the node compensation in the finer block gives

$$e^{ib(B)_c} = e^{iB_c} \operatorname{Ad}(V_{0,c}) v_z(B). \tag{15}$$

The expression for the other orientation is obtained by reversing the bond. The factor v_z reads only the interior rows of its block. Consequently $B \mapsto b(B)$ is triangular, holomorphic and locally invertible, with $\sup|b(B)| \leq C \sup|B|$. One first estimates the response to b and then differentiates through this map. The seam term in that chain rule is part of the response.

The distinction between potential data and group-angle data also matters at a seam. A group-angle perturbation on a row of length ξ_c corresponds to a potential of size ξ_c^{-1}. If m is multiplication by ξ_c and Q is the linear average per unit length, then $K_0 = \mathsf{m}Q$. The identities $Q\mathsf{H} = 1$ below refer to potential data; the response to b is $\mathsf{H}\mathsf{m}^{-1}$.

For the construction one first uses the comb slice, whose projection R_{comb} is defined by the corner-comb pillar averages. The subsequent response uses the projection R associated with the actual averaged contours. These slices describe the same critical orbit, but their potentials and projections need not coincide. The comparison uses bounds on the rows and columns, rather than an identification of the projections.

For one step those norms have leading sizes L and L^{-3}. Their relative errors at an intermediate level m are bounded by

$$C(\alpha_0 L^{-2(j-m)} + L^{m-j}\xi_j \sup|A'|).$$

The errors sum geometrically to a constant independent of j. The composed rows and the accumulated site factors consequently obey

$$\sum_b |K_{A'}(c,b)| \le C\xi_c, \qquad \sum_{c\in\Gamma_j} |K_{A'}(c,b)| \le C\eta_k^4\xi_j^{-3}, \qquad \sum_m |Y_m| \le C\|A'\|_{(-1)}, \quad v_j = \prod_m e^{iY_m},$$

with the column bound taken separately at each row level. Here $\|A'\|_{(-1)} = \sup \mathsf{L}|A'|$. These are the uniform bounds used for the transpose of the constraint in the multiplier equation. A comparison that introduced a fixed error at every level would not suffice.

Folding and local problems. A field known only on a box is placed in the local estimates by folding onto that box. In each coordinate the folding map is the identity inside, reflects across the adjacent face, and is constant farther out. Collapsed bonds receive the value one. Every plaquette of the folded field is either one or an original plaquette, possibly with reversed orientation. Hence folding preserves the real curvature bound exactly. It also preserves the local dependence of the averaged quantities on double blocks. Across a seam the change of scale permits the fixed loss L^2; it does not permit a loss repeated for every preceding level.

For a history the first determining member is $\Omega_0 = \Omega_1^\sim$, not the entire bare torus. This excludes uncontrolled old large fields from the regularity requirements. At its outer boundary there are no large blocks left to average: the rows there act at the bare scale. This geometric fact will be used in the local inverse construction.

2.3. **Propagators on separated regions.** We distinguish estimates for a whole inverse from a decomposition into local contributions. The first follows from a gap and an exponential weight. The second requires a local construction of the inverse, since later operations change the field and the regions outside the support of one contribution. Both must be uniform in the fine mesh and in the number of layers. The assembly follows the local-inverse and walk method of Bałaban[B84, B85]; the centred averaging rows and the scale-dependent gauge projections are retained throughout the estimates here.

The operator class and the current variable. For the complex estimates use pairs $(\mathbf{U}, \mathbf{J})$, where $\mathbf{U} = e^{i\eta_k A'} U$ and U is real. On a layer of scale ξ_j require

$$|\partial U - 1|, |\partial \mathbf{U} - 1| < \alpha_0 \eta_k^2 \xi_j^{-2},$$
$$\xi_j |A'|,\ \xi_j^2 |\nabla_U A'| < \alpha_1, \qquad |\mathbf{J}| < \gamma_0 \xi_j^{-3}.$$

In each local cube of side $O(M_1\xi_j)$ in two consecutive layers there must also be a real gauge with $\xi_j|A|, \xi_j^2|\nabla A| \le \alpha_0$. This is a bound on the entire cube, not $M_1\alpha_0$. When a regularity theorem produces only $CM_1\varepsilon$ in such a gauge, it is that larger quantity which must lie below the ceiling of this class. The field strengths of the construction are chosen after the cube sizes for this reason.

The current is an independent variable. Small curvature and small first differences alone do not bound the current uniformly in the mesh: a rapidly alternating field can satisfy those bounds while its second differences grow like a negative power of the mesh. Taking $\mathbf{J}$ independently with its own radius supplies the derivative control used by the propagators. The pair is on shell when $\mathbf{J} = J(\mathbf{U})$ at the bonds in question. At the outer boundary the current is that of the inner-plaquette action fixed above.

All operator transposes in a complex formula below are for the bilinear pairings. We continue to write * for them. They are not conjugate adjoints at complex data. Positivity statements are made only at real pairs.

For the site average Q' and the linear average Q per unit row length, put

$$G' = (\Delta_{\mathbf{U}} + Q'^* \mathsf{a} Q')^{-1}, \qquad R = I - G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G'. \tag{16}$$

Here a has size $a_0\xi_y^{-2}$ on tops and $a_0\xi_c^{-2}$ on rows, with a fixed $a_0 > 0$. At real data the subtracted operator is the orthogonal projection onto the range of $G'Q'^*$. Hence $R^2 = R = R^*$ and $Q'G'R = 0$. It projects onto $\Delta_U \ker Q'$, rather than onto $\ker Q'$ itself.

The uncorrected vector inverse is

$$\mathsf{G}_0(\mathbf{U}) = (\Delta_{\mathbf{U}}^{\mathrm{W}} + D_{\mathbf{U}} R D_{\mathbf{U}}^* + Q^* \mathsf{a} Q)^{-1}. \tag{17}$$

On one lattice $Q = Q_k^\sharp$ and the row mass is of order one in its units. On a layer it has size ξ_j^{-2}.

The flat scalar problem at all scales. At $\mathbf{U} = 1$ the site form has the lower bound

$$\langle \lambda, (\Delta + Q'^* \mathsf{a} Q')\lambda \rangle \geq \gamma \sum_j \xi_j^{-2} \|\lambda\|^2_{L^2(\mathcal{L}_j)}.$$

The local Poincaré inequality controls the difference from the block mean; the term $Q'^* \mathsf{a} Q'$ controls that mean. Conjugate the equation for $u = G'(f + \nabla^* F)$ by a regularized exponential weight. The slope of the weight is $O(\xi_j^{-1})$ in layer j, matching its mass. For a small fixed rate the perturbation is at most half the gap, and the weighted energy estimate is

$$\|\xi^{-1} e^{\varphi} u\|_2 + \|e^{\varphi} \nabla u\|_2 \leq C\gamma^{-1} \big(\|\xi e^{\varphi} f\|_2 + \|e^{\varphi} F\|_2\big).$$

No iteration in the number of layers occurs. Restricting this estimate to local cubes and using interior estimates at scale ξ_y gives the corresponding pointwise and first-difference estimates. In that step the averaging mass is treated as a source at its local scale.

The one-scale interior estimates must themselves be uniform in the number of fine sites in a block. They use the massive free resolvent and the factorization of $\partial(1 - R)\partial^*$ through the unit lattice. The weighted L^2 estimate is first improved to L^8 with the resolvent kernel, and then to a bound on the field and its first differences with the derivative kernel. The kernel exponents are respectively $8/5 < 2$ and $8/7 < 4/3$. The coarse-image forcing in the equation is controlled before the second step. The detailed flat calculation for the centred straight averages is given again at its curvature use below; it does not by itself address the seams between different averaging orders.

There is no Dirichlet problem on a fine lattice cut through large averaging blocks. The separated geometry places $\partial\Omega_0$ in layer zero. There $Q = Q' = 1$ and the gauge term vanishes. A local Dirichlet inverse therefore has

$$a_0 \leq \eta_k^2 G_\square^{-1} \leq a_0 + 16.$$

Its mass is comparable to its norm and finite range gives decay on arbitrary sets of bare bonds. Using instead a Dirichlet inverse of positive averaging order would require uniform boundary regularity that need not hold, even at a flat face or a re-entrant edge. Interior local inverses away from layer zero are formed by periodic one-scale or two-scale problems and cut off before their seams.

The two-scale problem at a seam. At a junction of orders j and $j + 1$, use the exact linear change of gauge

$$A = A_2 + H_j B - \partial H_j' \vartheta.$$

Here $A_2 \in \ker Q_j$, B is a one-form on the coarse lattice vanishing on the axial trees of the coarse blocks, and ϑ is a coarse site function of zero mean in each of those blocks. Its support lies in the coarse region. The operator H_j minimizes the one-form action at order j; H_j' is the scalar minimizer for $\|\Delta\lambda\|^2$, not for $\|\partial\lambda\|^2$.

With these choices the representation is unique. The form splits into a form in A_2, a form in B, and a form in $\vartheta - \vartheta_1(A_2, B)$. The reason for retaining the scalar square is that the Landau projection changes at the seam. The three resulting unit-lattice forms have gaps. In particular the axial part obeys

$$L^2 \|Q_1^{\mathrm{st}} B\|^2 + \|\partial_1 B\|^2 \geq c_L \|B\|^2, \qquad c_L \geq cL^{-C}.$$

One may retain the sharper negative power L^{-5} in the version with its specified axial coordinates. The fixed positive constant c_L suffices for the assembly here. The averages in this inequality are the straight one-step averages. The gauge-corrected plain average is a different operator.

The tree restriction is essential. A gradient of a function of zero block mean has vanishing straight average and zero curvature; it would be a null direction without the tree condition. For rows next to the coarse region, the norm of $Q_1^{\mathrm{st}}B$ includes outward rows with at least one end inside that region. Omitting them would leave the gradient of its indicator uncontrolled.

The two-scale representation is used for one junction. It is not iterated to express an order-k gauge propagator as a long product of lower-order ones. Different orders use different gauge projections, so such an iteration would lose the uniform constants. Uniform one-scale estimates, the local two-scale calculation, and the spatial parametrix are the three ingredients of the many-layer estimate.

Inverses on tops and rows. Two finite-dimensional lower bounds are needed in addition to the scalar gap. On tops they have the scale of $Q'G'^2Q'^*$; on rows they have the scale of $Q\mathsf{G}_0Q^*$. Their normalized forms are

$$\Sigma_{\mathrm{s}} = \xi^{-2}Q'G'^2Q'^*\xi^{-2}, \qquad \Sigma_{\mathrm{r}} = \mathsf{m}^{-1}Q\mathsf{G}_0Q^*\mathsf{m}^{-1}.$$

The desired lower bounds are $\Sigma_{\mathrm{s}}, \Sigma_{\mathrm{r}} \geq \gamma I$, with the row and top pairings inherited from their averaging operators. Changing from potential rows to angle rows $\mathsf{m}Q$ must be accompanied by the corresponding change of transpose and multiplier.

For the site matrix, test by block bumps ψ_y with $Q'\psi_y = \delta_y$, whose second differences, including those across block faces, have size $C\xi_y^{-2}$. Duality with these bumps gives $Q'G'^2Q'^* \geq \gamma\xi^4$. This is the scaled version of the unit-block bump calculation: the factor ξ^4 is the fourth power of its local length, not a constant independent of scale.

For the row matrix, the variational formula requires a right inverse for all rows simultaneously. On one lattice, bumps placed near the face of a coarse bond give an average matrix with its diagonal share greater than one half; the remaining shares are its off-diagonal part. Diagonal dominance inverts that matrix with bounded energy.

At a seam the correction is exact rather than perturbative. Start with the same-level right inverses E_0, chosen biorthogonal to the two chain-count profiles that neighbouring rows induce in a pillar. Their error has the form

$$Q\mathsf{E}_0 = I + \mathsf{N}.$$

It goes only from noncrossing rows to crossing rows one level higher. A row in that image has no further image, because the gap separates the next seam. Thus

$$\mathsf{N}^2 = 0, \qquad \mathsf{E} = \mathsf{E}_0(I - \mathsf{N}), \qquad Q\mathsf{E} = I.$$

Each coefficient of the cross-level error is at most L^{-4} and the coefficients in a row sum to at most one. The weighted overlap estimate required for its energy is $\|\mathsf{m}^{-1}\mathsf{N}\mathsf{m}\| \leq CL^{-1}$. With this estimate the energy of the correction is controlled without a loss depending on the number of levels.

The remaining local calculation in this row construction is to realize those biorthogonal profiles by the actual centred-contour rows, with bounded overlap and the stated weighted energy. Nilpotence gives the correction once this calculation is verified; it does not establish the bump estimate itself. The row normalization must be kept throughout that calculation. This is one of the local inverse verifications required by the layered construction.

Once these lower bounds and the decaying kernels have been obtained, their inverses are controlled by conjugation and the Schur test. If $\Sigma \geq \gamma I$ and $|\Sigma(y, y')| \leq Ce^{-\tau d_{\mathbf{B}}(y,y')}$, conjugation by a smaller exponential changes it by an operator of norm at most $\gamma/2$. A Neumann series gives

$$|\Sigma^{-1}(y, y')| \leq 2\gamma^{-1}e^{-\tau' d_{\mathbf{B}}(y,y')}.$$

Undoing the scale normalization gives, symmetrically, factors $\xi_y^{-2}\xi_{y'}^{-2}$ for the top inverse and $\xi_c^{-1}\xi_{c'}^{-1}$ for the row inverse. Spending a fraction of the rate on their ratio gives the useful forms

$$|(Q'G'^2Q'^*)^{-1}(y, y')| \leq C\xi_y^{-4}e^{-cd_{\mathbf{B}}(y,y')}, \qquad |(Q\mathsf{G}_0Q^*)^{-1}(c, c')| \leq C\xi_c^{-2}e^{-cd_{\mathbf{B}}(c,c')}.$$

No coarse matrix is inverted by a bound uniform in the unscaled entries alone.

The corrected Hessians. In (17) the curl controls nongauge high frequencies, the Landau term controls the remaining fine gauge modes, and the average mass controls the coarse modes.

Let $D = D_{\mathbf{U}}$ and $\pi = I - DG'RD^*$. The first correction $\Delta'_\pi[\mathbf{J}]$ is linear in the independent current. On shell it has the identity

$$\Delta^{\mathrm{W}}_{\mathbf{U}} - \Delta'_\pi[J(\mathbf{U})] = \pi^* \Delta^{\mathrm{W}}_{\mathbf{U}} \pi.$$

The gauge-corrected inverse and response are

$$\mathsf{G} = (\mathsf{G}_0^{-1} - \Delta'_\pi[\mathbf{J}])^{-1}, \qquad \mathsf{H} = \mathsf{G}Q^*(Q\mathsf{G}Q^*)^{-1}, \qquad \widetilde{\mathsf{G}} = \mathsf{G} - \mathsf{H}Q\mathsf{G}.$$

For the second correction, let $C^{(2)}$ be the quadratic part of the straightened average per unit row length and define $\Delta^{(2)}[j]$ by $\langle A, \Delta^{(2)}[j]A\rangle = 2\langle C^{(2)}(A), j\rangle$. Then

$$\mathsf{G}_1 = (\mathsf{G}^{-1} - \pi^*\Delta^{(2)}[\mathsf{H}^*\mathbf{J}]\pi)^{-1}, \qquad \mathsf{H}_1 = \mathsf{G}_1 Q^*(Q\mathsf{G}_1 Q^*)^{-1}.$$

It is this last inverse that differentiates the nonlinear background at a critical datum: the Hessian of the action must be corrected by the multiplier paired with the curvature of the constraint fibre. When the current vanishes the two corrections vanish.

The exact gauge identity starts from covariance of the straightened average,

$$QD = \bar{D}Q'.$$

It is not an identity for the plain logarithm of the average. Differentiating action invariance and retaining the independent current gives

$$\mathsf{G}^{-1}D\lambda = D\Delta_{\mathbf{U}}\lambda - \Delta'_\pi[\mathbf{J} - J(\mathbf{U})]D\lambda \quad (\lambda \in \ker Q').$$

Consequently

$$\mathsf{G}_{(1)}DR = DG'R + \mathsf{G}_{(1)}\Delta'_\pi[\mathbf{J} - J(\mathbf{U})]DG'R.$$

On shell the defect vanishes and $RD^*\mathsf{H}_{(1)} = 0$. Off shell it must be retained; positivity at real pairs and holomorphy do not remove it. In particular an exact slice identity cannot be transferred to G_0 at nonzero current.

Local inverses and the parametrix. For the flat many-layer operator choose $\sum_\square h_\square^2 = 1$ and local periodic one-scale or two-scale inverses $\mathsf{G}_\square$. The cutoffs vanish across the seams to finer layers before the local model is changed. In layer zero next to the outer boundary use the massive bare Dirichlet inverse.

The delicate term is $\mathcal{P} = \partial(1-R)\partial^*$. With a second cutoff $\zeta = 1$ near the support of h, its commutator is split as

$$\mathcal{P}h - h\mathcal{P}_\square = \zeta[\mathcal{P}_\square, h] + \zeta(\mathcal{P} - \mathcal{P}_\square)h + (1-\zeta)\mathcal{P}h.$$

The first term costs M_1^{-1}: one derivative of the cutoff has size $(M_1\xi_j)^{-1}$ and is paired with a local inverse gradient of size ξ_j. The second term is a change-of-domain error. Resolvent identities on the two domains, recombined through $R = I - (1-R)$, give decay across the collar and a factor e^{-cM_1}. The last term is a tail with the same decay. Inserting the partition of unity at its endpoint turns it into a step between local cubes. These three estimates are what make the spatial assembly possible despite the nonlocal gauge projection.

At a complex regular pair the corrections introduce additional nonlocal expressions. The local construction unfolds them, together with R and the average constraint, into an auxiliary system of finite range. Its unknowns are the potential, the site and row auxiliary fields, and the multiplier. The propagators are blocks of the inverse of that system. The local perturbation estimate to prove in the gauges of the class has size $C(\alpha_0 + \alpha_1 + \gamma_0)$ relative to the flat local system.

The needed verification is at this last local system: its auxiliary equations and scale norms must give a bounded flat inverse and the stated perturbation estimate without a loss of derivatives. In particular a source that is a divergence must be kept in that form, and no uncontrolled derivative

may be placed on its row auxiliary variable. The flat scalar gap alone proves none of these facts. This is the local analytic calculation still required for the full corrected layered inverse.

With those local inverse bounds, the gluing argument is explicit. Write $\mathbb{K}$ for the auxiliary operator, choose source cutoffs $\theta_\square$ with $\sum_\square \theta_\square = 1$, and larger cutoffs $\widetilde{h}_\square = 1$ near their supports. The local system $\mathbb{K}_\square$ agrees with $\mathbb{K}$ on the stencil neighbourhood of $\operatorname{supp} \widetilde{h}_\square$. Put $S_k = \sum_\square \widetilde{h}_\square \mathbb{K}_\square^{-1} \theta_\square$. The derivative of $\widetilde{h}_\square$ is separated from the support of $\theta_\square$ by a fixed fraction of the local cube side. Then

$$\mathbb{K} S_k = I - \mathcal{K}_k, \qquad \mathcal{K}_k = -\sum_\square [\mathbb{K}_\square, \widetilde{h}_\square] \mathbb{K}_\square^{-1} \theta_\square. \tag{18}$$

Here the identity is first formed for the finite-range system and then read in its propagator blocks. A local commutator has norm, between local cubes,

$$C M_1^{-1} e^{-c\delta_{\mathrm{G}} M_1 / L}.$$

The power comes from differentiating the cutoff; the exponential comes from the separation of its derivative from the source cutoff. At fixed L it outweighs the bounded number of adjacent cubes. In the weighted local norm $\|\mathcal{K}_k\| < 1$, uniformly in the regions, and

$$\mathbb{K}^{-1} = S_k \sum_{n \geq 0} \mathcal{K}_k^n.$$

Its propagator blocks give the required Green functions. The Neumann series is a consequence of the local calculation, not a replacement for it.

Walks, branches and local dependence. The terms of this series are indexed by overlapping local cells. A cell consists of $O(1)$ cubes of side $O(M_1 \xi_j)$ in two consecutive layers. For the corrected inverse G_1, the coefficient $\mathsf{H}^* \mathbf{J}$ itself has a walk expansion. Its zero-length part stays in the local inverse; the remainder is a small vertex bearing a branch. Thus the resulting index is a tree walk.

List the cells depth first, recording the parent of each new cell. For a fixed initial cell there are at most C^m lists of length m. Arbitrary parent lists would count the same tree factorially many times and would defeat the estimate. The two depth-first conventions, inserting a branch as soon as it is met or listing the trunk before its branches in reverse order, give the same tree, hull and length.

A typical propagator term satisfies

$$|\mathsf{G}_\omega(x, y)| \leq C (C M_1^{-1/2})^{|\omega|} M_1^{-|\omega|/2} e^{-\delta_{\mathrm{G}} d_{\mathbf{B}}(\omega; x, y)}, \tag{19}$$

with the appropriate local scale factors for its source and output. The first factor pays for $C^{|\omega|}$ possible lists. The second remains available when terms are transported through coverings. Spatial decay is supplied by the local inverses and does not deteriorate as M_1 grows.

The hull of a tree walk enlarges its trail by five local cube sides, taking each cube at its own scale. It is not cut at Ω_0. Its intersection with Ω_0 is the carrier of the field variables and can be disconnected. Each term is holomorphic in the pair on that carrier and reads the regions only through their traces on the uncut hull. This is the local dependence needed in the volume comparison.

The terms are chosen as the same functions when a local translation carries the hull, pair and traces of the regions between two settings. For the fluctuation form the translation carries also the next-level region and the conditioning set. Equality of terms under such transport is stronger than equality or decay of their sums. It is imposed in their construction: real gauge averaging and conjugation symmetrization preserve the sum and the local bounds. Translation averaging, push-forward from covering tori, and a normal-family limit fix a single assignment on all tori with room.

When a walk wraps under push-forward it can produce at most $|\omega| + 1$ additional pieces. The reserved factor $M_1^{-|\omega|/2}$ absorbs that loss. The entrywise bound used only to take a normal-family subsequence may depend on the level. It enters no decay rate, local norm or sum of weights. The assignment is made once, rather than choosing a new subsequence for every later comparison.

The fluctuation form. The inverse used for a fluctuation is not obtained by restricting a previously chosen global covariance. Let C_V eliminate the pivots in the one-step average at V, and let $\mathsf{G}_V^{(2)}$ be the quadratic soft-gauge form. The quadratic effective action in the datum is

$$\Delta_k = (Q\mathsf{G}_1Q^*)^{-1} - \mathsf{a} - \mathsf{E}^{(2)}[\mathsf{H}_1^*\mathbf{J}], \qquad \mathsf{A} = \mathsf{C}_V^*(\mathsf{G}_V^{(2)} + \Delta_k)\mathsf{C}_V.$$

Here $\mathsf{E}^{(2)}$ is the quadratic part of the plain one-step constraint. Its correction is not optional. Differentiating the multiplier system identifies the first derivative of the background action with its multiplier. A second differentiation gives the inverse coarse form minus the artificial mass. Along the nonlinear one-step fibre there is also the second derivative of that fibre; paired with the multiplier it gives exactly the last term in Δ_k.

At the critical pair of the step this yields $\mathsf{A} = \mathsf{C}^*\Delta^{(k)}\mathsf{C}$. One can add the average equation as a multiplier equation in the auxiliary system, so the fluctuation covariance is another inverse block. For the free bonds Y of a union of next-level blocks, its definition is

$$C^{(k)}(Y) = (\chi_Y \mathsf{A}\chi_Y)^{-1}.$$

The required real lower bound is $\chi_Y\mathsf{A}\chi_Y \geq \gamma^{\mathrm{pos}} I$. At the flat field it uses the axial unit-lattice gap and the contraction of the curvature norm under averaging. For a real regular pair the local inverse is continued to the flat one; the positive gap must be retained along this homotopy. The same local parametrix then has error $CM_1^{-1} < 1$ and yields exponential decay, at a rate allowed to depend on L. This positivity verification belongs to the local corrected-system estimate above. Mere continuity of a finite-volume inverse would not give its uniform gap.

With that verification, let x lie in the local cell at y and let f be supported in the local cell at y'. The estimates used later take the forms

$$|\mathsf{G}_k f(x)| + \xi_y|\nabla\mathsf{G}_k f(x)| \leq C\xi_y^2 e^{-\delta_{\mathrm{G}} d_{\mathbf{B}}(y,y')}\|f\|_\infty, \qquad |C^{(k)}(Y)(b,b')| \leq Ce^{-\delta'|b-b'|}.$$

There are first-difference and Hölder variants, with Hölder differences taken only over distances at most the local scale ξ_y. Complex parallel transports over larger distances need not remain uniformly bounded.

2.4. **The nonlinear background problem on a sequence of layers.** The inverse estimates are used twice. They first construct a regular critical orbit from a regular trial field, and then estimate its response. A contraction on a gauge slice alone does not prove the Euler–Lagrange equation in the full fibre. The constraint is therefore kept nonlinear and a multiplier is retained until the gauge condition has been recovered.

The trial field and the nonlinear equation. On one lattice lift a coarse configuration V by putting one on bonds within a block and $V(c)$ on the bonds crossing the face of c. The lift $V^\downarrow$ satisfies $\mathcal{M}(V^\downarrow) = V$; its nontrivial plaquettes are plaquettes of V. The preceding background $U_0 = U_{k-1}(V^\downarrow)$ is a trial field. Write

$$B = \frac{1}{i}\log[V(\mathcal{M}^k U_0)^{-1}].$$

This datum is zero for that particular trial field on one lattice. It is kept because a nearby trial field, a seam compensation, and a variation of V all produce nonzero B.

In the chart $U = e^{i\eta_k A'}U_0$,

$$F(A') = F(0) + \langle A', J(U_0)\rangle + \tfrac{1}{2}\langle A', \Delta_{U_0}^{\mathrm{W}} A'\rangle + \mathcal{N}(A'),$$
$$\mathcal{N}(A') = O(|J(U_0)|\,|A'|^2 + |A'|^3)$$

after the constraint is straightened. The quadratic term proportional to the current is the current paired with the quadratic change of coordinates. It remains even when the original action has already been expanded to second order.

For a potential on the slice write $A' = \mathsf{H}_k B + A_1$ with $QA_1 = 0$. Its regularity equation is

$$A_1 = -\widetilde{\mathsf{G}}_k J(U_0) - \widetilde{\mathsf{G}}_k \mathcal{N}'(\mathsf{H}_k B + A_1). \tag{20}$$

The constrained inverse maps a force into a potential and its first differences. The derivative of the nonlinear term has size $C(\varepsilon + |B|)$ in those norms. This proves the estimates once a solution on the slice is known; it does not alone prove that the resulting potential has the prescribed gauge.

For existence retain the multiplier equation

$$\begin{aligned} F'(A') + D_{U_0} R_{\mathrm{comb}} D_{U_0}^* A' &= [\mathcal{Q}_k^\sharp{}'(A')]^* \omega, \\ \mathcal{Q}_k^\sharp(A') &= b(B). \end{aligned} \tag{21}$$

In the first construction the reference rows and projection are those of the comb slice. The actual straightened constraint differs from these rows by a small linear term as well as by its quadratic remainder. Write it as $\mathsf{Q}A' + \mathsf{C}(A')$, and put

$$\mathsf{G}_0 = (\Delta_{U_0}^{\mathrm{W}} + D R_{\mathrm{comb}} D^* + \mathsf{Q}^*\mathsf{Q})^{-1}, \quad H_0 = \mathsf{G}_0 \mathsf{Q}^* (\mathsf{Q}\mathsf{G}_0\mathsf{Q}^*)^{-1}, \quad \widetilde{G}_0 = \mathsf{G}_0 - H_0 \mathsf{Q}\mathsf{G}_0.$$

Here the row lengths have been included in Q. Abbreviate $F'(A') = F'(0) + \Delta_{U_0}^{\mathrm{W}} A' + \mathcal{V}(A')$. The multiplier system has the fixed-point form

$$\begin{aligned} A' &= \widetilde{G}_0 \mathsf{N} + H_0(b(B) - \mathsf{C}(A')), \\ \mathsf{N} &= -F'(0) - \mathcal{V}(A') + \mathsf{C}'(A')^* \omega, \\ \omega &= H_0^* \big[F'(A') + D R_{\mathrm{comb}} D^* A' - \mathsf{C}'(A')^* \omega\big]. \end{aligned}$$

The current $F'(0)$ and the datum are the two inhomogeneous terms. The current is not discarded when the trial field fails to be critical.

Use

$$\|A'\|_X = \sup_y \max\{\xi_y |A'|, \xi_y^2 |\nabla_{U_0} A'|\}, \qquad |f|_{(-3)} = \sup_y \xi_y^3 |f|.$$

Summation by parts is needed in the multiplier component: $H_0^* \mathrm{d}^* \mathrm{d} A' = (\mathrm{d} H_0)^* \mathrm{d} A'$. It leaves first differences, which are controlled by $\|A'\|_X$; directly differentiating A' twice would lose the required uniformity in the mesh.

The Lipschitz matrix has entries $C(\alpha_0 + r)$ on a ball of radius r, except for one bounded entry. Weight the multiplier norm by a fixed Θ^{-1}. First choose Θ to reduce that bounded entry; then choose the radii so that the small entries multiplied by Θ remain small. This gives a contraction with constant at most one half. On a sequence one may eliminate the multiplier: $I + H_0^* \mathsf{C}'(A')^*$ is invertible, so $\omega = w(A')$. Its Lipschitz constant need not be small, but it reenters the first equation multiplied by the small derivative $\mathsf{C}'(A')^*$. The contraction can then be run in the potential norm alone.

Two balls serve different purposes. A fixed small ball gives uniqueness. A ball of radius $C(|F'(0)|_{(-3)} + \sup |B|)$ gives the linear bound. Uniform convergence of the holomorphic iterates gives a holomorphic solution. The same estimates bound the multiplier. The uniform comparison of the actual straightened rows with the comb rows is needed here; identifying their gauge projections would not give it.

Recovering the gauge condition. For a real solution, differentiate gauge transformations that preserve the nonlinear constraint. In their coordinate directions X_λ, action invariance gives $\langle F'(A'), X_\lambda \rangle = 0$, and the multiplier term vanishes because $K_{A'} X_\lambda = 0$. The remaining equation is

$$\langle R_{\mathrm{comb}} D^* A', R_{\mathrm{comb}} D^* X_\lambda \rangle = 0.$$

The local gauge construction makes the second factors span the range of the projection. Therefore $R_{\mathrm{comb}} D^* A' = 0$. The equation on the slice is then the constrained Euler–Lagrange equation. The transversality used in this step concerns the actual nonlinear average with its site factors; infinitesimal gauge covariance at zero alone is not its full statement.

The later response is constructed with the true straightened rows K_0 and their Landau projection R. It parametrizes the same orbit. In its multiplier equation the reference inverse is G_1, since

the linearization at a critical pair contains the multiplier paired with the second derivative of the constraint. This explains why the two corrected Hessians were retained separately above.

Representatives in the slice and uniqueness of the orbit. Uniqueness of a fixed point is useful only after every regular configuration in the fibre can be placed in its ball. On one lattice first align the contour variables of the preceding averages. The rigidity estimate for one step says that matching contours, average error α_1 and plaquette size ϖ imply bond error at most $\alpha_1 + C\varpi$. The coefficient of α_1 is one. Descending through the averages therefore adds the curvature errors rather than multiplying the datum error at every level.

A change to the gauge slice then restores the normalization at the nodes. It is obtained by a finite-dimensional fixed-point argument in the local gauge variables. Its output is an estimate, not a holomorphic choice of gauge. The analytic potential will instead be selected by the unique multiplier solution.

For a sequence, a single preliminary gauge equal to one at the tops aligns all contour variables at once. One uses rigidity with a defect: the mismatch of the contour variables at both endpoints is added to the bond estimate. Fix a bond and let j be the last level containing both its endpoints. The increments below that level have sizes $C\alpha_0 L^{2(n-j)}$ and sum geometrically. A seam incurs the factor $2L-1$ only once, because the gauge is frozen in the outer block. Thus

$$|\mathcal{M}^n(U^{\hat{g}})(b)\mathcal{M}^n U_0(b)^{-1} - 1| \le (2L-1)\alpha_1 + C\alpha_0$$

for the retained bonds. This estimate passes to the truncations of the determining sequence.

One next ascends through those truncations. Rescaling can lose L^2, but the estimate in the slice resets the potential:

$$\|A'\|_X + |\mathrm{d}^*\mathrm{d}A'|_{(-3)} \le \mathsf{B}_1\big(\alpha_0 + \sup|\mathcal{Q}_k^\sharp(A')|\big).$$

The constant B_1 is determined by the inverse norm, independently of the size before this reset. It suffices that the rescaled potential remain within the fixed small ball. This is the first mechanism preventing a loss proportional to the number of levels.

Let now U_a, U_b be two regular critical configurations with the same datum, and take U_a as reference. The preceding construction puts U_b in the slice as $e^{i\eta_k A'}U_a$, with zero reduced datum and potential in the uniqueness ball. Criticality supplies a multiplier of size $C\varepsilon_{\mathrm{reg}}$. The pairs (A', ω_b) and $(0, \omega_a)$ solve the same zero-datum system; hence they agree. Thus $A' = 0$, and the compensating node factor is one. This gives one orbit under $\mathcal{G}^{\mathrm{res}}(\mathbf{B})$.

The radius improvement. The trial lift and change of level may enlarge the regular radius by L^3. One cannot let this loss pass to the next level unchanged. The local estimate for a critical field improves its radius independently of the construction that produced it.

Choose a cube of side σ, trivialize there, cut off the reference field to one, and align its slice with a comb gauge of the datum. Splitting the equation into near and far contributions gives an estimate of the following form:

$$r(x_0) \le C v_R(x_0) + \big(C\sigma^2 e^{-c\delta_*\sigma} + C\sigma^4 r_R(x_0)\big) r_R(x_0).$$

Here r is the local radius of the critical field, v is the size of the datum tests, and the suffix R is a maximum over a weighted ball of radius $R \asymp L\sigma$. The first error is the far part of the inverse, including the length of the comb; the second is the local nonlinear error. The coefficient in front of v_R is independent of σ. This is why one can first choose σ large and then the regular radius small.

The remaining estimate in this local calculation is the uniform bound for the cutoff and nonlinear errors that gives the two displayed coefficients. It uses the local inverse at the trivial field and the actual averaging rows. Once it is verified, choose σ so that the exponential term is small, and then take $\varepsilon_{\mathrm{reg}} \ll \sigma^{-4}$. A maximum argument gives the global improvement

$$\varepsilon_{\mathrm{reg}} \longmapsto \max\{C\varepsilon, \varepsilon_{\mathrm{reg}}/2\}.$$

The local form uses a weight varying at rate $\delta \asymp (L\sigma)^{-1}$. The variation on the ball of radius R is then bounded, and the same absorption gives

$$r(x) \le C \max_{\tau} e^{-\delta d_{\mathbf{B}}(x,\tau)} |\partial \mathbf{V}(\tau) - 1|.$$

It concerns an arbitrary regular critical field and does not presuppose uniqueness.

Repeated halving restores $C\varepsilon$ after every trial step, closing $C\varepsilon \to CL^3\varepsilon \to C'L^3\varepsilon \to C\varepsilon$. There is also a separate reset when the datum is descended near a seam. A section of one step is exact away from the seam; near it the section loses a fixed factor $C_{\downarrow}$ in a collar. The gap keeps that collar away from the errors of preceding sections. Using a small radius in the collars and radius $C_{\downarrow}\varepsilon$ elsewhere therefore descends with the same loss at each stage, rather than its kth power. The representative estimate, the radius improvement, and this collar reset address different accumulations.

Decay of the response. At a critical U_0, solve once more with the actual rows K_0 and R. The multiplier linearization has the additional term

$$W_0\delta = \frac{d}{dt} K^*_{t\delta}\omega_0\Big|_{t=0},$$

which is included in G_1^{-1}. Because the constraint was not replaced by a global change of variables, the remaining nonlinear terms stay local. All nonlocality is in the already estimated inverse operators.

The fixed-point derivative is contractive in every norm with weight $e^{\theta d_{\mathbf{B}}(\cdot,y_0)}$ for θ below a fixed fraction of the inverse decay rate. If $\mathbf{A}(b)$ is its solution with free row datum b, differentiation gives

$$(I - \partial\Psi)(\mathsf{h}, \dot\omega) = (\mathsf{H}_1 \mathsf{m}^{-1} e_c, 0).$$

Invert $I - \partial\Psi$ by its convergent Neumann series in that same weighted norm. The source column already decays, and the norm of the inverse is at most two. Consequently

$$\xi_y|\mathsf{h}| + \xi_y^2|\nabla \mathsf{h}| \le Ce^{-c\delta_{\mathrm{G}} d_{\mathbf{B}}(y,y_c)}.$$

Finally compose with $b(B)$. Each input bond changes only $O(1)$ rows within $O(L)$ local spacings, so the chain rule retains the decay. The differentiated multiplier equation also gives the current and appropriate second-order members. They are obtained from that equation, not by differentiating a merely pointwise kernel bound twice.

At the trivial datum on one lattice the linear response is the periodization of its translation-invariant kernel on $\mathbb{Z}^4$. Summing its exponential tail gives error $Ce^{-c\delta_{\mathrm{G}}\ell_k/2}$ for lifts within half the torus. This is the response comparison later used when a local computation is placed in a different torus.

2.5. **Complex backgrounds and the domains of the activities.** The complex domain must survive taking averages, forming a local background, and inserting a fluctuation into an older activity. An estimate only for a real critical field is insufficient for these operations. The two radii serve different purposes: α_0 bounds curvature and current, while α_1 bounds the complex potential. In the application $\alpha_1 \gg M\alpha_0$, but α_1^2 remains small compared with α_0.

Continuation with a prescribed representative. Fix real regular datum $\mathbf{V}_0$ and the representative U_0 pinned axially outside its nodes. For sufficiently small r_B, the multiplier contraction defines a holomorphic map $\Phi_{\mathbf{V}_0}(B)$ on $\sup|B| < r_B$ agreeing at real B with the representative of the orbit over $e^{iB}\mathbf{V}_0$. There is a holomorphic Landau pair $(\mathbf{H}(B), u_B)$ with

$$\Phi_{\mathbf{V}_0}(B) = (e^{i\eta_k \mathbf{H}(B)} U_0)^{u_B}, \qquad RD^*_{U_0}\mathbf{H}(B) = 0,$$

and with $u_B v_j[\mathbf{H}(B)] = 1$ at the tops of level j. Its values at zero are $(0, 1)$. The fixed-point estimate gives $\|\mathbf{H}(B)\|_X \le C \sup|B|$. At real B the pair is real. The comb slice has its own holomorphic pair, with the same potential estimate and an all-site bound on its compensating gauge. At the trivial reference the two slices agree. Away from it, they are two gauge-related potentials, not equal maps.

The node condition is part of this statement. A Landau gauge transformation need not be one at the nodes; its effect on the plain averages is compensated by the site factor. Removing that condition would change the datum being prescribed.

Why the complex radius enters quadratically. Suppose $\mathbf{U} = e^{i\eta A'}U$, with U real, and

$$|\partial U - 1|, |\partial \mathbf{U} - 1| < \alpha_0'\eta^2, \qquad |A'| < \alpha_1'.$$

The plaquette expansion gives

$$\partial(e^{i\eta A'}U) = \big[I + i\eta^2 \mathrm{d}_U A' + O(\eta^2|A'|^2 + \eta|A'|\,|\partial U - 1|)\big]\partial U.$$

Using both plaquette bounds, rather than the size of A' alone, yields

$$|\mathrm{d}_U A'| \le C(\alpha_0' + \alpha_1'^2).$$

The real adjoint transports commute with taking the real and imaginary parts. Thus the real fields $e^{i\eta\,\mathrm{Re}\,A'}U$ and $e^{i\eta\,\mathrm{Im}\,A'}U$ have curvature bounded by $C(\alpha_0' + \alpha_1'^2)$ as well. A bound merely linear in α_1' would miss this gain.

For a holomorphic component ϕ bounded by M_ϕ on a ball of radius r, let $w = u + iv$ with u, v real and $\|w\| \le r/2$. Its first Taylor term is complex linear, so

$$\phi(w) = \phi(u) + i\phi(v) - i\phi(0) + O(M_\phi\|w\|^2/r^2).$$

Apply this to a rescaled plaquette or current of a tower background. The three real values are controlled by the real regularity theorem. The crude complex supremum occurs only in the quadratic remainder. Consequently the complex background curvature and current have the bound $K^\sharp(\alpha_0' + \alpha_1'^2)$ in their respective units.

The quadratic form of this error is preserved by a change of level. At level $n \le p$, the two input sizes become $L^{-2(p-n)}\alpha_0'$ and $L^{-(p-n)}\alpha_1'$. The square of the latter has the same factor as the former. A bound linear in α_1' would introduce L^{p-n} and could not be iterated through a tower. The condition $K^\sharp(\alpha_0' + \alpha_1'^2) \le \alpha_0$ also leaves room for the larger potential radius.

Towers and hereditary domains. The activities are evaluated on a triple $(U, A', \mathbf{J})$, where U is real and $\mathbf{J}$ is independent. Its pair is $(e^{i\eta_j A'}U, \mathbf{J})$. On a domain X at level j, the simple conditions are the two plaquette bounds, the bounds $|A'|, |\nabla_U A'| < \alpha_1$ and $|\mathbf{J}| < \alpha_0$, and real local gauges on the cubes of side $O(LM)$ with potential and gradient bounded by $CM\alpha_0$.

The domain must also be stable under localization to every smaller subregion. Given $Z \subset X$ at level $p \le j$, form its tower by successive erosion. One tower uses M_1-cubes and an outer gap exceeding ρ_{geom}; the other uses the larger localization cubes and a gap measured in the inner scale. Truncate either tower at any $n \le p$. Its background is continued from the real datum to the complex averages of $e^{i\eta_j A'}U$.

The core condition requires all these backgrounds to exist, and, after two boundary layers have been removed, to satisfy

$$\begin{aligned} |\partial U_n(Z;\mathbf{U}) - 1| &< \alpha_0 L^{-(j-p)}\eta_p^2, \\ |J(U_n(Z;\mathbf{U}))| &< \alpha_0 L^{-(j-p)}L^{2(n-p)}. \end{aligned}$$

The current in the second line is measured in level-n units. The same conditions are imposed on the real field. The quantifiers over Z, p, n make the core hereditary. They are needed when a domain is cut into windows and then evaluated at an earlier level; a condition on the background of X alone would not suffice.

At this stage the continued tower is defined through the real reference U, hence through the triple. The activity construction must give the same value at compatible splittings of the same pair. The core itself is not saturated under arbitrary complex gauge transformations. When gauge covariance is used, both representations must remain in the specified domain, with the required condition-number bound for the gauge.

With the uniform background and response bounds on their common complex domains, the quadratic complex estimate gives the core conditions from the simple conditions at smaller radii. The application retains fixed room in the radii and in the two removed boundary layers. The common-domain verification is still needed for the joined and truncated towers used in this passage; the quadratic Taylor estimate by itself does not prove it. The trivial triple belongs to every core.

To apply the inverse estimates to these domains, the gauge clause must be converted to the clause of a regular pair. A gauge on a cube of side $O(LM)$ has size $O(LM)\alpha_0$, so the regular-pair radius is $O(LM)\alpha_0$, not merely α_0. On a label sequence whose regions are unions of LM-cubes, the layered conversion costs $O(M)\max_n \alpha_{0,n}$. A cube meeting layer n uses that layer's gauge; a smaller cube lying in layer $n+1$ uses the containing cube and gauge at level $n+1$, restricted and rescaled. Thus no critical-point estimate is needed to place a pair of the activity domain in the local inverse class. The field radii are chosen after M so the converted radii lie below its ceiling.

Substitution from the next fluctuation integral. The outer substitution is $\mathbf{U} \mapsto e^{i\eta_j \mathbf{A}}\mathbf{U}$ with $|\mathbf{A}|, |\nabla_{\mathbf{U}}\mathbf{A}| < c\alpha_0$. A core at half the radii is mapped into the full core when $\alpha_0 \le \alpha_1$ and $\alpha_1^2 \le C\alpha_0$. Each localized tower changes by its response to the perturbed averages. The response bound is uniform through the tower; taking c smaller than the reciprocal of its constant leaves the unused half of the curvature and current radii. The first-difference bounds retain the potential radius. This is a statement about every tower of the core, not just its outer field.

The inner substitution is more delicate because it evaluates an old activity inside a new fluctuation integral. Work on a box $\square_0$ containing $O(1)$ localization cubes. Let $\mathbf{H}_{k+1}$ be the Landau potential of its background, and let e^{iB} be the averages of

$$e^{i\eta_k \mathbf{A}} e^{i\eta_{k+1}\mathbf{H}_{k+1}}, \qquad |\mathbf{A}| < c\alpha_{0,k}.$$

The fluctuation response with a cutoff is the relevant choice of $\mathbf{A}$. Take the background over the trivial reference at datum $tB + B'$, $0 \le t \le 1$, where

$$\sup |B'| < c \min\{\alpha_{0,j-1}, \alpha_{0,j}\}.$$

The input field has the simple conditions and the tower conditions on a slightly larger box, with radii larger by a fixed factor. Once the common-domain background and response bounds hold uniformly through this substitution, it returns, well inside the box, a triple in the core of every level $1 \le j \le k$. The response and agreement margins must be verified also for the joined towers; no unestimated remainder may be introduced at a join.

Both the level and its radius must be kept in this assertion. If $\nu = k + 1 - j$, the flow gives

$$\alpha_{\iota,k+1} \le C(1+\nu)^{1/2}\alpha_{\iota,j}, \qquad \iota = 0, 1.$$

The change of units supplies powers of $L^{-\nu}$, which dominate this polynomial loss. Discarding those powers, or replacing the two radii by one fixed radius, would not justify the evaluation of all the earlier terms. The resulting Landau triple is

$$(1, \mathbf{H}, J(e^{i\eta_j \mathbf{H}})), \qquad \mathbf{H} = \mathcal{H}(tB + B'),$$

and the actual background pair is in its complex gauge orbit. Evaluations are transferred only on the portions of that orbit allowed by the domain.

In a local box the complex gauge is controlled by its condition number,

$$|v|\,|v^{-1}| \le e^{CM\alpha_1}.$$

There is no bound $|v - 1| \le CM\alpha_1$ for an arbitrary real reference gauge: a pure gauge can already have large real values. The condition-number bound is the one used to continue the averages and the activities.

Restriction and comparison of backgrounds. The finer constraints fix the coarser ones. If a configuration is varied inside a smaller determining region while its finer averages are kept, the coarser datum outside that region does not change. The difference of the two inner-plaquette

actions is exactly the difference of their actions in the smaller region. Restriction therefore preserves criticality. The restricted datum is $CL^2\varepsilon$-regular, and uniqueness of the regular orbit gives the restriction identity. On one lattice this is

$$U_j(\mathcal{M}^j U_k(V)) = U_k(V) \quad \text{modulo gauge}, \qquad j \le k.$$

For locality, a collar boundary is required to stay within one layer. Align gauges there and compare the finer averages with the lift of the datum at the collar level. The defect-of-lifts reconstruction has coefficient one at every descent; the remaining plaquette errors sum as

$$\sum_{q<j} 2\varepsilon L^{2(q-j)} \le C\varepsilon.$$

The aligned mismatch is a real datum B_1 of size $C\varepsilon$ supported in the collar. Its response decays, so two backgrounds with matching determining data on the collared region W differ there, modulo gauge, by at most

$$C_\mu \varepsilon e^{-c\delta_{\mathrm{G}} d_{\mathbf{B}}(y,W^c)}.$$

The constant depends on the fixed collar width μ; the decay starts at the inner edge of that collar. One must not replace it by a constant independent of a variable collar width. Nor can a gap of one M-cube in a prepared nest be read as a gap of $M\check{R}_j$: that gap supplies fixed decay in M only.

2.6. **Complex decoupling and the localized response.** Localization of the next fluctuation integral uses a response whose propagators have been decoupled in space. This is a separate construction from taking the actual background of a field with altered averages. The decoupled operators retain their bounds and locality, but generally lose the algebraic identities of the original inverse. The fixed-point equation has to be arranged before that loss occurs.

The operator expansion as a datum. For each operator fix its tree-walk terms and weights once. The first variable of a kernel term lies near its first cell; the second lies near some cell of its tree. The local bound is its original scale-weighted estimate times

$$\mathrm{wt}(\omega) e^{-\delta d_{\mathbf{B}}(\omega;y,y')}.$$

The weights of all trees with a fixed first cell sum to a constant. A term is defined on the pair restricted to its carrier, with the same estimate on every intermediate region between the carrier and Ω_0. A separate crude holomorphic bound is used for convergence only. That bound may depend on the level; the local weights and rates do not.

Partition into cubes of side ℓ_{P} in local units, where ℓ_{P} is a multiple of M_1. Let $\mathsf{P}(\omega)$ be the cubes meeting the uncut hull of a tree. For $|s(\square)| \le e^{\bar\kappa}$ set

$$\mathsf{O}(s) = \sum_\omega s^\omega \mathsf{O}_\omega, \qquad s^\omega = \prod_{\square \in \mathsf{P}(\omega)} s(\square).$$

Each parameter occurs with degree one. This convention marks a cube once even when a walk returns to it. It is what gives the mixed Cauchy estimate used to localize an activity in a connected set of cubes.

A tree of weighted length t meets at most $16 + Ct/\ell_{\mathsf{P}}$ such cubes. The constant $16 = 2^4$ allows its first cell to lie at a corner of the partition. Thus

$$|s^\omega| e^{-\delta d_{\mathbf{B}}(\omega;y,y')} \le e^{16\bar\kappa} e^{-(\delta - C\bar\kappa/\ell_{\mathsf{P}}) d_{\mathbf{B}}(\omega;y,y')}.$$

Choose ℓ_{P} after $\bar\kappa$ so that the remaining rate is a fixed positive fraction of δ. The exponential in the tree length, rather than the reserved walk factor, pays for the parameters. The sum and all its restricted partial sums therefore have the same local estimates, with multiplier $Ce^{16\bar\kappa}$. The factor at the first corner does not disappear by making ℓ_{P} larger.

If parameters vary only in a subfamily σ_0, the same calculation leaves an additional factor

$$e^{-\bar\kappa \#(\mathsf{P}(\omega) \setminus \sigma_0)}.$$

In particular forcing a prescribed set of cubes into the hull gives exponential smallness in their number. The hull is connected through walls. A nonzero term therefore belongs to a single wall-connected component of $\{s \neq 0\}$. The operator splits exactly over these components. This assertion uses the uncut hull, not connectedness of its carrier inside a possibly rough Ω_0.

The same construction on a region W keeps only walks with $\mathsf{P}(\omega) \subset W$. Their sums use the pair only on $W \cap \Omega_0$. Agreement of the data for traces of regions on the hull gives agreement of these partial sums in two different settings. This is the form of consistency needed when a shift is first formed on a window and then transported into a larger box.

Which operator identities are lost. At $s = 1$ and on shell,

$$Q\mathsf{H} = I, \qquad RD^*\mathsf{H} = 0, \qquad \mathsf{G}^{-1}\mathsf{H} = Q^*(Q\mathsf{G}Q^*)^{-1}.$$

They are identities between full operators. They are not identities between their walk terms. If all parameters near one row are zero, the decoupled response has no term there, so $Q\mathsf{H}(s)$ is zero there and cannot be the identity. Even at $s = 1$ the slice identity has the off-shell defect described above.

The force in the localized response equation is therefore formed as follows. First rewrite the nonquadratic gradient at the genuine critical background, using the exact inverse identities to remove the linear expression $\Delta_\pi \mathsf{H}$, where $\Delta_\pi = \Delta^{\mathrm{W}}_{\mathbf{U}} - \Delta'_\pi[\mathbf{J}]$. Only then decouple the remaining local expression. Denote that expression by W_s. For general s and an independent-current pair this is its definition; it is an actual action gradient only at $s = 1$ on shell. The localized force calculation must establish its quadratic estimate on the whole complex ball. Reusing the original gradient and the false identity $Q\mathsf{H}(s) = I$ would give a different equation.

Use admissible weights ϖ satisfying

$$\varpi(y) \leq e^{\theta d_{\mathbf{B}}(y,y')}\varpi(y')$$

for θ below the retained rate, and define

$$\|A\|_\varpi = \sup_y \varpi(y) \max\{\xi_y|A|, \xi_y^2|\nabla_{\mathbf{U}}A|\}, \qquad |f|^\sharp_\varpi = \sup_y \varpi(y)\xi_y^3|f|.$$

The decoupled inverse maps the force norm into the potential norm with norm at most $\widehat{b} \leq Ce^{16\bar{\kappa}}$. The required local force estimate is

$$|\mathsf{W}_s(A_1) - \mathsf{W}_s(A_2)|^\sharp_\varpi \leq C\widehat{b}^2 r\|A_1 - A_2\|_\varpi \quad (\|A_i\| \leq r < c/\widehat{b}).$$

The term containing the independent current receives its factor r from the Schwarz estimate on the full complex ball. This is why the estimate is needed before restricting to real fields.

The response with the current correction included. Take walk data for the two compound operators $\mathsf{G}_1 DRD^*\mathsf{G}_1$ and $D(1-R)D^*$. Define the expansion of

$$\mathfrak{G} = \mathsf{G}_1 - \mathsf{G}_1 DRD^*\mathsf{G}_1 - \mathsf{H}_1 Q\mathsf{G}_1$$

from these data. The expression $\mathfrak{G}(s)$ is the decoupling of this full expansion. It must not be replaced by multiplying independently decoupled factors in the displayed algebraic formula.

Write the nonlinear average as $\mathcal{Q} = \mathsf{m}Q + \mathcal{C}$. The localized system has two unknown corrections $X, \mathbf{A}$:

$$\begin{aligned} A_\bullet &= \mathsf{H}_1(s)\mathsf{m}^{-1}B' + \mathbf{A}, \\ \mathbf{A} &= -\mathfrak{G}(s)\mathsf{W}_s(A_\bullet), \\ X &= \mathcal{C}(A_\bullet - \mathsf{H}(s)\mathsf{m}^{-1}X), \\ \mathbf{H} &= A_\bullet - \mathsf{H}(s)\mathsf{m}^{-1}X. \end{aligned}$$

For fixed $A_\bullet$, the equation for X is solved in a small ball using the quadratic bound on $\mathcal{C}$. The other equation is contractive provided $\sup|B'|$ is small compared with $\widehat{b}^{-4}$. Indeed its Lipschitz constant has four operator factors in size:

$$\widehat{b} \cdot \widehat{b}^2 \cdot \widehat{b} \sup|B'|.$$

They come from the inverse, the reduced-force bound, and the size of the linear response. Choosing the datum radius after these constants gives one small holomorphic solution, bounded by $C\widehat{b}\sup|B'|$.

The same contraction works in every admissible weight. Setting $\varpi = e^{\theta d_{\mathbf{B}}(\cdot,y_c)}$ therefore gives decay of the derivative with respect to $B'(c)$. Splitting, covariance and transport are proved by taking the separately constructed solutions on each component as a candidate for the full equation, and using uniqueness in the same ball. They are not consequences of a total norm bound alone.

For arbitrary complex parameters the controlled second-order object is a reduced current, not the actual current of a critical point. It is built from the multiplier, using the pairing

$$\langle \mathrm{d}^*\mathrm{d}A, \mathsf{H}e_c\rangle = \langle \mathrm{d}A, \mathrm{d}\mathsf{H}e_c\rangle.$$

Only first differences occur. This avoids claiming that first-order bounds for arbitrary operator data imply a uniform bound on $\Delta_{\mathbf{U}}\mathbf{H}$.

At $s = 1$ over a real critical reference, the constructed solution satisfies the constraint and Landau condition, and its Euler–Lagrange defect lies initially in the fine gauge directions. Testing it against the constraint-preserving gauge directions removes that defect on a real neighbourhood of zero. The identity theorem then extends the three equations over the whole connected complex datum polydisc:

$$\mathcal{Q}(\mathbf{H}) = B', \qquad RD^*\mathbf{H} = 0, \qquad F'(\mathbf{H}) = K^*_{\mathbf{H}}\lambda.$$

Only this undeformed on-shell identification uses the actual critical orbit. A small real neighbourhood used for the gauge test need not have a radius uniform in the volume, because no estimate is taken from that radius. All quantitative bounds were proved on the fixed complex ball first.

A reduced system and its exact scope. There is a second useful formulation, with unknowns the response T and its multiplier. Use angle rows $\widehat{Q} = \mathsf{m}Q$ with their unweighted row pairing, and set

$$H = \mathsf{H}\mathsf{m}^{-1}, \quad S = (\widehat{Q}\mathsf{G}\widehat{Q}^*)^{-1}, \quad H_0 = \mathsf{G}_0\widehat{Q}^*(\widehat{Q}\mathsf{G}_0\widehat{Q}^*)^{-1}, \quad \widetilde{G} = \mathsf{G}_0 - H_0\widehat{Q}\mathsf{G}_0.$$

Let a_0 denote the diagonal row mass in this convention, and put $\mathcal{N}(T) = F'(T) - F'(0) - \Delta^{\mathrm{W}}_{\mathbf{U}}T$. The reduced equations are

$$\begin{aligned} T &= H_0(s)(B' - \mathcal{C}(T)) - \widetilde{G}(s)[\mathcal{N}(T) - \mathcal{C}'^*_T\lambda],\\ \lambda &= H(s)^*\mathbf{J} + (S(s) - \mathsf{a}_0)(B' - \mathcal{C}(T)) + H(s)^*[\mathcal{N}(T) - \mathcal{C}'^*_T\lambda]. \end{aligned}$$

Each of the four operators has its own fixed walk data. Their exact identities are again used only at $s = 1$.

Center the multiplier at $H(s)^*\mathbf{J}$. Most terms are quadratic on a ball of radius $2\widehat{b}r$ and give the condition $\widehat{b}^2 r \ll 1$. There are also linear terms containing the current, notably the derivative of $\mathcal{C}'^*_T H(s)^*\mathbf{J}$. Their norm is $C\widehat{b}^2\gamma_0$. Thus this formulation requires a second current ceiling chosen after $\bar{\kappa}$. With that ceiling it is a contraction in the weighted norm of the pair, on a datum ball of size $c\widehat{b}^{-2}$.

The response splits over the parameter components. The multiplier itself does not: outside them it contains the diagonal term $-\mathsf{a}_0B'$. For the same reason the force is split only after this term is retained. Further seminorms pass from $H_0(s)$ and $\widetilde{G}(s)$ to the solution; they need not be bounds for $H(s)$.

At $s = 1$ over a critical real pair, the two response systems agree on their common small ball by uniqueness of the constrained Euler–Lagrange solution. Their separate force bounds then give both $\mathrm{d}^*\mathrm{d}\mathbf{H}$ and $DD^*\mathbf{H}$ at that value of the parameters. At a general off-shell pair or at $s \neq 1$ they are different functions. In particular no exact constraint is asserted for the first system at arbitrary parameters. At $s = 1$ the reduced system enforces $\mathcal{Q}(T) = B'$ even off shell, by $\widehat{Q}H_0 = I$ and $\widehat{Q}\widetilde{G} = 0$. The first system can instead retain the defect $\mathsf{m}Q\mathbf{A}$ there.

For fields given only on a region, the first system is formed by keeping the operator terms whose hulls lie in that region. The local reduced system is used on the strip sequences for which its four

partial sums have been constructed. It is not silently extended to all determining structures. This distinction is needed when the large-field operation localizes a shift in a boundary strip.

2.7. **The fluctuation integral and the local expansion.** Write $W = V_{k+1}$ and let $V_0 = \mathcal{M}^k U_{k+1}(W)$ be the level-k background. In this paragraph all bond potentials are dimensionless level-k variables. For every coarse bond c, choose the middle bond $b_0(c)$ among the L fine bonds on c. The oddness of L makes this choice commute with the isometries of the coarse lattice. Let $\mathfrak{X}$ be the set of the remaining fine bonds. A field on $\mathfrak{X}$ will always be extended by zero at the pivots before an operator on all bonds is applied to it.

The relative constraint has the expansion

$$q(B') := \frac{1}{i}\log\left[\mathcal{M}(e^{iB'}V_0)W^{-1}\right] = LQ_{V_0}B' + q_{\geq 2}(B').$$

The coefficient of the pivot variable in its linear part is an invertible colour matrix $L\mathsf{m}_c$. This matrix is a small perturbation of the corresponding coefficient at the identity. It is uniformly invertible for the real regular backgrounds under consideration. Denote by h the right inverse supported on the pivots, so that $LQ_{V_0}\mathsf{h} = 1$. If E is extension by zero from $\mathfrak{X}$, the map

$$\mathsf{C} = (1 - \mathsf{h}LQ_{V_0})E$$

identifies the free variables with the kernel of the linearised constraint. Its restriction to $\mathfrak{X}$ is the identity. In particular the use of free variables does not introduce a new constraint on their values.

For a real scale $g = \mathsf{t}^{-1/2}$, solve

$$D = q_{\geq 2}(g\mathsf{C}B - \mathsf{h}D), \qquad B' = g\mathsf{C}B - \mathsf{h}D. \tag{22}$$

The derivative of $q_{\geq 2}$ vanishes at zero. On a sufficiently small fixed ball the first map is therefore a contraction in D. Its solution is holomorphic in the background and gB, and $|D| \leq C_L g^2 |B|^2$. Applying q to the second formula gives $-D + q_{\geq 2}(B') = 0$. Thus $V_W(B) := e^{iB'}V_0$ lies on the exact fibre, rather than on its tangent approximation. All these assertions use the uniform relative-average estimates established before the fluctuation step. They do not follow from a formal expansion of the constraint.

The fluctuation test is imposed only on $\mathfrak{X}$:

$$|B(b)| < \mathfrak{b}(\mathsf{t}), \qquad \mathfrak{b}(\mathsf{t}) = \sqrt{\mathsf{t}}\,\delta(\mathsf{t}).$$

The eliminated variables satisfy $|B'| \leq C_L\delta(\mathsf{t})$, with C_L at most a fixed multiple of L^4. This weaker bound is enough. The block contour variables depend on bonds inside their block, and the same comparison with V_0 bounds their deviation by $C_L\delta(\mathsf{t})$. Choosing $\varepsilon(\mathsf{t})/\delta(\mathsf{t}) \geq C_L$ makes every gauge-fixing indicator equal to one on this fibre chart. The smooth gauge-fixing weight remains in the integral. Its indicators are not analytically continued.

There are two curved constraints in the quadratic action. The background $U_k(V)$ is critical under the k-fold average, and V is restricted by the next average. For a constrained critical point with $DF = \lambda Dq$, differentiating $q(V(z)) = q(V(0))$ twice gives the restricted Hessian

$$D^2F[V'(0), V'(0)] - \lambda D^2 q[V'(0), V'(0)].$$

The second term cannot be omitted. For the first constraint its multiplier is represented by the current of the fine background; for the second it pairs with the quadratic remainder of the one-step average. This explains both occurrences of the current in $\Delta^{(k)}$. At $k = 0$ the background map is the identity, so the first of these corrections is absent.

Criticality removes the first derivative on the fibre. The gauge-invariant action Hessian vanishes on its residual gauge directions, including its mixed pairing with any tangent direction. Split a tangent vector into a vector in the kernel of the linearised gauge fixing and an orbit vector generated

by λ. The action acts on the first vector and the quadratic gauge fixing acts on λ. With the normalisation of the gauge-fixing action this gives

$$\langle B, \Delta^{(k)} B\rangle = \Delta_k(B'', B'') + \sum_x |\lambda(x)|^2.$$

Positivity on the first summand is the constrained propagator estimate. The second summand supplies the missing gauge mass. Thus the free-bond form $\mathsf{K} = \mathsf{C}^{\mathsf{T}}\Delta^{(k)}\mathsf{C}$ is positive at real data. This positivity concerns the full fluctuation form. Retaining selected local pieces in a large-field operation requires the separate comparison in section 3.

Write GF_k for the soft gauge-fixing action, whose weight at the reference inverse coupling t is $e^{-\mathsf{t}\mathrm{GF}_k}$. Taylor expansion on the exact chart now gives

$$\mathsf{t}\{A(U_k(V_W(B))) + \mathrm{GF}_k(V_W(B))\} = \mathsf{t}A(U_{k+1}) + \tfrac{1}{2}\langle B, \mathsf{K}B\rangle + \text{terms of order at least three in } gB.$$

The Jacobian is kept at the same precision as the action. Disintegrating the group delta function leaves $L^{-3}|\det \mathsf{m}_c|^{-1}$ at each coarse bond. The determinants $|\det \mathsf{m}_c|^{-1}$ are put in the Gaussian normaliser. The logarithm of the nonlinear Jacobian and the Haar density in the exponential chart are put in $\mathbb{P}^J$. The action and gauge-fixing terms of order above two are denoted $\mathbb{P}^A$ and $\mathbb{P}^G$. Consequently

$$\mathbb{P}^{(k)} = \mathbb{P}^A + \mathbb{P}^G + \mathbb{P}^J.$$

This division by origin will matter when the action weight is complex while the gauge fixing remains real.

The constants can be counted without an error per bare bond. Put $\mathsf{V}_j = |\mathbb{T}^{(j)}|$, and let σ_0 be the Haar density at the identity in the chosen exponential coordinates. There are $4(\mathsf{V}_k - \mathsf{V}_{k+1})$ free bonds and $\mathsf{V}_k - \mathsf{V}_{k+1}$ independent residual gauge sites. If

$$\mathfrak{z}(\mathsf{t}, \epsilon) = \int_{\mathrm{SU}(2)} 1_{\{|u-1|<\epsilon\}} e^{-\mathsf{t}(1-\frac{1}{2}\operatorname{tr} u)}\, du,$$

the field-independent factor is

$$\mathsf{c}_k(\mathsf{t}) = \mathfrak{z}(\mathsf{t}, \varepsilon(\mathsf{t}))^{-(\mathsf{V}_k - \mathsf{V}_{k+1})} (\sigma_0 \mathsf{t}^{-3/2})^{4(\mathsf{V}_k - \mathsf{V}_{k+1})} L^{-12\mathsf{V}_{k+1}}. \tag{23}$$

The three factors come respectively from gauge averaging, scaling the free Haar variables, and the coarse delta functions. At real data the determinants in this calculation have their positive values. Their holomorphic continuations are fixed from those values; no absolute-value operation is applied at a complex pair.

For $\mathfrak{b}(\mathsf{t}) \geq 1$, comparison with the quadratic action near $u = 1$ gives $\log \mathfrak{z} = -\frac{3}{2}\log \mathsf{t} + O(1)$. Hence

$$\log \mathsf{c}_k(\mathsf{t}) = -\tfrac{9}{2}(\mathsf{V}_k - \mathsf{V}_{k+1}) \log \mathsf{t} + O(\mathsf{V}_k).$$

For real W the full Gaussian normaliser is

$$Z^{(k)}(W) = (2\pi)^{\frac{3}{2}|\mathfrak{X}|} \det(\mathsf{K}(W))^{-1/2} \prod_c |\det \mathsf{m}_c(W)|^{-1}.$$

Its complex continuation is the holomorphic branch fixed by this real formula. It has no coupling dependence at the trivial background. The action-dependent part of one step is therefore separated from its numerical normalisation before the local expansion begins.

The small-field integral therefore has the form

$$\mathsf{c}_k e^{-x_k A(U_{k+1}) + \mathbf{E}_k(U_{k+1})} Z^{(k)}(U_{k+1}) \int d\mu_{C^{(k)}}(B)\, \chi^{(k)} e^{\mathbb{P}^{(k)}(B) + \mathbf{E}_k(U_k(V_k)) - \mathbf{E}_k(U_{k+1})}. \tag{24}$$

The Gaussian normaliser is $Z^{(k)}$; c_k contains the Jacobian, Haar, delta-function and gauge-fixing normalisations. On the free-bond variables the covariance is $C^{(k)} = (\mathsf{C}^{\mathsf{T}}\Delta^{(k)}\mathsf{C})^{-1}$, as determined by the preceding quadratic form. Here $\chi^{(k)}$ is the product of the fluctuation indicators on the free bonds. The factorisation defines $\mathbb{P}^{(k)}$ as the remaining fluctuation exponent after the displayed

quadratic form has been put into the Gaussian measure. The exponent $\mathbf{E}_k$ is the accumulated regular interaction with its constant and quadratic action parts subtracted; its precise form is (56) below.

The local terms are indexed by connected unions X of M-cubes and by a site $z \in X$. Let $d_j(X)$ be the length of a shortest tree meeting these cubes, divided by M. Their norm is

$$\|E^{(j)}\|_{j,\kappa} = \sup_{X,z} e^{\kappa d_j(X)} \sup_{\text{analytic domain}} |E^{(j)}(X,\cdot,z)|.$$

The number of domains through a fixed cube with $d_j(X) \le n$ is at most $e^{C(n+1)}$. Thus a sufficiently large κ makes $\sum_{X \supset \square} e^{-\kappa d_j(X)}$ finite uniformly in the lattice. This elementary count is used throughout the local expansion.

To localise the exponent, attach a parameter $s(\Delta)$ to each cube traversed by a walk. Allowing $|s(\Delta)| \le e^{\bar\kappa}$, with $\bar\kappa \ge CL\kappa$, leaves the background equations contractive when M is sufficiently large. Long walks pay for these parameters by their length, since meeting m cubes costs at least cmM in the exponent. The fixed cost of short walks is absorbed by the small fluctuation amplitude. Cauchy estimates then give decay for the resulting local potentials.

Two small quantities occur. If the fluctuation test is $|B| < \delta_k/g_k$, then the cubic interaction is bounded by $C\delta_k|B|^2$. It is small under the Gaussian weight. Old interactions are evaluated at a shifted background. Their smallness instead contains the ratio $\mathsf{s}_k = \delta_k/\alpha_{0,k}$ of that shift to its analyticity radius. The running radius satisfies $\alpha_{0,k} \asymp g_k(\log x_k)^q$, with q chosen large after the threshold exponents. Here $\mathfrak{p}_0$ is the exponent of the threshold $p_0(g)$ in section 3. In particular, $q > 3\mathfrak{p}_0 + C\mathfrak{r}$, where $\mathfrak{r}$ is the fixed exponent of the large-field cube size and C covers the geometric losses in localisation. We also take $q > 21\mathfrak{p}_0$ for the two-point comparison. The coupling ceiling is chosen after q. On running radii this ratio is only a negative power of $\log x_k$; it should not be replaced by a power of g_k. After coupling subtraction, an old term of age n has the summable weight $\omega_n \asymp n^{3/2}L^{-\alpha_{\mathrm{H}}n}$. Writing N_ω for the corresponding weighted sum of old norms, the local old-field potential is bounded by $C\mathsf{s}_k N_\omega e^{-c\kappa d_k(Y)}$. The construction gives $N_\omega \le CE_0$; it is $\mathsf{s}_k N_\omega$, rather than N_ω, that is small. Remainders left by the large-field operation have no coupling counterterm and receive no such decreasing age weight. Once their cubes fit in single blocks, they enter a separate part of N_ω, weighted by the square of the ratio of radii. Their summation uses their own size $g_j^{\kappa_0}$ and the lower flow bound. Younger remainders remain separate, as in section 3.

The identity for the old interaction is a difference along the background response, integrated from the unperturbed background to the fluctuating one. The potential and current are transported together. At the actual endpoints they satisfy the field equation, but the affine differentiations and decoupling interpolations between them need not. This is why the interactions are analytic functions of the independent pair $(\mathbf{U}, \mathbf{J})$. The pointwise coupling subtraction described next is performed before the decoupling; the surviving brackets are rooted at the original domain. Their sum recovers exactly the difference of the old interaction in (24).

For completeness, the subtraction and the local interpolation are performed in the following order. For a term of level j, write the response curve as a curve of pairs

$$p_j(A, \mathcal{J}) = (e^{i\eta_j A}U_{k+1}, L^{-3(k+1-j)}J(U_{k+1}) + \mathcal{J}),$$

where $\mathcal{J}$ is the independent current displacement. At the actual response the potential and current displacements are related by the critical equation. In applying the fundamental theorem of calculus they are differentiated as independent coordinates. A partition of unity in the displacement localises its directions to windows. The response of the whole lattice is next replaced by that of a box, with their difference written as a second integral of a derivative. Finally the walks in the resulting box response are decoupled. The two removed differences and all differentiated directions therefore require the off-shell pair domain.

An equivalent formula starts with the original domain as a root. If $h_Y(s;B)$ is its localised response and F is the activity with its pointed counterterm already removed, set

$$\Phi_Y[F](s,\nu) = F(p_j(\nu h_Y(s;B))),$$

where h_Y contains the potential and current components. For a finite set K of decoupling cubes, the interpolation bracket integrates $\prod_{\Delta\in K}\partial_{s(\Delta)}$ over the corresponding unit cube of parameters, with the retained cubes fixed to one. Repeated one-variable fundamental theorems of calculus recover the original function. A connected piece that does not meet its root vanishes in the rooted decomposition. Cauchy estimates in s pay for the new cubes; the Schwarz estimate in ν pays for the response amplitude. These are algebraic sum rules before taking absolute values. The analytic estimate to check is that the decoupled responses stay in their independent field/current domains on one common polydisc, uniformly in the old level. The algebraic identities alone do not give that domain control.

Here is the bracket identity used in that construction. For disjoint finite cube sets $\mathcal{K},\mathcal{K}'$, define

$$\Psi[\mathcal{K}\mid\mathcal{K}'] = \int_{[0,1]^{\mathcal{K}}}\prod_{\Delta\in\mathcal{K}}\partial_{s(\Delta)}\Psi(s+1_{\mathcal{K}'})\,ds. \tag{25}$$

If $\mathcal{K}''$ is disjoint from the base $\mathcal{K}'$, repeated fundamental theorems of calculus give the exact resummation

$$\Psi(1_{\mathcal{K}'\cup\mathcal{K}''}) = \sum_{\mathcal{K}\subset\mathcal{K}''}\Psi[\mathcal{K}\mid\mathcal{K}']. \tag{26}$$

The bracket is applied to the decoupled response function $\Phi_Y[F]$, rather than to an arbitrary function for which no locality has been established. At a zero decoupling parameter no walk passes through its cube. A connected differentiated component separated from the base therefore does not affect this function, and its bracket vanishes. This is the rooting assertion.

The complex parameter disc has radius $e^{\bar\kappa}$. A derivative in a newly added cube pays $Ce^{-\bar\kappa}$. The response multiplier ν is allowed up to the ratio of analytic radius to response size. Subtracting its value at zero and applying Schwarz gives the independent factor s_k. Both estimates apply to a single holomorphic function on a common polydisc. They do not follow by separately bounding the response at real values zero and one.

There are two exact ways to organise the old-field identity. In the window organisation, the main response curve is generated by the datum

$$B_j(t) = \mathcal{Q}_j\big(U_{k+1};\eta_{k-j}t\mathbf{H}_k\big), \qquad 0\le t\le 1.$$

It is on shell when the decoupling parameters are one. Its endpoints are the pair of U_{k+1} and, up to gauge, that of $U_k(V_k)$. Differentiate along it, resolve the direction with $\sum_\square\zeta_\square = 1$, and replace the whole-lattice response by a box response near the window. This produces two removed integrals. The differentiating directions and the first removed straight segment are affine in their parameter and can leave the shell. Decoupling is performed after these replacements.

In the rooted organisation the bracket (25) uses the domain of the term as base. Its counterterm is removed at each base point before interpolation. Both organisations recover exactly

$$\sum_Y \mathsf{W}_k^{\mathrm{old}}(Y;B) = (\mathbf{E}_k+\mathbf{R}'_k)(U_k(V_k)) - (\mathbf{E}_k+\mathbf{R}'_k)(U_{k+1}), \tag{27}$$

where $\mathbf{R}'_k$ denotes the aged remainder sum. The younger remainder sum is passed separately to the next remainder slot.

The rooted form uses a stronger piece of information than invariance of the total interaction: the pointed quadratic tensor must cancel at the centre of every window. Once that estimate is available, one base point of level j contributes at most

$$C\varrho_j^3\eta_{k-j}^{4+\alpha_{\mathrm{H}}}\|E^{(j)}\|_{j,\kappa}, \qquad \varrho_j\le C(k+2-j)^{1/2}.$$

A current-scale window has $C\eta_{k-j}^{-4}$ base points. The factor ϱ_j^3 is the cost of up to three Cauchy estimates on the older, possibly smaller radius. After cancelling the site count, the contribution is $C\mathsf{s}_k\omega_{k+1-j}\|E^{(j)}\|_{j,\kappa}$. An aged remainder lacks the quadratic counterterm, and its analogous contribution has ϱ_j^2 in place of ω_{k+1-j}. Thus the actual old-data load is

$$N_\omega = \sum_{j\le k} \omega_{k+1-j}\|E^{(j)}\|_{j,\kappa} + \sum_{j \text{ aged}} \varrho_j^2\|R^{(j)}\|_{j,\kappa}. \tag{28}$$

The first sum uses Hölder decay and the second uses the lower coupling drift. This distinction survives every later localisation of the old interaction.

The local gap and the conditional Gaussian integral. The Gaussian localisation uses the form K itself as well as its inverse. The propagator construction supplies a local analytic expansion

$$\mathsf{K}(\mathbf{p}) = \sum_\omega \mathsf{K}_\omega(\mathbf{p}), \qquad |\mathsf{K}_\omega| \le C(CM_1^{-1/2})^{|\omega|}e^{-c\,\mathrm{length}(\omega)}.$$

The terms are covariant under real gauge transformations and agree in two regions whenever their carriers and the required background data agree there. This is a property of the actual operator construction. A decay bound for its full kernel alone would not define local terms with this agreement property.

Let γ_0 be a lower bound for $\mathsf{K}(1,0)$, and first consider a real regular pair. On cubes of a fixed side ℓ_0, choose the real gauges contained in the analytic-domain conditions. In each such gauge $U = e^{i\eta A}$ has $|A| + |\nabla A| \le a_{\mathrm{w}}$. The local walk expansion can be compared with its value at the identity. Schwarz's lemma on a fixed analytic ball bounds this difference by Ca_{w}. For a complex pair $(e^{i\eta A'}U, \mathbf{J})$, a second use of Schwarz on

$$z \longmapsto (e^{iz\eta A'}U, z\mathbf{J})$$

adds $C(b_0 + b_1)$ to the bound. Both comparisons use the independent current coordinate. They do not replace it by the current of a general complex perturbation.

Choose real cutoffs h_i with $\sum_i h_i^2 = 1$ and $|\partial h_i| \le C\ell_0^{-1}$. The exact localisation identity is

$$\mathsf{K} = \sum_i h_i\mathsf{K}h_i + \tfrac{1}{2}\sum_i [h_i, [h_i, \mathsf{K}]].$$

The twice-commuted walk kernel has two factors of cutoff variation. Its operator norm is bounded by $C\ell_0^{-2}$, using the exponentially weighted second moment of the kernel. Walks leaving the gauge cube contribute $Ce^{-c\ell_0}$. Thus the real quadratic form has lower bound

$$\mathsf{K}(U,0) \ge [\gamma_0 - C\ell_0^{-2} - Ce^{-c\ell_0} - Ca_{\mathrm{w}}]\,1. \tag{29}$$

The complex perturbation has operator norm at most $C(b_0 + b_1)$. Fix ℓ_0, then a_{w}, then M large enough to contain these cubes, and finally the radii. This leaves a fixed real gap and a smaller fixed complex perturbation margin. All full compressions $1_Y\mathsf{K}(U,0)1_Y$ inherit the real gap. At complex pairs their inverses are obtained by a norm Neumann series about the real compression.

The same argument gives decay for a compressed resolvent. Conjugation by $e^{\zeta\rho}$, with ρ a Lipschitz distance, changes the form by $O(|\zeta|)$ in the weighted operator norm. For $|\zeta|$ small relative to the gap,

$$(y + \mathsf{K}_Y)^{-1}, \qquad y \ge 0,$$

therefore has an exponentially decaying kernel, with uniform constants. Here and below K_Y is the compressed full form, not an arbitrary truncation of its walk terms. Integrating these resolvents in

$$\mathsf{K}_Y^{-1/2} = \frac{1}{\pi}\int_0^\infty y^{-1/2}(y + \mathsf{K}_Y)^{-1}\,dy \tag{30}$$

defines the inverse square root. The branch is the positive one at real data. The gap gives convergence near zero, and the resolvent bound gives convergence at infinity. Local resolvent expansions give the corresponding walk expansion of this root.

The decoupling parameters are inserted in these operator expansions. A carrier meeting m cubes gains at most $e^{m\bar{\kappa}}$. Apart from a bounded number of cubes at its starting point, each additional cube requires length of order M. Thus

$$e^{m\bar{\kappa}} \le e^{C\bar{\kappa}} e^{(C\bar{\kappa}/M)\mathrm{length}(\omega)}.$$

Taking $M \ge C\bar{\kappa}$ leaves a fixed fraction of the walk decay. The short-walk factor $e^{C\bar{\kappa}}$ is a fixed constant; its smallness is supplied by the fluctuation amplitude in the response equations, rather than by increasing the depth.

For conditioning on a region Z_0, parameters are put only on cubes outside a collar of Z_0. A carrier reaching a parameter from Z_0 must cross this collar. Its norm contains $e^{C\bar{\kappa}} e^{-cM}$, which is exponentially small after the preceding choice. This retains the real reference rows near the boundary of Z_0. Setting parameters to zero on a boundary without retaining this collar could instead remove whole rows and destroy invertibility.

Let $P = 1_{Z_0}$. Write

$$C = C(Z_0; s) = (P\mathsf{K}(s)P)^{-1}, \qquad \Gamma = P\mathsf{K}(s)(1-P)r(s).$$

Here $r(s) = (\mathsf{K}^{-1/2})(s)$ is the separately decoupled walk series of the inverse square root. It is not the inverse square root of $\mathsf{K}(s)$. At $s = 1$ the two original operators are recovered, so $r(1) = \mathsf{K}^{-1/2}$. For functions depending on the bonds of Z_0, define

$$\mathcal{J}(Z_0; s)[F] = \int d\mu_1(X)\, e^{-\frac{1}{2}\langle \Gamma X, C\Gamma X\rangle} \int d\mu_C(B)\, e^{-\langle B, \Gamma X\rangle} F(B). \tag{31}$$

The white noise X has a product measure. At real data, completing the inner square gives a Gaussian of mean $-C\Gamma X$ and covariance C. The outer quadratic factor cancels the normalising factor produced by that square. At $s = 1$ this is exactly the marginal of the original Gaussian, as checked in (34) below. In particular $\mathcal{J}(Z_0; s)[1] = 1$ at real admissible decoupling data, and the identity continues on the common admissible complex domain. For complex C, the symbol $d\mu_C$ means the analytic Gaussian density with its continued determinant; it is not a positive measure.

The estimate on this functional is local in the region where parameters or test factors occur. If $Z_0^\sharp \subset Z$ and $s = 0$ outside Z, all white-noise variables outside Z have integral one. Compare C, Γ with their real reference values, whose gaps and operator norms are uniform. Suppose their allowed perturbations have norm at most $\hat{\epsilon}$. For

$$|F(B)| \le F_0 e^{\rho |B|^2}, \qquad u = \hat{\epsilon} + \rho \le c,$$

the real reference square completion reduces the absolute integral to a Gaussian exponential moment with quadratic perturbation bounded by Cu. A reference Gaussian with covariance norm at most C_0 satisfies

$$\int e^{v|Y|^2}\, d\mu_{C_0'}(Y) = \det(1 - 2vC_0')^{-1/2} \le e^{Cvn}, \qquad 2v\|C_0'\| \le \tfrac{1}{2},$$

where n is its number of real components. The determinant normalisations in the comparison obey the same estimate, by $|\log\det(1+T)| \le Cn\|T\|$ for $\|T\| \le 1/2$. It follows that

$$|\mathcal{J}(Z_0; s)[F]| \le F_0 \exp[CuN(Z)], \qquad N(Z) \le C(LM)^4 \# Z. \tag{32}$$

The reference square completion includes the mixed B, X terms; they cannot be bounded separately at their original coefficients without spending the Gaussian gap. The perturbation bound above is applied after that completion.

The factor in (32) is bounded per coarse cube only when $u \le c(LM)^{-4}$. This condition applies to every contribution to u: the walk-decoupling error, the analytic radii, a complex change of the action coefficient, the quadratic growth of interaction potentials, and the Gaussian tail parameter.

Its dimensional factor is part of the argument. Using a bound exponential in the entire torus volume at this point would not give a local polymer estimate. Verifying the stated operator perturbations on the actual common pair domains remains part of the uniform one-step construction.

The covariance recovery can be checked directly at real data. Let $P = 1_{Z_0}$ and split the positive matrix K as

$$\mathsf{K} = \begin{pmatrix} A & B \\ B^{\mathsf{T}} & D \end{pmatrix} \quad \text{on } P \oplus (1-P), \qquad C_Z = A^{-1}.$$

For a fixed exterior value y, completing the square gives

$$\begin{aligned} \langle (x,y), \mathsf{K}(x,y)\rangle &= \langle x + A^{-1}By, A(x + A^{-1}By)\rangle \\ &\quad + \langle y, (D - B^{\mathsf{T}}A^{-1}B)y\rangle. \end{aligned} \tag{33}$$

Thus the conditional mean is $-A^{-1}By$, and the conditional covariance is C_Z. Draw the exterior value as $y = (1-P)\mathsf{K}^{-1/2}X$, with X standard Gaussian. Then $By = \Gamma X$. The law of total covariance gives

$$C_Z + C_Z\Gamma\Gamma^{\mathsf{T}}C_Z = P\mathsf{K}^{-1}P. \tag{34}$$

This is the identity used when the decoupling parameters return to one. It also fixes the sign of the conditional mean. At complex parameters the identity continues algebraically on the component where the matrices remain invertible; positivity and probability language apply only at real data.

The Gaussian perturbation bound has a volume restriction. If one integration cube has $O((LM)^4)$ field components, the allowed change per component is $O((LM)^{-4})$. The discarded tail must still obey

$$(\delta_k/g_k)^2 \geq C(LM)^4(\kappa + \log M). \tag{35}$$

One half of its coercive exponent is available for this tail, and the other half remains under the Gaussian integral. Both (35) and the smallness needed for short decoupled walks are obtained by choosing the coupling ceiling after L, M, κ.

From local potentials to a connected logarithm. We spell out the expansion because its local Gaussian bound, its connected summation and its pointed sum rule are used again in the source argument. Let $\mathfrak{V}$ denote the finite family of localised potentials at one step. Besides the small-box potentials, which are defined only where their field tests hold, it can contain the quadratic gauge potential that occurs when the action coefficient is varied. That potential is defined for all real B. Write

$$|V(Y;B)| \leq w_Y \left(1 + M^{-4}\sum_{b\subset Y} |B(b)|^2\right), \qquad w_Y \leq w\, e^{-\lambda_V d_k(Y)}.$$

The sum over Y is absolutely convergent after restriction to a finite torus. The uniform estimates will be independent of that restriction. In the present application

$$w \leq C\{\delta_k + \mathsf{s}_k N_\omega\}$$

for the interaction part; the extra quadratic gauge potential is bounded separately by $C|1-\tau|M^4$.

First use the finite identity

$$\exp\Big(\sum_Y V(Y;B)\Big) = \sum_{\mathcal{D}} \prod_{Y\in\mathcal{D}} (e^{V(Y;B)} - 1). \tag{36}$$

Here $\mathcal{D}$ ranges over subsets of the potential labels; two different origins on the same domain retain different labels. For a chosen $\mathcal{D}$, let Y_0 be the union of the domains of its small-box potentials. Their field tests remain in place. The tests outside Y_0 may be expanded as

$$\prod_b \chi_b = \prod_{b\in Y_0} \chi_b \sum_{P\subset\mathfrak{X}\setminus Y_0} (-1)^{|P|} \prod_{b\in P} \chi_b^c. \tag{37}$$

This order is necessary: expanding a test required by a potential would evaluate that potential outside its domain. The quadratic gauge potential contributes no test to Y_0.

An index $\iota = (\mathcal{D}, P)$ has a support $Z_0(\iota)$, formed from its potential domains and failed-test bonds. Apply the conditional representation (31) to its product F_ι. Then interpolate all coarse-cube parameters outside the collar $Z_0^\sharp$. For a region $Z \supset Z_0^\sharp$ the term is

$$T(Z,\iota) = \int_{[0,1]^{Z\setminus Z_0^\sharp}} \prod_{\Delta\in Z\setminus Z_0^\sharp} \partial_{s(\Delta)} \mathcal{J}(Z_0; s1_Z)[F_\iota]\, ds. \tag{38}$$

The parameters in the collar are one. Completeness of the conditional representation and repeated fundamental theorems of calculus show that the sum of these terms is the original integral. The statement is an identity before estimates.

At zero parameters, the operators, conditional integrals and interactions split over wall-components. Therefore $T(Z,\iota)$ splits over those same components. Regrouping its components gives a gas with polymers equal to connected unions of coarse cubes. Two polymers are compatible when they share neither a cube nor a wall. Denote their activities by $H(Z)$. The exact integral is

$$\Xi(H) = \sum_{\{Z_1,\dots,Z_n\}\ \text{compatible}} \prod_{i=1}^{n} H(Z_i),$$

including the empty family with value one. A factorisation into components here is justified by the decoupled operator structure, not by decay of correlations for the original Gaussian.

For each Mayer factor, introduce a numerical parameter $t_Y \geq 1$. The elementary inequality

$$|e^v - 1| \leq t_Y^{-1} e^{2t_Y|v|}$$

follows from $|e^v - 1| \leq |v|e^{|v|}$ and $t_Y|v| \leq e^{t_Y|v|}$. Taking t_Y proportional to the inverse small activity weight extracts one small factor for each vertex. The remaining exponential has a quadratic growth coefficient that can be made smaller than the local Gaussian margin. A failed test is bounded by

$$1_{\{|B(b)|\geq \mathfrak{b}\}} \leq e^{-\gamma_2 \mathfrak{b}^2/2} e^{\gamma_2 |B(b)|^2/2}.$$

Its second factor uses the same quadratic margin; its first is the tail gain. A differentiated decoupling parameter contributes $Ce^{-\bar{\kappa}}$ by Cauchy on the common complex parameter disc. These three gains are taken before applying (32).

The local Gaussian cost is at most $e^{C\#Z}$ after imposing the per-component restriction $u \leq c(LM)^{-4}$. The tail gain is useful uniformly only if

$$e^{-\gamma_2 \mathfrak{b}^2/2} (LM)^4 e^{C\kappa} \ll 1.$$

Since γ_2 itself is of order $(LM)^{-4}$, this is the floor (35). Half of the coercive exponent may be assigned to the failed test; the other half stays under the integral. No second use of the same half is made in the Mayer estimate.

Two geometric inequalities organise the sums. For a wall-connected union of carriers,

$$d\Big(\bigcup_i X_i\Big) + 5 \leq \sum_i (d(X_i) + 5). \tag{39}$$

For the collar of a fine-scale carrier,

$$d_{k+1}(Y^\sharp) \leq (C/L) d_k(Y). \tag{40}$$

The latter inequality includes the case of one fine cube: the coarse cubes in its collar share a corner, and their connecting length is zero. It is the passage in (40), rather than an estimate on an individual Gaussian integral, that restores the decay rate at the next scale. The domain count

through one cube is exponential in $d_k(Y)+1$ and is paid by a fixed fraction of the incoming rate. The remaining fraction, after rescaling, gives

$$|H(Z)| \le C\epsilon_k e^{-\lambda(d_{k+1}(Z)+5)}, \qquad \epsilon_k = \delta_k + \mathsf{s}_k N_\omega + e^{-c\gamma_2 \mathfrak{b}^2}, \tag{41}$$

with $\lambda \ge cL\kappa$. In including the free quadratic potential, its fixed small bound is added to ϵ_k. The constants are chosen so that $\epsilon_k \le 1$. The exact value of c in the tail exponent is immaterial after the floor has been fixed.

We use the Kotecký–Preiss criterion [KP86], with its pinned connected-graph bound, to take the logarithm. For a finite hard-core polymer set, put

$$f(Z,Z') = -1_{\{Z \nsim Z'\}}, \qquad \rho^T(Z_1,\dots,Z_n) = \sum_{G \text{ connected}} \prod_{\{i,j\}\in G} f(Z_i,Z_j).$$

The incompatibility relation is symmetric and reflexive, so repeated polymer labels in the tuple are treated correctly. Suppose nonnegative numbers $a(Z)$ obey

$$\sum_{Z' \nsim Z} |h(Z')| e^{a(Z')} \le a(Z). \tag{42}$$

Delete vertex one from a connected incompatibility graph. Each remaining connected component has to be attached to the root by a nonempty set of allowed edges. Summing the signs over all those nonempty sets gives -1 if at least one edge is allowed, and zero otherwise. Taking absolute values after this cancellation and discarding the remaining restrictions bounds the rooted connected sum by the exponential of smaller rooted sums. If $t_N(Z)$ is the resulting pinned majorant truncated at N other vertices, the recursive upper bound is

$$t_{N+1}(Z) \le \exp\Big(\sum_{Z' \nsim Z} |h(Z')| t_N(Z') \Big).$$

Starting with $t_0 = 1$, (42) proves $t_N(Z) \le e^{a(Z)}$ by induction. Monotone passage in the nonnegative majorants gives the pinned estimate

$$\sum_{n\ge 1} \frac{1}{(n-1)!} \sum_{Z_2,\dots,Z_n} |\rho^T(Z,Z_2,\dots,Z_n)| \prod_{i=2}^{n} |h(Z_i)| \le e^{a(Z)}. \tag{43}$$

The cancellation concerns hard-core edge factors $0,-1$. Replacing it by an arbitrary sum of absolute graph weights would lose this recursion.

The exponential formula for finite labelled graphs identifies the formal logarithm of Ξ with the connected series. The preceding bound makes this series absolutely convergent on the admitted activity polydisc. Along the path rh, $0 \le r \le 1$, the same majorant is uniform. The identity with Ξ holds near zero by the finite algebraic formula and then throughout this path by analytic continuation and continuity. Thus

$$\Xi(h) = \exp\left[\sum_{n\ge 1} \frac{1}{n!} \sum_{Z_1,\dots,Z_n} \rho^T(Z_1,\dots,Z_n) \prod_{i=1}^{n} h(Z_i) \right], \tag{44}$$

and in particular $\Xi(h) \ne 0$. The pinned estimate contains more information than this nonvanishing: it controls the response to a marked factor and will provide the local activity bound and the source derivatives.

Apply the criterion with

$$h(Z) = H(Z) e^{\mu(d_{k+1}(Z)+5)}, \qquad a(Z) = \#Z,$$

where μ is the desired output rate. A polymer incompatible with Z contains a cube adjacent to one of the cubes of Z. There are a bounded number of such neighbours. Also $\#Z' \le 16(d_{k+1}(Z')+1)$.

Hence the left side of (42), divided by $\#Z$, is bounded by a fixed multiple of

$$\epsilon_k \sum_{Z' \ni \Delta} e^{-(\lambda-\mu)(d_{k+1}(Z')+5)+16(d_{k+1}(Z')+1)}.$$

The sum is at most one once the gap $\lambda - \mu$ is fixed large enough. Its part proportional to $d_{k+1}(Z')$ pays the domain entropy, while the spare constant $e^{-5(\lambda-\mu)}$ pays the fixed neighbour and volume factors. Only $\epsilon_k \le 1$ is used to verify the criterion. The smallness of ϵ_k remains available in the bound on the output.

Assign each connected cluster in (44) to the union of its carriers. Inequality (39) transfers the weight $e^{\mu(d+5)}$ to its vertices. Mark one vertex whose carrier meets a prescribed cube, and apply (43) to all the other vertices. The marked vertex still has its actual activity H, so the bound is linear in ϵ_k:

$$|E_{\mathrm{int}}(X)| \le C\epsilon_k e^{-\mu d_{k+1}(X)}.$$

This is why the result does not merely assert an order-one convergent logarithm. Choosing the incoming rate and L with a reserve permits $\mu > (1+c)\kappa$, after every volume and domain sum has been performed.

The same marked sum gives the differential of this logarithm with respect to a potential. If a new potential $\epsilon v(Y;B)$ is added, differentiate the expansion at $\epsilon = 0$ and keep its marked carrier. Summing its local outputs gives the normalised expectation of $\sum_Y v(Y;B)$. A difference of two inputs is obtained by integrating this derivative along their segment. The list enters through the linear old-potential rule with norm at most $C\mathsf{s}_k\|\mathfrak{f}\|$, so

$$\|E_{\mathrm{int}}[\mathfrak{f}] - E_{\mathrm{int}}[\mathfrak{f}']\| \le C\mathsf{s}_k\|\mathfrak{f} - \mathfrak{f}'\|.$$

All these series have the same nonnegative majorant. Each integrated term is holomorphic on the common complex domain; uniform convergence on smaller domains preserves holomorphy. No restriction depending on the number of preceding levels is introduced by taking these derivatives.

Bond densities and the exact pointed sum rule. The logarithm just constructed must be assigned base points before applying the marginal subtraction. Its unpointed norm does not prescribe this assignment. A convenient construction uses three paths from the unit integral to the desired one, keeping a bond label in every derivative.

First turn on the free quadratic gauge potential from zero. Differentiation produces its normalised expectation with its original bond label. Second lower each threshold from infinity to $\mathfrak{b}$. With parameters $t_b \in [0,1]$, use $1_{\{t_b|B(b)|<\mathfrak{b}\}}$, and integrate the sum of its partial derivatives along the diagonal $t_b = t$. A derivative at a single bond can be defined after integration by rescaling that one real variable. The test is then fixed and the derivative acts on the Gaussian and the remaining smooth integrand. This also proves holomorphy in the pair variables. Its local bound contains the failed-test Gaussian tail.

Third turn on the physical fluctuation and the interaction. The cubic action and gauge remainders are interpolated as $t^{-2}V^{A,G}(tB)$, with their continuous values at zero; the Jacobian terms have their own analytic expansion. Differentiation retains the bond of the physical fluctuation argument. Marked cluster expansions localise each resulting normalised expectation. Denote the three bond-labelled outputs by $G_\tau(X,b)$, $E^\chi(X,b)$ and $E^\bullet(X,b)$. Each vanishes unless $b^\sharp \subset X$. Telescoping along the three paths gives the original logarithm, with the branch fixed by continuity from the unit integral.

Put the tents at bond midpoints and define

$$G_\tau(X,z) = \sum_b h_z(b) G_\tau(X,b), \qquad F(X,z) = \sum_b h_z(b)[E^\chi(X,b) + E^\bullet(X,b)].$$

Their reach is smaller than the collar, so only $z \in X$ occur. Summing over X first recovers the exact bond densities obtained before localisation, and $\sum_z h_z(b) = 1$ then recovers the logarithm. These

two exact sum rules preserve the covariance of the pointed totals. They do not require an individual polymer term to be invariant under a unit translation that changes the auxiliary partition.

An isometry acts on the free variables by the orthogonal change of variables induced on the fibre. On a reversed bond, the potential is transported by $-\mathrm{Ad}(V_0(b)^{-1})$. The action, gauge fixing, Haar density and tests are invariant under this change. The bond density is therefore covariant before localisation. The midpoint identity $h_{\mathsf{r}z}(\mathsf{r}b) = h_z(b)$ transfers this covariance to its pointed total. When this argument evaluates the older families, their backgrounds are nested, and their full pointed sums include all domains, including wrapping ones.

For a domain that does not wrap, all preceding operations read only the pair and older terms in that domain and its prescribed collar. Transport between two such domains on different tori commutes with the construction. Increasing the torus therefore defines one infinite-lattice companion. Its coefficient is extracted from that companion. Agreement of individual finite-volume integrals alone would not select this common coefficient.

The Gaussian normaliser is localised separately. Up to local elimination factors, $\log Z^{(k)} = -\frac{1}{2}\log\det \mathsf{K}$. Its diagonal trace gives a bond label; a midpoint tent partition $\sum_z h_z(b) = 1$ converts this into the base point z. The tents satisfy $h_{\mathsf{r}z}(\mathsf{r}b) = h_z(b)$ for lattice isometries. Localising the diagonal logarithm by inclusion–exclusion over regions leaves only closed walk chains meeting every cube of the domain. Their length is at least $cMd_{k+1}(X)$, so their decay dominates both the domain count and the subsets in inclusion–exclusion. The resulting Gaussian family has size $O(1)$ and decay rate of order M; the interaction family is small with the coupling. Together their norm obeys

$$\|E^{(k+1)}\|_{k+1,\kappa} \le \tfrac{1}{2}E_0 + C\{\delta_k + \mathsf{s}_k N_\omega + e^{-cp_0(g_k)^2}\} \le E_0$$

after the constants are chosen in the stated order.

One exact localisation formula makes the two uses of a walk expansion particularly clear. Let K_Y be the fluctuation form constructed in a region Y, compressed to the bonds on which it is invertible. For a bond b well inside the region write $f_b(Y) = \mathrm{tr}[(\log \mathsf{K}_Y)(b,b)]$, with the interior convention giving zero otherwise. The diagonal block is taken after the operator logarithm. On the finite set of localisation cubes define

$$e(X;b) = \sum_{Y\subset X} (-1)^{|X|-|Y|} f_b(Y). \tag{45}$$

The same conventions are used in the regions and their subsets. Summing this formula over $X \subset \mathcal{C}$ and interchanging the two finite sums gives

$$\sum_{X\subset\mathcal{C}} e(X;b) = \sum_{Y\subset\mathcal{C}} f_b(Y) \sum_{Y\subset X\subset\mathcal{C}} (-1)^{|X|-|Y|} = f_b(\mathcal{C}).$$

This is an exact sum rule; it uses no smallness. For its bound one expands the diagonal logarithm by resolvents of the local form. A chain missing a cube of X occurs in two subsets with opposite signs and cancels. A surviving closed chain meets every cube of X, and therefore has length at least $cMd_{k+1}(X)$. Its walk decay pays for the subset entropy as well as the domain entropy.

The pointed Gaussian activity is consequently formed as

$$N(X,z) = -\frac{1}{2}\sum_b h_z(b)[e(X;b) - e(X;b)|_{(1,0)}] + \text{local elimination terms}.$$

The minus sign is the sign of the Gaussian normaliser. Each elimination term is placed at its pivot bond. For bond midpoints x_b, the tents in the units of level k can be chosen as

$$h_z(b) = \prod_{\mu=1}^{4} \max\{1 - L^{-1}|(x_b - z)_\mu|, 0\}, \qquad z \in \mathbb{T}^{(k+1)}.$$

They sum to one and commute with the lattice isometries. The localisation domain retains a margin wider than their reach, so only $z \in X$ occur. The bond density, summed over its domains before

applying these tents, is independent of the auxiliary partition. This is how covariance is returned to the pointed totals even though the individual terms depend on the partition.

The decay of the Gaussian family uses two different expansions, and the distinction prevents a loss at large domains. At the real factor of an admitted pair, each regional compressed form has spectrum in a fixed interval $[\gamma, B]$. The perturbation at a complex pair has norm smaller than a fixed fraction of γ. Choose a fixed $c_N > 0$ such that

$$R_Y = 1 - c_N^{-1} \mathsf{K}_Y, \qquad \|R_Y\| \le q < 1$$

throughout that complex domain. The operator logarithm is then

$$\log \mathsf{K}_Y = (\log c_N) 1 - \sum_{m \ge 1} \frac{1}{m} R_Y^m.$$

The centre c_N and q depend only on the real spectral window and the admitted perturbation margin. They are independent of Y, the torus and the depth.

Apply the regional inclusion–exclusion formula (45) to this diagonal series. A term of R_Y^m is a closed chain of local operator carriers. When the chain does not reach a cube of X, adjoining or removing that cube leaves the term unchanged. The two subset signs cancel. This uses the common assignment of regional operator terms, including their fixed interior margins. A chain that survives must reach within that margin of every cube of X. Consequently its total length is at least $cMd_{k+1}(X)$.

Put $d = d_{k+1}(X)$. For the powers $m \le \theta M d$, the walk estimate for a closed chain gives a bound of the form

$$C Q_N^m e^{-c' M d}, \qquad Q_N > 1.$$

The walk norm need not be smaller than one. Choose θ after Q_N, c' so that $Q_N^{\theta M d} \le e^{c' M d/2}$. Summing these powers leaves an exponential in $-Md$. For the powers $m > \theta M d$, do not use the walk norm. The operator estimate $\|R_Y^m\| \le q^m$ gives instead

$$\sum_{m > \theta M d} \frac{q^m}{m} \le C q^{\theta M d}.$$

There are at most $2^{\#X}$ subsets in the localisation, with $\#X \le C(d+1)$. The number of relevant bonds and tent base points is polynomial in M and $d+1$. Both factors are absorbed by a smaller fraction of the rate proportional to M. The elimination determinant is local at its pivot and obeys the same domain convention. Hence

$$|N(X, \mathbf{p}, z)| \le E_N e^{-c'_N M d_{k+1}(X)}. \tag{46}$$

The constants here contain no coupling. In particular the choice $E_0 \ge 2E_N$ is made before the smallness conditions on the interaction. Increasing the coupling weight cannot supply smallness for this determinant family.

Möbius inversion recovers the whole diagonal logarithm before the tents are applied. Therefore the pointed sum recovers $\log Z^{(k)}(W) - \log Z^{(k)}(1)$ exactly, rather than up to a small error per bond. Its prelocalisation bond density is independent of the auxiliary partition. The common regional construction transports this family to a nonwrapping domain of any larger torus; this also supplies its infinite-lattice companion. The gap, regional agreement and walk estimates used above are the uniform operator inputs. The two-range series estimate explains how those inputs yield the required determinant decay.

At the real action weight, an equivalent single interpolation combines the threshold and interaction paths. Insert it by multiplying the fluctuation field by t, from zero to one, including its occurrence in the tests. Its derivative is a sum over bonds. Localising each such summand first preserves its bond label, and the tents then convert it to a site label. The procedure must retain the pointed sum rule, rather than distributing an already estimated unpointed activity arbitrarily among its sites.

More explicitly, for the interacting integral use

$$I(t) = \int d\mu_C(B) \prod_b \chi(tB(b)) e^{\mathsf{W}_k(tg_k, B)}.$$

On the nonvanishing component supplied by the local expansion, its logarithmic derivative is a sum of two bond-labelled types:

$$\frac{d}{dt} \log I(t) = I(t)^{-1} \int d\mu_C(B)\, e^{\mathsf{W}_k(tg_k,B)} \Big[g_k \prod_{b'} \chi(tB(b')) \sum_b \operatorname{tr} \mathsf{W}_k'(tg_k, B, b)(\mathsf{C}B)(b) + \sum_b \partial_t \chi(tB(b)) \prod_{b' \neq b} \chi(tB(b')) \Big]. \tag{47}$$

The prime is the derivative in the physical fluctuation argument, with its component at b. Products of tests and the test-derivative sum range over free bonds; the functional derivative retains the bond label of the physical field $\mathsf{C}B$. Derivatives of the characteristic functions in this formula are understood after integration over B, as derivatives of the moving integration domains. They are not pointwise functions of a sharp test. The Gaussian tail estimates control this operation. Integrating in t and localising each bond summand before forgetting its label gives the pointed density. The sums over all local domains recover these exact bond densities; the tent conversion therefore recovers their full lattice covariance. This is the reason that an arbitrary division of the unpointed logarithm over its sites would not be sufficient.

2.8. **Extracting the marginal action.** A bounded norm of each family is not yet a bound on their sum at a late background. One unit volume at level k contains η_{k-j}^{-4} sites of level j. The interaction at level j must therefore be evaluated more accurately than its original supremum bound. The two subtractions are its value at the trivial field and its quadratic curvature contribution.

The coefficient of the latter is defined on a single family $\widehat{E}^{(j)}$ on $\mathbb{Z}^4$ that agrees with all torus terms on domains not wrapping around the torus. Let $\mathfrak{h}_\infty$ be the linear response at the trivial field in both potential and current: a unit coarse-bond perturbation δ_c is mapped to the pair consisting of the resulting fine potential and fine action current. For bonds c, c', define the following absolutely convergent kernel, with each activity Hessian evaluated at the trivial pair $(\mathbf{U}, \mathbf{J}) = (1, 0)$:

$$\Pi^{(j)}(c, c') = \sum_z \sum_{\widehat{X} \ni z} \operatorname{Hess} \widehat{E}^{(j)}(\widehat{X}, \cdot, z)[\mathfrak{h}_\infty \delta_c, \mathfrak{h}_\infty \delta_{c'}].$$

Response decay and the tree bound give convergence. Translation and hypercubic covariance apply to the pointed totals, and linearised gauge invariance makes this kernel transverse. With $c = (x, \mu)$, $c' = (y, \nu)$ and $u = y - x$, translation covariance identifies it with the directional kernel $\Pi_{\mu\nu}^{(j)}(u)$. With $\partial_\mu(p) = e^{ip_\mu} - 1$, its small-momentum expansion is

$$\widehat{\Pi}_{\mu\nu}^{(j)}(p) = b_j \left(\delta_{\mu\nu} \sum_\rho |\partial_\rho(p)|^2 - \overline{\partial_\mu(p)} \partial_\nu(p) \right) + O(|p|^3), \quad b_j = \sum_u \Pi_{\mu\nu}^{(j)}(u) u_\mu u_\nu \quad (\mu \neq \nu). \tag{48}$$

Simplicity of the Lie algebra makes the colour factor a scalar. The expression in parentheses is the Hessian of the plaquette action. The coefficient is independent of the chosen distinct directions μ, ν.

The coefficient and the symmetry of its pointed tensor. The argument uses Bałaban's differentiated gauge identities [B87, equations (4.14)–(4.15), (4.33)] and the comparison of pointed and total moments [B88, equations (3.58)–(3.64)]. Here the pointing uses bond midpoints, so that it commutes with the lattice isometries. The Hessian defining b_j is taken in independent potential and current coordinates at the trivial pair. Let $v = (A, J)$ and put $\|v\|_X = \max\{\sup|A|, \sup|\nabla A|, \sup|J|\}$. On a ball of radius a_j' mapped into the activity core, Cauchy's estimate gives

$$|D^2 F(0)[v, v']| \leq C(a_j')^{-2} \sup|F|\, \|v\|_X \|v'\|_X.$$

For each direction, the first derivative is a real-adjoint invariant linear functional on $\mathfrak{su}(2)$ and therefore vanishes. Differentiating real gauge invariance once more gives

$$DF(0) = 0, \qquad D^2F(0)[(\nabla\lambda, 0), v] = 0.$$

The additional term from differentiating the coordinate map vanishes because $DF(0) = 0$. Thus the Hessian is intrinsic at this point and is unchanged by adding a linear gauge direction to either input.

The linear response $\mathfrak{h}_\infty$ is determined modulo precisely these directions. A pure gauge is the background over its own pure-gauge averages, and uniqueness identifies the response to a gradient as a gauge direction. It also makes colour rotations and real conjugation commute with the response modulo those directions. Hence

$$D^2F(0)[\mathfrak{h}_\infty(\delta_c t), \mathfrak{h}_\infty(\delta_{c'} t')] = \langle t, t'\rangle \Pi_F(c, c')$$

for one scalar kernel Π_F. Here t, t' belong to the colour algebra and δ_c is the scalar unit bond field. Equivalently Π_F is one third of the sum with $t = t'$ over an orthonormal colour basis. This specifies the scalar colour convention in the definition of $\Pi^{(j)}$ above.

For a term on X the response bound and Cauchy give

$$|\Pi_F(c, c')| \le C(a'_j)^{-2}\|E^{(j)}\|_{j,\kappa} e^{-\kappa d_j(X)} e^{-\delta_0[\mathrm{dist}(c,X)+\mathrm{dist}(c',X)]}.$$

The distance between the bonds is bounded by the sum of these two distances and $M(d_j(X) + C)$. Half of the exponents consequently gives decay in $|c-c'|$ at rate $\frac{1}{2}\min\{\delta_0, \kappa/M\}$. The other half pays the domain entropy and the sum of base points in X. Therefore the kernel and all its polynomial coordinate moments are absolutely summable, uniformly in the torus size in their periodised versions. The bound for its coefficient has the two-radius-derivative cost $C(a'_j)^{-2}\|E^{(j)}\|_{j,\kappa}$.

Full lattice symmetry is passed to this kernel through the totals, rather than assumed for a term with its auxiliary cube partition. On a finite torus differentiate the covariance identity for the pointed total twice at $V = 1$. This point lies in an invariant ball of block data, so all the differentiations are legitimate on both sides. Fix a box of radius R about the base point. Terms contained in that box agree with their companions for every sufficiently large torus; their response kernels converge by periodisation. The total of all other terms has an exponentially small bound in R/M, uniform in the torus. First increase the torus at fixed R, then increase R. Dominated convergence gives the covariance identity for the companion Hessian and its moments.

Translation covariance of the total makes its kernel a function of the bond directions and $c'_- - c_-$. Signed permutations act on its quadratic moment tensor. At a fixed base point they act on the pointed tensor as well, because the full pointed total was used in the preceding limit. Differentiated gauge invariance applied to linear and quadratic gauge parameters shows that a linear potential contributes only through its curl. These tests may have polynomial growth: their boundary errors tend to zero against the exponential response kernel.

For completeness, a pointed tensor can be written as

$$T_{\mu\nu,\rho\tau}(z) = \sum_{c,c'} \Pi^{(j)}(c, c'; z)(c_- - z)_\rho (c'_- - z)_\tau,$$

where the sum restricts c to direction μ and c' to direction ν, and $\Pi^{(j)}(c, c'; z)$ is the sum over all companion terms pointed at z. Its antisymmetries are the curl identities just obtained. Reflection of one coordinate eliminates a pairing of two different coordinate plaquettes, and permutation of coordinates identifies their diagonal coefficients. The invariant colour pairing is already fixed. This gives the tensor in (50).

To identify its number with (48), apply the Ward identity also to cubic gauge parameters. Define

$$P_{\mu\nu,\rho\tau} = \sum_u \Pi^{(j)}_{\mu\nu}(u) u_\rho u_\tau.$$

The moment form of the cubic identity is

$$P_{\rho\nu,\tau\sigma} + P_{\tau\nu,\rho\sigma} + P_{\sigma\nu,\rho\tau} = 0.$$

The relation between the two moment conventions is $T_{\mu\nu,\rho\tau} + T_{\mu\nu,\tau\rho} = -P_{\mu\nu,\rho\tau}$. It relates the pointed linear-potential tensor to this second moment; the scalar left after the signed-permutation identities is exactly $\sum_u \Pi^{(j)}_{\mu\nu}(u) u_\mu u_\nu$, independent of $\mu \neq \nu$. In passing from bond anchors to midpoint coordinates, the extra terms multiply lower moments. The zeroth moment vanishes by the linear Ward identity, and inversion removes the first moment. No additional action coefficient comes from this change of anchors.

This argument also explains why numerical normalisations do not affect the coefficient. Two local logarithms of the same nonzero integral differ by a locally constant value. Their Hessians and coefficients agree. In particular the total of G_τ in (61) is field independent, so its coefficient is zero even when its individual local terms vary with the field. The coefficient is extracted after the exact pointed sum rule, before discarding any wrapping tail in a finite volume.

The gauge-first subtraction makes the power counting explicit. In an axial gauge centred at the base point, write $F_{\rho\mu}(z)$ for the background curvature component at z, measured in the units of level j. Then

$$B^{\mathrm{ax}}_\mu(x) = \sum_{\rho<\mu} (x_\rho - z_\rho) F_{\rho\mu}(z) + O\Big(|x-z|^2 \eta_{k-j}^{2+\alpha_{\mathrm{H}}}\Big).$$

Thus this potential has size η^2_{k-j} on the local domain. Terms of third and higher order already have a summable gain. The linear term vanishes by gauge invariance and the absence of an invariant vector in the Lie algebra. The remaining quadratic curvature form has the same coefficient as $b_j A$. The covariance of the pointed totals is needed here: it gives this cancellation at each base point, rather than only after summing over a window. Individual localised terms need not be covariant under all unit translations. The tent h_z is evaluated at plaquette centres in $A(h_z, U)$, using the same partition of unity.

For a regular background of level k, curvature size ε, and curvature radius $\alpha_{0,j}$, the resulting one-lattice estimate is

$$\begin{aligned} &\Big| \sum_{X \ni z} [E^{(j)}(X, U_k, z) - E^{(j)}(X, 1, z)] - b_j A(h_z, U_k) \Big| \\ &\qquad \le C \|E^{(j)}\|_{j,\kappa} \Big\{ (\varepsilon/\alpha_{0,j})^2 \eta_{k-j}^{4+\alpha_{\mathrm{H}}} + e^{-c\kappa/\eta_{k-j}} \Big\}, \qquad \varepsilon \eta_{k-j} \le c\alpha_{0,j}. \end{aligned} \tag{49}$$

Long domains supply the exponential tail. On non-wrapping local domains the field factor is retained; domains that wrap may lose it, and are paid for by their size. On nested regions the same estimate has boundary tails and the counterterm carries exactly the sum of the tents of the retained base points. These terms are included in section 3.

The pointed cancellation can be stated at the quadratic-tensor level. Let $T_z F$ be the linear axial potential at z with constant curvature tensor F:

$$(T_z F)_\mu(x) = \sum_{\rho<\mu} (x_\rho - z_\rho) F_{\rho\mu}.$$

For each pointed companion term take its quadratic form on the free-boundary response to this potential. Summing the terms through z gives a quadratic form $Q_z(F)$ on antisymmetric two-forms with colour indices. Absolute convergence follows from response decay and the exponential domain norm, since the coordinate moments grow only polynomially.

At z the covariance of the pointed total acts on this form. A sign change of one coordinate reverses the curvature components containing that coordinate. Consequently a mixed term between two different coordinate plaquettes changes sign under some such reflection and vanishes. Permuting axes makes all the remaining diagonal coefficients equal. The simple colour algebra supplies one

invariant colour form. With the normalisation in (48), the resulting tensor is the Wilson quadratic tensor multiplied by b_j. In all ordered antisymmetric indices its coordinate part is proportional to

$$\delta_{\mu\rho}\delta_{\nu\tau} - \delta_{\mu\tau}\delta_{\nu\rho}. \tag{50}$$

Equivalently, on the components $\mu < \nu$ it is the identity tensor. Thus the leading $F(z)$ in the pointed interaction and in $b_j A(h_z, U)$ is exactly the same quadratic form. The passage from a local box response to the free-boundary response is made inside this gauge-invariant quadratic form. Its exponentially small error is retained separately.

Taylor's formula in the axial block datum then has three relevant contributions. The zeroth order is removed. The first order vanishes by constant gauge invariance. The displayed quadratic tensor is removed by the action counterterm. Because the axial potential is built from curvature, the third-order contribution has an extra two powers of η_{k-j} before the analytic-radius factors are taken. The remaining loss in the quadratic part is the Hölder remainder of the potential beyond $T_z F$, which gives precisely the factor $\eta_{k-j}^{\alpha_{\mathrm{H}}}$. This is the cancellation needed by the rooted interpolation. It cannot be applied to a single unpointed domain term.

The pointed cancellation at complex window fields. The subtraction is also needed inside the local fluctuation expansion, where the arguments are independent complex pairs. The real-background estimate alone does not cover these arguments. We describe the additional window calculation, retaining its domain requirement.

Fix an older level j and a window at the current scale. Put $\varsigma = L^{-(k-j)}$ and write $\bar{a}$ for a fixed multiple, depending on M, of the output curvature radius. The box response estimates give a block datum e^{iB} in level-j units, with

$$|B| \le C\bar{a}\varsigma, \qquad |\partial B| \le C\bar{a}\varsigma^2, \qquad [\partial B]_{\alpha_{\mathrm{H}}} \le C\bar{a}\varsigma^{2+\alpha_{\mathrm{H}}}.$$

The last estimate is a Hölder estimate for first differences; it supplies no bounded second derivative. The substituted potential and reduced current must together belong, with a fixed margin, to the core of every older term that is evaluated. That membership uses the nonlinear background and domain construction. The calculation below applies on that common admitted family of pairs.

Choose an axial gauge about a base point z. The nonabelian Stokes expansion on the comb gives

$$\widehat{B}_\mu(x) = \sum_{\rho<\mu} (x_\rho - z_\rho) F_{\rho\mu}(z) + R_\mu(x), \qquad |R(x)| \le C\bar{a}(1+|x-z|)^2\varsigma^{2+\alpha_{\mathrm{H}}}. \tag{51}$$

Here $F(z)$ is the curvature of this block datum, with $|F(z)| \le C\bar{a}\varsigma^2$. It is the same curvature in the activity and in its pointed action counterterm. A different strength defined from an unrelated bare gauge would not give an exact marginal cancellation.

For small real data, uniqueness of the regular critical orbit identifies the window field with the box Landau response to $\widehat{B}$, up to a real gauge. Gauge invariance identifies their activity values. Both expressions are holomorphic along the specified small complex datum family. The identity theorem therefore gives the same identity on its connected complex continuation. It does not assert invariance under every complex gauge transformation on an unrestricted orbit.

Let $\mathcal{H}$ denote that box response, including its current entry, and let a'_j be the radius used for Taylor estimates near the trivial pair. The weighted response bound controls its size on the domain of a term by

$$\|\mathcal{H}(\widehat{B})\|_X \le N_X, \qquad N_X = CM(d_j(X)+2)\bar{a}\varsigma^2, \qquad \frac{N_X}{a'_j} \le C(M)(d_j(X)+2)\varrho_j\varsigma^2. \tag{52}$$

The factor ϱ_j is the ratio of output to older radius, not a number of differentiating variables. The decay of the response permits the linear growth in (51); it is not estimated by the supremum of that growing datum on an arbitrarily large torus.

There are two elementary cases before Taylor expansion. If $N_X \geq a'_j/2$, the original norm bound is bounded by a fixed multiple of its product with $(N_X/a'_j)^3$. This is bounded by a polynomial in $d_j(X)+2$ times $\varrho_j^3\varsigma^6$. If the age is so small that $C\varrho_j\varsigma > 1$, use the independent norm bound. There are only boundedly many such ages, since $\varrho_j \leq C(k+2-j)^{1/2}$ and ς decreases geometrically. Enlarging the fixed constant covers them. In the remaining case $N_X < a'_j/2$ and $C\varrho_j\varsigma \leq 1$, Taylor expansion is inside the analytic ball and its cubic remainder obeys the desired power count.

Absorb the pointed action counterterm into the family and write its terms as $\widetilde{E}^{(j)}(X,\cdot,z)$. The absorbed family has the same decay norm up to a fixed constant: its coefficient is bounded by $Ca_{0,j}^{-2}\|E^{(j)}\|_{j,\kappa}$ and each pointed plaquette action is $O(a_{0,j}^2)$ on the core. Constant gauge invariance removes the first derivative. The Hessian annihilates gradient directions, as follows by differentiating real gauge invariance. Therefore replacing a box response by the free-boundary response is allowed inside this quadratic form. Its discrepancy is exponentially small in the distance to the window boundary.

For the numerical tensor $\widetilde{T}^{X,z}$ formed from that Hessian, the resulting expansion is

$$\begin{aligned}\widetilde{E}^{(j)}(X,\cdot,z)^{\square} &= \widetilde{E}^{(j)}(X,(1,0),z)\\ &\quad + \tfrac{1}{8}\sum_{\mu,\nu,\rho,\tau}\widetilde{T}^{X,z}_{\mu\nu,\rho\tau}\langle F_{\rho\mu}(z),F_{\tau\nu}(z)\rangle + \mathcal{R}_X,\\ |\mathcal{R}_X| &\leq C(M)\|E^{(j)}\|_{j,\kappa}e^{-\kappa d_j(X)}(d_j(X)+2)^C\\ &\quad\cdot[\varrho_j^2\varsigma^{4+\alpha_{\mathrm{H}}}+\varrho_j^3\varsigma^6].\end{aligned} \tag{53}$$

The first error comes from pairing the linear axial potential with the Hölder remainder. The second is the cubic Taylor remainder. Commutator and reduced-current errors have an additional full power and obey the same bound. Polynomial coordinate moments are paid by a fraction of κ. This accounts for the two distinct radius powers in the estimate.

The cancellation is a sum over all companion domains through z:

$$\sum_{\widehat{X}\ni z}\widetilde{T}^{X,z}=0.$$

The tensors are taken at the trivial pair; they depend on neither the current field nor the system of radii used to bound it. In a window only the domains lying inside its fixed enlargement are used. Their tensor sum is the negative of the missing tail. Every omitted domain reaches at least $2/\varsigma-2$ units in the domain-length convention, so its tensor norm has an exponential tail. The quadratic contribution of that tail is bounded by

$$C\|E^{(j)}\|_{j,\kappa}\varrho_j^2\varsigma^4e^{-c\kappa/\varsigma}.$$

It is absorbed in the Hölder error. This is an equality of the same numerical tensors before estimating a partial sum; there is no claim that each individual local term has zero marginal tensor.

After summing domains through z, and then the $O(\varsigma^{-4})$ base points in a current-scale window, (53) is bounded by

$$C\|E^{(j)}\|_{j,\kappa}[\varrho_j^2\varsigma^{\alpha_{\mathrm{H}}}+\varrho_j^3\varsigma^2]\leq C\|E^{(j)}\|_{j,\kappa}\varrho_j^3\varsigma^{\alpha_{\mathrm{H}}}.$$

Thus the full age sum has precisely the summable weight $(k+2-j)^{3/2}L^{-\alpha_{\mathrm{H}}(k-j)}$. The gain is α_{H}; the supplied first-difference regularity does not justify replacing it by a full derivative.

The cancellation is performed before decoupling the old interaction. Each bracket then acts on a window function whose marginal tensor has already been removed. Its response multiplier ν has analytic radius

$$\rho_F\geq c\varsigma^{-1}a_{0,j}\delta(\mathsf{t})^{-1}e^{-C\bar{\kappa}}\geq c\varsigma^{-1}(\varrho_j\mathsf{s}_{\mathfrak{a}})^{-1}e^{-C\bar{\kappa}}.$$

The exponential scale factor dominates the square-root radius ratio. Choosing $\mathsf{s}_{\mathfrak{a}}$ small places $\nu=1$ in this disc at every age. Schwarz in ν, after subtracting its value at zero, supplies the additional

factor $Ce^{C\bar{\kappa}}\mathsf{s}_{\mathfrak{a}}$ in the old potential estimate. Cauchy in the cube parameters supplies the carrier decay on the same common polydisc.

An aged remainder has no pointed action counterterm and therefore has only the first, quadratic Taylor estimate. After the base-point count its load is $\varrho_j^2\|R^{(j)}\|$, with no geometric gain in age. The lower coupling drift sums that load. The regular and remainder estimates cannot be combined into one artificially decreasing age weight. For general fields on a torus, the wrapping-domain tail also retains the norm-only treatment of (55). The window calculation does not remove flat holonomy from such domains.

For layered regions, split the retained base points as $\Xi = \bigcup_{n\geq j}\Xi_n$ by the local layer. The estimate takes the quantitative form

$$\begin{aligned}\Big|\sum_{z\in\Xi}\sum_{X\ni z}[E^{(j)}(X,U_k,z)-E^{(j)}(X,1,z)] - b_jA\Big(\sum_{z\in\Xi}h_z,U_k\Big)\Big| \\ \leq CE_0\sum_{n\geq j}(1+n-j)L^{-c(n-j)}L^{-4(n-j)}\#\Xi_n, \qquad c\leq\alpha_{\mathrm{H}}.\end{aligned} \tag{54}$$

The retained points must lie a cell inside the allowed summation region; this controls the missing-domain tail at the boundary. Here $L^{-4(n-j)}\#\Xi_n$ is current volume, $1+n-j$ is the squared radius ratio, and $L^{-c(n-j)}$ is the summable surplus. If the tent sum is replaced by a smoother profile, the difference is supported on the layer's rim. That rim costs a constant per site at its own scale, without the surplus age factor.

The exponential tail can be kept sharper than the version in (49). Put $\eta=\eta_{k-j}$. There are three different classes of domains through z. The local class stays in a nonwrapping window of size of order η^{-1} in level-j units. On it the axial potential is controlled by curvature, and Taylor's formula gives the first term in (49). A second class leaves this window but still lies in a nonwrapping box. Such a domain has $d_j(X)\geq c/\eta$ and remains gauge equivalent to the identity when the curvature vanishes. Its estimate retains the quadratic field factor. Only the third class, consisting of domains in no such box, can detect flat holonomy. For this class $d_j(X)\geq c\ell L^{K-j}/M$, and one uses the norm alone. Thus the required sharper estimate is

$$C\|E^{(j)}\|_{j,\kappa}\left[(\varepsilon/\alpha_{0,j})^2\eta^{4+\alpha_{\mathrm{H}}}+e^{-c\kappa\ell L^{K-j}/M}\right]. \tag{55}$$

There is no factor of ε in its last summand.

For the second class the field factor is obtained without a Taylor expansion over the whole torus. On a box containing X, axial gauge bounds the potential by $C(d_j(X)+1)\varepsilon\eta^2$. If its ratio to the analytic radius is small, a quadratic Cauchy estimate applies; if that ratio is large, the norm bound is at most a constant times the square of the ratio. In either case the polynomial in $d_j(X)+1$ is absorbed by part of the decay. The remaining factor $e^{-c\kappa/\eta}$ is smaller than any prescribed power of η and is absorbed in the first term of (55). This argument uses a nonwrapping gauge box. It does not restore the field factor on the third class.

For a restricted set Ξ of retained base points, the same pointed subtraction gives the counterterm $b_jA(\sum_{z\in\Xi}h_z,U)$. Replacing this tent sum by an arbitrary smooth profile changes the action near the rim of Ξ. That difference is a boundary activity and is paid there. It has no decreasing age weight merely from changing the name of the profile.

The factor η_{k-j}^4 is cancelled by the number of base points per current volume. It is precisely marginal in four dimensions. The additional Hölder factor is summable after the quadratic action has been subtracted. Even with running radii the estimate $(\varepsilon_k/\alpha_{0,j})^2\leq C(1+k-j)$ leaves a summable series. Hence the old interaction is kept in the form

$$\mathbf{E}_k(U)=\sum_{j\leq k}\left\{\sum_z\sum_{X\ni z}[E^{(j)}(X,U,z)-E^{(j)}(X,1,z)]-b_jA(U)\right\}. \tag{56}$$

The norms of previously formed terms are unchanged. Their subtracted values at later backgrounds decrease; this is the estimate used when the local expansion weights their ages.

With these definitions the small-field integral returns an identity of the same form at the next level:

$$\int dV_k\, \delta(\mathcal{M}(V_k)V_{k+1}^{-1})\mathfrak{G}_k\chi^{(k)}e^{-x_kA(U_k)+\mathbf{E}_k(U_k)} = e^{\mathsf{E}^{(k)}}e^{-x_{k+1}A(U_{k+1})+\mathbf{E}_{k+1}(U_{k+1})}.$$

The new coupling is determined from the output family, while $\mathsf{E}^{(k)}$ collects the field-independent constants. At the trivial field their logarithmic part includes

$$\log \mathfrak{z}(x_k,\varepsilon_k) = -\tfrac{3}{2}\log x_k + O(1), \quad \mathfrak{n}_k = -\left(\tfrac{9}{2}\log x_k + O(1)\right)(|\mathbb{T}^{(k)}| - |\mathbb{T}^{(k+1)}|)$$

for SU(2). The remaining local Gaussian numbers and vacuum values are counted once in $\mathsf{E}^{(k)}$. This exact normalisation is used by the source bounds: an arbitrary error per bare bond would produce a cutoff-dependent exponential factor in the partition function.

2.9. **The one-loop sum and the closed flow.** The coefficient of the output separates into the Gaussian normaliser, the remaining small-field interaction, and terms descending from large-field regions:

$$x_0 = \beta, \qquad x_{k+1} = x_k - b_{k+1}, \qquad b_{k+1} = b^0_{k+1} + r_{k+1} + q_{k+1}. \tag{57}$$

The required coefficient estimates are

$$\left|\sum_{l=1}^{k} b^0_l - c_0 k\right| \le C_0, \qquad |r_{k+1}| \le C g_k (\log g_k^{-2})^{\mathfrak{p}_0}, \qquad |q_{k+1}| \le C g_k^{\kappa_0 - 4}, \tag{58}$$

where, in the action normalisation fixed above, $c_0 = \frac{11}{3\pi^2}\log L$ for SU(2).

Fix positive induction constants $B_\sharp, \lambda_\sharp, C_\sharp$. A prefix through level k is *admissible* if $g_j \le \gamma$ for $j \le k$, the two-sided bound for $x_i - x_j$ in (69) holds for $0 \le i \le j \le k$, and the torus condition (70) holds for $j \le k$. These are requirements on the already constructed levels. The coefficient estimates below must preserve them at the next level, with one fixed choice of these constants; no next-level bound is part of the premise.

Uniform estimates in the renormalisation step. The induction must establish eqs. (14), (17), (19) and (49) on the specified independent field/current domains for every admissible prefix, on the allowed tori and regular layered regions. The constants must be independent of the cutoff, level, torus side and localisation position. Local agreement on nonwrapping domains then gives one coefficient sequence on all tori. The quantitative flow calculation is eqs. (57) and (58); the contribution of healed regions is estimated in section 3. Only the already constructed prefix enters the next increment, which is formed before the next large-field operation.

Two analytic domains enter the estimate of the errors. For a family of size ϵ on radius α, taking the two derivatives that define its coefficient costs $C\epsilon\alpha^{-2}$. Response decay and the domain count also give exponential decay of its Hessian kernel. Applying this estimate only on the running radius, which is of order g_k times a logarithm, would destroy the smallness of the ordinary small-field interaction. Instead, the small-field chain is constructed on fixed absolute radii and carries its own action counterterm. It gives r_{k+1} in (58). The terms left by the large-field operation are treated on running radii. Their summed size is $Cg_k^{\kappa_0-2}$ under the lower flow bound, and two derivatives cost g_k^{-2} up to logarithms; this gives q_{k+1}. For an input list bounded on both systems of radii, the coefficients from the two determinations agree on their common domain, since their logarithms differ only by a field-independent constant. Thus this split describes one increment, rather than two different coupling prescriptions.

The two chains must also accept lists which were not produced by an earlier small-field step. This occurs when a healed region is replaced by local activities, and again when the action is varied while the real tests are held fixed. The one-step construction is therefore formulated on its analytic

input families, rather than on their history of formation. Each family must still have the covariance and the common infinite-lattice companion used in the marginal subtraction.

To describe that input, write $b[S^{(j)}]$ for the coefficient extracted from the companion of a pointed family $S^{(j)}$. Its contribution to the interaction is

$$\sum_{j\le k}\Big\{\sum_{z}\sum_{X\ni z}[S^{(j)}(X,\cdot,z)-S^{(j)}(X,1,z)]-b[S^{(j)}]A\Big\}+\sum_{j\in\mathcal{J}}\sum_{X}[R^{(j)}(X,\cdot)-R^{(j)}(X,1)]. \tag{59}$$

The second sum contains the aged remainders, which have no action counterterm. The first sum has both subtractions for each member separately. The requirement is stronger than asking that the sum have one quadratic coefficient: the rooted estimates use the cancellation in each older family at its own scale. The load is (28), with $S^{(j)}$ in place of $E^{(j)}$. Thus the pointed families carry the summable weights ω_{k+1-j}, whereas the families in $\mathcal{J}$ carry only the squared radius ratios ϱ_j^2. The lower coupling drift, rather than geometric decay in age, is what makes that latter sum small for the actual remainders. Younger remainders are kept outside this input until their cubes fit in single blocks, as described in section 3.

Freeze the tests, field scaling and gauge fixing at a real number t, while the Wilson action has a complex weight x with $|x-\mathsf{t}|<\sigma_\star\mathsf{t}$, and put $\tau=x/\mathsf{t}$. The relative radius $\sigma_\star$ is fixed after the geometric constants so that the Gaussian coercivity margin is retained. The fluctuation threshold is $\delta(\mathsf{t})\asymp\mathsf{t}^{-1/2}(\log\mathsf{t})^{\mathfrak{p}_0}$. Choose radii $(a_{0,j},a_{1,j})$ so that

$$CMa_{0,j}<a_{1,j},\qquad a_{1,j}^2\le Ca_{0,j},\qquad a_{i,j}\le(1+c)a_{i,k+1}\le C(k+2-j)^{1/2}a_{i,j}.$$

The norms of the list are taken on these domains. In the running system their comparability is supplied by the flow bound. On absolute radii it is part of the fixed analytic construction. In either case put $\mathsf{s}=\delta(\mathsf{t})/a_{0,k+1}$, allowing the fixed comparison factors between consecutive radii. If $\mathsf{s}(1+N_\omega)\le c$, the required one-step output besides the Gaussian family obeys

$$|F(X)|\le C\big[|1-\tau|+\delta(\mathsf{t})+\mathsf{s}N_\omega+e^{-c\mathsf{t}\delta(\mathsf{t})^2}\big]e^{-(1+c)\kappa d_{k+1}(X)}. \tag{60}$$

The output has radii $(1+c)a_{i,k+1}$ and decay rate strictly larger than κ. The large-field operation subsequently spends this reserve. At fixed absolute radii s is of order $\delta(\mathsf{t})$; at running radii it is only a negative logarithmic power. The same analytic map must supply these enlarged domains with independent potential and current variables. A bound on the physical slice alone would not allow the later interpolations.

The complex action weight and the Gaussian count. The term $|1-\tau|$ in (60) has a special origin. Let Δ_A be the Hessian of the composite action on the actual constrained fibre, in fluctuation coordinates. It includes the current-dependent corrections from the constraint and the critical field. It is not the ambient Wilson Hessian at nonzero current. First work at a real regular constrained critical point. Let K be the tangent space of the residual gauge orbit, of dimension $n_{\mathcal{O}}$. Gauge invariance makes the intrinsic action Hessian Δ_A vanish on K. The quadratic gauge-fixing form is a sum of squares of the $n_{\mathcal{O}}$ mean-contour coordinates. Its restriction to K is positive definite by the gauge-slice property. Thus Δ_{GF} has rank $n_{\mathcal{O}}$: its rank is at most the number of these coordinates and at least the dimension of K. In particular,

$$\text{fluctuation space}=\ker\Delta_{\mathrm{GF}}\oplus K.$$

In coordinates adapted to this decomposition, Δ_A has only its $\ker\Delta_{\mathrm{GF}}$ block and Δ_{GF} only its K block. The change of coordinates is independent of τ, so its determinant factors cancel in the ratio. Consequently

$$\det(\tau\Delta_A+\Delta_{\mathrm{GF}})=\tau^{n_{\mathrm{phys}}}\det(\Delta_A+\Delta_{\mathrm{GF}}),\qquad n_{\mathrm{phys}}=\dim\ker\Delta_{\mathrm{GF}}.$$

These dimensions are fixed by the fluctuation and gauge-fixing coordinates and are independent of the regular background. The gauge-slice construction must retain the invertibility of these

coordinates on its chosen complex domain. There the same complementary blocks are holomorphic, and the displayed identity continues holomorphically; positivity was used only at the real data.

For the fluctuation expansion one instead uses a Gaussian with covariance $\tau^{-1}C^{(k)}$. The remainder of the quadratic form is the free potential

$$-\tfrac{1}{2}(1-\tau)\langle B, \Delta_{\mathrm{GF}} B\rangle.$$

It is already a sum of block terms and requires no further spatial localisation. Its size restricts the relative complex disc for x; that disc is fixed after the geometric constants so that the Gaussian coercivity margin remains available. The construction separates the local family produced by this potential, denoted here by G_τ, from the interaction family F_{int}. Their respective bounds are

$$|G_\tau(X)| \le C|1-\tau| e^{-(1+c)\kappa d_{k+1}(X)},$$
$$|F_{\mathrm{int}}(X)| \le C\big[\delta(\mathsf{t}) + \mathsf{s}N_\omega + e^{-c\mathsf{t}\delta(\mathsf{t})^2}\big] e^{-(1+c)\kappa d_{k+1}(X)}.$$

Their sum is the family estimated in (60).

Let n_{f} be the number of free fluctuation variables and $n_{\mathcal{O}}$ the number of residual gauge directions, so that $n_{\mathrm{phys}} = n_{\mathrm{f}} - n_{\mathcal{O}}$. The determinant identity gives the two contributions

$$\sum_{X,z} G_\tau(X,U,z) = \tfrac{1}{2} n_{\mathcal{O}} \log\tau, \qquad -\tfrac{1}{2} n_{\mathrm{f}} \log\tau + \tfrac{1}{2} n_{\mathcal{O}} \log\tau = -\tfrac{1}{2} n_{\mathrm{phys}} \log\tau. \tag{61}$$

The logarithm is the branch near $\tau = 1$. The first equality concerns the total family. Individual local terms can depend on the field. The second term on the left of the second equality is combined with the normalization of the complex Gaussian. Adding the original count $-\frac{1}{2} n_{\mathrm{phys}} \log \mathsf{t}$ leaves $-\frac{1}{2} n_{\mathrm{phys}} \log x$. Thus the auxiliary number t disappears from this logarithmic normalization, and the total G_τ has zero action coefficient. The increment still contains the Gaussian coefficient and the coefficient of the interaction family. Assigning a coefficient to a separate local term of G_τ would break this cancellation.

Derivatives of a finite prefix. This formulation is needed for more than real coupling estimates. Hold the real choices at numbers $\theta_0, \dots, \theta_k$ and allow the action weights to vary on $|x_i - \theta_i| < \sigma_\star \theta_i$. Iterating a holomorphic one-step map on the common domains gives a holomorphic ordinary correction $f_{k+1}(x_0, \dots, x_k)$ in the absolute-radius chain. The required bound is

$$\sup |f_{k+1}| \le C\theta_k^{-1/2} (\log\theta_k)^{\mathfrak{p}_0}.$$

The remainder contribution is estimated separately on running radii; it must not be included in this bound merely because it contributes to the same final increment.

For a point in the half-polydisc and a complex direction v, restrict this function to the line $x + zv$. If $m(v) = \max_{i\le k} |v_i|/(\sigma_\star\theta_i) > 0$, the line stays in the full polydisc for $|z| < (2m(v))^{-1}$. Cauchy's one-variable estimate therefore gives

$$\left| \sum_{i\le k} v_i \partial_i f_{k+1}(x) \right| \le C\theta_k^{-1/2} (\log\theta_k)^{\mathfrak{p}_0} \max_{i\le k} \frac{|v_i|}{\sigma_\star\theta_i}. \tag{62}$$

For the zero direction the inequality is immediate. The estimate costs no factor equal to the number of levels: it uses a single complex line, rather than adding separate Cauchy estimates in all coordinates. This is the form useful when a source perturbs a whole coupling prefix at once.

There is no geometric decay in $k - i$ in (62). Boundedness on a polydisc alone cannot give it; a function proportional to x_0/θ_0 is a counterexample. Any assertion that the chain forgets a particular old perturbation has to use its recursive weighted dependence, in addition to holomorphy. The present first-passage construction uses the ordinary size bound and the separate remainder sum; it does not require the stronger tuned-cutoff comparison.

One local construction on all volumes. The input to a step includes its local realization as well as its norm. It consists of the decoupled covariance and background response on each region, the

localized new interactions, the rule evaluating older families at the fluctuating background, and the localized Gaussian logarithm. These four choices must be transported together. For example, an old potential on Y uses only family members on subdomains of Y and their coefficients; changing a representative outside Y must not change that potential.

Fix these local formulas once, with their margins and constants. On a nonwrapping region with a common lift to $\mathbb{Z}^4$, the same formulas then define the corresponding term on either torus. The realization on smaller analytic radii is the restriction of the one on larger radii. Evaluating those formulas on $\mathbb{Z}^4$ defines the companion of the new family and permits the next coefficient to be extracted before a volume is selected. Uniqueness here is relative to the fixed local realization. In particular the Gaussian pointing can depend on that realization even though its sum is the same Gaussian logarithm. No equality of every individual local term under unrelated localization choices is needed or asserted.

The uniform analytic map and its compatible local realization are therefore parts of the same construction. Once their estimates hold, the preceding identities explain how it handles arbitrary input lists, frozen real choices and common coefficients. They do not derive the required background or walk estimates from holomorphy alone.

The two chains and the remainder load. The separation of absolute and running radii can be carried through the iteration quantitatively. Denote the fixed real inverse-coupling sequence by θ_m; the real tests and field scaling of step $m+1$ use θ_m. The ordinary chain contains only pointed families and their own counterterms. On absolute radii its one-step small factor is $C\delta(\theta_m)$, and the Gaussian family contributes an order-one norm. The complex gauge correction contributes $K_{\mathbf{M}}\sigma_\star$. Choose

$$E_0 \geq \max\{2E_N, 4K_{\mathbf{M}}\sigma_\star\}, \qquad N_\star \geq E_0 \sum_{n\geq 1} \omega_n + C_{\mathrm{slot}}.$$

Then the output budget is one half of E_0 for the Gaussian family, one quarter for the fixed gauge correction, and one quarter for the interaction, after the coupling ceiling has been decreased. The geometric age weights make the bound on the old pointed load independent of the number of levels. The constants $K_{\mathbf{M}}, \sigma_\star$ are chosen before $E_0, N_\star$. Thus decreasing the subsequent ceiling on s does not change the constants it has to overcome.

The formula of a local term contains no analytic radius. Its construction on smaller cores is the restriction of the same construction on larger cores. Restrict the ordinary chain to the running cores, and on those cores form a second chain containing the aged remainders as well. Both chains use the same coefficient x, the same real choices, the same normaliser and the same free gauge correction. Their difference therefore consists precisely of terms with at least one marked input descended from a remainder, including the counterterm of each marked pointed member. The Gaussian and free gauge families cancel from the difference. The Lipschitz bound furnished by the marked connected sum applies to this difference.

Here the hypotheses on the real sequence are the already constructed prefix bounds

$$\theta_j \geq \theta_m - c_5 + \lambda_\flat(m-j), \qquad \theta_j \leq \theta_m + C(m-j), \qquad j \leq m,$$

with a fixed lower floor. The radius ratio obeys

$$\varrho_{jm}^2 \leq C\frac{\theta_j}{\theta_m}\left(\frac{\log\theta_m}{\log\theta_j}\right)^{2q} \leq C\frac{\theta_j}{\theta_m}.$$

The bounded reverse drift is used in the last inequality when $\theta_j < \theta_m$. A remainder of level j has size at most $C\theta_j^{-\kappa_0/2}$. Put $p = (\kappa_0 - 2)/2 > 1$. Its total aged load at step $m+1$ is bounded by

$$\begin{aligned}\sum_{j \text{ aged}} \varrho_{jm}^2 \theta_j^{-\kappa_0/2} &\le C\theta_m^{-1} \sum_{j \le m} \theta_j^{-p} \\ &\le C\theta_m^{-1} \sum_{n \ge 0} (\theta_m - c_5 + \lambda_\flat n)^{-p} \le C_{\mathrm{h}} \theta_m^{-p}.\end{aligned} \tag{63}$$

The last inequality follows by comparison with the integral of t^{-p}; its constant is bounded by a fixed multiple of $1 + [\lambda_\flat(p-1)]^{-1}$. This is the loss of one inverse-coupling power caused by summing remainders over levels. No decay with age has been assigned to an individual remainder. If the inverse coupling were held constant, that load would grow with the number of levels.

Let u_i be the running-core norm of the marked output formed at step i. The one-step Lipschitz estimate gives

$$u_{m+1} \le K_{\mathbf{M}} \mathsf{s}(\theta_m) \left[\sum_{i \le m} \omega_{m+1-i} u_i + C_{\mathrm{h}} \theta_m^{-p} \right]. \tag{64}$$

Suppose inductively $u_i \le A\theta_{i-1}^{-p}$. The reverse-drift bound and the floor give $\theta_{i-1}^{-p} \le 2\theta_m^{-p}$, after a fixed adjustment of the floor. Thus the bracket in (64) is at most $[2A\sum_n \omega_n + C_{\mathrm{h}}]\theta_m^{-p}$. Choose $A \ge 1$ and then the ceiling so that

$$K_{\mathbf{M}} \sup \mathsf{s}(2\sum_n \omega_n + C_{\mathrm{h}}) \le 1.$$

This proves $u_{m+1} \le A\theta_m^{-p}$ and closes the induction. The factor s may be only a negative power of a logarithm; it need only be uniformly small, since the convolution kernel is summable.

Extracting a coefficient costs two derivatives on the radii of creation. For a marked term that cost is at most $C\theta_m$, up to favourable logarithms. Consequently

$$|b_{m+1}^{\mathrm{marked}}| \le C\theta_m u_{m+1} \le C\theta_m^{-(\kappa_0-4)/2}.$$

On absolute radii the derivative cost is fixed, and the ordinary correction has the bound

$$|r_{m+1}| \le C\theta_m^{-1/2} (\log \theta_m)^{\mathfrak{p}_0}.$$

These are the two different estimates for q_{m+1} and r_{m+1} in (58). Applying the running radius derivative cost to the ordinary chain would instead lose its smallness.

A remainder is admitted to the second chain only after its own partition is preserved by the next lattice. Its cube side is $M\check{R}_j$, a power of L. Once $M\check{R}_j \le L^{m+1-j}$, the next lattice is a sublattice of its cube-centre lattice, and all required translations preserve the remainder's partition. Before that age it is kept outside the regular exponent and passed through the separate remainder and boundary slots. This delay is a symmetry requirement. It is distinct from the waiting depth before a large-field component is healed.

The induction also retains its order of construction. The regular output and its coefficient at step $m+1$ use only $\theta_0, \dots, \theta_m$. The coefficient fixes the next inverse coupling before the next large-field cube size and tests are chosen. The created family has a fixed enlargement of the radius at θ_m, sufficient for both adjacent running radii. Hence neither the coefficient nor its definition depends on a next-level construction whose admissibility it is supposed to prove.

Varying the whole coupling prefix. There is a sharper way to state the dimension-independent directional estimate. Suppose a holomorphic scalar F is bounded by B on $\prod_i D(0, r_i)$. For an interior point ξ, choose disc automorphisms ϕ_i with $\phi_i(0) = \xi_i$. Their derivatives at zero have

moduli $r_i(1-|\xi_i/r_i|^2)$, and their phases can be chosen so that all terms in

$$\frac{d}{dz}F(\phi_1(z),\dots,\phi_n(z))\Big|_{z=0}$$

have the same phase. Cauchy's estimate on that one unit disc gives

$$\sum_i r_i(1-|\xi_i/r_i|^2)|\partial_i F(\xi)| \le B. \tag{65}$$

For a family-valued map, apply this calculation at a fixed domain, base point and pair, and then take its weighted supremum when estimating a direction. It introduces no factor equal to the number of earlier couplings.

In the half-polydisc of an ordinary coefficient, take $r_i = \sigma_\star\theta_i$ and $B = C\theta_m^{-1/2}(\log\theta_m)^{\mathfrak{p}_0}$. The slope sum is then summable along the lower-drift sequence after one further inverse-coupling factor is taken in a fixed-size perturbation direction. This is the analytic estimate used when a source shifts all the earlier action weights simultaneously. It does not prove an individual derivative decreases geometrically with its age; such a conclusion would require additional recursive information.

Finally a constant source can be carried in two ways. It may change the coefficient x to $x-\zeta$, or it may be inserted as a weight member ζA while the real tests are held fixed. Where both constructions are admitted, the exact integral depends on them through the same exponent $-xA+\zeta A$. Their Gaussian conventions give totals differing only by the explicit number

$$\tfrac{3}{2}|\mathfrak{X}|[\log(x/\mathsf{t})-\log((x-\zeta)/\mathsf{t})].$$

Continuity along $t\zeta$, starting at zero, fixes the logarithm branch. The coefficients of the two totals agree, since the difference is field independent. Individual local terms need not agree. The source induction uses this coefficient identity together with the specified local construction, rather than an identity between unrelated localisations.

Constrained determinants and the one-loop sum. The passage from many normalisers to one determinant uses an exact finite-dimensional identity. Let K be a symmetric form and m a full-rank linear map, with K positive on $\ker m$. Choose orthonormal columns L spanning $\ker m$ and Z spanning its orthogonal complement. The constrained Gaussian integral is

$$\int_{\mathbb{R}^n}\delta(mv)e^{-\langle v,Kv\rangle/2}\,dv = \frac{(2\pi)^{(n-r)/2}}{\det(L^{\mathsf T}KL)^{1/2}\det(mm^{\mathsf T})^{1/2}}, \qquad r=\operatorname{rank} m.$$

Indeed $v = La + Zb$ is an orthogonal change of variables, $mL = 0$, and integrating $\delta(mZb)$ sets $b = 0$ with Jacobian $|\det(mZ)|^{-1}$. The corresponding saddle matrix is

$$\mathbb{S}(K,m) = \begin{pmatrix} K & m^{\mathsf T} \\ m & 0 \end{pmatrix}.$$

In the same coordinates its determinant is

$$|\det\mathbb{S}(K,m)| = \det(L^{\mathsf T}KL)\det(mm^{\mathsf T}). \tag{66}$$

The mixed blocks of K cancel in this determinant. Thus the constrained integral is a constant times $|\det\mathbb{S}(K,m)|^{-1/2}$. This formula keeps the constraint local and avoids solving it by a nonlocal projection before taking determinants.

Near the trivial field, construct the critical branch $p_j(W)$ for one step in the axial gauge. The composite $c_k = p_1\circ\cdots\circ p_k$ is a critical configuration over the k-fold average. Its Hessian on the remaining fibre is positive: a closed axial form with zero average is zero. The implicit function theorem supplies this branch on a neighbourhood that may depend on the depth and torus. The uniqueness of the regular critical orbit identifies it with the background branch where both are defined. For the one-loop coefficient only derivatives at the identity are used; the uniform estimates required below concern their kernels, rather than a common nonlinear neighbourhood for all depths.

Successive Gaussian integrations on this branch compose exactly. For a positive block form,

$$\det \begin{pmatrix} A & B \\ B^{\mathsf{T}} & D \end{pmatrix} = \det A \, \det(D - B^{\mathsf{T}} A^{-1} B),$$

and the second factor is also the quadratic form obtained by minimizing over the first variables. Repeating this identity shows that the product of the first k normalisers is one constrained Gaussian integral in the hierarchical axial gauge. If Φ_j is the logarithm of the jth normaliser relative to its value at the identity and Ψ_j is the combined logarithm, then

$$\Psi_j = \Phi_j + \Psi_{j-1} \circ p_j.$$

The coefficient of Ψ_k is $\sum_{j \le k} b_j^0$. In this identity the action coefficient is exactly marginal: each curvature has scale L^{-2}, so the two curvature factors give L^{-4}, cancelled by the L^4 site count. The nonzero aliases in its symbol begin at higher momentum order and do not change the quadratic coefficient.

To compare with a translation-invariant massive form, change the gauge and the constraint together. Let Q' be the site average and P_U the orthogonal projection onto $\ker Q'(U)$. The gauge condition $P_U D_U^* B = 0$ removes the residual directions $D_U \ker Q'(U)$. The exact relation

$$Q_k^\sharp(U) D_U = D_{\mathcal{M}^k U} Q'(U)$$

puts those directions in the kernel of the straightened constraint. The finite-dimensional gauge Jacobian and (66) give the gauge and constraint replacement in one formula:

$$\begin{aligned} \Psi_k = \Big[& -\tfrac{1}{2} \log |\det \mathbb{S}(\Delta_U^{\mathrm{W}} + \mathfrak{J}_U + D_U P_U D_U^*, Q_k^\sharp(U))| \\ & + \log |\det \mathbb{S}(\Delta_U, Q'(U))| \Big]_{U=1}^{U=c_k(W)} . \end{aligned} \tag{67}$$

Here $\mathfrak{J}_U$ includes the current correction to the off-critical action Hessian and the multiplier paired with the curvature of the constraint. These corrections are required for the constrained Hessian represented by the left side. They cannot be discarded by calling the ambient Wilson Hessian gauge invariant. The projection P_U is also different from the propagator projection whose range is $\Delta_U \ker Q'(U)$.

The comparison of (67) with the massive determinant has two sources of uniformly bounded terms. A term containing a constraint perturbation factors through the unit lattice. Its trace therefore runs over a bounded number of dimensions per unit cell, even though a Bloch fibre at bare spacing η_k has $O(\eta_k^{-4})$ components. The remaining resolvent expansion is stopped only after three propagator factors have occurred. The Schatten inequality then uses the finite four-dimensional integral $\int (1 + |q|^2)^{-3} \, dq$. Terms with one or two propagators are retained as the mass derivatives in $\mathcal{L}^\sharp$; their traces are not uniformly bounded by this integral.

All shifted resolvents in this calculation must lie in one fixed complex-momentum strip. In that strip, bounded normalised symbols and the real coercivity give uniformly bounded inverses by the conjugation argument below. It follows that the comparison remainder is holomorphic in the external momentum on one fixed polydisc. Its quadratic coefficient is bounded by Cauchy on that polydisc. The actual factorisation and resolvent expansion with these uniform bounds are the substantive determinant comparison input. The displayed saddle identities establish the exact constrained object to which it must apply.

The Gaussian part is handled as a sum, not by evaluating each step independently. Along a background, the product of the first k normalisers is a single Gaussian integral over the tangent space to $\{U : \mathcal{M}^k U = V\}$. Its logarithm has coefficient $\sum_{l \le k} b_l^0$. Changing gauge in that Gaussian integral expresses it through the gauge and site Laplacians with the average constraints. The

comparison replaces those constraints by a mass. If $\mathcal{L}(m)$ is the massive determinant functional in level-k units, use

$$\mathcal{L}^\sharp = \mathcal{L}(1) - \mathcal{L}'(1) + \tfrac{1}{2}\mathcal{L}''(1).$$

This is its Taylor polynomial of order two at mass one, evaluated at zero. Modulo a polarisation term $O(|p|^3)$, the difference from the constrained Gaussian functional is analytic and bounded on a momentum polydisc of radius independent of k. Cauchy's estimate therefore bounds the change in its quadratic coefficient by a constant independent of k.

The uniform polydisc is obtained from the flat-background operators. Their Bloch symbols are coercive after normalisation by $(\Delta+1)^{-1/2}$. Adding a small imaginary momentum is equivalent to exponential conjugation in position space. The symbols remain invertible on one fixed strip, and the resolvent comparison is uniformly bounded there. Three propagator powers are trace class per unit volume in dimension four. The two lower mass-derivative terms are kept explicitly in $\mathcal{L}^\sharp$, rather than estimated by a trace-class bound that does not apply to them.

The uniformity in the momentum strip has a concrete operator meaning. Let $T_k(\varkappa)$ be a flat-background Bloch symbol of an inverted operator in the comparison and normalise it by the two factors of $(\Delta+1)^{-1/2}$. Write the result as $\mathcal{T}_k(\varkappa)$. For real momentum the required coercivity estimate is

$$cI \le \mathcal{T}_k(\varkappa) \le CI,$$

uniformly in k. On the initial fixed strip the symbols are holomorphic and bounded in operator norm. Cauchy's estimate on a smaller strip gives

$$\|\mathcal{T}_k(\varkappa+iq) - \mathcal{T}_k(\varkappa)\| \le C_1|q|.$$

Take $|q| < c/(2C_1)$. The inverse exists there by a Neumann series and has norm at most $2/c$. This is one fixed strip for every level, including levels at which the Bloch fibre has more components. It uses the normalised operator norm, rather than estimates on individual matrix entries.

After the lower mass derivatives have been retained explicitly, the remainder comparison has three resolvent powers. Their trace per unit volume is uniformly bounded because

$$\int_{\mathbb{R}^4} (1+|q|^2)^{-3}\, dq < \infty.$$

The same majorant bounds the rescaled lattice integrals, whose Brillouin regions grow as the bare spacing decreases. The first and second resolvent powers would instead diverge in that comparison. Thus the two lower mass-derivative terms cannot be hidden in this uniform trace estimate.

For external momentum p in a still smaller polydisc, the shifted factors in the polarisation comparison remain in the common inverse strip. The trace majorant gives a bound C_{rem} on the analytic difference symbol, modulo $O(|p|^3)$. Its quadratic action coefficient is consequently at most $C_{\mathrm{rem}} r_\pi^{-2}$ by Cauchy in two distinct momentum directions. The reduction of the constrained determinant to these remainders and the massive determinant, with this uniform majorant, is the substantive comparison input. The strip and trace argument explains the cutoff-independent constant once that reduction is available.

In bare-lattice units the determinant functional to be estimated is

$$\log\det(\Delta_U + z) - \tfrac{1}{2}\log\det(\Delta_U^{\mathrm{W}} + D_U D_U^* + z), \qquad z = \eta_k^2.$$

Evaluating its quadratic action coefficient in a constant abelian field gives a function $\widetilde{\beta}(z)$. The determinant estimate used here is

$$\left|\widetilde{\beta}(z) - \frac{11}{6\pi^2}\log\frac{1}{z}\right| + |z\widetilde{\beta}'(z)| + |z^2\widetilde{\beta}''(z)| \le C_{\mathrm{uv}}, \qquad 0 < z \le 1. \tag{68}$$

The coefficient of $\mathcal{L}^\sharp$ is $\widetilde{\beta}(\eta_k^2) - \eta_k^2\widetilde{\beta}'(\eta_k^2) + \frac{1}{2}\eta_k^4\widetilde{\beta}''(\eta_k^2)$. Since $\log(\eta_k^{-2}) = 2k\log L$, this estimate and the bounded comparison error give the first bound in (58). The mass coefficient estimate and

the fixed-strip comparison are substantive analytic inputs; the summation of the error terms is performed only after these uniform estimates are available.

For the drift calculation, choose the coupling ceiling so that the small-field errors are smaller than a fixed fraction of c_0 and the remaining errors are uniformly small. The bounds on sums of b_l^0 then give constants $B_\sharp, \lambda_\sharp > 0$ and $C_\sharp < \infty$ such that

$$b_{k+1} \le B_\sharp, \qquad \lambda_\sharp(j-i) - C_\sharp \le x_i - x_j \le B_\sharp(j-i) \quad (i \le j). \tag{69}$$

To see the drift estimate directly from (58), put $\mathsf{B}_n^0 = \sum_{l=1}^n b_l^0$. The first estimate in (58) implies

$$|\mathsf{B}_j^0 - \mathsf{B}_i^0 - c_0(j-i)| \le 2C_0.$$

Choose the coupling ceiling so that $|r_l| + |q_l| \le \epsilon_\gamma < c_0/2$ at every constructed level. Summing (57) then gives

$$x_i - x_j \ge (c_0 - \epsilon_\gamma)(j-i) - 2C_0.$$

Also $|b_l^0| \le c_0 + 2C_0$, so $b_l \le c_0 + 2C_0 + \epsilon_\gamma$. These two consequences provide fixed admissibility constants. They use no sign of an individual b_l^0.

The lower estimate uses sums of Gaussian increments and does not assert that every individual Gaussian increment is positive. One may take $C_\sharp = 2C_0$ and, after decreasing the coupling ceiling, $\lambda_\sharp \ge c_0/4$. The large-field induction uses these bounds to sum its remainders. Conversely, the next increment is formed from the terms at levels already constructed, before the next large-field operation uses the updated coupling. This order closes the induction without invoking ultraviolet stability as a proof of its own flow bound.

2.10. **One coupling flow on all tori and first passage.** The coefficient in (48) is defined from local families on arbitrarily large tori. The local choices in the background and walk constructions are made by one rule before choosing the torus. On a domain that lifts injectively with a margin to $\mathbb{Z}^4$, the formula for a term is the same formula in either setting: it uses local backgrounds, walks, connected polymers, earlier terms and the already determined coefficients. Induction therefore defines one family $\widehat{E}^{(j)}$ that serves all tori. In the large-field representation the same assertion holds inside the regular region; the history there has empty standing trace after healed components have been summed into local terms, as explained in section 3.

It is important that the induction is simultaneous over all tori. The family defining b_j includes domains larger than any one torus. There is no convergent total interaction on the whole infinite lattice; its Hessian kernel is instead the absolutely convergent sum defining $\Pi^{(j)}$. The coefficient agrees with the large-volume limit of the coefficient of torus totals. Non-wrapping terms agree locally, wrapping terms decay exponentially in the torus side, and the background response converges to its infinite-lattice response. These three estimates justify that identification. In particular, b_j depends only on the previous couplings and the fixed choices, rather than on ℓ.

The geometric restriction on the torus is also uniform in the cutoff. Let $\check{R}_j = R(\min_{i \le j} x_i)$ be the large-field cube factor, where $R(x)$ is the least power of L not smaller than $(\log x)^{\mathfrak{r}}$ for a fixed exponent $\mathfrak{r}$. The levelwise torus condition is

$$\ell L^{K-j} \ge 2L^{11} M \check{R}_j. \tag{70}$$

The flow bound gives $\check{R}_j \le L\check{R}_{j+1}$, so the condition at the last level implies the preceding ones. In the terminal reference window it is enough to take

$$\ell \ge \ell_\star(g) := 2L^{11} M R(g^{-2} + B_\sharp). \tag{71}$$

There is no dependence on K in this lower bound. This is the volume uniformity needed before taking the infinite-volume limits.

For $0 < g \le \gamma$ and $\beta > g^{-2} + B_\sharp$, stop the flow at

$$K(\beta, g) = \min\{k : x_k \le g^{-2} + B_\sharp\}. \tag{72}$$

Before exit, the upper increment bound implies $x_{k+1} > g^{-2} \geq \gamma^{-2}$, so the next step stays within the small-coupling domain. The positive lower drift ensures that exit occurs. At exit, $g^{-2} < x_K \leq g^{-2} + B_\sharp$, and

$$\frac{\beta - g^{-2}}{B_\sharp} - 1 \leq K \leq 1 + \frac{\beta - g^{-2} - B_\sharp + C_\sharp}{\lambda_\sharp}.$$

Thus increasing β gives unbounded exit depths. The physical unit is chosen so that this terminal scale has length one; the bare spacing is then $L^{-K(\beta,g)}$. This selects the ultraviolet sequences used by target theorem A. It does not fix the last inverse coupling at a single point of the reference window.

At depth s below the last level, the flow has the cutoff-independent envelope

$$g^{-2} + \lambda_\sharp s - C_\sharp < x_{K-s} \leq g^{-2} + B_\sharp(s+1).$$

The errors in (58) can therefore be summed in depth rather than bounded by their value at the largest coupling. Their sum gives

$$|x_{K-s} - x_K - c_0 s| \leq C + C\sqrt{s}\,\log^{\mathfrak{p}_0}(g^{-2} + s) = o(s) \qquad (s \longrightarrow \infty, g \text{ fixed}).$$

In particular, $\beta/\log(1/a_K) \longrightarrow 11/(3\pi^2)$ and $g_0 = \beta^{-1/2} \longrightarrow 0$. At a prescribed physical separation the same envelope controls the coupling at the corresponding depth. This is the use of the flow in the two-point comparison. The subsequence construction and the subsequent phase selection use uniform bounds and do not require tuning or uniqueness at fixed renormalised coupling.

3. Large fields, stored variables and the sum over histories

The fluctuation expansion applies only while both the background curvature and the fluctuation are small. The Wilson measure is not supported on that set. At each scale the construction therefore inserts local decompositions of unity, retains the variables where a test fails, and continues the small-field integration elsewhere. This section describes how the retained regions are controlled. The estimates are also needed with sources: a small probability for one bad region is insufficient unless it remains small under later integrations and complex changes of the exterior field. The removal and exponentiation follow Bałaban's large-field method [B89], whose Theorem 1 assumes small effective couplings. The estimates must remain local in the recorded history, control exterior suprema inside the component integrals, persist on the complex-source domains, and pay a prescribed weight for the age of every live region. We construct the operations in this order so that these four requirements enter at the estimates where they are used.

3.1. Tests and nested regions.

Write $x_j = g_j^{-2}$ and choose thresholds

$$p_0(g) = A_0(\log g^{-2})^{\mathfrak{p}_0}, \qquad \varepsilon_j = g_j p_0(g_j), \qquad \delta_j = (A_1/A_0)\varepsilon_j.$$

The constants A_1/A_0, and later the upper bound on the couplings, are small. A plaquette test bounds the localised background curvature by ε_j in the appropriate scale. An approximate fluctuation test compares the old block field with the average of that localised background. Where these tests pass, the true multiscale background is formed, and the logarithmic fluctuation can be tested as well.

An indicator includes existence of all the objects used to define it. If a logarithm or regular background is undefined, the indicator is zero and its complementary indicator is one. Thus each test gives an identity on the entire configuration space. Small-field regularity is used to estimate the terms that result, rather than assumed before the decomposition is made. Gauge-fixing insertions are likewise normalised identities on every configuration, with their specified extension where local coordinates do not exist.

Let Ω_j be the region where the background and fluctuation are defined and let $\Lambda_j \subset \Omega_j$ be the region on which the fluctuation is integrated. The complements are enlarged around failed tests and

around previously retained regions. Their boundaries are separated by several layers of cells. The resulting nesting has the form

$$T_\ell \supset \Omega_1 \supset \Lambda_1 \supset \Omega_2 \supset \Lambda_2 \supset \cdots .$$

At scale j the cells have side $M\check{R}_j$, where $\check{R}_j$ is the smallest power of L at least $(\log \min_{i\le j} x_i)^{\mathfrak{r}}$, with $\mathfrak{r} \ge 2$. Taking powers of L keeps the partitions nested. The running minimum makes $\check{R}_j$ nonincreasing; the bounded change of x_j ensures that consecutive drops in $\check{R}_j$ do not occur at sufficiently small coupling.

The boundary layers have an analytic purpose. Let $\mathcal{A}_j$ be the unscaled logarithmic fluctuation. Conditionally on its stored part outside Λ_{j+1}, its mean inside is

$$m_j = -C^{(j)}(\Lambda_{j+1})\Delta^{(j)}\mathcal{A}_j|_{\Omega_{j+1}\setminus\Lambda_{j+1}}, \tag{73}$$

where $C^{(j)}(\Lambda)$ is the inverse of the restricted quadratic form. A large stored field close to the boundary could shift this mean past an interior threshold. The construction therefore also tests a collar of stored bonds and the rim touching the integration region. The bounds there are respectively $\delta_j/(4\Theta_1)$ and $\theta_c\delta_j$, with $0 < \theta_c < 1$. Exponential decay of the conditional covariance gives

$$|m_j| \le C(\Theta_1^{-1} + \theta_c e^{-c\varpi_T} + e^{-cM\check{R}_{j+1}})\delta_j < \tfrac{1}{2}\theta_c\delta_j. \tag{74}$$

Here ϖ_T is the collar width. First choose $\theta_c\Theta_1$ and ϖ_T sufficiently large, and then reduce the coupling ceiling. Away from the collar and rim, the retained tests give $|\mathcal{A}_j| \le C\delta_j$; this bound is needed for the far-field tail in (74). Each failed collar or rim test enlarges the retained region. Every round examines new bonds or cells, so this enlargement terminates on a finite torus and is still a decomposition of unity.

The geometry is formed before an estimate is applied. At step $t = j + 1$, let P_t, Q_t, R_t be the failed plaquette, approximate-fluctuation and true-fluctuation test sets. The first two are defined using localised backgrounds. The set Ω_t^c is obtained by enlarging their union with the hull of the old live region, and Λ_t^c by enlarging $\Omega_t^c \cup R_t$. A boxing rule replaces sufficiently small components by boxes. Collar and rim tests can require further enlargement. Each new round examines previously unexamined cells, so the rule is total on a finite lattice. At a point where the true background is not yet defined, its failed existence test has already placed that point in the retained region.

The distinction between approximate and true fluctuations is needed for locality. The approximate test determines whether the global regular background can be formed. The true test, once that background exists, determines which Gaussian variables are integrated. The two cannot be interchanged when a history is compared on a local window.

On a strip next to $\partial\Lambda_t$, a bondwise change of variables replaces the true background in the kept tests by its localised version. In local coordinates this change has the form

$$W' \longmapsto W' e^{-i\theta_{\mathrm{co}}(W')H} e^{iH} V^{\mathrm{loc}}, \qquad |H| \le C\varepsilon_j e^{-\check{R}_j}.$$

The cutoff θ_{co} equals zero in the small-fluctuation range and one beyond the kept range. For constant cutoff this is a translation with unit Jacobian. Its nonconstant part is a local boundary Jacobian. It belongs to the boundary interaction. The construction requires that after this substitution the retained tests depend only on the stored variables. An old test reading a pivot bond uses the value obtained from that bond's individual averaging constraint, rather than the value at the averaged background.

The new tests and the carried constraints imply the old small-field tests through the estimate

$$CL^{-2} + C\delta_j/\varepsilon_j + Ce^{-\check{R}_j} < 1. \tag{75}$$

The three terms are, respectively, the curvature scaling, the fluctuation displacement, and the effect of the artificial boundary of a localised background. Choose L first, the ratio A_1/A_0 second, and the

coupling ceiling last. This order concerns the tests that remain on the support of a term, including its stored constraints. It does not assert that a new plaquette test alone implies every old test.

3.2. **Exact measures and effective actions.** A history records the outcomes of the tests, the regions at every scale, and the operations performed on their components. Stored variables are part of the history's state. At the measure level each move is a test, a change of variables, an integration, or a draw from a normalised auxiliary law. If $K_h(U, d\xi)$ is the positive kernel associated with a complete sequence of outcomes, exactness gives

$$\sum_h K_h(U, \mathcal{V}_k) = 1, \qquad \tau_h[\nu_0](d\xi) = \int K_h(U, d\xi)\, \nu_0(dU). \tag{76}$$

In particular,

$$\sum_h |\tau_h[\nu_0]|(\mathcal{V}_k) \le |\nu_0|(\mathcal{V}_0), \qquad |\tau_h[w\mu_0]| \le \|w\|_\infty \tau_h[\mu_0] \quad (\mu_0 \ge 0).$$

Here $\mathcal{V}_k$ is the level-k state space of the current block field, its history and all retained stored data. These identities do not depend on convergence of an expansion. The expansions provide formulas and estimates for the measures on the supports of the tests. Freezing the real construction freezes the kernels as well, which is the reason the same procedure can later transport a complex source.

For a test the two kernels are multiplication by χ and by $1 - \chi$. A measurable map supplies a deterministic kernel, and a reference draw supplies a probability kernel. Integration and removal of an auxiliary coordinate are marginalisations. Composition of these kernels shows (76) by induction, because the sum of the outcome masses is one at each move. Undefined analytic objects cause a complementary outcome, not a missing mass.

The total-variation inequality follows at the same level. For every bounded measurable F with $|F| \le 1$,

$$\left| \int F\, d\tau_h[\nu_0] \right| \le \int K_h(U, \mathcal{V}_k)\, |\nu_0|(dU).$$

Taking the supremum over such F, then summing over h, gives the first inequality following (76). Replacing $|\nu_0|$ by $|w|\mu_0$ gives the domination of a bounded weight. For a finite bare lattice the source weight $e^{A(w,U)}$ is bounded, so the same frozen kernels transport it. No source enters their tests, normalisers or auxiliary probability laws. Later holomorphic bounds concern the resulting complex integrals, not complex probability kernels.

These identities also construct the observable that must be retained when a reference law changes a block field. The joint measure of bare field, history and current block state is

$$\mu_0(dU) K_h(U, d\xi).$$

Conditional expectation with respect to (h, ξ) gives the transported version $F^{(h)}(\xi)$ of a bounded bare observable. It satisfies $|F^{(h)}| \le \|F\|_\infty$ and the exact insertion formula

$$\langle F \rangle_{\mu_0/\mu_0(1)} = \frac{\sum_h \int F^{(h)}\, d\tau_h[\mu_0]}{\sum_h \tau_h[\mu_0](\mathcal{V}_k)}. \tag{77}$$

Where all the operations act away from the bonds on which an averaged observable depends, its transported version is that same function of the block field. Inside a replaced box this identification is generally false. Gauge invariance permits the usual representative changes; it does not make a function of a replaced plaquette independent of the replacement.

The kernel construction and the expansion describe different aspects of the same measure. The kernel statement is exact before estimates; the expansion gives the following presentation of its

terms. Let $\mathcal{S}$ record the action regions and their standing marks. Then

$$\begin{aligned} \rho_k(V_k) &= \sum_{\mathcal{S}} \mathbf{T}'_k(\mathcal{S}) \left[\chi_{k,\mathcal{S}} e^{A_{k,\mathcal{S}}(U_k(\mathbf{V}))}\right], \\ A_{k,\mathcal{S}}(U) &= -A(x_k(\cdot), U) + \mathbf{E}_k(U) + \mathbf{R}_k(U) + \mathbf{B}_k(U, \underline{\mathfrak{f}}) - \mathcal{E}_{k,\mathcal{S}}. \end{aligned} \tag{78}$$

Here $\underline{\mathfrak{f}}$ denotes the stored fluctuation fields. The operator $\mathbf{T}'_k(\mathcal{S})$ integrates the fields stored in the live components, with their delta constraints, tests, gauge fixing and conditional Gaussian forms. It retains the integral operations of islands. Except for their core Wilson factors it contains no Wilson weight. The weight of the label is inside the exponential on which the operator acts. In particular it cannot be discarded in an estimate of the operator on one.

The local sums in (78) are

$$\begin{aligned} \mathbf{E}_k(U) &= \sum_{j\le k} \sum_{(z,X)\in \mathcal{Z}_j^{\mathrm{pt}}(\mathcal{S})} [E^{(j)}(X, U, z) - E^{(j)}(X, 1, z)] - \sum_{j\le k} b_j A(\varphi_j, U), \\ \mathbf{R}_k(U) &= \sum_{j\le k} \sum_{X\in \mathcal{R}_j(\mathcal{S})} [R^{(j)}(X, U) - R^{(j)}(X, 1)], \\ \mathbf{B}_k(U, \underline{\mathfrak{f}}) &= \sum_{j\le k} \sum_{X \text{ meeting the level-}j \text{ boundary}} B^{(j)}(X, U, \underline{\mathfrak{f}}). \end{aligned} \tag{79}$$

Here $\mathcal{Z}_j^{\mathrm{pt}}(\mathcal{S})$ is the retained pointed range, with base point in Λ_j and domain in its prescribed enlargement. The remainder range $\mathcal{R}_j(\mathcal{S})$ stays one level-j cell inside Λ_j. The precise enlargements and boundary origins are those fixed by the localisation rule; the ranges are not reselected from the value of the density. The boundary sum includes its distinct creation origins, whose constants are controlled separately below.

The same functions $E^{(j)}$ and $R^{(j)}$ serve all labels. The arguments, retained ranges and boundary functions can depend on the label, as can $\mathcal{E}_{k,\mathcal{S}}$. For the never-large label the boundary sum is absent and the expression is

$$\chi_k \exp[-x_k A(U_k) + \mathbf{E}_k(U_k) + \mathbf{R}_k(U_k) - \mathcal{E}_{k,\varnothing}].$$

This is the term retained for the floor. It also includes the effects of histories whose large regions were all healed and summed into remainder terms.

The net Wilson coefficient is preserved pointwise. Indeed,

$$-x_k(\cdot) - \sum_{j\le k} b_j \varphi_j = -\beta.$$

The regular action coefficient and its counterterm have been separated to make the next small-field step possible. They do not constitute two independent changes of the bare action. The same net-coefficient identity must hold through partial preparations, with their separate Wilson pieces.

The small-field integration stores V_j outside Ω_{j+1}. Inside Ω_{j+1} it integrates the scaled fluctuation $A = g_j^{-1}\mathcal{A}_j$ on Λ_{j+1}, conditionally on the stored part A'. Completing the square leaves the form

$$\Delta^{(j)} - \Delta^{(j)} C^{(j)}(\Lambda_{j+1}) \Delta^{(j)} \tag{80}$$

on A'. The Wilson action is not included in this conditional operation; it remains in the function on which the operation acts. This distinction locates the action that will pay for a failed plaquette test.

Factorisation over disconnected retained components requires a local choice of the form kept on their stored fields. The tests are first recentered at localized backgrounds by a change of variables; its Jacobian is assigned to a boundary interaction. The pivot bonds are then solved from their individual averaging constraints. The conditional form is cut only after these operations.

Chains and their assignment. Write $\mathsf{A} = \mathsf{C}^{\mathsf{T}}\Delta^{(j)}\mathsf{C}$ on the free variables, and split them into integrated and stored sets I, E. For a stored field a, completing the square gives

$$\langle B, \mathsf{A}B\rangle = \langle B_{\mathrm{I}} + \mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}}a,\ \mathsf{A}_{\mathrm{II}}(B_{\mathrm{I}} + \mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}}a)\rangle + \langle a, \mathsf{S}a\rangle, \qquad \mathsf{S} = \mathsf{A}_{\mathrm{EE}} - \mathsf{A}_{\mathrm{EI}}\mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}}.$$

Expand A in walks and its restricted inverse by the small-cube parametrix. Then $\mathsf{S} = \sum_{\varpi} \mathsf{d}_{\varpi}$, where a chain is either one walk term or a walk, a resolvent walk and a walk. Its entire connected carrier belongs to one term of the expansion.

A chain is retained when the level-$j+1$ cells it meets intersect the stored strip $\Omega_{j+1} \cap Z_{j+1}$ in a nonempty connected set and stay within one layer of that set. It is assigned, as a whole, to the strip component D containing that set. The decision uses only the carrier of the chain. Selecting all matrix entries whose endpoints belong to one global component would not have this property: a distant joining could change that selection while leaving the local field and history unchanged.

Every rejected chain has length at least that of one large-field cell, $W_{j+1} = LM\check{R}_{j+1}$ in level-j units. At $|a(b)| \le Cp_0(g_j)$, the tail of the chain expansion consequently gives

$$|\mathsf{B}^{\times}(Y)| \le Cp_0(g_j)^2 (LM)^8 e^{-c\delta_\star W_{j+1}} e^{-\kappa' \# Y}. \tag{81}$$

The rate κ' is prescribed before taking $\delta_\star LM$ large. The exponential in $\check{R}_{j+1}$ pays for the stored field squared and any fixed logarithmic enlargement of its range. With $\tau_j = x_j/\theta_j$, the exact decomposition of the remaining conditional exponent is

$$-\tfrac{1}{2}\tau_j\langle a, \mathsf{S}a\rangle = -\tfrac{1}{2}\sum_D \langle a, \Delta_D^{\mathrm{loc}} a\rangle + \sum_Y (\mathsf{B}^{\times} + \mathsf{B}^{\tau})(Y), \qquad \sum_Y \mathsf{B}^{\tau}(Y) = -\tfrac{1}{2}(\tau_j - 1)\sum_D \langle a, \Delta_D^{\mathrm{loc}} a\rangle.$$

The companion term has only

$$|\mathsf{B}^{\tau}(Y)| \le C(M)|1 - \tau_j|\, \mathfrak{b}^2 e^{-\kappa' \# Y}$$

when $\mathfrak{b}$ bounds the stored field. It has no long-chain factor; its smallness comes from the frozen run's weight tube. In the unsourced run $\theta_j = x_j$, it vanishes.

The local coercivity comparison. The required bound is

$$\langle a, \Delta_D^{\mathrm{loc}} a\rangle \ge c\gamma_0 \sum_{b \in D} |a(b)|^2. \tag{82}$$

For the full form $\mathsf{A} \ge \gamma_0$, its genuine Schur complement satisfies this lower bound: minimize $\langle (B_{\mathrm{I}}, a), \mathsf{A}(B_{\mathrm{I}}, a)\rangle$ over B_{I}. This argument does not apply to Δ_D^{loc}, a partial chain sum. Nor can it require a global regular extension of the pair if the stored factor is defined only near D.

The local construction uses a quadratic partition on small cubes, a regular gauge and cutoff extension on each cube, and comparison with a genuine Schur complement inside $D^{\sim}$. The algebra of the partition is explicit. For a symmetric kernel K and real functions with $\sum_\alpha \chi_\alpha^2 = 1$,

$$\sum_\alpha \langle \chi_\alpha a, K\chi_\alpha a\rangle - \langle a, Ka\rangle = -\tfrac{1}{2}\sum_{b,b'} \langle a(b), K(b,b')a(b')\rangle \sum_\alpha (\chi_\alpha(b) - \chi_\alpha(b'))^2.$$

Thus its error is controlled by the weighted kernel row sums and the small-cube cutoff differences. On each localized cube the real field is gauged near one and cut off, so the full torus form has its already established gap. The Schur comparison is made with the weight $e^{c\,\mathrm{dist}(\cdot, D)}$, suppressing the influence of the remote artificial boundary. At a globally regular real pair the proposed comparison reads

$$\Delta_D^{\mathrm{loc}} \ge \gamma - Ce^{-c\delta_\star W_{j+1}}$$

if the full form has gap γ. For a pair given only on $D^{\sim}$, the same argument must be checked after the local gauge and cutoff, with a fixed fraction of the gap left after localization. This uniform comparison, including the partition error, is the analytic verification needed for (82). The small

scalar tail bound (81) at bounded stored fields is a different estimate; it is not by itself an operator lower bound.

Carrying a form to later backgrounds. The retained form and the elimination factor are formed once as functions of the pair near their carrier. At a later level they are evaluated at the background reached then. Their tests, gauge factors and delta constraints do not move with that background. A move therefore creates the potential

$$-\tfrac{1}{2}\langle a, [\Delta_D^{\mathrm{loc}}(\mathbf{p}^\sharp) - \Delta_D^{\mathrm{loc}}(\mathbf{p}^\flat)]a\rangle,$$

not a newly chosen truncation rule. For a piece with carrier R, its pair oscillation is at most

$$C(M)p_0(g_{l-1})^2 e^{-\kappa d_l^\sharp(R)}.$$

On the enlarged far-field discs and near-field switches this has the additional factor $(\lambda^{\mathrm{st}})^2$, since the piece is quadratic in those variables.

Localize the difference with the same cube parameters as the response. Cauchy gives a factor per added cube; Schwarz gives the relative response amplitude; the old factor already has decay across its own carrier. At depth p, the number of old cubes per current cube is L^{4p}, and

$$L^{4p}e^{-cL(p-1)_+} \le CL^4.$$

The response therefore pays this multiplicity. After also charging the switches, the resulting stage piece has the form

$$|V(Y)| \le C(\log\theta_j)^{-(q-3\mathfrak{p}_0-C\mathfrak{r})}e^{-cd_k^\natural(Y)}.$$

One power $\mathfrak{p}_0$ comes from the physical fluctuation, two from the stored form. This is why the radius exponent must satisfy $q > 3\mathfrak{p}_0 + C\mathfrak{r}$. The smallness of this carried difference is compatible with a large negative stored-only exponent; no absolute bound on that exponent is required.

The state carries two region sequences. One determines the averages and the current background. The other determines the already produced action terms, their profiles, tests and vacuum constants. They agree at ordinary step boundaries. During a partial preparation they need not agree, and both belong to the recorded history. A completed integration changes the determining sequence without retroactively changing where an old term was created or where its counterterm was assigned.

More precisely, a history trace on a window records the intersection with each region at that region's own scale. It also records the unions of the marked components, their stored regions and nests, separately for each mark level, kind and completed stage number. Three stored-bond sets are recorded separately: collar, rim and cutoff-zone bonds. The regions alone do not determine these sets. This trace, together with the local terms and coefficients entering the window, determines the new term formed there. Transport between two settings requires a local translation by their cell lattice, a common nonwrapping lift, and equality of this entire trace.

At a real pair every stored operation is positive. Reality of the individual analytic activities needs its own induction. The involution is $\mathbf{U} \mapsto (\mathbf{U}^*)^{-1}$, $\mathbf{J} \mapsto \mathbf{J}^*$, together with complex conjugation of scalar parameters. The real local choices, series, integrals, unique critical solutions and the logarithm branches based at positive real integrals commute with it. Starting from real β gives real activities at real pairs. This does not identify a complex Gaussian with a positive measure.

The averaging constraints also leave elimination factors when their pivots are solved. If $\mathsf{m}_c(W)$ is the derivative of the average at the pivot of a coarse bond, the real input required by this chart is

$$\det(L\mathsf{m}_c(W)) \ge \frac{4}{\pi^2}L^{-15} > 0 \tag{83}$$

where no contour loop of that average is -1. Its elimination factor is $|\det \mathsf{m}_c(W)|^{-1}$. The bound must apply on real large-field configurations in this chart as well as at the regular background. It is a property of the actual symmetric average, included in the pivot construction; it is not inferred from the sign of a Gaussian quadratic form. Outside such charts the total Borel rules and complementary tests are used.

On a regular chart the unique fixed points, means and gauges are limits of their continuous iterations and hence are Borel. The local tests, deterministic maps and normalised reference draws therefore give Borel kernels there. Their specified total continuations and the sets on which the regular objects are declared to exist must be Borel as well. These must be checked before assigning the history law or integrating its witnesses. Local analyticity on a regular chart does not settle this total Borel continuation.

Positivity of the spatial Wilson weights is separate again. The profiles φ_j are fixed by the action sequence and defined in section 4. Away from the final prepared layers, the telescoping array is either β or a convex combination of two successive x_j. In the final thin shells its additional vertex is $x_k + b_{k-1}$; the one-step lower flow bound makes this vertex positive. The full profile minimum and its proof are given in (179). These facts give positive reference denominators on the supports of the real construction.

3.3. **Interrupted operations and superposition of histories.** An interrupted operation retains more than its current regions. Suppose an operation at level t stops after $n(X)$ stages, and write $t_\flat = t - N_t$. Its mark stores the regions $\Omega_l^\circ, \Lambda_l^\circ, S_l^\circ$ that were present before it began, as well as the prepared nest $Z_l''(X)$. For $0 \le \nu(X) \le n(X)$ define the intermediate determining regions on X by

$$\Omega_l^\nu \cap X = \begin{cases} X \setminus Z_l''(X), & t_\flat < l \le t_\flat + \nu(X), \\ \Omega_l^\circ \cap X, & t_\flat + \nu(X) < l \le t_\flat + n(X). \end{cases} \tag{84}$$

Other entries are unchanged. The choice $\nu = n$ describes the fields that remain and determines the background and its propagators. The choice $\nu = 0$ describes where the old action terms, tests, profiles and constants were made. The hereditary domains include the intermediate choices as well.

A mark at level s inside a class component at level t has $s < t - N_t$. Its window is disjoint from the present waiting window. Marks that overlap in space are therefore nested with disjoint windows, and at one location and level only one interrupted mark is active. The intermediate choice in (84) is realised by the history in which that operation stops at ν stages. The later geometry is unchanged: creations, component classes and nests do not read how far those interrupted stages had progressed. Thus the intermediate domains are tied to actual histories, not to fictitious interpolations of regions.

The local superposition rule has two ingredients. First, the margin around a non-touching component exceeds the reach of every bond test, pivot and collar. Taking a hull and adding layers commute with unions; boxing is done one component at a time. An ordinary step therefore combines two histories whose regions never touch by acting separately on their components.

Second, the operation class looks back through a waiting window. Nothing created within that window can already have been healed. Indeed, suppose something created at step $s \ge t - N_t$ were healed at $i < t$. Healing would require $s \le i - N_i$, or the stricter $s < i - N_i$ for a mark. Since $N_i \ge N_t$,

$$i - N_i \le i - N_t < t - N_t \le s,$$

which is a contradiction. The creations and marks needed to decide membership still lie in their respective live regions. This window argument is necessary because the overall evolution is not monotone: a healed place can become large again later.

Consequently each history is a union of clusters. A cluster is rooted either at a live component of the last level or at a component at the level where it is healed. It contains all its predecessor components and everything recorded in them. Every component has one last successor and hence one cluster root. Non-touching clusters can be interchanged between histories. In particular, deleting the entire clusters of healed components leaves a history with the same final label. This is its standing part $h^\flat$. Deleting only the last healed box and retaining its old tests would not give this history.

Fix standing data $\bar{h}$ at level k. The histories that reduce to it after the current healed clusters are erased are parametrised by a family $\mathfrak{X}$ of pairwise non-touching possible healed components,

disjoint from the standing live region, and by one internal history for each $X \in \mathfrak{X}$ in which X is healed and none of its predecessors was healed earlier. The local choices of these internal histories are independent. Their stored integrals remain coupled through the exterior argument, which is why the integrated supremum majorant is needed later. This parametrisation supplies the hard-core component sum used by exponentiation.

The trace on a window must remember the action sequence, interrupted nests, stage counts and the separate collar, rim and cutoff-zone bond sets. For complete equality of the natural analytic domains, two further geometric facts must match under a chart: the sets of bare bonds carrying the fields, and whether two bare unit cubes belong to one drop-out mark. A nonconvex domain can cut a mark into two visible pieces, so its visible union does not determine that second relation. On unmarked domains inside the first region these extra clauses are vacuous.

If a current window lies a cell inside the small-field region, its standing trace is empty. The regular term formed there is therefore the term of the never-large standing data, already including the summed healed clusters. This is the local identity that makes the interior families independent of labels and, under a common nonwrapping lift, of the torus and cutoff. The all-tori companion and the coefficient sequence use this identity, not only a bound on a label-dependent term.

3.4. **Removing a small old component.** Without a further operation, a retained region would remain in every later label. Even very small probabilities would then be summed over the cells of all finer scales. The large-field operation acts on a component that has become a small box and has had a sufficiently long interval with no new failure, no joining to another component, and no intervening operation. Let this interval have length $N_k = \check{R}_k$, ending at level k. These restrictions ensure that the component has a single nested chain of boxes throughout the interval.

The stored fields are first integrated in successive shells. This reduces a box of side $M\check{R}_k$ to a prepared box of side $O(M)$, with multiscale shell data on its boundary. The preparation uses the sharper fluctuation threshold

$$p_1(g) = (\log g^{-2})^{\mathfrak{p}_1}, \qquad \delta_j' = g_j A_1 p_1(g_j), \qquad \mathfrak{p}_1 < \mathfrak{p}_0.$$

One needs both the accumulated implications between successive tests and enough action to pay for a failed sharper test. The exponent conditions have the form

$$\mathfrak{p}_1 < \mathfrak{p}_0 - C\mathfrak{r}, \qquad 2\mathfrak{p}_1 - C\mathfrak{r} > \mathfrak{p}_0. \tag{85}$$

The exponents are chosen to leave these margins before the coupling ceiling is fixed.

For a component that passes the preparation tests, integrate the block field in the prepared box and replace it by a normalised local reference law. This law contains a tree gauge, a cutoff, and the local Wilson weight with coupling x_k. Its denominator is the integral of exactly that numerator over the box. Multiplying by the previous mass preserves the integral. A finite average over cubic frames keeps the operation covariant; the numerator, denominator, trees and tests must all use the same frame in each summand.

There are three outcomes. If all tests pass, the component is healed and removed from the label. If only the final small-field test fails, the operation is kept as an island with no inner regions. If a preparation test fails, the component remains with the integrations already performed; this is a drop-out. Islands and drop-outs are the marked components. Each records the failed test and its extracted gain. A component still retained after these operations is called live. Write $\mathcal{Z}_k$ for Z_k with the components healed at step k removed; it is the input region of step $k+1$. The standing label consists of the unhealed regions and their component marks, after histories differing only in healed regions have been summed. Islands and drop-outs retain their local measures and factors. The regions used to form the current background and the regions used to index the already constructed action terms must then be recorded separately. A partial shell integration changes the former without authorising a new sorting of the latter.

The reference-law replacement preserves total mass and observables independent of the replaced variables, while changing the coarse-field distribution inside the box. Observables involving those variables are transported by the kernels in (76) and the exact integral identities.

We describe the membership and the main integral more precisely. There are five requirements. The depth condition is $k \geq N_k$ and availability of the compatible intermediate cube scales. The size condition puts X in a cube of side $CM\check{R}_k$. Quiet means that no creation or joining occurred during $k - N_k < t \leq k$. Quarantine excludes marks placed in the preceding waiting interval. Finally, each $X \cap Z_j$, for $k - N_k \leq j \leq k$, must be one component and one box; these boxes form its nest. Before the depth condition is met the class is empty. The free evolution of a quiet component contracts its linear size over two levels. It acquires the geometric conditions after a contraction period and the waiting interval; a new event restarts this wait.

Let ϕ_X be the construction's cutoff supported in X and equal to one well inside it. Let Λ^{pr} be the prepared box and V_Λ the critical orbit of its local action, with the fixed exterior data, and let $U_{k,X}$ be the background localized in X. Put $V'_k = V_k V_\Lambda^{-1}$. In a fixed cubic frame the reference law is

$$\mathfrak{s}_X(V_k) = \frac{\delta_{\mathrm{ax}}(V'_k)\chi(V'_k)e^{-x_k A(\phi_X, U_{k,X}(V'_k V_\Lambda))}}{\displaystyle\int dV'|_{\Lambda^{\mathrm{pr}}} \delta_{\mathrm{ax}}(V')\chi(V')e^{-x_k A(\phi_X, U_{k,X}(V' V_\Lambda))}}. \tag{86}$$

The cutoff restricts the free tree-gauge variables to $|\frac{1}{i}\log V'(b)| < M_0\varepsilon_k$. Both occurrences of ϕ_X, both local backgrounds and all tree choices in this quotient are identical. If its denominator is not finite and positive, the construction declares a complementary outcome. At real data on the specified supports it is finite and positive. The analytic estimates must also keep the denominator nonzero on the required complex domain. This latter assertion does not follow merely from real positivity.

The cubic frame average is inserted after integrating the old box and before imposing the new tree gauge. Subsequent choices are made in that frame. A failure after this point retains its frame. For an earlier drop-out, the identity $1 = |W_4|^{-1}\sum_{r\in W_4} 1$ may be used in its mass bound, but no tree draw has taken place. This placement keeps the reference replacement linear in the input measure and preserves every outcome mass.

The preparation integrates shells in the order $j = k - N_k, \dots, k-1$. Every shell receives the coupling increments missed while it was stored. The remaining multiscale datum consists of thin shells around the old core, and the main integral uses one scaling g_k on all of them. The old core operation is not unfolded; it retains the earlier failed tests and their paying action. The least-action continuation is taken only in the prepared box of side $O(M)$, among its regular critical orbits. Its constant loss depends on M and is independent of $\check{R}_k$.

3.5. **Preparing the shells and restoring their couplings.** Put $k_\flat = k - N_k$. On a component in the operation class the quiet geometry has an explicit form. If H_i is the hull in level-i cells of $X \cap Z_{i-1}$ and $H_i^{[r]}$ is the box with each face expanded by r such cells, then

$$\Omega_i^c \cap X = H_i^{[8]}, \qquad Z_i \cap X = H_i^{[10]}, \qquad k_\flat < i \leq k. \tag{87}$$

The boxes at a level are concentric. The single-box condition excludes a joining within this window even if the joined region is small and no new test fails. Without it the preceding layer could have two components and no nested shell tree of this form.

The prepared box is the final nest box with one layer of M-cubes added. Its diameter is CM in level-k units. The old core has diameter at most

$$CM\check{R}_{k_\flat} L^{-N_k}$$

in those units. Shrinking the core is a consequence of waiting; the stages themselves thin the outer shells. The admissible torus condition puts every box and all its needed neighbourhoods in a nonwrapping region.

The gauge tree on the prepared datum uses bonds of all the shell lattices. For a pair of boxes $P_1 \supset P_2$, take axial paths issuing from one site and going around P_2. The family is stable under enlarging P_2: delete the paths of the removed sites and obtain the chosen tree on the smaller shell. This stability is why the geometric class uses boxes. A shell tree fixes every site gauge except one. For each inner box add one bond crossing its boundary, and add one at the outermost box. They join the shell trees into one tree T_0, rooted at the exterior endpoint of its entering bond. Haar invariance gives unit Jacobian for this tree gauge.

The bare averaging chain is resolved from the highest level down:

$$\int dV_i(c)\, \delta(\mathcal{M}(V_{i-1})(c)V_i(c)^{-1}) = 1.$$

It leaves a free integral over the prepared multiscale datum and the unopened old core. Its schematic measure, with every shell at its own scale, is

$$\chi_k\chi_{k,\Lambda} \int dV_{k_\flat}|_{C_{k_\flat}}\, \chi\, \mathbf{T}_{k_\flat}(C_{k_\flat}) \int dV''|_{\mathbf{B}''\setminus C_{k_\flat}}\, \chi''_k e^{A''_k(U''_k)}. \tag{88}$$

Here $C_{k_\flat} = X \cap Z_{k_\flat}$, $\chi_{k,\Lambda}$ is the regular-extension test, and the primes denote the data, tests and action after the stages. The field expanded in the main integral is the free datum V'', not the stored bare core field. Its background datum is

$$V_0 = (\mathcal{M}^j U_0 \text{ on the shell of level } j)_{k_\flat \le j \le k}, \qquad V'' = e^{ig_k B}V_0, \qquad |B| < \delta'_k/g_k.$$

One scale g_k and one test control all these components of B.

The extension test also has a paying alternative. For exterior curvature bounded by ε, choose axial gauge on the boundary of the prepared box and continue by one inside. This gives a continuation with curvature $CM^2\varepsilon$. Thus absence of an extension in the required radius forces an exterior plaquette of size at least $cM^{-2}\varepsilon_k$. The failed test is controlled by the same local action payment as a plaquette failure. It must not be declared impossible from exterior smallness at an unspecified radius.

The critical extension on the prepared box. The stages have reduced the free box to side $O(M)$. This permits a fixed-radius argument, rather than an inverse estimate growing with $\check{R}_k$. Let $P = \Lambda^{\mathrm{pr}} \cap \mathbb{T}^{(k)}$, let $E(P)$ be the bonds having an endpoint in P, and let G be its tree rooted at one exterior site. Write $\mathcal{X}$ for the real Lie-algebra fields on $E(P)$ that vanish on G. All constants in this box argument may depend on M, fixed before the coupling ceiling.

Start with a real regular extension $V^\sharp$. An exterior increment b vanishes on $E(P)$; a free increment δ is supported there. If $\mathsf{W}_{V^\sharp}$ denotes the background continued from this datum, set

$$F(\mathsf{b},\delta) = \sum_{p\subset\Omega_0} \left[a_p(\mathsf{W}_{V^\sharp}(\mathsf{b}+\delta)) - a_p(\mathsf{W}_{V^\sharp}(\mathsf{b}))\right].$$

The continued constrained-background estimate gives holomorphy on a fixed polydisc and

$$|F(\mathsf{b},\delta)| \le K_M\big(\varepsilon + \max\{|\mathsf{b}|_\infty, |\delta|_\infty\}\big)|\delta|_\infty. \tag{89}$$

For real arguments this is the difference of the regular critical-background actions at the two data. Its derivative in δ is therefore the variation of that regular critical-background action over the box datum.

Transverse Hessian. Let $\mathsf{H}_1\delta$ be the linearized background response to a free datum variation. The coercivity of the fine action, the linear averaging estimate and the tree Poincaré inequality give the chain

$$\text{fine Hessian} \;\gtrsim\; \|\mathrm{d}\mathsf{H}_1\delta\|^2 \;\gtrsim\; \|dQ_k^{\mathrm{st}}(1)\mathsf{H}_1\delta\|^2 \;\simeq\; \|d\delta\|^2 \;\ge c_T|\delta|_2^2.$$

The averaging estimate in the middle has constant one at the trivial field in four dimensions: averaging the plaquette squares by Cauchy–Schwarz and then summing fine plaquettes gives

$$\sum_{p'\in S} |dQ_k^{\mathrm{st}}(1)A(p')|^2 \le \sum_{p\in S_0} \eta_k^4 |dA(p)|^2.$$

The factors from length, averaging and plaquette count cancel. This is a free-action estimate; no mass term is added at the bare scale.

For a regular centre the response reproduces the coarse variation only up to the coarse node gauge. To justify the middle comparison, differentiate the gauge-invariant coarse Wilson action in two ways: directly at its coarse datum, and through the fine response and its average. The gauge factor drops out of that action. The remaining perturbation terms have size $C_M\varepsilon|\delta|_2^2$, with the box bond count included in C_M. Taking the regularity radius small compared with c_T/C_M gives

$$\partial_\delta^2 F(0,0)[\delta,\delta] \ge \tfrac{1}{2}c_T|\delta|_2^2, \qquad \delta\in\mathcal{X}. \tag{90}$$

The tree condition is essential: gradients would otherwise remain in the kernel. Its Poincaré constant is fixed because the free box has $O(M^4)$ bonds.

Solving the box equation. Let $\mathsf{L} = \partial_\delta^2 F(0,0)|_{\mathcal{X}}$. It has a bounded inverse by (90). Decompose the derivative as

$$\partial_\delta F(\mathsf{b},\delta) = g_{\mathsf{b}} + \mathsf{L}\delta + \mathcal{N}_{\mathsf{b}}(\delta), \qquad \mathcal{N}_{\mathsf{b}}(0) = 0.$$

On a smaller fixed polydisc, Cauchy estimates from (89) give

$$|g_{\mathsf{b}}| \le C_M(\varepsilon + |\mathsf{b}|_\infty), \qquad \|\partial_\delta \mathcal{N}_{\mathsf{b}}(\delta)\| \le C_M(\varepsilon + |\mathsf{b}|_\infty + |\delta|_\infty).$$

Choose first a transverse ball of radius r_P, then an exterior radius a_P and the regularity ceiling so that $\|\mathsf{L}^{-1}\partial_\delta\mathcal{N}_{\mathsf{b}}\| \le 1/2$ and $\|\mathsf{L}^{-1}g_{\mathsf{b}}\| \le r_P/2$. The map

$$\delta \longmapsto -\mathsf{L}^{-1}[g_{\mathsf{b}} + \mathcal{N}_{\mathsf{b}}(\delta)]$$

is a contraction of that ball. Its fixed point $B_\Lambda(\mathsf{b})$ is holomorphic in the exterior increment, real at real increments, and obeys

$$|B_\Lambda(\mathsf{b})|_\infty \le K_P(\varepsilon + |\mathsf{b}|_\infty).$$

If the centre is already critical then $B_\Lambda(0) = 0$, and the bound improves to $2K_P|\mathsf{b}|_\infty$. Uniform convergence of the contraction iterates proves this holomorphic dependence on the same fixed disc.

Stationarity has so far been imposed on the transverse tree variables. The action is invariant under gauges at sites of P. The remaining directions are tangent to those gauge orbits, so its derivative vanishes in all directions supported in $E(P)$. In particular the basic background U_0 has zero current in the free box; this is the exact criticality used later to cancel the linear Taylor term.

The uniqueness statement concerns regular continuations, not arbitrary box data. In the tree gauge a continuation whose plaquettes in the prescribed neighbourhood of P have size less than a_P lies in the preceding fixed chart, after reducing a_P using the box Poincaré constant. The Hessian stays positive there by the same perturbation estimate. Thus its real critical points form one gauge orbit and attain the minimum on that regular class. Its tree representative is V_Λ, with

$$|\partial V_\Lambda - 1| \le C(M)\varepsilon.$$

The argument requires the constrained-background holomorphy and response bounds uniformly over the stated exterior class. It is not a claim about least action over unrestricted rough continuations. The preparation to a fixed box is what makes all constants above independent of the number of shells and the waiting depth.

The main-step tree Poincaré bound for a single bond is weaker:

$$|B(b)|^2 \le C(M\check{R}_k)^C \sum_p |\partial B(p)|^2.$$

One power loss comes from the diameter of a shell in its own lattice, at most $CML\check{R}_k$, and another from the number $N_k + 1$ of shells. This is the loss used when paying for a failed sharper test. It is distinct from the fixed-M bound used to construct V_Λ.

To check the implications of successive stage tests, let $U_{k,X}^{(n)}$ be the local background after n stages and put $j = k_\flat + n$. On a shell of level $i \le j$, the comparison required of the response is

$$\begin{aligned} |\partial U_{k,X}^{(n)} - 1| \le (1 + c\beta_{\mathrm{s}} 2^{-(j-i)})|\partial U_{k,X}^{(n+1)} - 1| \\ + c\beta_{\mathrm{s}} 2^{-(j-i)} \varepsilon_i \eta_i^2. \end{aligned} \tag{91}$$

Indeed, its error has the form

$$C\frac{\delta'_j}{\varepsilon_j}(1 + \sqrt{j-i})e^{-c(M/M_1)(j-i)}\varepsilon_i\eta_i^2.$$

The fixed geometric separation makes the exponential dominate the square root. The small threshold ratio pays the remaining constant, leaving $\beta_{\mathrm{s}} 2^{-(j-i)}$. Here $\beta_{\mathrm{s}} > 0$ is a fixed fraction reserved for this operation.

The stage-n shell threshold is therefore chosen as

$$[1 - \beta_{\mathrm{s}}(1 - 2^{-(j-i+1)})]\varepsilon_i\eta_i^2.$$

Summing the lost margins gives a fixed fraction rather than N_k times that fraction. For outer shells whose collars overlap, choose an integer $2 \le L_\flat < L/2$ and relax the shell-i bound by $L_\flat^{2(i-k_\sharp-1)_+}$. Here $k_\sharp + 1$ is the last level whose cell has side M in level-k units. This factor depends on the shell, not on the stage. A passage to the next finer scale gains L^{-2} and loses only $L_\flat^2$, so the relaxed bound still implies the earlier one. Prepared nests with gaps of only one M-cube require this calculation; an estimate using $e^{-cM\check{R}_k}$ at every such gap would be false.

The old fluctuation tests are implied by $CL\delta'_j \le \delta_j/2$. At the main integral the sharper threshold must also be small at the oldest core level. The combined requirement is

$$C\sqrt{N_k}\frac{p_1(g_{k_\flat})}{p_0(g_{k_\flat})} + C(M)L^{-N_k}\sqrt{N_k} \le c. \tag{92}$$

The first term gives $\mathfrak{p}_1 < \mathfrak{p}_0 - C\mathfrak{r}$; the second is made small by the wait $N_k = \check{R}_k$. These implications hold on the common support of the carried tests and averaging constraints, and only one current cell inside X. They do not extend to ∂X, where the local background is pinned bond by bond to the exterior.

Each completed stage restores the missed increments on the shell it integrates. Let $\varphi[\Lambda]$ denote the profile of a region and set

$$\begin{aligned} x^{(m)} &= \beta - \sum_{i \le k} b_i \varphi[\Lambda_i^{(m)}], \\ \Lambda_i^{(m)} &= \begin{cases} \Lambda_i \cup (X \setminus Z_i''), & k_\flat < i \le k_\flat + m, \\ \Lambda_i, & \text{otherwise.} \end{cases} \end{aligned} \tag{93}$$

The Wilson difference between $x^{(m)}$ and $x^{(m-1)}$ is kept separately, before localisation. At each bare plaquette the total of the Wilson weights remains $-\beta$. After all stages, $x_k'' = x^{(N_k)}$ equals x_k outside the prepared box and equals $x_k + O(1 + k - j)$ on its shell of level j. This restoration supplies the net weights in the following cancellation.

The backgrounds used in the main cancellation. The cancellation compares several backgrounds, each solving its own constrained problem. We describe their data before estimating the resulting differences. Write $Z = X$, $\Lambda = \Lambda^{\mathrm{pr}}$, $N = N_k$, and $\xi_j = L^{j-k}$ in level-k units. The prepared determining set $\mathbf{B}''$ has level k outside its final box, descends through the thin shells, and retains all

the old levels in the innermost core I_2. The healed determining set $\bar{\mathbf{B}}$ instead has level k throughout Z.

There are four local determining sets besides these two. The towers $\mathbf{B}^Z, \mathbf{B}^\Lambda$ are those of Z, Λ. Join $\mathbf{B}''$ to $\mathbf{B}^Z$ in the two layers next to ∂Z, obtaining $\mathbf{B}^{\mathrm{f}}$. Join it to the tower of the innermost prepared box, obtaining $\mathbf{B}^{\mathrm{in}}$. A joined determining set agrees with the first set in the interior and with the tower in its joining layers. In particular $\mathbf{B}'', \mathbf{B}^{\mathrm{f}}$ agree well inside Z, and $\mathbf{B}'', \mathbf{B}^{\mathrm{f}}, \mathbf{B}^{\mathrm{in}}$ agree well inside the innermost box. These equalities of data and determining sets are what permit local background comparisons.

Let V^{e} be the exterior datum and let V_Λ be its critical extension, in the tree gauge already chosen. Its basic background U_0 has zero current on the free box. In a mixed axial and Landau gauge it has a potential $\mathbb{A}_0$ on the bulk of Λ, with

$$|\mathbb{A}_0| + |\nabla \mathbb{A}_0| + |\mathrm{d}^* \mathrm{d} \mathbb{A}_0| \le C(M)\varepsilon.$$

For an exterior increment b, the same estimate holds with $\varepsilon + |\mathsf{b}|_\infty$. The averages of this representative on a shell of level $j < k$ are $e^{i\mathsf{v}_0(\mathsf{b})}$, where

$$|\mathsf{v}_0(\mathsf{b})| \le C\xi_j(\varepsilon + |\mathsf{b}|_\infty), \qquad \left|\tfrac{1}{i}\log(e^{i\mathsf{v}_0(\mathsf{b})}e^{-i\mathsf{v}_0(0)})\right| \le C\xi_j|\mathsf{b}|_\infty. \tag{94}$$

The factor ξ_j is a potential scaling, rather than a curvature scaling. The formula applies to every average whose corner blocks lie in the bulk, including levels below $k_\flat$.

Glue this basic datum to the real stored datum V^{in} at the inner rows and to the lifted exterior datum at the outer rows. Call the result V^0 on $\mathbf{B}^{\mathrm{f}}$. The mixed gauge is needed here: at the inner seam its basic averages are $O(L^{-N}\varepsilon)$ from one. The other side is controlled by the last-shell test on $(1, V^{\mathrm{in}})$, with threshold $\varepsilon_{k_\flat}/2$. Thus both the interior plaquettes and those crossing the seam are regular. Gauge-invariant small-field tests on the two unglued data would not by themselves imply this fact.

Let U^0 be the background of V^0. Locality and (94) give, on an inner test cube separated from its comparison boundary by D_2,

$$|\partial U^0 - 1| \le \left[\tfrac{1}{2}\varepsilon_{k_\flat} + C\varepsilon(L^{-N} + e^{-\delta_{\mathrm{G}} D_2})\right] \eta_k^2 \xi_{k_\flat}^{-2}. \tag{95}$$

The first margin is numerical; the other two terms tend to zero at the chosen wait. This verifies why a half-threshold test can supply an unchanged full threshold at the glued field.

For complex arguments use Landau forms centred at these real data. We denote the healed background by $\bar{W}$, its tower localization in Z by W^Z, the basic background by W^0, the glued local background by $W^{\mathrm{f}}(\mathsf{b}, \mathsf{f})$, the prepared background by $W''(\bar{\mathsf{b}}, \mathsf{f})$, and the inner one by $W^{\mathrm{in}}(\mathsf{f})$. Here $\bar{\mathsf{b}}$ is the increment of the exterior healed datum, b its restriction to the exterior rows of Z, and $\mathsf{f} = g_k B$ the single scaled increment on every free shell row. At real arguments these forms represent the corresponding critical orbits. At complex arguments they are the holomorphic continuations with these fixed real centres.

There is one inner increment that is not quite f. The basic shell averages enter the inner joining datum; call the resulting increment $\gamma^\flat$. On the inner rows read by this comparison,

$$|\gamma^\flat - \mathsf{f}| \le CL^{-N}(\varepsilon + |\mathsf{b}|_\infty). \tag{96}$$

This difference need not vanish at zero exterior increment: the basic datum itself is generally not one.

The ten Wilson differences. Choose $\zeta_0 = \phi_X$, equal to one away from the outer collar, and $\zeta_1 = \phi_{\mathrm{core}}$, equal to one on the old core, with $\operatorname{supp}\zeta_1 \subset \{\zeta_0 = 1\}$. Put $\zeta = \zeta_0(1 - \zeta_1) = \zeta_0 - \zeta_1$. Thus $1 = (1 - \zeta_0) + \zeta + \zeta_1$. The support of $1 - \zeta_0$ is at weighted distance $cM\check{R}_k$ from the free box. The support of ζ is that far from the unknown inner core, in the local units of its layers.

For this calculation let θ be the inverse coupling used by the reference law and let w be the restored spatial Wilson weight. Set $\bar{\mathsf{w}} = \mathsf{w}$ outside Z and $\bar{\mathsf{w}} = \theta$ in Z. Write H_1 for the linear response of U^0 and define the bilinear continuation

$$\mathsf{q}_0(\mathsf{f}) = \sum_p \eta_k^4 \zeta(p) \langle \mathrm{d}_{U^0} \mathsf{H}_1 \mathsf{f}(p), \mathrm{d}_{U^0} \mathsf{H}_1 \mathsf{f}(p) \rangle.$$

It is nonnegative for real f. No such sign is asserted at arbitrary complex f.

The exact remainder consists of the following differences:

$$\begin{aligned}
T_1 &= A(\mathsf{w}(1-\zeta_0), \bar{W}) - A(\mathsf{w}(1-\zeta_0), W''),\\
T_2 &= A((\bar{\mathsf{w}} - \mathsf{w})(1-\zeta_0), \bar{W}),\\
T_3 &= \theta[A(\zeta_0, \bar{W}) - A(\zeta_0, W^Z)],\\
T_4 &= \theta A(\zeta_1, W^0),\\
T_5 &= \theta[A(\zeta, W^0) - A(\zeta, W^{\mathrm{f}}(\mathsf{b}, 0))],\\
T_6 &= -\theta[A(\zeta, W^{\mathrm{f}}(\mathsf{b}, \mathsf{f})) - A(\zeta, W^{\mathrm{f}}(\mathsf{b}, 0)) - \tfrac{1}{2}\mathsf{q}_0(\mathsf{f})],\\
T_7 &= \theta[A(\zeta, W^{\mathrm{f}}(\mathsf{b}, \mathsf{f})) - A(\zeta, W'')],\\
T_8 &= -A((\mathsf{w} - \theta)\zeta, W''),\\
T_9 &= A(\mathsf{w}\zeta_1, W^{\mathrm{in}}(\mathsf{f})) - A(\mathsf{w}\zeta_1, W^{\mathrm{in}}(\gamma^\flat)),\\
T_{10} &= A(\mathsf{w}\zeta_1, W^{\mathrm{in}}(\gamma^\flat)) - A(\mathsf{w}\zeta_1, W'').
\end{aligned}$$

Adding them, and using the two partitions of the Wilson weight, gives

$$\begin{aligned}
&-\theta A(\zeta_0, W^Z) - A(\mathsf{w}, W'') + \theta A(\zeta_0, W^0)\\
&\quad = -A(\bar{\mathsf{w}}, \bar{W}) - \tfrac{1}{2}\theta \mathsf{q}_0(\mathsf{f}) - A(\mathsf{w}\zeta_1, W^{\mathrm{in}}(\mathsf{f})) + \sum_{\nu=1}^{10} T_\nu.
\end{aligned} \tag{97}$$

No estimate has been used in this identity. In particular the excess $(\mathsf{w} - \theta)\zeta$ and the whole old core weight remain visible.

In the unsourced run $\theta = x_k = g_k^{-2}$ and $\mathsf{f} = g_k B$. Equation (97) is the precise form of

$$\begin{aligned}
&-x_k A(\phi_X, U_{k,X}(V_k' V_\Lambda)) - A(x'', U'') + x_k A(\phi_X, U_0)\\
&\quad = -A(x_k(\cdot), U_k) - A(x''\phi_{\mathrm{core}}, U^{\mathrm{in}}(g_k B)) - \tfrac{1}{2}\|\zeta^{1/2} \mathrm{d}_{U^0} \mathsf{H}_1 B\|^2 + \mathrm{Err}_X.
\end{aligned} \tag{98}$$

The replacement of the prepared core by the real inner background is $T_9 + T_{10}$, not an implicit removal of its Wilson weight. Bounding the three original actions separately would leave the reference denominator's positive term of order $p_0(g_k)^2 (M \check{R}_k)^4$.

Sizes of the errors and the surviving core action. Let

$$\mathsf{s} = \varepsilon + \mathsf{a} + \delta, \qquad E_1 = e^{-c\delta_{\mathrm{G}} M \check{R}_k}, \qquad \mathsf{n} \le C \check{R}_k^5,$$

where a, δ bound the exterior and shell increments and n counts prepared unit cells outside the core, each at its own scale. If y is such a cell in layer j_y, set

$$\|\mathsf{w}\|_\sharp = \sup_y \frac{\sup_{p \in \Delta(y)} |\mathsf{w}(p)|}{1 + k - j_y}, \quad \mathsf{w}_2 = \sup_{\zeta \neq 0} |\mathsf{w} - \theta|, \quad \mathsf{w}_2' = \sup_{Z \cap \{\zeta_0 \neq 1\}} |\mathsf{w} - \theta|.$$

The background response estimates give

$$|T_1| \le C\|\mathsf{w}\|_\sharp \mathsf{s}^2 E_1, \qquad |T_2| \le C\mathsf{w}_2' \mathsf{s}^2 \check{R}_k^4, \qquad |T_3| \le C\theta \mathsf{s}^2 \check{R}_k^4 E_1, \tag{99}$$

$$|T_4| \le C\theta \mathsf{s}^2 \mathsf{n} L^{-4N}, \qquad |T_5| + |T_7| \le C\theta \mathsf{s}^2 \mathsf{n} E_1, \qquad |T_8| \le C\mathsf{w}_2 \mathsf{s}^2 \mathsf{n}, \tag{100}$$

$$|T_9| \le C\|\mathsf{w}\|_\sharp N \mathsf{s}^2 \mathsf{n} L^{-N}, \qquad |T_{10}| \le C\|\mathsf{w}\|_\sharp N \mathsf{s}^2 \mathsf{n} E_1. \tag{101}$$

Here and below a fixed M-dependent constant is harmless. For T_1, before summing cells, the bound is

$$C\mathsf{w}_y \mathsf{s}^2 e^{-c\delta_{\mathrm{G}} d_{\bar{\mathbf{B}}}(y,\Lambda)}.$$

This also pays the linear growth of the old coupling in $k - j_y$.

The estimates express three distinct comparisons. For $T_1, T_3, T_5, T_7, T_{10}$, the backgrounds have the same data and determining set near the action being compared, and differ only across a long collar. For T_4, the basic field in the inner region is a level-k field viewed N levels finer; its action there has the fourth power L^{-4N}. For T_9, it is the increment difference (96) that supplies the small factor. In dimension four the count of bare plaquettes in a unit cell, $6(\xi_j/\eta_k)^4$, cancels the action factor $\eta_k^4 \xi_j^{-4}$. Thus the sums are over prepared cells, not bare volume.

The Taylor term has the bound

$$|T_6| \le C\theta \mathsf{n}[\,\mathsf{s}\delta(E_1 + L^{-N}) + \mathsf{s}\delta^2\,]. \tag{102}$$

Its linear coefficient vanishes for the full action: minimizing first under fixed averages and then over the free datum removes precisely those constraints. The derivative is therefore zero in all free box directions. After restricting the action by ζ, the omitted outer part is separated by the long collar and the omitted inner part has the scaling L^{-2N}. These give the first two small factors in (102). The quadratic coefficient at the perturbed exterior differs from that at the real centre by at most

$$\tfrac{1}{2}\theta |q_2(\mathsf{b})[\mathsf{f}, \mathsf{f}] - \mathsf{q}_0(\mathsf{f})| \le C\theta \mathsf{n}\mathsf{s}\delta^2.$$

Higher terms have at least a third field factor.

In the application $\mathsf{a} = O(\alpha_{0,k} + \alpha_{1,k})$, $\delta = \delta_k'$ and $\varepsilon = O(\varepsilon_k)$. Hence $\theta\mathsf{s}^2$, n and N are only logarithmic powers. Both E_1 and L^{-N} beat all these powers, while $\theta\mathsf{s}\delta^2$ has a positive power of g_k times logarithms. The weight errors T_2, T_8 have no such geometric factor. They are charged to their displayed local volume bounds, using the restored profiles and, in a frozen run, its weight tube.

For clarity, the complex core comparison is the cellwise response estimate

$$\sum_j \sum_{x \in \Gamma_j} Ce^{-\delta_{\mathrm{G}} d(x)} L^{-N} \alpha_{1,k}[x_k + C(k-j)] \check{R}_j^2 \varepsilon_j \le (\log x_k)^C L^{-N}. \tag{103}$$

Response decay is measured in the layers' own units, and dominates both their finer-site count and polynomial growth of the old coupling. The real term $A(x'' \phi_{\mathrm{core}}, U^{\mathrm{in}}(g_k B))$ therefore stays in Φ_ω°, nonnegative at real stored data, while its exterior dependence belongs to the small error. This is the separation needed by the later supremum under the integral.

A quantitative denominator estimate. The same box calculation controls the reference denominator, which cannot be treated merely as a bounded phase. Let $E \setminus \mathsf{G}$ be the non-tree free bonds, $n_{\mathsf{G}} = |E \setminus \mathsf{G}| = O(M^4)$, and let $\sigma(Y)\,dY$ be Haar measure in the chart $Y \mapsto e^{iY}$, with $\sigma_0 = \sigma(0) > 0$. Centre at the continued critical point $B_\Lambda(\mathsf{b})$ and write the free variable as $\theta^{-1/2} B'$. After subtracting the basic action, the denominator is

$$\mathcal{N}(\mathsf{b}) = \int_{\substack{B'|_{\mathsf{G}} = 0 \\ |B'|_\infty < \theta^{1/2} \mathsf{r}}} \prod_{b \in E \setminus \mathsf{G}} \frac{\sigma(\theta^{-1/2} B'(b))}{\sigma_0}\, dB'(b)\, e^{-\Psi(\mathsf{b}, B')}.$$

Here Ψ is θ times the localized action difference from its critical value. Its quadratic reference form is strictly positive by the fixed-box tree Poincaré estimate. The linear term of the full action is zero

even at complex exterior data. Taylor's formula, the variation of this Hessian, the Haar density and the distant cutoff error are controlled by

$$\theta^{1/2}\mathsf{r} \geq 1, \qquad C\theta\mathsf{r}[\,\mathsf{r}(\mathsf{r}+\mathsf{a})+(\mathsf{s}+\mathsf{r})E_1\,] \leq 1. \tag{104}$$

Take C large enough for the comparison error to be at most one fifth. Comparison with the positive real Gaussian reference integral $\mathcal{N}_0$ gives

$$|\log \mathcal{N}_0| \leq C(M), \qquad \left|\frac{\mathcal{N}(\mathsf{b})}{\mathcal{N}_0}-1\right| \leq \tfrac{1}{5}.$$

Thus the integral has no zero on the prescribed exterior disc. The branch $I(\mathsf{b}) = -\operatorname{Log}\mathcal{N}(\mathsf{b})$ is holomorphic, real at real data, and satisfies $|I| \leq C(M)$, $|\operatorname{Im} I| \leq 1/4$. Restoring the removed action and chart scaling gives

$$\int dV'\,\mathbf{1}_{\{|\frac{1}{i}\log V'|<\mathsf{r}\}} e^{-\theta A(\zeta_0, U_{k,Z}(V'V_\Lambda))} = e^{-\theta A(\zeta_0,U_0)}(\theta^{-3/2}\sigma_0)^{n_{\mathsf{G}}} e^{-I(\mathsf{b})}.$$

With $\mathsf{r} = M_0\varepsilon_k$, the first condition in (104) holds and the second tends to zero: its terms are positive powers of g_k times logarithms, or logarithms times E_1. The box has $O(M^4)$ bonds, independent of N; this fixed size is essential to the uniform comparison.

3.6. **Localising a change on a history.** The same response device handles an ordinary fluctuation integral, a stage, and the main step. The data change by $\mathbf{V} \mapsto e^{ig_k B}\mathbf{V}$. At each old level this gives a pair shift $(\mathbf{H}, \mathsf{j})$ with

$$e^{i\eta\mathbf{H}}U^{\text{new}} = (U^{\text{old}})^u, \qquad J(U^{\text{new}}) + \mathsf{j} = \operatorname{Ad}(u)J(U^{\text{old}}).$$

The response is nonlocal before it is expanded: changing a datum near a box changes the background at every old level. Localisation must therefore be performed in the equations that determine the response, before an old activity is evaluated. Every propagator in those equations is replaced by

$$\mathsf{G}(s) = \sum_{\omega}\left(\prod_{\Delta \text{ met by } \omega} s(\Delta)\right)\mathsf{G}_\omega, \qquad |s(\Delta)| < e^{\bar\kappa}. \tag{105}$$

The fixed choice of $\bar\kappa$ costs a factor $e^{C\bar\kappa}$ in the analytic estimates. At $s(\Delta) = 0$ no walk can pass through Δ, so the fixed-point equations decouple exactly. The cube parameters are inserted in every propagator entering the coupled field and current equations; inserting them only in the last displayed response would not give that decoupling. A potential and its stored data are then evaluated through the same decoupled response. Only the pair is shifted; the stored data stay fixed under the pinned compensation of (126). The complex potential in a triple is not one of these stored real variables. Its analytic variation is performed at fixed stored data.

Let an old term f on X have bound b_f. Suppose its relative analytic room for this shift is a^{-1}, so the multiplier λ is allowed in $|\lambda| < a^{-1}$. At $(s,\lambda) = (\mathbf{1}, 1)$ the substitution gives the old background; when all response walks are cut it gives the new one. Subtracting the value at $\lambda = 0$ makes the difference vanish there. The bracket resummation in all cube parameters, followed by Cauchy in s and Schwarz in λ, gives

$$f|_{U^{\text{old}}} - f|_{U^{\text{new}}} = \sum_{Y \supset X} f_Y, \qquad |f_Y| \leq 2ab_f e^{-(\bar\kappa-1)\mathfrak{m}(X,Y)}. \tag{106}$$

The length $\mathfrak{m}(X,Y)$ counts the added cubes and the least number of cubes of X needed to connect them to a cube carrying an integrated field. Parameters are placed on cubes of X as well as on the added cubes. Keeping a parameter equal to one does not give its cube free: apply Schwarz, Cauchy and the maximum principle in the common scale of the kept parameters. This rooted length is needed because ordinary joining through an exempt hull is unavailable.

The three relative amplitudes are

move	relative amplitude
$\mathbf{T}$	$\mathsf{s}_k = \delta_k / \min_\iota \alpha_{\iota,k}$
stage at j	$N_k \mathsf{s}_j$
main step	$g_k^{-1} e^{-cM\check{R}_k}$.

(107)

The factor N_k is the analytic margin spent per stage, not a count of independent small masses. The conditions on $q - \mathfrak{p}_0$ make $N_k \mathsf{s}_j$ small. For an island exponent the bound applies to its oscillation, since constants cancel in a difference. The product of this amplitude with $(\log x_j)^{2\mathfrak{p}_0}$ must be bounded as well. This is why the radius exponent is chosen above the threshold exponents before the coupling ceiling is fixed.

The analytic room through repeated moves is recorded layer by layer. The coefficient of $\alpha_{\iota,n}$ contains the baseline $1 - \beta_{\mathrm{s}}(1 - 2^{-(k-n)})$, a reserve

$$\mathsf{g}_k(n) = \begin{cases} \theta_{\mathrm{s}} \beta_{\mathrm{s}} 2^{-(k-n)}, & n < k, \\ \beta_{\mathrm{s}}, & n = k, \end{cases} \qquad 0 < \theta_{\mathrm{s}} < 1,$$

and charges for remade levels in interrupted operations. Remaking level l costs $c\beta_{\mathrm{s}} 2^{-|n-l|}$ at layer n. The mark of a drop-out therefore records the number of completed stages; the geometry alone does not determine the room.

One ordinary level uses only the fraction $1 - \theta_{\mathrm{s}}$ of its geometric loss. A stage uses $\mathsf{g}_k(n)/(2N_k)$ outside the component and the remade-level charge inside it. The main step uses half the reserve, except on components whose marks are forgotten in the final label. There the returned domain may become larger, and no monotonicity argument is available. The response estimates fit the successive gaps because

$$2^\nu (1+\nu)^{1/2} e^{-c\delta_{\mathrm{G}} ML(\nu-1)} \le C, \qquad \nu \ge 1.$$

Thus the geometrically small margin at a distant layer is still larger than the response that reaches it.

Boundary terms have no coupling subtraction, so their sum across levels uses two different decays. If p layers are not reached, response decay gives $e^{-c\delta_{\mathrm{G}} MLp}$. If q' layers are crossed by the domain, its own length gives $e^{-c\kappa L^{q'}}$. Outside drop-outs, $p + q' = k - j$, which pays for the $L^{C(k-j)}$ positions at level j per current cube. A drop-out can have a lag l between its two region sequences, so $p + l + q' = k - j$ instead. The separate factor $e^{-cL\check{R}_k}$ pays for this lag. Neither response decay alone nor domain decay alone accounts for all positions.

Substitutions next to a prepared box. The old analytic domain contains conditions on more than the plaquettes of the field at which an activity is evaluated. It also requires regularity of the backgrounds determined by averages on subdomains. A subdomain can begin near a prepared box and extend past the component being integrated. Evaluating the old activity only at the background of that component would then require an unavailable continuation of this background outside the component. The substitution must be defined on the whole activity domain.

For an activity f of level j on $\mathfrak{Q}$, write $P = (U, A', \mathbf{J})$ for the triple representing $(e^{i\eta_j A'} U, \mathbf{J})$, and let ξ denote the stored variables of the operation. Its evaluation is made at

$$\mathfrak{S}_{\mathfrak{Q}}(s, \lambda; P, \xi).$$

The domain $\mathfrak{Q}$ determines the rule. The real member of the triple must remain SU(2)-valued and depend only on U and the group-valued variables of ξ. Holomorphy is required in $s, \lambda, A', \mathbf{J}$ and the complex fluctuation variables. Keeping this distinction is necessary when a real stored variable appears under a sharp test.

Domains away from the boxes. If $\mathfrak{Q}$ misses the kernels $\Omega_k^c \cap X$ and keeps away from the boxes, use the restriction of the new triple, shifted by λ times the response to the adjacent datum. That

datum uses the collar values of P and the appropriate variables of ξ. The response is determined by its fixed-point equations with (105); thus cutting the parameters also cuts the dependence on distant data. This rule applies to any old level $j \le k$.

The old core and its innermost shells. Here the real member is fixed by the stored data. Denote it by $U_\star$. It is the regular critical configuration, in the chosen gauge, for the action over the plaquettes of the component inside the first region Ω_0 of its determining set. On an averaging bond that carries a group-valued variable of ξ, its datum is that variable; on the other bonds of the determining set the datum is $\mathbf{1}$. A bare bond stored outside Ω_0 is not added to this datum. On the test support its plaquettes obey the prescribed smallness bound in the units of their own levels.

Start from $(U_\star, 0, J(U_\star))$ and add λ times the response, computed at $U_\star$, to the difference between the old datum of the activity and this fixed datum. This is the anchored rule. At $\lambda = 0$ it has no dependence on the exterior triple P. This property permits its possibly large constant part to remain under the stored-variable integral, rather than to be bounded by an exterior supremum.

Other domains inside a component. If $\mathfrak{Q}$ lies wholly inside a component after removing a margin of width CM, use the triple generated by that component's datum. Its base point depends on U. The generated configuration is put in the analytic gauge constructed below; its real factor is not selected by a complex change of the stored variables.

Domains crossing the transition region. The remaining domain meets a kernel or comes within $M\check{R}_k$ of a box, and also reaches a cube outside the component with its margin removed. Use the generated triple near the kernel and the shifted triple at distance at least $M\check{R}_k$ from it. In the transition region their potentials and currents are interpolated using covariant forms. A plain bondwise interpolation of arbitrary representatives does not preserve the relation between two splittings of one complex pair. The covariant choice is needed for the equality at the actual evaluation points.

Such a level-j domain crosses a distance at least $cM\check{R}_k$ measured at level k. In its own units this is $cM\check{R}_k L^{k-j}$. Hence its length satisfies

$$d_j(\mathfrak{Q}) \ge c\check{R}_k L^{k-j}. \tag{108}$$

The activity's own exponential decay can therefore pay for this case even where the response is not small. The long-domain estimate has a different role from the small relative amplitudes in (107).

The anchor, its analytic room and its inner response. We verify the old-core rule more quantitatively. This is also where one must distinguish the support of a single operation from that of a completed effective term. Set $h = k - N_k$. Localize the prepared sequence to the component C, using its M_1-tower:

$$\Omega_m^C = \Omega_m'' \cap C_m, \qquad \mathbf{B}^C = \mathbf{B}[\Omega_\bullet^C].$$

The first member can have holes where bare fields were stored outside Ω_0. Those bonds are absent from the constrained problem, rather than assigned an arbitrary regular extension. Let $\mathbf{V}_\star$ equal the real stored datum on its inner rows and one elsewhere. The local operation tests make this datum regular. Denote its critical background by $U_\star$, with the representative adapted to the stored sites. Let ϱ_0 be the separation constant of this localized sequence:

$$\operatorname{dist}_\infty((\Omega_m^C)^c, \Omega_{m+1}^C) > \varrho_0 \xi_m.$$

This is the separated class used by the local regularity estimate below. It need not have the larger gaps of the earlier inverse construction with constant ρ_{geom}.

Write $l(x)$ for the prepared layer index and let $\mathsf{e}_m^\sharp = C g_m (\log x_m)^{\mathfrak{p}_0 + O(\mathfrak{r})}$. Write e_j for the bound on the scaled test-loop defects of $\mathbf{V}_\star$ in layer j. The local regularity estimate for this constrained problem is

$$\eta_k^{-2} \xi_{l(x)}^2 |\partial U_\star - 1|, \quad \xi_{l(x)}^3 |J(U_\star)| \le C \sup_j \mathsf{e}_j e^{-c\delta_{\mathrm{G}} d_{\mathbf{B}^C}(x, \text{layer } j)} \le \mathsf{e}_{l(x)}^\sharp. \tag{109}$$

The last inequality uses the square-root bound on ratios of thresholds across levels and the separation $d(x, \text{layer } j) \geq \varrho_0(|j - l(x)| - 2)_+$. Since

$$\frac{\mathsf{e}_m^\sharp}{\alpha_{0,m}} \leq C(\log x_m)^{-(q-\mathfrak{p}_0-O(\mathfrak{r}))} \longrightarrow 0,$$

this anchor is much smaller than the analytic radii.

Membership in every old inner domain. Fix an old target domain $\mathfrak{Q}$ in the inner region, with its structure preceding the last stage and its largest prepared layer index at most $h+1$. Its determining sequence is finer than the prepared sequence. Thus its index m at a point is at most $l(x)$. The scaled curvature and current at this index have the additional factors $L^{-2(l-m)}$, $L^{-3(l-m)}$. The inequalities for thresholds and radii imply

$$L^{-a}\mathsf{e}_{m+a}^\sharp \leq \mathsf{e}_m^\sharp, \qquad \mathsf{e}_{m-1}^\sharp \leq C_L \mathsf{e}_m^\sharp.$$

Every old radius coefficient is bounded below by $c_* > 0$, and its lift is at least one. Consequently all the pointwise bounds at $(U_\star, 0, J(U_\star))$ hold with half their right-hand sides once the coupling ceiling is small.

The potential bounds concern small cubes, some touching a layer boundary. For an index $m \geq 1$, construct the regular gauge one level below: sequence separation places the needed neighbourhood in Ω_{m-1}^C. The local gauge theorem gives

$$\xi_m|A| + \xi_m^2|\nabla A| \leq C_L M \mathsf{e}_m^\sharp.$$

The neighbourhood may meet the boundary of a layer but does not meet a hole. At index zero use a comb inside the bare cube itself. In a cube of at most $s+1$ sites per direction, a bond in direction μ is obtained from the comb by at most $3s$ plaquette multiplications. Hence

$$|U_\star^v(b) - 1| \leq 3s\bar{\mathsf{e}}, \qquad \eta_k|A(b)| \leq \tfrac{3\pi}{2} s\bar{\mathsf{e}}, \qquad \eta_k^2|\nabla A| \leq 3\pi s\bar{\mathsf{e}},$$

where $\bar{\mathsf{e}} = \sup_m \mathsf{e}_m^\sharp$. Here $2\sin(\varphi/2) \geq 2\varphi/\pi$ supplies the logarithm bound in SU(2). All plaquettes of the comb lie inside the cube, so this argument still applies to cubes next to a hole.

The tower clauses require another observation, beyond these pointwise estimates. A regular critical configuration is its own background over any finer subordinate system of its averages. Its restriction is admissible for that system, remains critical, and lies in the regular class; uniqueness of the critical orbit identifies the new tower background with a gauge copy of the restriction. Its curvature and current are therefore those in (109), rescaled to the index read by the clause. This checks every subdomain tower clause, rather than just towers of the whole component. For a single-scale clause inside an old small-field region, truncate the same sequence at that scale and apply the same argument, with its prescribed margin. Thus the pointwise, good-gauge, tower and single-scale groups all hold at half radii. This proof uses separated sequences and their local regularity theorem; it does not apply a membership criterion that requires the stronger wide-sequence condition at the tower along ∂C.

Regularity at holes and the pins. At a bond next to a hole, all adjacent present plaquettes are in the bare layer. The current of the action on Ω_0^C uses only those plaquettes, at most six, and consequently

$$\eta_k^3|J(U_\star)(b)| \leq 6\bar{\mathsf{e}}.$$

At a bond with all its plaquettes present the identity

$$(\mathrm{d}_{U_\star}^* \partial U_\star)(b) = a(b)\mathbf{1} + i\eta_k^2 J(U_\star)(b), \qquad |a(b)| \leq \tfrac{3}{2}\bar{\mathsf{e}}^2\eta_k^2\xi_{m-1}^{-3}$$

converts the current bound to the input of the local gauge theorem. This follows by writing $\partial U(p) = c_p\mathbf{1} + i\eta_k^2 F_U(p)$, where $1 - c_p = \frac{1}{2}|\partial U(p) - 1|^2$. For a top of level $m \geq 1$, separation places a fixed ξ_m-neighbourhood of its pillar inside Ω_0^C, provided ϱ_0/L exceeds the fixed pinning reach. All such tops are therefore available to the pinned construction. At a bare inner row the datum increment is zero. The constraint then fixes the configuration on that bond exactly, and both

endpoint node factors are one. The Landau response is zero there by continuity from zero increment. These checks are needed before applying the pinned response theorem to a component with holes.

A slowly varying room. Let $\mathsf{s}(x)$ be a nonnegative profile, varying at rate $\theta_\flat$ in the prepared metric, with

$$\mathsf{s}(x) \le \kappa_{\mathrm{R}} \min_{\iota} \alpha_{\iota,l(x)}.$$

A shift of this size in the prepared units has size at most $\widehat{\mathsf{s}}_m(x) = L^{-(l(x)-m)}\mathsf{s}(x)$ at an older index m. The radius-ratio estimate gives

$$\widehat{\mathsf{s}}_m(x) \le C\kappa_{\mathrm{R}} L^{-(l-m)}(1+l-m)^{1/2} \min_{\iota} \alpha_{\iota,m} \le C'\kappa_{\mathrm{R}} \min_{\iota} \alpha_{\iota,m}. \tag{110}$$

Potential, curvature and current changes then spend at most half of each pointwise radius if κ_{R} is chosen after these fixed constants. The real factor and its local gauges have not changed.

For a tower quantity G, its Lipschitz bound with decay gives

$$|\mathsf{G}(U_\star, \mathbf{A}; x) - \mathsf{G}(U_\star, 0; x)| \le K_{\mathrm{T}} \sup_b e^{-\delta_{\mathrm{T}} d^\flat(x,b)} \widehat{\mathsf{s}}_{m_b}(b).$$

The old localized metric satisfies $d^\flat \ge d''$. Layer indices change by at most $1 + d^\flat/\varrho_0$, so

$$\widehat{\mathsf{s}}_{m_b}(b) \le L^2 e^{(\theta_\flat + 2\log L/\varrho_0) d^\flat(x,b)} \widehat{\mathsf{s}}_m(x).$$

Choose $\theta_\flat + 2\log L/\varrho_0 \le \delta_{\mathrm{T}}$. The last two displays and (110) then spend less than half the tower radius. Applying this at each index also checks the single-scale tower clauses. Thus s is actual room in the whole old domain, not merely a bound on its pointwise coordinates.

The inner datum and its depth profile. The increment $B^\natural$ over the anchor is zero on all real inner rows. On a free shell row of level $n < k$ it obeys

$$e^{iB^\natural(c)} = e^{ig_k B(c)} e^{i\nu_P(c)}, \qquad |\nu_P(c)| \le C_1' L^{n-k} a_C,$$

where $a_C = \max_\iota \alpha_{\iota,k}$. The remaining outer and tower rows have

$$|B^\natural(c)| \le \epsilon_{\mathrm{out}}, \qquad \epsilon_{\mathrm{out}} = C_1' M^2 \check{R}_k a_C \longrightarrow 0.$$

The shell estimate is (94); entrance rows are bounded by closing their bond with the tree and $O(M)$ exterior bonds. The closed path spans $O(M^2)$ plaquettes of the critical extension. For invariance under a change of splitting, the exterior comb is formed from the complex pair. It acts at the nodes of the datum; it is not a complex transformation of the stored real variables.

For $h \le n \le k$ define

$$\mathsf{D}_n(x) = \inf\{d''(x,z) : z \in C \setminus I_2,\ l(z) \ge n-1\},$$

and, taking a fixed $\theta_\flat < \delta_{\mathrm{G}}$, set

$$\mathsf{s}_\star(x) = K_{\mathrm{S}} \Big[\delta_k' e^{-\theta_\flat \mathsf{D}_h(x)} + C_1' a_C \sum_{n=h}^{k-1} L^{n-k} e^{-\theta_\flat \mathsf{D}_n(x)} + \epsilon_{\mathrm{out}} e^{-\theta_\flat \mathsf{D}_k(x)} \Big].$$

The response and current have this majorant, up to a fixed constant. Each distance is one-Lipschitz, so the profile varies slowly. For $l = l(x) \le h+1$, separation gives $\mathsf{D}_n(x) \ge \varrho_0(n-2-l)_+$, and

$$\sum_{n=h}^{k-1} L^{n-k} e^{-\theta_\flat \varrho_0 (n-2-l)_+} \le 6 L^{3-N_k} e^{-\theta_\flat \varrho_0 (h-2-l)_+}.$$

Use $Le^{-\theta_\flat \varrho_0} \le 1/2$ here. Both $a_C L^{3-N_k}/\delta_k'$ and $\epsilon_{\mathrm{out}} e^{-\theta_\flat \varrho_0 (N_k - 3)}/\delta_k'$ tend to zero. Consequently

$$\mathsf{s}_\star(x) \le 3K_{\mathrm{S}} e^{\theta_\flat \varrho_0} \delta_k' e^{-\theta_\flat \varrho_0 (h-1-l(x))_+}. \tag{111}$$

Let $n_{\mathfrak{Q}}$ be the largest layer index of the target and $u_{\mathfrak{Q}} = (h - 1 - n_{\mathfrak{Q}})_+$. The pinned response costs a fixed profile-transfer constant. Combining (111) with the analytic room above gives the interpolation disc $|\lambda| < 2/a^\star_{\mathfrak{Q}}$, where

$$\begin{aligned} a^\star_{\mathfrak{Q}} &\le C(2+N_k)^{1/2}\frac{\delta'_k}{a_*}e^{-\delta_\flat \varrho_0 u_{\mathfrak{Q}}} \\ &\le C'(\log x_k)^{-(q-\mathfrak{p}_1-\mathfrak{r}/2)}e^{-\delta_\flat \varrho_0 u_{\mathfrak{Q}}}. \end{aligned} \tag{112}$$

Here $a_* = \min_\iota \alpha_{\iota,k}$; $\delta_\flat < \theta_\flat$ is chosen so that $(1+u)^{1/2}e^{-\theta_\flat \varrho_0 u} \le e^{-\delta_\flat \varrho_0 u}$. The factor $(2+N_k)^{1/2}$ comes from $1 + k - l \le (2+N_k)(1+(h-1-l)_+)$. Choose the separation so that $e^{-\delta_\flat \varrho_0} \le (4L^8)^{-1}$. The resulting decay is faster than the growth of old stored sites and their fixed multiplicities.

Comparison with a real stored-only point. One can retain more than the zero-response anchor. Let b_ξ equal $g_k B$ on the free inner shell rows and zero elsewhere. It depends only on the real variables of the operation. The interpolation at this datum defines a real triple $\mathfrak{q}_{\mathfrak{Q}}(\xi)$ whose pair is the pair of the inner background $W^{\mathrm{in}}(g_k B)$.

To bound the exterior dependence, set

$$B(z) = \tfrac{1}{i}\log(e^{ig_kB}e^{iz\nu_P}) \text{ on the free rows below } k, \qquad B(z) = zB^\natural \text{ on the other non-inner rows.}$$

It is zero on the fixed inner rows, $B(0) = b_\xi$, and $B(1) = B^\natural$. It stays in the background's fixed datum polydisc on $|z| < r_{\mathrm{B}}/(2\epsilon_{\mathrm{out}})$. Split $\mathsf{s}_\star = \mathsf{s}^{(1)} + \mathsf{s}^{(2)}$, where $\mathsf{s}^{(1)}$ is its first summand. The crude response bound on that disc and Schwarz' lemma give

$$|\mathfrak{h}(B^\natural) - \mathfrak{h}(b_\xi)|_b \le \tfrac{3}{2}K_1\left[\frac{4\epsilon_{\mathrm{out}}}{r_{\mathrm{B}}}\mathsf{s}^{(1)} + \mathsf{s}^{(2)}\right](x_b).$$

Apply the scalar Schwarz estimate to each linear matrix functional defining the shift norm; taking their supremum gives this same bound for the norm. Thus the large δ'_k contribution acquires the extra small factor ϵ_{out}.

Interpolate between these two shifts, with the real factor fixed at $U_\star$. Its analytic room gives an amplitude bounded by

$$\frac{C(2+N_k)^{1/2}}{a_*}\left[\epsilon_{\mathrm{out}}\delta'_k + a_C L^{3-N_k} + \epsilon_{\mathrm{out}}e^{-\theta_\flat \varrho_0(N_k-3)}\right]e^{-\delta_\flat \varrho_0 u_{\mathfrak{Q}}}. \tag{113}$$

For a bounded old activity f, Schwarz' lemma therefore bounds its change from $\mathfrak{q}_{\mathfrak{Q}}(\xi)$ by this amplitude times $\sup|f|$. The value at that real point can remain under its own stored-variable integral. This refined comparison explains how the paying real inner core action is retained while the complex exterior dependence becomes a small local interaction.

Checking the conditions on subdomain backgrounds. The domain of an old activity includes bounds on the backgrounds of all its subdomain averages. We give the localization and gluing argument for those conditions, isolating the uniform comparison it uses. A bound on the plaquettes of the substituted field alone would not check these inherited conditions.

At a layer of index m, write γ_m for the additive room reserved between the prepared and output analytic domains. If t is the level of the operation and $\alpha^\vee_m$ is the smallest radius occurring in that clause, the room obeys

$$c\beta_{\mathrm{s}}2^{-(t-m)}\alpha^\vee_m \le \gamma_m \le C\alpha^\vee_m.$$

Inside a prepared region the endpoint estimates instead leave a fixed relative fraction of their right sides. The distinction is essential at older layers, where the additive room can decrease with $t - m$.

Cutting a subdomain at a distant boundary. Let $\widetilde{Z}$ be a subdomain appearing in one of the old background conditions, and let x be the point where its tower curvature or current is tested. Here a

tower is the sequence of backgrounds obtained from its successively localized averaging data. Choose a union R of current cubes with

$$\operatorname{dist}_\infty(x, \widetilde{Z} \setminus R) \geq D + 3M_1, \qquad D \geq CM.$$

Let $\widetilde{Z}_R$ be the connected component of $\widetilde{Z} \cap R$ containing x. The partition at the old level refines the current partition. Thus the cut is again an admitted subdomain. Membership in a localized layer through the depth being read is decided by the complement within $2M_1$ spacings. The original and cut domains therefore have identical layers and averaging rows in the required neighborhood of x. Other components of the cut are separated by a full old cube. All rows changed by the cut lie at least distance D from x; the layer index used to normalize the tested quantity is unchanged.

For the two resulting towers, the analytic estimate needed here is

$$\Big|\, |G^{\mathrm{full}}(U, A'; x)| - |G^{\mathrm{cut}}(U, A'; x)| \,\Big| \leq C_{\mathrm{tw}} a_*^{-1} e^{-c\delta_{\mathrm{G}} D} \gamma_m. \tag{114}$$

The quantities G are the curvature or current with the layer normalization of its analytic-domain clause; a_* is the smallest top-layer radius. The constant C_{tw} depends only on the fixed construction parameters, independently of D, the levels, the subdomain and its position. This is a comparison of their norms on the whole admitted complex domain. It must have no additional quadratic remainder in A'. Such a remainder need not fit into γ_m at deep layers. The geometric cutting argument reduces the inherited condition to this estimate, but does not prove its uniform analytic bound.

Gluing the domain conditions. Cover the activity domain by overlapping pieces on each of which one endpoint construction, or the controlled interpolation, already has its complete domain bounds. Each point x is assigned a piece containing its $D + 3M_1$ neighborhood. Require also that the cubes outside the prepared region and their adjacent cubes belong to one common exterior piece. Suppose the local endpoint estimates leave relative room $r\beta_{\mathrm{s}}$ inside the prepared region and additive room $r\gamma_m$ elsewhere. The conditions for the real member $(U, 0)$ have their own rooms, denoted by κ_3 and $r'\gamma_m$. These do not follow solely from the conditions for (U, A').

There are four kinds of condition to check. Pointwise bounds, potentials on small cubes and the local good-gauge conditions are read within distance CM of their base point, so they are conditions of its assigned piece. For a tower condition, cut the subdomain to that piece as above. Its right side and layer index are unchanged. Apply (114) to compare the full and cut quantities and spend half the saved room. For the old target spaces used by this substitution, a current-top single-scale domain lies in Λ_t of the old label, while each treated component lies in its complement. Those domains are therefore wholly in the common exterior piece. This assertion is about the old targets; a newly healed output domain can include the treated component. For an older single-scale condition, its local parts belong to the corresponding cut components; its subdomain-tower parts use the same cutting argument, both at (U, A') and at $(U, 0)$.

More quantitatively, the right side b_m of a tower clause obeys $\gamma_m \leq C b_m$. It suffices to choose the fixed parameters so that

$$C_{\mathrm{tw}} a_*^{-1} e^{-c\delta_{\mathrm{G}} D} \leq c \min\{r\beta_{\mathrm{s}}, \kappa_3, r, r'\}.$$

Then each inherited clause holds with at least half its original room. The proof compares one clause at its own base point; it does not sum the truncation errors over all subdomains or layers.

The prepared-box geometry provides the required cover by an interior piece, a generated-field piece, an interpolation strip and an exterior piece. The strip cutoff is constant near its two ends, with its transition confined to a shell of width comparable to $M\check{R}_t$. The overlap margins can be chosen to be at least $M\check{R}_t/16 - M$. For $\check{R}_t$ above a fixed constant this exceeds $M\check{R}_t/32 + 3M_1$. Thus one can use $D = M\check{R}_t/32$, uniformly through a seam of stored layers. The exterior piece contains the cubes outside the prepared region together with their adjacent cubes, as required by the top-level single-scale clause.

The factor a_*^{-1} is at most an inverse coupling times a fixed logarithmic factor. With $\check{R}_t \geq (\log(x_t - C_\sharp^{\prime}))^{\mathfrak{r}}$ and $\mathfrak{r} \geq 2$, the exponential in D beats every fixed power g_t^N. Choose the required power and all room fractions before the coupling ceiling. This makes the displayed numerical condition possible once the uniform analytic estimate and endpoint rooms have been established.

This also explains where a fixed enlargement of the radii can be used. On a collar the saved room is a relative fraction, and a small common enlargement of the two radii is harmless before returning to the true target. Away from it the room is the layer-dependent γ_m; a fixed relative enlargement would exceed it at sufficiently old layers. The first remaining analytic verification is therefore the uniform no-remainder estimate (114), together with the endpoint bounds furnishing the stated rooms on every permitted generated triple. The cutting and gluing deduction applies once those estimates hold, including at the seams.

Oscillations and compatibility of the substitutions. The response construction is to give an amplitude $a_{\mathfrak{Q}} \leq 1$ for which $\mathfrak{S}_{\mathfrak{Q}}(s, \lambda)$ remains in the domain of f on $|\lambda| < 2/a_{\mathfrak{Q}}$. On anchored domains the amplitude contains a negative power of $\log x_k$ and decays in $(k - N_k - 1 - n)_+$, where n is the highest level of a field stored on $\mathfrak{Q}$. The ratio is at most $(4L^8)^{-1}$ per deeper stored level. This decay is stronger than the L^4 growth of stored sites when that level is lowered.

Apply the Schwarz estimate in λ to the difference from its value at zero. Once the common domain and enlarged radius have been verified, this gives

$$|f \circ \mathfrak{S}_{\mathfrak{Q}}(s, 1) - f \circ \mathfrak{S}_{\mathfrak{Q}}(s, 0)| \leq a_{\mathfrak{Q}} \sup |f|. \tag{115}$$

At zero the rule is independent of s. For an anchored domain it depends only on ξ; that value is absorbed into Φ_ω°. Comparing the values at two exterior arguments therefore costs at most $2a_{\mathfrak{Q}} \sup |f|$. It is this oscillation, rather than the stored-only exponent, that is bounded uniformly in the exterior pair. The same amplitudes enter (106).

The rules must also agree when a localisation domain is viewed inside a larger one. This is consistency with restriction of the local construction, not dependence on s cube by cube: a parameter carried by one cube of a component can affect a function of the component's whole collar. The rooted length in (106) was chosen to retain that cost. Complex conjugation and the local charts of a history must transport the rules together with their stored data. Real gauge transformations act on the stored variables only through the value at the root of their gauge tree. Thus an old term invariant under this action retains that invariance under the substitution.

At the evaluation points, s is an indicator, $\lambda = 1$ and ξ is real. The substituted pair must then be the pair of the critical field for the old datum. Equality for two splittings of that pair is proved using the pinned gauge and the common holomorphic family in the discussion following (126). Its applicability still requires the uniform spare radii identified above. This is a statement at the evaluations used in the component integrals; no saturation of the whole analytic domain by complex gauges is required.

Near a stored boundary the field is subject to sharp tests and need not enter analytically. Each such bond b of level l has a switch μ_b, with radius a large power of $\log x_l$. The boundary function is holomorphic jointly in these switches, the pair and far stored fields; it is Borel in near stored fields. At $\mu_b = 0$ it is independent of the field at b, and at zero field it is independent of μ_b. Its actual value is at $\mu_b = 1$. This structure is used to control the successive stage integrals and their logarithms. Uniform nonvanishing of those complex integrals must be established with these switches and bounds independent of depth.

3.7. **The stage expansion and the generated field.** Each stage moves a background over the whole torus, although its change is largest at the collar being integrated. Its exponent must therefore contain the difference of every previously formed factor between the old and new backgrounds. Besides the regular and boundary interactions and the terms of earlier stages, there are seven kinds: the forms integrated in this stage; island factors near the treated component; the remaining island

factors; forms of levels still waiting inside the component; forms on untreated components; forms below the whole waiting window; and logarithms of pivot-elimination factors. The first kind belongs to the Gaussian quadratic form. The other kinds enter as localized differences. Characteristic functions in stored operations are not differenced: their genuine independence of the moving background is necessary here.

Used variables, direct terms and Gaussian tails. Partition the variables into finite blocks. For a potential V and a subset S of its variable blocks put

$$V_S(B) = \sum_{T \subset S} (-1)^{|S \setminus T|} V(B_T),$$

with unlisted blocks zero. Summing over S recovers $V(B)$. For $S \neq \emptyset$, the summand vanishes when any listed block is zero. Schwarz' lemma in each block therefore gives a factor

$$q_v = \frac{\text{range of block } v}{\text{analytic radius of block } v}$$

per block actually used. Its skeleton consists of those blocks, together with the connecting walks. The full localization domain of V is not included merely because V was initially defined on it.

For a new field on the integrated shell the ratio is δ_j'/δ_j. A far stored field has a polydisc λ^{st} times its real range. A near stored field is real and may stand under a sharp test. There the analytic variable is a switch μ_v, not that field: at $\mu_v = 0$ the function is independent of the field, and at zero field it is independent of μ_v. Evaluating at $\mu_v = 1$ gives the same factor $(\lambda^{\mathrm{st}})^{-1}$ without differentiating the characteristic function. Set

$$\lambda^{\mathrm{st}} = (\log x_j)^{6\mathfrak{r}+1}. \tag{116}$$

This switch property is preserved when the terms of earlier stages re-enter the next integral.

The part $V_\emptyset$ does not use the current integration variables. Keep it outside that integral as a direct term. The other parts are expanded into a gas of configurations. A configuration records its used blocks, its connecting walk chains, and any blocks outside the preferred Gaussian range. Write H for the latter blocks. Reserving a small Gaussian weight γ_2 assigns the tail

$$\mathsf{p} = \exp[-\tfrac{1}{2}\gamma_2 \mathfrak{b}_{\mathrm{cut}}^2]$$

to each such block, where $\mathfrak{b}_{\mathrm{cut}}$ is the auxiliary field threshold. The retained part of the quadratic estimate must still control the remaining Gaussian integral.

The general configuration bound has the form

$$|\operatorname{val}(\mathbf{c})| \leq e^{c_{\mathrm{v}} \# Z(\mathbf{c})} (C\mathsf{p})^{|H|+|P_1|+|P_2|} e^{-(\kappa_1 - 1)\#\tau} \prod_{(\iota, S) \in \mathbf{D}} e^{\bar{u}} u_\iota(S) \prod_{v \in S} q_v.$$

Here P_1, P_2 are the additional large-block lists created by the Gaussian expansion, τ its nonlocal walk cubes, and $Z(\mathbf{c})$ its integration support. The number $u_\iota(S)$ bounds the block component of its potential. The full potential amplitude is bounded by u_ι, and $\bar{u} \geq \sup_\iota u_\iota$. This majorant is independent of the interpolation parameters and operator entries. Its extensive factor counts variables of the configuration, rather than the volume of every potential domain.

One residual dependence on a potential's used blocks must be included. A large block in one configuration can affect a potential in another. If $\mathfrak{V}_\iota^{\mathrm{h}}$ is the set of its blocks that can be large, charge that potential $e^{2a_1 |\mathfrak{V}_\iota^{\mathrm{h}}|}$. It suffices to choose a_1 as small as the large-block activity. The configuration criterion is that both the sum of weighted used-block activities through one block and

$$C\mathsf{p}(e\mathsf{K})^{\mathsf{n}_{\mathrm{t}}} e^{2\mathsf{u}_{\mathrm{hot}}}$$

are small, with the latter at most a_1. The fixed $\mathsf{K}, \mathsf{n}_{\mathrm{t}}$ account for local variables and their multiplicity, and $\mathsf{u}_{\mathrm{hot}}$ is the load of potentials that can meet a large block. These are the quantities controlled by

the Gaussian expansion; a bound on the phase of an arbitrary Borel potential would not imply this criterion.

The recursion through the stages. There are two useful norms. The action and primary interactions are summed over their domains through a fixed cell, with their exponential length weight. On a graded cell of level μ, use the normalization $L^{-4(\mu-j-1)}$ for its number of level-$j+1$ sites. Let J^{a} be this action load, J^{pr} the primary load, and J^{Σ}_{A} the action load summed over the anchor cells of one component. For direct terms retain separately their largest single-stage norm $\bar{D}_1$, their sum over stages $\bar{D}^{\Sigma}_1$, and their accumulated load $\bar{D}^{\circ}$ through a fixed cell. A normalizer can have a controlled summed direct norm without being assigned a small independent constant at every stage.

At stage $n+1$, its input includes both direct and cluster parts from every stage $\nu \le n$. For $N = N_k$ the induction uses

$$\bar{\mathsf{J}} = J^{\mathrm{a}} + CN + \bar{D}^{\circ}, \qquad \mathsf{u}_{\mathrm{hot}} \le C(\bar{\mathsf{J}} + J^{\mathrm{pr}}).$$

Write $N^* \le C\check{R}_k^4$ for the number of possible anchor cubes in a component. Let $\hat{q}, \hat{q}^{\mathrm{h}}$ be the largest used-variable ratios, including their fixed local entropy factors, for all blocks and for stored blocks respectively. The component load to control is

$$\begin{aligned} \epsilon_{\mathrm{comp}} &= CN^*\mathsf{p}\, e^{2\mathsf{u}_{\mathrm{hot}}} \\ &\quad + C\left[\hat{q}^{\mathrm{h}}(J^{\Sigma}_{\mathsf{A}} + N^*CN) + \hat{q}N^*(\mathsf{a}_{\mathrm{st}}\bar{\mathsf{J}} + J^{\mathrm{pr}})\right]. \end{aligned} \tag{117}$$

Here a_{st} is the small relative shift amplitude entering the stage expansion. Every previous-stage term is assigned the crude bound one when checking this load. Thus its accumulated pinned norm is $O(N)$; the required smallness comes from its used-variable ratio, rather than an independent small mass assigned to that term.

The largest stored-field contribution, after summing over the N stages, is

$$CN^2\check{R}_k^4(\lambda^{\mathrm{st}})^{-1} \le C(\log x_k)^{-1}. \tag{118}$$

There is one N for the old stage load and one for the sum of the newly made terms. For new-shell variables the separate ratio

$$N^*\frac{\delta'_j}{\delta_j} \le C\check{R}_k^4(\log x_j)^{-(\mathfrak{p}_0-\mathfrak{p}_1)}$$

must be small. The source of smallness in these two expressions is different. The first explains the exponent $6\mathfrak{r}+1$ in the switch; the second explains the additional margin between $\mathfrak{p}_0$ and $\mathfrak{p}_1$. The logarithms throughout a waiting window are comparable, so one coupling ceiling works for all its stages.

The large-block term in (117) must be controlled at the same time, including its $e^{2\mathsf{u}_{\mathrm{hot}}}$ factor. Put $\mathsf{L}_k = \log x_k$. The crude bound one for each earlier stage gives a pinned norm $O(N)$; after the other inputs are included, the bound used at a block which can be hot is

$$\mathsf{u}_{\mathrm{hot}} \le C\mathsf{L}_k^{4\mathfrak{r}}.$$

At the characteristic functions of the integral there is no used-variable ratio multiplying this exponent. It must be paid by the Gaussian tail itself. The required tail bound is

$$\mathsf{p} \le \exp\left[-c\mathsf{L}_k^{2\mathfrak{p}_1-4\mathfrak{r}}\right].$$

Since $N^* \le C\mathsf{L}_k^{4\mathfrak{r}}$, it gives

$$CN^*\mathsf{p}\, e^{2\mathsf{u}_{\mathrm{hot}}} \le C\mathsf{L}_k^{4\mathfrak{r}} \exp\left[-c\mathsf{L}_k^{2\mathfrak{p}_1-4\mathfrak{r}} + C\mathsf{L}_k^{4\mathfrak{r}}\right]. \tag{119}$$

The right side is smaller than every negative power of L_k once $2\mathfrak{p}_1 > 8\mathfrak{r}$ and the coupling ceiling is sufficiently small. It therefore also pays the fixed logarithmic entropy and variable-ratio factors in the stage expansion. This is a separate use of the sharper exponent $\mathfrak{p}_1$, in addition to the

small new-shell ratio δ'_j/δ_j. The tail estimate still requires the uniform decoupled covariance and mean bounds for the actual collar integral, on its stated characteristic functions. Neither (118) nor smallness of the stored-variable ratios proves that estimate.

Closing the free source has its own analytic cost. During the expansion an auxiliary analytic source keeps the local parts of the background shift independent. Closing this source substitutes their actual values. If $R_{b,W} > 1$ is the available radius for the closing variable attached to a bond b and carrier W, and $\mathsf{c}(b)$ is the designated cell carrying b, put

$$\epsilon_{\mathrm{cl}} = \sup_{\Delta} \sum_{(b,W):\,\mathsf{c}(b)=\Delta} \frac{e^{C\rho_\natural \varrho_{\mathrm{a}} \# W}}{R_{b,W}-1}.$$

The stage criterion also requires

$$C\hat{q}N^* \le 1, \quad C\mathsf{p}e^{2\mathsf{u}_{\mathrm{hot}}} \le \hat{q}, \quad C\epsilon_{\mathrm{comp}} \le 1, \quad CN^*\epsilon_{\mathrm{cl}} \le 1, \quad N \ge C\varrho_{\mathrm{a}}, \quad \kappa_1 \ge C\rho_\natural \varrho_{\mathrm{a}},$$

where ϱ_{a} is the fixed geometric anchoring ratio. These conditions control closing the free analytic source as well as the used-variable activities.

Spatial decay is spent in units $1/N$. If $\rho_\natural$ is the input rate, set $\rho_n = \rho_\natural - C_\rho n/N$. Supports of separate component operations are separated on the scale NM. Consequently the count of components met by a domain of length d is $1 + Cd/N$, and the entropy costs have the form $e^{Cd/N}$. The budget per stage includes one $1/N$ for used-block support entropy, one for passing from a summed norm to a pinned norm, and fixed further multiples of $1/N$ for component sums and closing the free source. The total loss after N stages is the fixed C_ρ.

On the configuration criterion above, the connected expansion gives a nowhere-zero stage integral and its logarithm. The direct part is carried outside the integral; the source part is closed by the same small analytic shift. The resulting bounds are

$$|\mathbb{C}_{\mathrm{d}}^{(n+1)}(Y)| \le \bar{D}_1 e^{-\rho_\natural d_k^\natural(Y)}, \qquad |\mathbb{C}_{\mathrm{cl}}^{(n+1)}(Y)| \le 2\epsilon_{\mathrm{comp}} e^{-(\rho_{n+1}-C/N)d_k^\natural(Y)}.$$

Each cluster part contains an anchor cell of its stage. The stronger summed norm near one component is at most $C\epsilon_{\mathrm{comp}}$, without the extra pointwise component-count loss. Induction includes the old direct and closed parts in the next input, so these are bounds on the actual recursion.

Choose the input rate with the budget just described. The sum of the direct terms and the cluster estimate is bounded by

$$(\bar{D}_1^\Sigma + 2N\epsilon_{\mathrm{comp}}) e^{-(\rho_\natural - C_\rho - C)d_k^\natural(Y)}.$$

Thus the application must check $\bar{D}_1^\Sigma + 2N\epsilon_{\mathrm{comp}} \le C_{\mathrm{S}}$; smallness of ϵ_{comp} alone would not suffice. Equation (118) is the largest stored-field contribution to this check. With the remaining action, primary and direct loads controlled in their stated norms, the summed output is

$$\begin{aligned} \mathbb{I}_k^{(n+1)} &= \exp\left[\mathrm{const} + \sum_Y \mathbb{C}_k^{(n+1)}(Y)\right], \\ \sum_{n<N_k} \sup |\mathbb{C}_k^{(n+1)}(Y)| &\le C_{\mathrm{S}} e^{-(1+3\beta_{\mathrm{s}})\kappa d_k^\natural(Y)}, \qquad C_{\mathrm{S}} \text{ independent of } N_k. \end{aligned} \tag{120}$$

If the operation stops after n stages, the one function $\sum_{\nu\le n} \mathbb{C}_k^{(\nu)}(Y)$ is retained as its boundary contribution. No earlier stage integral is undone.

The analytic verification in this application is the uniform configuration criterion for its actual decoupled collar covariance, mean, switches and primary terms on the successive common domains. The displayed load, rate and recursion estimates give its quantitative reduction and the resulting depth-independent bound. Nonvanishing is supplied by the connected expansion under that criterion, rather than by boundedness of a complex phase.

Healing changes the domains in a further way. It erases the mark of the remade levels and returns their analytic charge. The new space can therefore be larger than the space of an old stage term. Moreover a unit pure gauge near a node has zero action but unit potential, so the old orbit cannot always be reached by a small potential shift. The following generated field construction addresses both issues.

Constructing a gauge for the generated configuration. A critical configuration is its own background over a finer system of its averages. This identity, and its holomorphic continuation, make a generated configuration suitable for the old tower clauses. In the middle zone of C, use the field $\mathbf{G}_C$ generated on C itself from the new datum and the inner variables. At a finer index n, its curvature gains $L^{-2(k-n)}$, whereas its potential gains only $L^{-(k-n)}$. The generating domain cannot be enlarged arbitrarily into old layers: a fixed overshoot of a potential there would exceed the available layer-dependent room.

There is a separate difficulty at the top index. The target has only $c\beta_{\mathrm{s}}\alpha_{1,k}$ additional room. The required estimate is

$$\max\{|A^{\mathrm{g}}|, |\nabla_U A^{\mathrm{g}}|\} < (1 + c\beta_{\mathrm{s}})\alpha_{1,k}, \qquad A^{\mathrm{g}} = \tfrac{1}{i\eta_k}\log(\mathbf{G}_C^{\mathsf{k}_C} U^{-1}). \tag{121}$$

A bound with an unspecified multiple of $\alpha_{1,k}$ would not put the substituted triple in the old domain. The construction therefore arranges its leading term as an average of the argument's potential A'.

Real local frames and the reduced background. Work first in level-k units. For a coarse site y in the middle zone, let $\mathsf{Q}_y, \mathsf{Q}_y^+$ be the unions of the $3^4, 5^4$ surrounding blocks. They lie in the top layer, away from both boundaries. For a real field W define a frame with bound ϱ to be a real gauge f, equal to one at y, for which

$$W^f = e^{i\eta_k a_f}, \qquad \max_{\mathsf{Q}_y^+}(|a_f|, |\nabla a_f|) < \varrho.$$

The regular-gauge bound makes this open set of frames nonempty for $\varrho_{\mathrm{f}} = C(M)\alpha_{0,k}$. Average their values with normalized Haar measure and normalize in the quaternion algebra:

$$\mathsf{f}[W](x) = \mathcal{N}(\langle f(x)\rangle_W), \qquad \mathcal{N}(m) = (\det m)^{-1/2} m.$$

When the box diameter times ϱ_{f} is small, the mean is nonzero. The frame estimate gives

$$W^{\mathsf{f}[W]} = e^{i\eta_k a}, \qquad \|a\|_{\mathsf{Q}_y^+} \le 2\varrho_{\mathrm{f}}, \qquad \mathsf{f}[W^u] = u(y)\mathsf{f}[W]u^{-1}.$$

Here $\|a\|$ includes one difference. The frame depends on the real field alone, so its formation does not interfere with holomorphy in A'.

Use the reduced determining sequence on $C \setminus \Lambda_C$, descending to the bare scale at both boundaries. Its real datum is the average of U; call its background $U^\natural$ and its continuation $\Phi^\natural(B)$. Set

$$\mathsf{f}_y = \mathsf{f}[U|_{\mathsf{Q}_y^+}], \qquad \mathsf{f}_y^\circ = \mathsf{f}[U^\natural|_{\mathsf{Q}_y^+}].$$

The construction is made on a product domain in (A', B): these two analytic variables are independent before the actual datum is substituted. The increment B is $O(\alpha_1)$ on top rows and lies in a fixed small polydisc on the other rows.

Removing the node part locally. Fix a large constant radius R_{g}. For coarse sites y' in that box use combs from y to y' and set

$$v_y(y') = V^\natural(\mathrm{cb}_{y'})^{-1}(e^{iB}V^\natural)(\mathrm{cb}_{y'}).$$

Take $v_y = 1$ at the other nodes. Its lift $\widetilde{v}_y$ at a fine site is the value of v_y at the first node reached by ascending through the block centres. The transformed increment B_y is defined by

$$e^{iB_y(c)} = v_y(c_-)e^{iB(c)}V^\natural(c)v_y(c_+)^{-1}V^\natural(c)^{-1}.$$

Let $\mathcal{H}_y, u_y$ be its Landau response and node gauge. Removing the comb part leaves a response controlled by the curvature of the coarse datum and by the exponentially weak influence of rows outside this fixed box.

To restore the desired leading potential, put $A'_y = \mathrm{Ad}(\mathsf{f}_y)A'$. Average its bare translates over a small fixed distance δ_{g}, obtaining $a_y = \mathsf{S}_{\mathrm{g}}A'_y$, and let ϕ_y be its comb integral from y. If $P_\nu(x)$ replaces the first ν coordinates of x by those of y, and I_ν is the signed sum along that comb segment, then

$$\phi_y = \sum_\nu I_\nu[(a_y)_\nu], \qquad \nabla_\mu \phi_y = (a_y)_\mu - r_\mu, \qquad r_\mu = \sum_{\nu<\mu} I_\nu[(da_y)_{\nu\mu}].$$

Thus the failure to be a primitive is controlled by da_y. Mollification gives

$$\|\mathsf{S}_{\mathrm{g}}X\| \le \|X\|, \qquad \sup|\mathsf{S}_{\mathrm{g}}X - X| \le C\delta_{\mathrm{g}}\|X\|, \qquad \sup|\nabla \mathsf{S}_{\mathrm{g}}F| \le 2\delta_{\mathrm{g}}^{-1}\sup|F|.$$

These formulas retain control of a difference of the primitive defect without demanding an extra derivative of the original potential.

With $n_y = e^{-i\phi_y}$, the local gauge is

$$\widehat{\mathsf{k}}_y = \mathsf{f}_y^{-1} n_y u_y(y) \mathsf{f}_y^{\circ} u_y^{-1} \widetilde{v}_y.$$

From right to left, its factors remove the node comb, pass to the Landau form, put its real background near one, normalize at y, insert the primitive, and return to the gauge of U. All factors are holomorphic in the independent complex variables. They are real gauges at real variables, and $\widehat{\mathsf{k}}_y(y) = 1$.

The quantitative local defects. The estimates needed from this construction can be written with four sizes. Let ϱ_{a} bound A' and its first difference, ϱ_{d} bound $\mathrm{d}_U A'$, ϱ_{p} bound the coarse plaquette increment in the node box, and ϱ_{e} bound the weighted influence of the lower rows. Set

$$\begin{aligned}
\varrho_{\mathrm{s}} &= C\mathsf{R}_{\mathrm{g}}(\varrho_{\mathrm{p}} + \alpha_0) + C\mathsf{R}_{\mathrm{g}} e^{-\delta_{\mathrm{G}}\mathsf{R}_{\mathrm{g}}/2}\varrho_{\mathrm{a}} + \varrho_{\mathrm{e}},\\
\varrho_{\mathrm{E}} &= \frac{C}{\delta_{\mathrm{g}}}(\varrho_{\mathrm{d}} + \varrho_{\mathrm{f}}\varrho_{\mathrm{a}}) + \frac{C}{\delta_{\mathrm{g}}^2}\varrho_{\mathrm{a}}^2 + C(\varrho_{\mathrm{s}} + \varrho_{\mathrm{f}}),\\
\Delta &= C\delta_{\mathrm{g}}\varrho_{\mathrm{a}} + \varrho_{\mathrm{E}}.
\end{aligned}$$

For

$$A^{(y)} = \tfrac{1}{i\eta_k}\log(\Phi^\natural(B)^{\widehat{\mathsf{k}}_y}U^{-1}),$$

the local estimate has the specific leading term

$$\mathrm{Ad}(\mathsf{f}_y)A^{(y)} = a_y + \mathsf{E}_y, \qquad \|a_y\| < (1 + C\varrho_{\mathrm{f}})\varrho_{\mathrm{a}}, \quad \|\mathsf{E}_y\| \le \varrho_{\mathrm{E}}, \quad \sup|A^{(y)} - A'| \le \Delta. \tag{122}$$

The first two terms of ϱ_{E} are the primitive defect and its nonlinear correction; the remaining terms are the reduced response and the small real gauges. This is the quantitative local estimate to establish, together with its node identity, rather than an assertion that some gauge has a bounded potential.

Joining the gauges without losing the leading constant. Use the $C^{1,1}$ partition of unity

$$\psi_y(x) = \prod_{\mu=1}^{4}\psi(x_\mu - y_\mu), \qquad \psi(s) = \begin{cases} 1 - 3s^2 + 2|s|^3, & |s| \le 1,\\ 0, & |s| > 1. \end{cases}$$

It has bounded first and second differences, $\sum_y \psi_y = 1$, $\sum_y \nabla\psi_y = 0$, and $\psi_y(y') = \delta_{yy'}$ at the coarse sites. Set

$$\mathsf{k}_C(x) = \mathcal{N}\left(\sum_y \psi_y(x)\widehat{\mathsf{k}}_y(x)\right). \tag{123}$$

On the real quaternion span, $m^*m = (\det m)1$ and $\mathcal{N}(m) = m/|m|$. It therefore takes real combinations back into SU(2). Near one local gauge the argument is that gauge times 1+ a small

matrix, so the branch of $(\det m)^{-1/2}$ is fixed by its value one. The same formula is holomorphic at complex arguments and is bi-covariant under determinant-one factors.

The local joining calculation gives

$$\|A^{\mathrm{g}}\|_U < \varrho_{\mathrm{a}} + K_{\mathrm{j}}(C\delta_{\mathrm{g}}\varrho_{\mathrm{a}} + \varrho_{\mathrm{E}}). \tag{124}$$

Its leading part is the average of the $A^{(y)}$. In its gradient, the differentiated weights have sum zero, so only differences of neighbouring leading potentials remain. These are bounded by Δ, rather than by ϱ_{a} once for every box. Normalization produces quadratic errors; an explicit factor η_k costs no new difference because $\|\eta_k X\| \le 2\sup|X|$. The node identity of the local gauges and $B = \mathcal{Q}_k(U; A')$ on the top rows give $\mathsf{k}_C = 1$ at the coarse sites. The construction thus preserves the attached stored data there.

For the actual generated datum, write $\mathcal{B}^{\mathrm{o}}$ for the exterior datum increment on the tower rows of the reduced determining sequence, and set $\alpha_i = gC_i^{\mathrm{rad}}(\log x)^q$, $i = 0, 1$, at the scale in question. Then

$$\varrho_{\mathrm{a}} = \alpha_1, \quad \varrho_{\mathrm{d}} \le C(\alpha_0 + \alpha_1^2), \quad \varrho_{\mathrm{p}} \le C\alpha_0, \quad \varrho_{\mathrm{e}} \le C[e^{-\delta_{\mathrm{G}} M}\sup|\mathcal{B}^{\mathrm{o}}| + e^{-c\delta_{\mathrm{G}} M\check{R}_k}].$$

The boundary datum is restricted by $Ce^{-\delta_{\mathrm{G}} M}\sup|\mathcal{B}^{\mathrm{o}}| \le c\beta_{\mathrm{s}}\alpha_0$. Choose constants in their operative order: first K_{j}; then δ_{g} with $K_{\mathrm{j}}\delta_{\mathrm{g}} \ll \beta_{\mathrm{s}}$; then R_{g} with $K_{\mathrm{j}}\mathsf{R}_{\mathrm{g}}e^{-\delta_{\mathrm{G}}\mathsf{R}_{\mathrm{g}}/2} \ll \beta_{\mathrm{s}}$; then M larger than this node box; then the ratio $C_1^{\mathrm{rad}}/C_0^{\mathrm{rad}}$; and finally the coupling ceiling. The ratio of the radii pays every fixed multiple of α_0. The ceiling pays $C\alpha_1^2$ and the distant inner response. Inserting these choices in (122) and (124) yields (121). The quantitative local and joining estimates, including their covariance and analyticity, are the analytic content needed for this last deduction; the quaternion normalization alone does not supply them.

The generated field also supplies the complex regularity needed to sum old marginal terms. A generic pair in the full analytic domain has no Hölder control of adjacent curvatures. It gives only the exponent zero in the following local bound. For a pointed family G of level $j = k - \nu$ with coefficient e, suppose on a window of l cube layers the pair has a potential regular at scale L^ν with amplitude a_{p} and Hölder exponent $\beta' > 0$. The required derivative bounds are

$$|\mathbf{A}| \le a_{\mathrm{p}}L^{-\nu}, \quad |\nabla\mathbf{A}| \le a_{\mathrm{p}}L^{-2\nu}, \quad [\nabla\mathbf{A}]_{\beta'} \le a_{\mathrm{p}}L^{-(2+\beta')\nu}, \quad |\mathbf{J}| \le a_{\mathrm{p}}L^{-3\nu}.$$

Here the current is still independent. For the sum of domains through z that stay inside that window and inside Y, the pointed subtraction gives

$$C\|G\|(a_{\mathrm{p}}/\alpha_0)^2L^{-4\nu}\left[L^{-\beta'\nu} + e^{-\kappa D_Y/2} + e^{-\kappa(l-1)/2}\right], \tag{125}$$

where D_Y is the shortest domain length from z to Y^c. The $L^{-4\nu}$ pays the number of base points; only the positive Hölder exponent leaves age decay. Domains leaving the window are paid separately by $e^{-c\kappa L^\nu}$. Using the full analytic domain without a generated configuration would set $\beta' = 0$ and let the scale sum grow with k. This is why the extension used in the component integrals is chosen through the generated field.

3.8. **Keeping stored gauges and passing between splittings.** A stored fluctuation is a real integration variable in the axial gauge in which it was created. A later response matches new averages only up to a node factor v. Acting on the stored variable by v^{-1} would move its real integration set when the response parameters are complex. We instead compensate the triple, prescribing the gauge at both the nodes and the sites where stored data are read.

Stored sites and the prescribed values. Let y be an inner top of level $m \ge 1$, with pillar $\Pi(y) = B^m(y)$. The L^4 sites of level $m-1$ in that pillar form its set of stored sites $\mathsf{A}(y)$. Their blocks $\Pi'(a) = B^{m-1}(a)$ partition the pillar; for $m = 1$ they are single sites. Write

$$\tau[\mathbf{U}](a) = (\mathcal{M}^{m-1}\mathbf{U})(y, a), \qquad \tau[\mathbf{U}](y) = 1.$$

This is the group average of the shortest coordinate-path transports from y to a. Its transformation law is

$$\tau[\mathbf{U}^g](a) = g(y)\tau[\mathbf{U}](a)g(a)^{-1}.$$

The condition $\tau[U](a) = 1$ says that the corresponding averaged real field is in the block axial gauge in which the level-$m-1$ fluctuation was stored.

For a shift $(\mathbf{A}, \mathbf{K})$, let $v[P; \mathbf{A}](y)$ be its accumulated node factor. Its smallness is measured by $\varepsilon_y(\mathbf{A}) = \sup_{\Pi^*(y)} \xi_m |\mathbf{A}|$, where ξ_m is the length of a level-m block in the current units and $\Pi^*(y)$ is the determining neighbourhood of that pillar. The needed node-factor estimate is

$$|v^{\pm 1} - 1| \le c_{\mathrm{r}} \varepsilon_y(\mathbf{A}).$$

At a stored site define the pin

$$\begin{aligned} u^\delta(y) &= v[P; \mathbf{A}](y)^{-1}, \\ u^\delta(a) = \mathsf{p}[P; \mathbf{A}](a) &:= \tau[\mathbf{U}](a)^{-1} v[P; \mathbf{A}](y)^{-1} \tau[e^{i\eta \mathbf{A}} \mathbf{U}](a). \end{aligned} \tag{126}$$

In particular $\mathsf{p}(y) = v(y)^{-1}$. The pins depend on the triple only through its complex configuration on $\Pi^*(y)$. Their required bound is

$$|\mathsf{p}^{\pm 1} - 1| \le c_{\mathrm{m}} \varepsilon_y(\mathbf{A}), \qquad c_{\mathrm{m}} = c_{\mathrm{m}}(L).$$

They are one at zero shift and are conjugated by $g(a)$ under a real gauge transformation. On the domain of the complex averages and their logarithms, they are holomorphic in the complex configuration and shift. Continuity in the real factor is required as well.

The depth-one geometry of stored bonds is important here. An endpoint of a stored bond is either a top itself or one of the sites of $\mathsf{A}(y)$ one level below its top. The pillar, transport and axial-gauge condition agree in all the stage and main-step structures while that field remains stored. Restricting to a finer determining sequence can lower the endpoint's layer index; it does not turn it into a site deeper than one level below a top. Thus these prescribed values cover the sites at which the stored variables are read.

Extending the pins inside their cells. Choose the tops whose stored sites are to be pinned. Every inner top of level at least two is eligible. A top of level one is eligible only when the bare sites within 2η of its pillar lie in pillars of level at most one. For a pinned pillar the turning cells are its blocks $\Pi'(a)$, centered at the stored sites. For every remaining inner pillar there is a single turning cell, centered at its top.

On a turning cell $B^q(a)$, take the covariant frame $T_a[U]$ associated with its local regular-gauge class. The frame is normalised at a and transports a Lie-algebra value at a to a Lie-algebra value at x. Its required covariance is

$$T_a[U^g](x) = \mathrm{Ad}(g(x)) T_a[U](x) \,\mathrm{Ad}(g(a))^{-1}.$$

It is this regular frame, with controlled differences, that enters the construction. Transport along a chosen coordinate path does not in general have the second-difference bound needed to control the derivative of the turned potential.

Fix an even C^2 bump equal to one at zero and supported in $[-1/3, 1/3]$. Its four-coordinate product, at the scale of the cell, is denoted by $\mathrm{bp}_a(x)$. For independent complex parameters $\zeta(a)$, put

$$\mu_\zeta(x) = \mathrm{bp}_a(x) T_a[U](x) \zeta(a), \qquad u_\zeta = e^{-i\mu_\zeta} \tag{127}$$

on that cell, and put $\mu_\zeta = 0$ elsewhere. On a cell consisting of one site use $\mu_\zeta(a) = \zeta(a)$. The inner thirds are disjoint. The turn is therefore defined without choosing an order for its factors.

The pinned parameters are

$$\zeta^{\mathrm{pin}}(a) = \frac{1}{i} \log \mathsf{p}(a)^{-1}, \qquad u^\delta = u_{\zeta^{\mathrm{pin}}}.$$

At an unpinned top this reads $\zeta^{\mathrm{pin}}(y) = \frac{1}{i}\log v(y)$. We use these formulas on a common open small-shift domain where all the tops involved are levelled: their local shift size is strictly below the fixed logarithm threshold. At a pinned level-one top the size is measured on the neighbourhood extending 2η beyond its pillar. Unused pillars have the identity turn. Analyticity and the Lipschitz estimates below are used with a margin inside this domain; they do not assert continuity of a rule that switches abruptly to the identity at the levelling threshold.

Define the turned shift by

$$e^{i\eta\mathbf{A}^\zeta}\mathbf{U} = (e^{i\eta\mathbf{A}}\mathbf{U})^{u_\zeta}, \qquad \mathbf{J}+\mathbf{K}^\zeta = \mathrm{Ad}(u_\zeta)(\mathbf{J}+\mathbf{K}). \tag{128}$$

The first identity defines $\mathbf{A}^\zeta$ by the local logarithm, keeping the original real factor fixed. On a cell of scale ξ_q, the triple must satisfy the local regular-gauge condition together with $\xi_q|A'|, \xi_q^2|\nabla_U A'| < c$. The local estimates to verify are

$$\begin{aligned}
\xi_q|\mathbf{A}^\zeta| &\le 2\xi_q|\mathbf{A}| + C|\zeta|,\\
\xi_q^2|\nabla_{\mathbf{U}}\mathbf{A}^\zeta| &\le 4\xi_q^2|\nabla_{\mathbf{U}}\mathbf{A}| + C(\varepsilon+|\zeta|),\\
\xi_q^3|\mathbf{K}^\zeta| &\le (1+C|\zeta|)\xi_q^3|\mathbf{K}| + C|\zeta|\mathsf{j}_0.
\end{aligned}$$

Here $\xi_q|\mathbf{A}| \le \varepsilon$, and $\xi_q^3|\mathbf{J}| \le \mathsf{j}_0$ is the local current bound. For a one-site cell the same bounds are used at scale η, without a smoothness requirement on its site value.

The matrix calculation behind the potential estimate is the local map

$$\mathsf{F}(w,X,Z) = \log(e^{-iw}e^{X}e^{iw+Z}).$$

For small arguments it is holomorphic, vanishes when $X = Z = 0$, and satisfies

$$|\mathsf{F}| \le 2(|X|+|Z|), \qquad |\Delta\mathsf{F}| \le C\max(|X|+|Z|)|\Delta w| + 4(|\Delta X|+|\Delta Z|).$$

Applied bond by bond, the covariant differences of the bump and frame give the first two estimates. The current estimate follows from the second identity in (128). The uniform regular-frame and local-logarithm estimates are the analytic inputs to this calculation; prescribing the values at the centers alone would not give the derivative bound.

Let $\|\cdot\|_y$ denote the local scaled shift norm controlling these potential, first-difference and current quantities. Let $\mathrm{nb}(y)$ consist of y and the pinned level-one pillars within 2η of its pillar, and set $\|\cdot\|_y^+ = \max_{y'\in\mathrm{nb}(y)}\|\cdot\|_{y'}$. For a top of level at least two this neighbourhood consists just of that top. Passing from a pinned pillar to its turning cells costs a fixed change of scale by L. The bound needed for the pinned shift is consequently

$$\|(\mathbf{A}^\delta,\mathbf{K}^\delta)\|_y \le C_{\mathrm{m}}\|(\mathbf{A},\mathbf{K})\|_y^+, \qquad C_{\mathrm{m}} = O(L+L^2c_{\mathrm{m}}). \tag{129}$$

It is required uniformly in the label, level and number of pillars. The same estimate holds in Lipschitz form on a half ball in the common small-shift domain. It also bounds every free turn with $|\zeta(a)| \le 2c_{\mathrm{m}}\varepsilon'_{y(a)}$, using the larger of the shift size and the local bounds ε' on the right. For an unpinned pillar the corresponding constant is $O(1+c_{\mathrm{r}})$.

Locality, domain room and preservation of the stored gauge. The turn at a bond over y reads only the pillars in $\mathrm{nb}(y)$. It is the identity outside the inner thirds of the turning cells, and on every pillar where the shift is zero. In particular, shifts whose nonzero local neighbourhoods are disjoint are compensated separately. For fixed U, the construction is holomorphic in the complex potential, current, shift and free turn parameters on their common domain. It is real at real data, commutes with conjugation, and is carried along by translations and the lattice isometries preserving the pillar geometry. These statements use the same properties of the node factors, local frames and logarithms.

The local domain test uses the bound for every free turn, before the pinned point is substituted. If s'_y denotes the available room expressed in the local shift norm, require

$$C_{\mathrm{m}}\|(\mathbf{A},\mathbf{K})\|_y^+ \le \mathsf{s}'_y$$

at each pillar meeting the target. Then the compensated triple and all the free turns in that polydisc remain in the old term's domain. For a slowly varying room profile the maximum over $\mathrm{nb}(y)$ costs only $e^{C\delta}$, where δ is its variation rate. The enlarged test that keeps a level-one pinned top levelled costs an additional factor L: its size is bounded by $Le^{C\delta}$ times the local room profile. This test must stay strictly below its logarithm threshold throughout the parameter polydisc. Checking the room inequality in the pointwise, gauge, tower and core clauses is still necessary; it is not a claim that arbitrary complex gauge copies belong to the same domain.

Finally the prescribed values give the exact identity

$$\begin{aligned}\tau[(e^{i\eta\mathbf{A}}\mathbf{U})^{u^\delta}](a) &= v(y)^{-1}\tau[e^{i\eta\mathbf{A}}\mathbf{U}](a)\mathsf{p}(a)^{-1}\\ &= \tau[\mathbf{U}](a).\end{aligned}$$

At an adapted real representative the last transport is one. The stored field is therefore read in its original gauge and its real coordinate has not moved. Although the frame extending the turn can depend on the real splitting, the values prescribed at tops and pinned stored sites depend only on the pair and the shift. Equality of the interpolated terms for two splittings uses the common holomorphic family below, together with this agreement at the sites carrying data.

Some terms are first formed on an axial slice. If $\mathfrak{u}[\mathbf{U}]$ is the holomorphic gauge carrying the pair to that slice, their extension has the form

$$F^\natural(\mathfrak{e}) = F(\mathfrak{u}\star\mathfrak{e}).$$

In $\mathfrak{u}\star\mathfrak{e}$, the complex pair is transformed by $\mathfrak{u}[\mathbf{U}]$, but the real factor and stored data use the real gauge $\mathfrak{u}[U]$. The extension is defined only where this transformed point belongs to the original analytic domain. It need not contain a fixed complex neighbourhood of every real field.

For example, a constant imaginary pure-gauge potential of size α can have its contour transport over the entire tower concentrated on one bond by the axial compensator. Its resulting potential there has size αL^k. Along this direction the allowed radius can be as small as $C\alpha_{1,k}L^{-k}$. Thus a full complex orbit cannot be used to argue that every gauge representative is in the same analytic domain. The component construction uses its explicit pinned points and domain margins.

For the noncollar terms, equality at two splittings of one pair is proved along a common holomorphic family. Take one splitting and write

$$A'_z = A'_{\mathrm{R}} + zA'_{\mathrm{I}}, \qquad \mathbf{J}_z = \mathbf{J}_{\mathrm{R}} + z\mathbf{J}_{\mathrm{I}}.$$

For $z = it$, $0 \le t \le 1$, the current is $\frac{1+t}{2}\mathbf{J} + \frac{1-t}{2}\mathbf{J}^*$. It stays in its convex current ball. In the other splitting choose the potential holomorphically so that its real factor gives this same pair. At real z both potentials are real.

The substituted-triple rule is first formed for free gauge-turn parameters ζ in a polydisc; the actual rule evaluates them at the pinned point ζ^{pin}. At real z and real ζ the two substituted pairs differ by a real gauge function equal to one on the datum. The old term, already a function of its pair and invariant under real gauges, takes the same value on them. The constructions remain defined on radii increased by a fixed factor and land inside the true domains. This margin gives an open complex neighbourhood of the chord and of the parameter disc. The identity theorem first in z and then in ζ proves equality at $z = i$ and at the pinned parameters. This argument does not require the entire analytic domain to be connected; it uses this specified common family. The technical point is to verify the common-domain margins along this family; the identity theorem then gives the equality.

At an evaluation point, with real stored variables, $\lambda = 1$ and decoupling parameters equal to indicator functions, the substituted pair is the pair of the old regular critical field. Real gauge invariance and the just established splitting identity therefore identify the localised formulas with the original integrals. For collars, the construction keeps the explicit triple and its stored-data integral as a single operation; a global complex-gauge extension is not substituted for it.

3.9. **Extracting the factors for failed tests.** The gain assigned to a failed test must come from the weight that is still present at that place in the history. In particular, a plaquette test is paid for by every real configuration with the recorded averages, not only by the regular background used to define the test. At a creation of step t, the paying Wilson coefficient is x_{t-1}. During a later healing, the old action already spent on the component must be excluded from the new payment.

Energy of the averages and the three tests. For a set S of level-t plaquettes, let $\mathcal{N}_0(S)$ be the union, over $p' \in S$, of the bare plaquettes whose corners lie in the blocks beneath the corners of that single plaquette p'. The estimate is

$$\sum_{p\in\mathcal{N}_0(S)} a_p(U) \geq \min\left\{ \tfrac{1}{4}\sum_{p'\in S} |\partial\mathcal{M}^t U(p') - 1|^2, \tfrac{1}{2}e_* \right\}, \qquad e_* = e_*(L) > 0. \tag{130}$$

It applies to every real U, without a regularity assumption. In particular its use under a constrained integral does not restrict the paying configuration to the regular minimizer.

We give the nonlinear iteration. In the comb gauge the derivative of one averaging step is the straight average Q, and its plaquette coboundary is given by the discrete Stokes formula

$$(Qa)(\partial p') = L^{-4} \sum_{z\in B(y_1)} a(\partial S_z).$$

Here y_1 is one corner of p', and S_z is the L-by-L plaquette square starting at z. Taking absolute values gives a nonnegative plaquette matrix P. Each square has L^2 plaquettes, so a row sums to L^2. For a fixed fine plaquette, the starts in its two plane directions have L^2 possibilities in total across the adjacent coarse blocks; the transverse coordinates are fixed. Its column therefore sums to $L^2/L^4 = L^{-2}$. Consequently

$$\|P\|_{2\to 2} \leq 1, \qquad \|P\|_{1\to 1} \leq L^{-2}.$$

All these norms use counting measure at the respective levels.

Put $f_l = |\partial\mathcal{M}^l U - 1|$. Whenever the plaquettes in the four corner blocks are sufficiently small, the comb-gauge formula and analyticity of $\mathcal{M}$ give

$$f_{l+1} \leq P_l f_l + q_l, \qquad q_l \geq 0, \qquad \|q_l\|_1 \leq C(L)\|f_l\|_2^2.$$

The correction is quadratic because the displayed Stokes average is the linear term. Its single-plaquette remainder is bounded by the square of the largest curvature in those blocks; replacing that square by the sum of squared curvatures and summing uses only a bounded number of block overlaps.

These estimates can be restricted to the ancestors of S. Starting from S, include at each preceding level all plaquettes read by the comb-and-square formula for the next set. The bare ancestor set is contained in $\mathcal{N}_0(S)$: every path and square stays in the blocks below the corners in question. Restricting P_l discards nonnegative entries, so the same row, column and operator bounds hold. Let $E_l = \|f_l\|_2^2$ on these ancestor sets. For a product of m restricted matrices,

$$\|P_{l+m-1}\cdots P_l v\|_2 \leq \|P_{l+m-1}\cdots P_l v\|_1 \leq L^{-2m}\|v\|_1.$$

Iteration of the nonlinear inequality therefore yields

$$E_t^{1/2} \leq E_0^{1/2} + C(L)\sum_{l=0}^{t-1} L^{-2(t-1-l)} E_l. \tag{131}$$

The linear part is marginal in dimension four. The nonlinear corrections have an ℓ^1 norm, and propagation of that norm is strictly contracting.

Choose e_* so small that

$$\frac{2C(L)}{1-L^{-2}}\sqrt{e_*} \le \sqrt{2}-1$$

and that $\sqrt{2e_*}$ lies inside every local small-curvature chart used above. Suppose $E_0 \le e_*$. If $E_l \le 2E_0$ at the preceding levels, (131) gives

$$E_t^{1/2} \le E_0^{1/2} + \frac{2C(L)}{1-L^{-2}}E_0 \le \sqrt{2E_0}.$$

Starting at $l=0$, this proves $E_l \le 2E_0$ for every level, and simultaneously justifies all the small-curvature formulas used in the iteration. The constants do not grow with t.

Finally $E_0 = 2\sum a_p(U)$ on the bare ancestor set. If the action on $\mathcal{N}_0(S)$ is below $e_*/2$, the preceding bound gives

$$\sum_{p'\in S} |\partial \mathcal{M}^t U(p') - 1|^2 \le 2E_0 \le 4 \sum_{p\in \mathcal{N}_0(S)} a_p(U).$$

If that action is at least $e_*/2$, the other alternative in (130) applies. This proves the estimate on the full real constraint set, including configurations which were not assumed regular at the outset.

Each payment is most conveniently checked in contrapositive form. If its paying quantity is below both the required threshold cost and a fixed small constant, the data must be regular, the localised orbit and contour variables must exist, and the test must pass. This formulation also treats the complementary outcome where a tested object is undefined.

For a plaquette test, failure at a test cube of step t requires a nearby coarse plaquette with $|\partial V_t - 1| \ge c\varepsilon_t$. Otherwise the local background estimate places the orbit inside the plaquette test. Together with (130), the required contrapositive gives paying action at least $c\varepsilon_t^2$ for every real field with these averages.

For the approximate fluctuation test, the Wilson action must be combined with the retained gauge-fixing term. Write $\mathrm{GF}_j(Y;V_j)$ for the part of the nonnegative soft gauge-fixing action located at the blocks in Y, so its weight is $\exp[-x_j \mathrm{GF}_j(Y;V_j)]$. A gauge transformation at a site of level $t-1$ which is not a level-t site can change V_{t-1} relative to the axial representative of the localised background without changing its action or V_t. Action alone therefore cannot control that difference. The local implication to establish is

$$\sum_{p \text{ near the test cube}} a_p(U) + \mathrm{GF}_{t-1}(\text{near the test cube}; V_{t-1}) \ge c\delta_{t-1}^2$$

on the failed approximate test, again for all compatible real fields. The gauge fixing supplies the directions invisible to the Wilson action.

For the true fluctuation, put $\mathcal{A}_j^{\mathrm{s}} = g_j^{-1}\mathcal{A}_j$, where $\mathcal{A}_j$ is the unscaled logarithmic fluctuation. Coercivity on the constrained fluctuation space gives the payment

$$\frac{1}{2}\langle \mathcal{A}_{t-1}^{\mathrm{s}}, \Delta^{(t-1)}\mathcal{A}_{t-1}^{\mathrm{s}}\rangle \ge c\frac{\delta_{t-1}^2}{g_{t-1}^2}. \tag{132}$$

The corresponding stored-field estimate uses (82), which must be proved for the locally retained conditional form. Its validity is not inherited from Schur positivity after arbitrary truncation. Collar and rim failures have the same payment, with constants determined by the fixed fractions $1/(4\Theta_1)$ and θ_c of their thresholds.

The contrapositive must also be checked after the change of variables on the strip. There its shift vanishes when the underlying unscaled integration variable is small. Consequently an undefined tested logarithm cannot be counted as a failed small-field test while that variable remains below its threshold. The lower bound is then applied to the integration variable, with its actual kept

Gaussian form. This is the local existence and coercivity check needed to include the undefined outcomes in the same estimate.

The one-step flow bounds compare x_{t-1} and x_t by a fixed additive constant, without requiring an individual increment to be positive. Thus

$$x_{t-1}\varepsilon_t^2 = \frac{x_{t-1}}{x_t} p_0(g_t)^2 \asymp p_0(g_t)^2, \qquad x_{t-1}\delta_{t-1}^2 = (A_1/A_0)^2 p_0(g_{t-1})^2 \asymp p_0(g_t)^2.$$

The threshold ratios are fixed before the coupling ceiling. Once the three local implications above have been established, each created cell therefore supplies $\exp[-cp_0(g_t)^2]$. The coefficient used for this conclusion is the actual x_{t-1} in the paying weight; comparison with x_t is only a subsequent estimate.

Disjoint shares of the action and the integration reserve. At a later level k, a piece (t, D) consists of a component of the region produced by step t on which no subsequent large-field operation has acted. Give it the bare plaquettes within one cell of its failed plaquette or approximate-fluctuation test cells. The buffers between new tests and the hull of an older region make these shares disjoint. On the share of (t, D) the spatial coefficient is exactly x_{t-1}: the older profiles have completed their transition and the newer profiles have not begun it. Extending U_k beyond its first determining region by the stored bare field, the allocation is

$$\begin{aligned} A(x_k(\cdot), U_k) &\geq A_{\mathrm{cr}}(U_k) \geq 0, \\ A_{\mathrm{cr}}(U_k) &:= \sum_{(t,D)} x_{t-1} \sum_{p \in \text{share of } (t,D)} a_p(U_k). \end{aligned} \tag{133}$$

All other plaquettes retain their nonnegative action. The allocation uses one Wilson weight, together with disjoint sets of retained gauge-fixing factors and Gaussian variables.

The fields entering the tests and the paying action are related by the averaging constraints. On the recorded set Γ_{t-1},

$$\mathcal{M}^{t-1} U_k = V_{t-1}, \qquad V_t = \mathcal{M}(V_{t-1}).$$

The first identity is part of the background datum and the second is imposed by the averaging delta function. Hence the payment estimate for every real field with those averages applies to U_k, although the test itself was formed with a localised background of step t. An estimate only at that background would not justify (133) in the component integrals.

Split each paying exponent into fixed halves: one half gives the small factor and the other remains for integration. In particular the coupling-dependent compact-group normaliser satisfies

$$\int e^{-\frac{1}{2}x(1-\frac{1}{2}\operatorname{Re}\operatorname{tr} u)}\, du \leq C \int_{|u-1|<\epsilon} e^{-x(1-\frac{1}{2}\operatorname{Re}\operatorname{tr} u)}\, du, \qquad 0 < \epsilon \leq 1, \quad x\epsilon^2 \geq 1.$$

Both integrals have order $x^{-3/2}$. Thus using half an exponent for payment leaves an integrable weight against its own normaliser. The stored Gaussian and gauge-fixing factors are treated with the same reserve. This allocation, followed by the pointwise product majorant in (143), is what permits gains at different places and levels to be multiplied.

New failures during a healing. Inside an operation of level m, the tests use the sharper thresholds δ'_j. A tree gauge there has length at most $C\check{R}_j^C$. A large bond in that gauge implies only that some nearby plaquette exceeds $c\check{R}_j^{-C}\delta'_j$. The resulting gain has the weaker form

$$\exp[-c\check{R}_j^{-C} p_1(g_j)^2] \leq \exp[-p_0(g_j)].$$

The second condition in (85) pays this power loss. It does not give another full creation deposit of order p_0^2.

Put $m_\flat = m - N_m$. The old *floor* is the union of the level-$m_\flat$ cells in $\Omega_{m_\flat}^c \cap X$ that do not touch $\Omega_{m_\flat}$. It is only the interior part of that old core, not all of $X \cap Z_{m_\flat}$. The plaquettes there, and

the gauge-fixing and stored Gaussian factors at levels below $m_\flat$, have already paid for the history preceding the wait. They are unavailable for a new failure.

Suppose now that the operation has a failed preparation test and leaves the drop-out X. Let X^+ be X enlarged by two layers of level-m unit cubes, and consider real stored variables with the recorded averages and test values. The old-core datum on Γ_l must have plaquette size at most $C\check{R}_l^2\varepsilon_l$ in its own level units. On these compatible evaluations the required fresh payment is

$$cx_m \sum_{\substack{p\subset X^+\\ \text{no corner in the floor}}} a_p(U) + \sum_{j=m_\flat}^{m-1} x_j \mathrm{GF}_j(X;V_j) + c\sum_{j=m_\flat}^{m-1} \|\mathcal{A}_j^{\mathrm{s}}\|_X^2 \geq (\log x_m)^{2\mathfrak{p}_1-C\mathfrak{r}} \geq p_0(g_m). \tag{134}$$

Here $\|\mathcal{A}_j^{\mathrm{s}}\|_X^2$ is the sum of squared Euclidean norms over the level-j stored bonds in X carrying the retained Gaussian form, with one orientation per bond and counting measure. The fields are expressed in their respective level units. The Wilson sum excludes every plaquette with a corner in the floor, and both other sums start at $m_\flat$. The estimate must hold for every real U compatible with the stored variables on X^+. This is the uniform version needed when a later containing drop-out is evaluated at those data.

For the contrapositive, let the fresh quantity on the left of (134) be small, and put

$$\varsigma = Cg_m(\log x_m)^{\mathfrak{p}_1-4\mathfrak{r}} \asymp C\delta'_m\check{R}_m^{-4}.$$

The smallness threshold has order

$$g_m^{-2}\varsigma^2 \asymp (\log x_m)^{2\mathfrak{p}_1-8\mathfrak{r}}.$$

Apply (130) to the stored plaquettes whose bare ancestors lie outside the floor. Their plaquette deviations are then bounded by $C\varsigma$. The kept gauge fixing and Gaussian coercivity control the remaining contour and fluctuation coordinates. At each stage the datum used by the test is thus regular on the side of the waiting collar where that test is made. The inverse couplings remain comparable throughout this one wait, since its length is $O(\check{R}_m) = o(x_m)$.

The background estimate required at this point retains the distance from the old floor. If $\hat{U}$ is any of the stage backgrounds and $j(p)$ is the greatest layer index whose region contains a corner of p, it has the form

$$|\partial\hat{U}(p) - 1| \leq C\left(\varsigma + \check{R}_m^2\varepsilon_m e^{-\delta_{\mathrm{G}}M\check{R}_m}\right) L^{-2j(p)} \tag{135}$$

on the region where the operation acts, outside the old core. Local regularity supplies the first term; locality across the waiting collar supplies the second. The collar has at least one cell of width in the local units and separates the tested region from the unspecified old field in the floor. The exponential makes the second term smaller than ς at small coupling. This is a local background estimate: it does not assert small action or small curvature in the excluded floor.

We spell out its use in the most demanding test, the main-step test $|\frac{1}{i}\log V'(b)| < \delta'_m$. Let $W_l = \mathcal{M}^l U$ be the averages of the compatible real configuration and $W_l^0 = \mathcal{M}^l\tilde{U}_0$ those of the basic background. On the shells the two fields have plaquette deviations at most $C\varsigma$, by the preceding estimate. They agree on the imposed exterior datum and, in the selected tree gauge, on the tree bonds. Close a bond b by a walk ℓ_b of tree and exterior bonds. The variable $V'(b)$ is conjugate to the quotient

$$W(\ell_b)\,W^0(\ell_b)^{-1}.$$

For unitary holonomies,

$$|V'(b) - 1| \leq |W(\ell_b) - 1| + |W^0(\ell_b) - 1|.$$

The shell-loop estimate applies to both terms. A loop made of n straight pieces in a shell of side $s \le CM\check{R}_m$ satisfies

$$\left|\prod_{b\in\gamma} W(b) - 1\right| \le C\mathsf{n}s^2 \sup_{p \text{ in the shell}} |\partial W(p) - 1|.$$

The shell is a box with a strictly interior box removed; in at least three dimensions this is the quantitative loop-filling estimate used here. It is applied within the shells of the chosen tree, rather than across the unknown floor. Consequently

$$\left|\tfrac{1}{i}\log V'(b)\right| \le CM^2\check{R}_m^2\varsigma \le C'M^2\check{R}_m^{-2}\delta'_m < \delta'_m$$

after reducing the coupling ceiling. Two powers of $\check{R}_m$ pay for the loop area and the remaining two powers absorb its fixed constants.

The earlier approximate tests use the one-step averaging rigidity, and the fluctuation, collar and rim tests use the kept Gaussian form. Thus the same chain is action of a compatible field, regularity of its stored averages, local regularity of the test background, and finally its contour or bond variable. Verification of (135) for the actual stage backgrounds, the stated shell-loop bound for their chosen trees, and the local kept-form lower bound are the model-specific inputs to this contrapositive. No gain is extracted from the main quadratic form q_X. The weakest scale obtained from the loop argument is

$$\check{R}_m^{-8}p_1(g_m)^2 \asymp (\log x_m)^{2\mathfrak{p}_1 - 8\mathfrak{r}};$$

the strict inequality $2\mathfrak{p}_1 - 8\mathfrak{r} > \mathfrak{p}_0$ makes it larger than $p_0(g_m)$. The number of places at which the operation can fail costs only its logarithm and is paid by a fixed part of this surplus.

The allocation separates nested operations. Every Wilson plaquette used by an earlier operation underneath X lies in the floor, and the gauge-fixing and Gaussian factors used underneath it have levels below $m_\flat$. The new Wilson sum has no corner in that floor, and the new sums of forms begin at $m_\flat$. This is why a new failure cannot be paid by the old creation deposit.

For the mass of the drop-out X itself, the indicator of the test that made X a drop-out is left unused. Its own-mass estimate below is obtained from its earlier creations and the carrying Wilson action on real compatible evaluations. The fresh failure estimate is used when X is a mark inside a later component, where the compatible field is defined on X^+ and the required fresh plaquettes and forms are available.

3.10. **The genealogy of a retained component.** Each unhealed component at one scale has a unique successor containing its coarse hull. The predecessors of a fixed terminal component form a tree. Write n_t^{cr} for its number of created cells at level t and n_t^{mk} for its number of marked components. If $C(u)$ is the region at a node u, let $|C(u)|_{n(u)}$ count its cells at that node's level. With a nonincreasing cost sequence $\widehat{r}_t$, the required combinatorial estimate is

$$\sum_u \widehat{r}_{n(u)}|C(u)|_{n(u)} \le C\sum_t (n_t^{\mathrm{cr}} + n_t^{\mathrm{mk}})N_t\widehat{r}_t. \tag{136}$$

Increasing N_t by a fixed additive constant is understood here. The bound is independent of the depth of the genealogy.

The count separates large components from stretches with only boundedly many cells. Coarsening contracts the occupied-cell skeleton of a large component over a fixed number of steps. A light component can persist without an event for only $O(N_t)$ steps, because it then enters the class on which the large-field operation acts. The events are creations, successors of marks, and joinings. A joining is assigned to a leaf below one of its nonpreferred predecessors. Fixing a first predecessor at each branching makes this assignment injective: the first departure from that predecessor on the upward path identifies the branching. Every leaf contains a creation. Thus joinings do not require a new, unpaid gain.

For the analytic estimates take

$$\widehat{r}_t = CM^4\check{R}_t^5, \qquad N_t\widehat{r}_t \le CM^4\check{R}_t^6.$$

The four powers from cell volume, the logarithmic normalisation per site, and the possible waiting time are all included. Choosing $\mathfrak{p}_0 > 6\mathfrak{r}$ ensures that even a mark's weaker gain $p_0(g_t)$ exceeds this cost. More explicitly, the excess of gains over the summed local costs has the lower bound

$$\sum_t cp_0(g_t)^2 n_t^{\mathrm{cr}} + \sum_t p_0(g_t) n_t^{\mathrm{mk}} - \sum_u C\log x_{n(u)}(M\check{R}_{n(u)})^4 |C(u)|_{n(u)} \ge 2\sum_t p_0(g_t) n_t^{\mathrm{cr}}. \tag{137}$$

The constants in this inequality must be chosen once for the whole run. It is the quantitative margin used both to sum histories and to retain decay in their age.

The combinatorial estimate separates a geometric input from the ensuing summation. Let o_n be the occupied cell count in the large part of the genealogy at level n, and let l_n count its light components. For a fixed number s_0 of steps, the occupied-skeleton estimate required by the construction is

$$o_n \le \tfrac{1}{2} o_{n-s_0} + Cl_{n-s_0} + C\sum_{t=n-s_0+1}^{n} n_t^{\mathrm{cr}}.$$

The factor $1/2$ concerns occupied cells, not just a linear diameter. This contraction is the geometric estimate to verify. Multiply it by the nonincreasing $\widehat{r}_n$ and sum in n. The delayed occupied sum is bounded by the original one, so its half is absorbed on the left. The creation sum involves only s_0 levels. This bounds the total weighted occupied volume by weighted light stretches and creations.

A light stretch ends at an event or at an operation after at most CN_t steps. Monotonicity bounds its cost by $CN_t\widehat{r}_t$ at its first event level. Joinings can be assigned to distinct creation leaves by choosing one preferred predecessor at each branch and following a nonpreferred one down to its first leaf. A leaf is charged a bounded number of times, and its earlier event cost is no smaller. Summing these stretch costs gives (136). This derivation explains why an estimate of diameter alone or a cost sequence increasing with level would not give that conclusion.

It also gives the credit for a component which has not yet completed its wait. Continue its hulls fictitiously, with no new creation, until it enters the operation class. Apply the same count to this extended genealogy, within the levels at which the flow bounds are available. The unspent part of the creation gains bounds every future volume cost of that continuation. A credit cannot be spent twice: on a later joining it is assigned through the new genealogy, and on a marked outcome the new failed test supplies the weaker p_0 gain for the next wait.

The deposit constant in $cp_0(g_t)^2$ is chosen after the fixed fractions in the collar and rim tests. Shrinking g_t cannot fix a deposit constant larger than the available gain of one such test: both have the same logarithmic degree. After this constant has been fixed, the strict inequality $\mathfrak{p}_0 > 6\mathfrak{r}$ makes

$$(C + c_a)M^4\check{R}_t^6 = o(p_0(g_t)) \quad (g_t \longrightarrow 0)$$

for each fixed age reserve c_a. A fraction $\mathsf{w} < 1$ of creation deposits can also be withheld, since $(1-\mathsf{w})cp_0^2$ still exceeds all these costs and $2p_0$. The ceiling is chosen after c_a and w.

3.11. **Uniform masses and local exponentiation.**

The integrated history estimate. Consider an operation of level i on a component X. Let ω be its local history and cubic frame, ξ its stored variables and $\mathbf{p}$ the exterior field and current pair. Once the local test-payment, retained-form and background estimates above hold, the mass estimate is

$$\frac{1}{|W_4|}\sum_\omega e^{cv(\omega)} \int^* \sup_{\mathbf{p}\in\mathcal{U}_i(X)} |e^{\Phi_\omega(\xi;\mathbf{p})}|\, d\mu_\omega(\xi) \le e^{-(2-c')p_0(g_i)}, \qquad 0 < c' < 1. \tag{138}$$

The loss c' is chosen sufficiently small to leave the reserve used in the subsequent local exponentiation, and the number $c \ge 0$ is fixed before the coupling ceiling. The volume v counts a stored site of level j with weight $\log x_j$, counting each pair consisting of a level and a site only a bounded number of

times. It is additive when the local history is cut at its inner operations. The upper integral $\int^*$ permits a supremum that is not known to be Borel. The construction below gives Borel majorants depending only on the variables of this component. The positive measure μ_ω contains the fixed stored-variable and auxiliary reference measures, together with the history's test factors. The exponential weights are included in Φ_ω. This separation is retained before taking the exterior-pair supremum.

A majorant at fixed stored variables. For a healed component or an island, the main-step formula separates the exponent as

$$\Phi_\omega(\xi;\mathbf{p}) = \Phi^\circ_\omega(\xi) - \tfrac{1}{2} q_X(\xi;\mathbf{p}) + V_X(\xi;\mathbf{p}), \qquad q_X = \|\zeta^{1/2}\mathrm{d}\mathsf{H}B\|^2 \geq 0. \tag{139}$$

Here $\zeta = \phi_X(1-\phi_{\mathrm{core}})$ is the nonnegative shell cutoff in (97), at the level of this component. The norm square is formed at the real glued background belonging to the G-valued factor of a splitting of $\mathbf{p}$. This is not a positivity assertion for a quadratic form at a complex background. The last two summands can depend on the splitting; their sum is the invariant function of the pair. Every other factor belongs to Φ°_ω. These factors include the old core Wilson action, the kept gauge-fixing and Gaussian weights, chart densities and elimination factors, and inner operations evaluated at real pairs determined by ξ.

The quadratic factor of an island. For a second-class component X of level i, retained in a label of level $k' \geq i$, the two pair-dependent functions in (139) have different carriers and different bounds. The quadratic factor can be kept as one function with carrier X. Indeed, on the recorded support the sharper main-step test gives

$$|B(b)| < x_i^{1/2}\delta_i' = A_1(\log x_i)^{\mathfrak{p}_1}.$$

The quadratic background estimate needed here is $q_X \leq C\mathsf{n}\sup|B|^2$, where the number of bonds entering the form satisfies $\mathsf{n} \leq C\check{R}_i^5$. Consequently

$$0 \leq q_X \leq C(\log x_i)^{2\mathfrak{p}_1+5\mathfrak{r}} \leq C(\log x_i)^{2\mathfrak{p}_0-3\mathfrak{r}-2}. \tag{140}$$

The last inequality uses $\mathfrak{p}_1 \leq \mathfrak{p}_0 - 4\mathfrak{r} - 1$. Here, as above, the background used to define the norm square is the real glued background of the chosen splitting. Its exterior datum on the bonds inside X is $\mathcal{M}^i(U)$, determined by $U|_X$. The quadratic function reads no other member of the triple. Its uniform bound is required for the regular real factors in the evaluation domains, including $|\partial U - 1| < 2\alpha_{0,i}\eta_i^2$.

This statement concerns the sum of the walk terms. In the expansion

$$q_X = \sum_{\omega_1,\omega_2} \langle \mathrm{d}\mathsf{H}_{\omega_1}B, \zeta\,\mathrm{d}\mathsf{H}_{\omega_2}B\rangle,$$

each summand is formed at a glued minimizer on all of X. Even if its displayed walks occupy a smaller set, that minimizer can read the exterior datum and stored variables throughout X. Thus one cannot assign that smaller set as the carrier of the summand. After the absolutely convergent walk sum is taken, (140) bounds the single carrier-X factor.

The component remains outside the small-field region of level $i+1$. The waiting rule prevents a later operation from remaking that scale while the component remains standing: an operation meeting its enlargement would contain X and would have to wait past level i. Hence every cube of $X^\sim$ stays outside the action-sequence region $\Omega^{\mathrm{a}}_{i+1}$. The hull of X among its stored components has a tree with one corner, so its reduced hull length is zero. This permits the bound for $-q_X/2$ to be filed as an oscillation of order $(\log x_i)^{2\mathfrak{p}_0}$ on X.

The function V_X instead has carrier $X^\sim$. A hull for this carrier must also traverse its collar. More importantly, its available bound is proportional to the number of stored sites, as in (141); that number is not bounded by a function of x_i alone. The main step's Wilson contribution to V_X consists of differences between its backgrounds. The new label's Wilson action at the new background, including the plaquettes in X, is already part of the new action. It is not an additional term of

V_X. This distinction is needed both when summing the Wilson differences and when using the mass estimate: the quadratic factor has a levelwise bound, whereas V_X is paid by the stored-history volume.

The collar estimate for the localized terms is

$$\sup_{\mathbf{p}\in\mathcal{U}_i(X)} |V_X(\xi;\mathbf{p})| \le C_V N_{\mathrm{st}}(\omega), \tag{141}$$

where stored sites are counted at their own levels. In this estimate older regular terms are summed as pointed totals with both subtractions. At age ν, their current-volume count $L^{4\nu}$ is canceled by the quadratic field factor $L^{-4\nu}$. The remaining weights have a convergent sum of the form

$$\sum_{\nu\ge 0}(1+\nu)^C \left[L^{-\beta_{\mathrm{H}}\nu} + L^{-\nu} + L^{4\nu}e^{-c\kappa L^{\nu}}\right] < \infty.$$

These terms come from the regular potential on a generated configuration, the surface of its localization box, and domains leaving that box. In the old core, the part at zero response is put in Φ°_{ω}; the remaining anchored response decreases by at least $(4L^8)^{-1}$ per level, exceeding the L^4 growth of the number of finer sites. The Wilson terms are combined with their net profiles before this estimate. Separate Wilson differences would have a cost increasing with the age. These are the main-step estimates behind (141).

Discarding only the nonnegative q_X gives the explicit bound

$$\sup_{\mathbf{p}} |e^{\Phi_\omega(\xi;\mathbf{p})}| \le b_\omega(\xi) := \exp(\Re\Phi^{\circ}_{\omega}(\xi) + C_V N_{\mathrm{st}}(\omega))\,.$$

This is a Borel function of ξ, independent of every other component's variables. Thus the supremum has been bounded before any integral is taken. Taking the supremum of the integral would not supply this function for the later substitutions.

The real inner background and the available action. At this point ξ lies only in the support S_ω of a single local operation. It need not be part of the support of a whole effective term. The inner background must therefore be constructed from the local tests still present in that operation. Up to the depth $i_\flat$, these tests have not been altered by later preparations. At the innermost box, the half-threshold plaquette tests of the last shell replace the tests that were removed. On the remaining datum the field is e^{ig_iB}, with $|g_iB| < \delta_i' \ll \varepsilon_i$. The local background theorem then gives the real $U''(\xi)$, compatible with the stored averages on the enlarged old core and regular in each layer on one further enlargement.

Keep its old-core Wilson factor in Φ°_{ω}. On every share used by the outer genealogy its weight is at least cx_t at the level t of that event. Indeed, the core cutoff equals one there, and the spatial inverse coupling is a convex combination of the corresponding running coefficients. At a point entering the large-field region the coupling stops running within $O(\check{R}_t)$ levels; the drift over that interval is $O(\check{R}_t) = o(x_t)$. One cannot replace this observation by a lower bound on all later running couplings.

Let $G_j(\xi)$ be the nonnegative gauge-fixing action left at level j, and let γ_{p} be the retained-form lower bound. The disjoint payment allocation gives

$$A(\upsilon, U'') + \sum_j x_j G_j(\xi) + \tfrac{1}{4}\gamma_{\mathrm{p}} \sum_j \|\mathcal{A}^{\mathrm{s}}_j\|^2 \ge c_{\mathrm{cr}} \sum_t p_0(g_t)^2 n^{\mathrm{cr}}_t + \sum_t p_0(g_t) n^{\mathrm{mk}}_t. \tag{142}$$

Here υ is the retained core Wilson weight; the genealogy is cut at the inner islands. The creation and mark gains in this formula use disjoint Wilson plaquettes, gauge-fixing factors and stored Gaussian bonds. A newly failed operation uses no plaquette with a corner in its old floor, and no form below its waiting depth. Thus its gain is distinct from the older creation gains.

Before splitting any paying exponent, bound all remaining local terms, including terms quadratic in bounded stored fields, by their costs per site. Then one fixed part of the paying exponent gives (142), and the remaining part is used for integration. In particular a gauge-fixing normalizer is

integrated against a fixed fraction of its own exponent:

$$\int e^{-\frac{1}{2}x(1-\frac{1}{2}\Re\operatorname{tr} u)}\,du \le C\int_{|u-1|<\epsilon} e^{-x(1-\frac{1}{2}\Re\operatorname{tr} u)}\,du, \qquad x\epsilon^2 \ge 1.$$

Both sides have order $x^{-3/2}$. Stored Gaussian coercivity similarly leaves a product Gaussian envelope. No further analytic factor is charged to this integration reserve after it has been used.

For pieces (t, D) not yet reached by an operation, this gives the pointwise product

$$e^{-A_{\mathrm{cr}}}\prod_{(t,D)} w_{t,D} \le \prod_{\text{creations}} e^{-cp_0(g_t)^2}\prod_{(t,D)} \bar{w}_{t,D}(\xi_{t,D}), \qquad \int \bar{w}_{t,D} \le e^{C\#_t(D)}. \tag{143}$$

The variables $\xi_{t,D}$ of different pieces are disjoint, and $\#_t(D)$ counts sites, rather than large-field cells. The functions $\bar{w}_{t,D}$ contain no background parameter. Their integral cost has no additional $\log x_t$ per site; the separate vacuum and scaling constants are included in the numerical history account.

Induction over the inner islands. For an inner island Y of level $m < i$, the pair $\mathbf{q}_Y(\xi)$ at which its factor is evaluated is real and belongs to its fixed analytic core. The margin follows from

$$\frac{C\check{R}_m^2\varepsilon_m}{\alpha_{0,m}} \le C(\log x_m)^{\mathfrak{p}_0+2\mathfrak{r}-q} \longrightarrow 0.$$

Replace this factor by its Borel majorant at fixed inner stored variables. Its sum and integral are at most the already proved mass $\mathsf{M}(m, Y) \le 1$. Additivity of v makes these inner mass sums separate. In this induction the estimates at real pairs are the inputs; the complex-pair supremum is a consequence of (139), rather than a new real positivity assumption.

The final island test χ_m^c is outside that inner mass. It supplies, in the parent operation, a failed ordinary plaquette test of the new field on Y, away from its boundary margin. Its gain $cp_0(g_m)^2$ is drawn from the parent's core Wilson weight. Thus the island is a creation leaf in the cut genealogy, while its entire interior is paid by its own mass. There is no second use of its interior deposits.

Use (136) and (137) on this cut genealogy. All constants from normalizers, chart Jacobians, integration and the unrecorded outcomes are charged as a fixed multiple of the stored-site volume. The weight cv increases that multiple. The strict inequality $\mathfrak{p}_0 > 6\mathfrak{r}$ makes the event gains pay the volume and its entropy at a coupling ceiling chosen after c. The remaining gain is at least

$$2\sum_t p_0(g_t)n_t^{\mathrm{cr}}.$$

Every leaf of the cut genealogy contains a created cell, including the creation just supplied by an inner island's final test. The entropy is summed from the terminal region down: for example,

$$\sum_{Z_{n-1}\subset \mathsf{pa}^{-1}Z_n} L^{-4|Z_{n-1}|_{n-1}} \le (1+L^{-4})^{L^4|Z_n|_n} \le e^{|Z_n|_n}.$$

Between events the geometry is deterministic. The possible place where a preparation is interrupted costs its logarithm and is paid by a fixed part of the weakest mark gain. The ledger already includes these entropy costs. The lower flow bound gives, uniformly for every creation level $t \le i$, $2p_0(g_t) \ge (2-c')p_0(g_i)$ after the coupling ceiling is reduced. This proves (138).

The own mass of a drop-out. A drop-out has no real inner background of its own and its operation contains no outer Wilson factor. Define its envelope instead over real evaluations U compatible with ξ. The evaluations have the recorded layer regularity, the stored averages, all recorded tests except the one that made this component a drop-out, and a real core pair at every inner island. Include the carrying Wilson factor $e^{-A(v\mathbf{1}_X,U)}$ on bare plaquettes whose corners all lie in X. The free form and Jacobian parameters range only over their one-sided coercivity and upper bounds.

The allocation (142) holds for every such evaluation, by the all-real-field version of (130). Its right-hand side and the remaining Gaussian and compact-group envelopes are independent of U.

They therefore give a Borel majorant depending only on ξ, before the supremum over evaluations is taken. Integrate this envelope by the same island induction, entropy and ledger. This proves the same own-mass bound for the drop-out without using its defining failed test.

The supremum must remain fibred over ξ: an arbitrary independent real field would include a flat field and remove the Wilson payment. Complex evaluations of this Wilson factor are not used. Nor is the root's fresh failure estimate applied after discarding its indicator. That failure is paid when the drop-out stands inside a later containing component, whose compatible field is defined on X^+ and whose fresh paying window is available.

Uniformity at later levels and multiplication of masses. The domains $\mathcal{U}_i(X)$, and the real evaluation fibres for a drop-out, are fixed from the mark and the data through level i. To see why they contain later arguments, retain the tests even at an interruption between their insertion and the following integration. On a belt the new plaquette and approximate-fluctuation tests control the nonpivot bonds; the averaging constraint supplies the pivot bond. The layer estimates are

$$|\partial U_k - 1| \le C\check{R}_j^2 \varepsilon_j \eta_j^2, \qquad |J(U_k)|_{\text{level-}j} \le C\check{R}_j^2 \varepsilon_j.$$

There is no loss in $k-j$: a more distant layer's influence decreases exponentially while its threshold grows at most as a square root, and $\sup_{a\ge 0}(1+a)^{1/2}e^{-(a-1)^+} < \infty$. The preceding radius margin places each pair in its fixed core. For a drop-out the carrying configuration, its spatial Wilson weight and the actual forms and Jacobians are consequently members of the prescribed evaluation fibre.

For a full term, cut at its outermost marks. Replace each marked operation by the own-variable Borel envelope above; use (143) on the pieces outside those marks. All remaining variable groups are disjoint after the averaging constraints have been solved. Tonelli now integrates the product and sums each inner history into its own mass. This establishes the product estimate under the actual background substitutions. Bounds on the integral at each fixed exterior argument alone would not justify this step.

After healing, X belongs to the small-field region at every earlier level. The new action must contain all regular terms there, and all remainders already available to this step, even on domains that previously met $Z_j \cap X$ and had never been integrated. Complete the action by adding those terms at the new background. Subtract precisely the same terms inside the local functions $V_{\mathfrak{X}}(Y)$. The terms actually present before healing enter there at their substituted triples. The part of $R^{(k)}$ to be made by the present exponentiation is not included in this completion. This makes the operation noncircular.

Terms made by the ordinary step and by the stages are treated in the same way. Far from a treated component they are carried unchanged. Near it use the corresponding term formed for the history with that component's past erased, adding it to the new action and subtracting it in $V_{\mathfrak{X}}$. A domain touching a healed collar is enlarged to contain the entire collar. This convention makes every such domain contain each collar wholly or not at all.

It is essential to combine the Wilson pieces before this localisation. With the weights of (93), their sum is

$$-\sum_{i\le k}\varphi_i b_i + \sum_{m=1}^{N_k}(x^{(m)} - x^{(m-1)}) = -\sum_{i\le k}\varphi_i'' b_i. \tag{144}$$

The last profile φ_k has disappeared from the total on the right. The coupling difference, counterterm and final stage difference must use one cover so their response to a change of this sorting profile cancels. If they were localised individually first, erasing X could leave a trace a fraction of a large cell away from its M-cube collar. The subsequent family would no longer be a polymer family in the healed collars.

For noncollar domains the resulting bound is

$$|V_{\mathfrak{X}}(Y)| \le \alpha' e^{-(1+2\beta_{\mathrm{s}})\kappa d^{\mathrm{out}}(Y)}, \qquad \alpha' = C\max\{B_{\mathbf{T}}, M^4E_0, C_{\mathrm{S}}\} + C \ge 1. \tag{145}$$

The length d^{out} exempts the standing island hulls and the healed collars. At a collar only the volume bound is used, and that term stays inside the component operation. A level-k interaction crossing a collar can enter in full, with no counterpart at the other background. Its size is of order E_0, which does not go to zero with the coupling. This is the reason for the bounded amplitude $\alpha' \geq 1$.

The old boundary bound B does not enter α'. Outside a collar its contribution is multiplied by

$$C(\log x_k)^C \left[g_k^{-1} e^{-cM\check{R}_k} e^{CL\check{R}_k} + e^{-c\kappa L\check{R}_k} + e^{-c\bar{\kappa}\check{R}_k} e^{CL\check{R}_k} \right]. \tag{146}$$

The three summands come from a background difference, a long domain crossing the component, and decoupling across its width. Lower-level position entropy gives the two factors $e^{CL\check{R}_k}$. Choose M and $\bar{\kappa}$ after L, so that the negative exponents dominate that entropy. Since $\check{R}_k \geq (\log(x_k - C_\sharp))^{\mathfrak{r}}$ with $\mathfrak{r} \geq 2$, they then dominate the inverse coupling and all logarithmic factors as well. For fixed B, decreasing the coupling ceiling makes its entire contribution bounded independently of B. Inside the collar the old terms are charged as volume instead.

The faithful localisation rules give the exact locality relation

$$V_{\mathfrak{X}'}(Y) = V_{\mathfrak{X}}(Y) \tag{147}$$

whenever $\mathfrak{X} \subset \mathfrak{X}'$ have the same outside standing data and Y meets no added healed collar. The rules are linear in classes of terms, depend only on the nearby history, and do not separately read the sorting data whose cancellation was used in (144). This is the local identity used to extract the empty family. A decay estimate alone would not replace it.

The sum over components is next converted back to an exponential of local terms. Some terms attached to islands have bounded, rather than small, amplitude. They must be extracted before the expansion. Keep the collar interactions inside their integral operations. If $\mathfrak{X}$ is a family of healed components, enlarge each remaining interaction domain Y by every healed collar touching it and call the result $Y^{\mathfrak{X}}$. Subtract the empty-family terms only after grouping them on these enlarged domains:

$$W_{\mathfrak{X}}(Y') = V_{\mathfrak{X}}(Y') - \sum_{Y^{\mathfrak{X}} = Y'} V_{\varnothing}(Y), \qquad W_{\varnothing} = 0.$$

The local exponentiation identity then has the form

$$\begin{aligned} &\sum_{\mathfrak{X}} \Big(\prod_{X \in \mathfrak{X}} T'_k(X) \Big) \exp \sum_{Y} V_{\mathfrak{X}}(Y) \\ &\quad = \exp \sum_{Y} V_{\varnothing}(Y) \sum_{\mathfrak{X}} \Big(\prod_{X \in \mathfrak{X}} T'_k(X) \Big) \exp \sum_{Y'} W_{\mathfrak{X}}(Y'). \end{aligned} \tag{148}$$

The product of integral operations acts on the entire exponential following it. At retained exterior argument p, one operation is $[T'_k(X)F](p) = |W_4|^{-1} \sum_\omega \int e^{\Phi_\omega(\xi;p)} F(\xi;p)\, d\mu_\omega(\xi)$. In (148), $V_\varnothing$ is evaluated at the retained argument, and the grouped differences remain inside the component integrals. The resummation must retain the domain enlargement, localisation and these evaluation conventions. Every difference on the right touches a healed collar. Each remaining polymer then contains a healed component and has its small mass available. The cost of nearby bounded interactions and position counting is absorbed by that mass, using a bound of order $C\check{R}_k^4 \ll p_0(g_k)$. Expanding before extracting $V_\varnothing$ would create polymers made only of bounded island terms and would not give a convergent small-activity expansion.

The resulting exponential is a formula for sums of histories with the same standing regions and marks. It is not the action of each individual history. The regular and remainder interactions in the interior are constructed once on the infinite lattice and restricted to each nonwrapping domain. Boundary terms may retain the label and its stored data.

More quantitatively, the mass left after the empty family is extracted must pay the expansion cost

$$1 + \left[C\kappa + e^{C\alpha' + C\kappa} \right] (C\check{R}_k)^4 \leq c p_0(g_k). \tag{149}$$

The fourth power counts the current M-cubes in the collared component, the tree that traverses the region ignored by its hull length, and the possible positions of its attachments. The amplitude α' is fixed before the coupling ceiling, which therefore depends on it. The bound is possible because $\mathfrak{p}_0 > 4\mathfrak{r}$.

After the Mayer and polymer sums, the new local exponent decomposes as

$$\mathbf{R}'^{(k)} = V_\varnothing + \mathbf{R}'_\sharp, \qquad |\mathbf{R}'_\sharp(Y)| \le C e^{-(1+c)p_0(g_k)} e^{-(1+\beta_{\mathrm{s}}/2)\kappa d_k^\natural(Y)}. \tag{150}$$

The empty-family function vanishes except near a new island. One current cell inside the new small-field region, the result is the new remainder $R^{(k)}$; the rest is boundary data. The stated exponent belongs to the sum over histories with the same standing part. A history containing a particular healed component and the history with its whole past removed are different summands in this sum. Neither summand by itself has the final exponent.

At a new island the relevant bound is an oscillation, not an absolute bound on its exponent. With its factor supports R_f, it has the form

$$\sum_{R_f = R} \operatorname{osc}_{\mathbf{p}} \Phi_{\omega,f} \le C(\log x_k)^{2\mathfrak{p}_0} e^{-\kappa d_k^\natural(R)}. \tag{151}$$

For the whole collar the length is zero. Away from the old core there are only $(\log x_k)^{C\mathfrak{r}}$ sites at their own scales. The regular terms there are grouped as pointed totals with both subtractions, and each Wilson term is taken at its net weight. The quadratic form of the main integral is at most $CA_1^2 \check{R}_k^5 (\log x_k)^{2\mathfrak{p}_1}$. The exponent margins make both contributions at most $C_V(\log x_k)^{2\mathfrak{p}_0}$. The old core can contain arbitrarily many stored levels. There the anchored amplitude decreases by at least $(4L^8)^{-1}$ per level, more than paying the L^4 site count. Its zero-response value is in Φ°_ω, and only its oscillation is summed. Choosing the constant in (151) after C_V closes this part of the island presentation.

Boundary activities use a hull length rather than ordinary domain length. Its exempt sets are the complete level-j hulls of islands still standing in the label. A tree must meet every ordinary cube and must touch each chosen exempt hull contained in its domain, but it need not fill that hull. The minimum length divided by M is $d_j^\natural$. For a collar of a new island it can be zero, consistently with an interaction of bounded amplitude on the whole collar.

This length is not freely subadditive under joining across an exempt hull. Two terms meeting opposite sides of a large exempt box can each have zero hull cost, while a joined tree connecting both sides has positive length of order its side. The construction uses an anchored joining estimate: the domains being joined must each contain a cube meeting the integration region. Entropy is pinned at such a cube outside the exempt hulls. These anchor conditions belong to the local expansion used in (148).

An island of bounded activity is kept as an integral operation. Normalising it a second time and expanding its logarithm would consume its small mass and leave an order-one activity. Likewise, a boundary slot is either carried unchanged or replaced by a newly made slot. Its bound is the maximum of the carried and new bounds, not their sum at each level. If a later component absorbs an island, its donor history removes that island's entire earlier record before the new hull length is assigned. These conventions are needed for a single boundary constant to remain uniform in depth.

3.12. **Closing the common induction.** The object carried by the induction is a presentation of a measure, rather than the measure alone. Write it as the array

$$\mathfrak{P}_k = \big((x_j), (E^{(j)}, \widehat{E}^{(j)}, b_j), (R^{(j)}, \widehat{R}^{(j)}), (B_{\mathcal{S}}^{(j)}), (\mu_\omega, \Phi^\circ_\omega, (\Phi_{\omega,f})_f)\big)_{j \le k}. \tag{152}$$

The companion families are part of the array on all admissible tori at once. So are the origin, domain and old analytic space of every boundary term, and the measures and factors of standing islands. Two different arrays can represent the same density. The gain at later backgrounds and

the local dependence on a history cannot be recovered from that density or from a bound on each family alone.

The inductive clauses have distinct mathematical roles. First, the regular families are identified with the families produced from the earlier array by the full-lattice small-field step, on both radius systems. Their coefficients are $b_j = b[\widehat{E}^{(j)}] = x_{j-1} - x_j$. Their pointed totals have the lattice covariance needed for subtraction. Second, every activity has its specified domain, reality and gauge invariance. It is a function of its pair at the admissible splittings; near stored tests it has the switch structure already described. The actual evaluation triple of each carried term belongs to that domain when its tests pass.

Third, the array is coherent under local charts matching the history trace. For an island this means matching its Borel measure and set of stored variables, its factors and its stored-only exponent, and the factorwise identity

$$\Phi_{\iota\omega,\iota f}(\iota\xi;\iota\mathbf{p}) = \Phi_{\omega,f}(\xi;\mathbf{p})$$

whenever ι matches the data on R_f. Its mass is bounded at the level where the island was made; only its oscillation in the outer pair is bounded. No bound on the large negative quantity Φ°_{ω} is imposed. The drop-out mass is obtained from its compatible real evaluation set, not by assigning it these island clauses. Finally, each term together with its stored variables is a positive measure at real data.

For the source construction one separates these weights from the real choices used to define tests and reference measures. Fix a real frame θ_j with the same weak two-sided drift bounds and nested cube sizes. Make all thresholds, scaling, gauge fixing and auxiliary measures with θ_j, and form the geometric size with $\min_{i\le j}\theta_i$. The weight x_j is allowed in a tube

$$|x_j - \theta_j| \le t_j, \qquad t_j \asymp c\theta_j(\log\theta_j)^{-C}.$$

This is wide in absolute terms but makes the weight discrepancy a small potential at the chosen radii. In the unsourced run of this section $\theta = x$, so the deviation is zero. Freezing these real choices is the interface used in section 4; it does not change the exact mass-preserving kernels.

The regular input load can now be checked quantitatively. If $w_{n,j}$ is the squared ratio of output to input radius, then

$$\begin{aligned} w_{n,j} &\asymp \frac{\theta_j}{\theta_n}\left(\frac{\log\theta_n}{\log\theta_j}\right)^{2q} \le C(n+2-j),\\ \sum_{j\le n}\omega_{n+1-j}\|E^{(j)}\|_{j,\kappa} &+ \sum_{j\le n} w_{n,j}\|R^{(j)}\|_{j,\kappa} \le N_\star. \end{aligned} \tag{153}$$

The two ratios in the first line are not simultaneously harmful. If $\theta_j \ge \theta_n$, the logarithmic ratio is at most one, while the flow bounds θ_j/θ_n by a constant times $1+n-j$. If $\theta_j < \theta_n$, the lower flow bound gives $\theta_n \le \theta_j + C_\sharp$, so the logarithmic ratio is uniformly bounded. The first sum in the second line uses pointed subtraction; the second uses the remainder power and the lower drift. This is the bounded input required by the arbitrary-list step.

Every new cluster is classified as marked if one of its vertices descends from a remainder, an earlier marked vertex or its counterterm. Otherwise it is unmarked. This gives

$$E^{(j)} = E^{(j),\mathrm{u}} + E^{(j),\mathrm{m}}, \qquad b_j = b_j^{\mathrm{u}} + b_j^{\mathrm{m}}, \qquad b_j^{\mathrm{u}} = b_j^0 + r_j, \qquad b_j^{\mathrm{m}} = q_j. \tag{154}$$

Each family carries its own counterterm. At level k, let k_1 be the last index with $L^{-(k-j)}M\check{R}_j \le L$, or zero if no such index exists. The regular action passed to the next integral is

$$\begin{aligned} \widetilde{\mathbf{E}}_k = &\sum_{\sharp=\mathrm{u},\mathrm{m}}\sum_{j\le k}\left[\sum_X (E^{(j),\sharp}(X,U) - E^{(j),\sharp}(X,1)) - b_j^{\sharp}A(U)\right]\\ &+ \sum_{j\le k_1}\sum_X [R^{(j)}(X,U) - R^{(j)}(X,1)]. \end{aligned} \tag{155}$$

The unmarked chain is autonomous on fixed radii. It supplies r_j without the inverse coupling cost of a shrinking radius. The marked chain has running radii and the small remainder load; it supplies q_j. The constant-total weight-deviation family has coefficient zero in either chain. This implements one coupling prescription on both systems of radii.

The remainder made at a step has two origins. The ordinary step moves the young remainders not yet included in (155). Their number is $O(\log\log x_k)$ and each carries $g_j^{\kappa_0}$. The displacement ratio s_k pays their expansion constants and the square of this number, leaving at most $\frac{1}{2}g_k^{\kappa_0}$. The healed-component expansion contributes at most $e^{-p_0(g_k)} \le \frac{1}{2}g_k^{\kappa_0}$. Their sum therefore satisfies the same remainder bound at level k.

Boundary data are assigned to slots by their level, domain and origin: the ordinary step, regular points next to regions, remainders near islands, stages, and exponentiation. A carried slot is literally unchanged. A made slot is bounded by the constants of the present expansion. Thus the common bound is a maximum,

$$B = \max\{1, B_{\mathbf{T}}, CM^4E_0, C_{\mathrm{S}}, \alpha' + C\}.$$

The made-term constants do not involve this B: in the ordinary step it enters through $B\mathsf{s}_k$; in the stages through the negative logarithmic switch gain; outside a main-step collar through (146); inside it only through the volume term paid under its component mass. The coupling ceiling is chosen after all these constants.

An island exemption ends if a later operation absorbs it. The old hull length can then increase. If a domain contains that absorbed hull, one of its cubes meets the treated region, so its old term is consumed in $V(Y)$. Its completed slot uses the term of the donor history with the entire past of the treated components erased. Superposition of histories makes this a valid history with the stronger hull weight. Terms not consumed are carried with exactly their old length. This accounts for the apparent increase of an old weight without asserting a stronger bound for an unchanged old function.

The spatial rate is spent in a fixed order,

$$(1+4\beta_{\mathrm{s}})\kappa > (1+3\beta_{\mathrm{s}})\kappa > (1+2\beta_{\mathrm{s}})\kappa > (1+\tfrac{1}{2}\beta_{\mathrm{s}})\kappa > \kappa. \tag{156}$$

These are the rates for the ordinary step, stages, completed differences, exponentiation and the returned array. Each gap pays an entropy sum. It is not spent afresh at every old level of a carried term.

The actual induction order is

$$E^{(k+1)}, \widehat{E}^{(k+1)} \longrightarrow b_{k+1} \longrightarrow x_{k+1} \longrightarrow R^{(k+1)} \text{ and the new boundary operations.}$$

The first formation uses only the old prefix and its aged remainders. The coefficient and flow check are therefore made before the next large-field geometry uses x_{k+1}. Along a fixed frame the additional tube condition for x_{k+1} must be checked separately; the local step does not supply it. For $\theta = x$, the flow and nested-size estimates give that check with zero deviation. The local expansion and the history estimates thereby form one induction, with the same couplings and analytic domains.

3.13. **Why the final bounds are uniform in the cutoff.** The ordinary remainder of level j is marginal after subtracting its constant term. Its scale sum uses the running coupling, not a geometric scaling gain. In an old layer n the analyticity radii introduce the two-level factor

$$\varpi_{jn} = \max\{1, (x_j/x_n)^2\}, \qquad \sum_{j\le n} \varpi_{jn} g_j^{\kappa_0} \le C g_n^{\kappa_0 - 2}.$$

This is one reason the flow bound and the small-field induction must be established together. The positive drift towards the ultraviolet makes these inverse powers summable. The regular terms, after the coupling subtraction, also have a geometric gain in $n - j$.

Here the remainder sum is an elementary consequence of the flow once the stated norm bound is available. Put $p = \kappa_0/2 > 3$ and $h = n - j$. For x_n sufficiently large compared with $C_\sharp$, (69) gives $x_j \geq c(x_n + h)$. Therefore

$$\begin{aligned}\sum_{j\leq n} \varpi_{jn} g_j^{\kappa_0} &\leq \sum_{j\leq n} x_j^{-p} + x_n^{-2} \sum_{j\leq n} x_j^{-(p-2)} \\ &\leq C x_n^{1-p} + C x_n^{-2} x_n^{3-p} \leq C g_n^{\kappa_0 - 2}.\end{aligned}$$

The comparison with an integral is valid since $p - 2 > 1$. Thus the choice $\kappa_0 \geq 7$ suffices for this sum. No geometric forgetting of remainders has been used. At fixed coupling the same argument would give a factor equal to the number of levels.

After paying for the local interactions, normalisations and entropy, each terminal retained component still has a factor $e^{-cp_0(g_k) - \kappa_1 |C|_k}$. The constant κ_1 dominates the exponential count of connected sets through a cell. Healed components have disappeared from the label, so the final sum is over cells of the current scale. Summing independently over unhealed labels at all earlier scales would instead introduce a factor of the form

$$\exp\left[|\mathbb{T}^{(k)}| \sum_{t\leq k} L^{4(k-t)} e^{-cp_0(g_t)} \right],$$

which need not remain bounded as k grows. Removing small old components is what avoids this accumulation.

The stronger history estimates retain some of the excess in (137). For a live lineage define its weighted volume by summing $2M^4 \check{R}_j^5$ per occupied level-j cell, and let its creation sum be the total of the factors $cp_0(g_j)^2$ allocated to its creations. A numerical history weight Θ_{age} is a fixed fraction of that creation sum plus c_a times the weighted volume. Inside each retained operation the corresponding part $\Theta_{\mathrm{age},X}(\omega)$ must already be included in the mass bound:

$$\frac{1}{|W_4|} \sum_{\omega} e^{cv(\omega) + \Theta_{\mathrm{age},X}(\omega)} \int \sup_p |e^{\Phi_\omega(\xi;p)}| \, d\mu_\omega(\xi) \leq e^{-(2-c')p_0(g_i)}.$$

Increasing only the coefficient of v in (138) does not give this assertion: v charges $\log x_j$ per stored site, whereas the volume mark has an additional factor of order $\check{R}_j$. The creation-versus-volume estimate must leave this extra margin before it is used later.

The precise numerical mark uses the unhealed lineage. For a component X live at level k, let its lineage regions be $\mathsf{Z}_k(X) = X$ and, recursively, the unions of live predecessors contained in the next region. For a creation component W, let $d'_j(W)$ be its tree length measured in level-j large-field cells. With $|C|_j$ counting cells, set

$$\begin{aligned}\mathrm{Vol}_k(X) &= \sum_{(j,C)\text{ in the lineage}} 2M^4 \check{R}_j^5 |C|_j, \\ \mathrm{Dep}_k(X) &= \sum_{(j,W)\text{ created in the lineage}} cp_0(g_j)^2 (d'_j(W) + 1), \\ \Theta^{(k)}_{\mathrm{age}}(h) &= \sum_{X\text{ live at }k} [\mathsf{w}\mathrm{Dep}_k(X) + c_a \mathrm{Vol}_k(X)].\end{aligned} \tag{157}$$

The lineage continues under marked operations. Deposits used in the mass of such an operation cannot be withheld again at a later stage. This is why its share of the numerical mark stands inside its original mass sum.

The volume of a prescribed trace. The weighted genealogy count suffices here; a separate choice of potential at every lineage member is unnecessary. Put $\mathsf{r}_i = M^4 \check{R}_i^5$, with $\check{R}_i = R(\min_{j\leq i} x_j)$ as above. Thus r_i, unlike the individual x_i, is nonincreasing. For a member (i, C), let $\mathrm{pred}(i, C)$ be its

live predecessors and let Act_{i-1} record predecessors on which the preceding operation acted but left a live component. The event count is

$$e(i,C) = (|\mathrm{pred}(i,C)| - 1)^+ + |\mathrm{pred}(i,C) \cap \mathrm{Act}_{i-1}| + \sum_{W \text{ created in } C} (d_i'(W) + 1).$$

These are, respectively, merges, failed operations and creations. Cut the genealogy at its creations and at its failed operations. Between two cuts its occupied size contracts at an ordinary change of partition. A drop in $\check{R}_i$ does not contract the partition, but two consecutive drops are excluded by the lower flow bound at the chosen coupling ceiling. Once the descendant has bounded size, its remaining wait is at most $C\check{R}_i$. Applying (136) with the nonincreasing weight r_i therefore gives

$$\sum_{(i,C)\in\mathfrak{l}} \mathsf{r}_i |C|_i \le c_G \sum_{(i,C)\in\mathfrak{l}} \mathsf{r}_i \check{R}_i e(i,C). \tag{158}$$

This is the geometric estimate used below. In particular the price assigned to one event is of order $M^4\check{R}_i^6$, including the wait. Counting only the volume of its first level would miss this factor.

The analytic majorant with the trace fixed. Record a trace $\mathfrak{l}$ by its occupied regions, creation sets and members of Act_i. Its volume and deposit are those in (157). Retain the numerical weight

$$\mathrm{wt}(\mathfrak{l}) = \exp\left[C_{\mathrm{chg}} \mathrm{Vol}(\mathfrak{l}) - \mathrm{Dep}(\mathfrak{l}) - 2 \sum_{(i,W) \text{ created}} p_0(g_i) - \sum_i |\mathrm{Act}_i| p_0(g_i) \right]. \tag{159}$$

The constant C_{chg} is fixed before the coupling ceiling. Normalizations and the local vacuum constants cost $C \log x_i$ per stored site; compact-group and Gaussian integrations cost C per stored site; and data not recorded in the trace have exponential entropy per cell. These costs are bounded by the volume term in (159). The creation deposits and the two displayed linear gains are retained, rather than absorbed into that term.

There is an important order to this estimate. First sum the histories of components that have already healed and absorb them into the local exponent by the exact operation $\mathbf{R}$. Only then take a majorant of the terms with the prescribed live traces. Majorizing every healed component separately would leave an independent sum of small weights at every old scale; that is precisely the nonsummable accumulation excluded above.

For the remaining marked components, unfold their mass estimates as far as their creations. At an inner island use the Borel envelope in the proof of (138); outside the islands use (143). Keep the trace fixed during this induction, so the gained deposits, the failed-operation gains and the occupied-site costs remain numerical factors. Each failed test uses its assigned Wilson, gauge-fixing or Gaussian share. The shares belong to disjoint sets of plaquettes, factors and bonds. The product envelopes depend on disjoint groups of the remaining integration variables; their Tonelli integrals thus multiply. At no stage is a product of separate conditional integral norms taken.

For an unmarked small-field piece, the supported exponent is bounded above by minus its Wilson action plus C per site. Its Haar integral is at most one after that extensive factor has been extracted. Applying this to all pieces not in a prescribed trace supplies $e^{C\mathsf{V}_k}$, where $\mathsf{V}_k = |\mathbb{T}^{(k)}|$. All the other factors have just been assigned either to their trace or to the common vacuum constant. Thus, writing $\mathsf{t}(h) = \tau_h(\mathcal{V}_k) \ge 0$ for the full unsourced mass, the finite induction gives

$$e^{-E_{(k)}} \sum_{h:\, \text{live traces } \mathcal{L}} \mathsf{t}(h) \le e^{C\mathsf{V}_k} \prod_{\mathfrak{l}\in\mathcal{L}} \mathrm{wt}(\mathfrak{l}). \tag{160}$$

The identity $\sum_h \mathsf{t}(h) = Z$ uses the exact history kernels. The estimate uses their exact resummation together with the local envelope and payment estimates above. It does not assert independence of the component measures.

Summing traces without spending the mark. Fix the top X at level k. Given Z_i, there are at most $2^{|\mathsf{Z}_i|_i}$ choices of its creation set, $2^{|\mathsf{Z}_i|_i}$ choices of Act_i, and $2^{L^4|\mathsf{Z}_i|_i}$ choices of Z_{i-1}. Set

$$\mathrm{ent}(\mathfrak{l}) = (L^4+2)\log 2 \sum_i |\mathsf{Z}_i|_i.$$

Starting at X and summing these choices successively shows

$$\sum_{\mathfrak{l}:\,\mathrm{top}(\mathfrak{l})=X} e^{-\mathrm{ent}(\mathfrak{l})} \le 1.$$

Consequently it suffices to bound

$$\mathrm{bal}(\mathfrak{l}) = \log \mathrm{wt}(\mathfrak{l}) + \mathsf{w}\mathrm{Dep}(\mathfrak{l}) + c_a \mathrm{Vol}(\mathfrak{l}) + \mathrm{ent}(\mathfrak{l}).$$

Fix $0 \le \mathsf{w} < 1$, $c_a \ge 0$, and a desired cell penalty c_{cell}. Equation (158), with its constants enlarged to include the entropy per cell, gives event prices

$$c_i^{\mathrm{ev}} = C_* \mathsf{r}_i \check{R}_i, \qquad (C_{\mathrm{chg}} + c_a)\mathrm{Vol} + \mathrm{ent} + c_{\mathrm{cell}}|X|_k \le \sum_{(i,C)} c_i^{\mathrm{ev}} e(i,C).$$

At sufficiently small coupling these satisfy

$$c_i^{\mathrm{ev}} \le \tfrac{1}{2}(1-\mathsf{w}) c\, p_0(g_i)^2, \qquad c_i^{\mathrm{ev}} \le \tfrac{1}{2} p_0(g_i). \tag{161}$$

The first inequality pays for new creations from the free part of their quadratic deposit. The second pays for events without a quadratic deposit. It uses the full condition $\mathfrak{p}_0 > 6\mathfrak{r}$, since $\mathsf{r}_i \check{R}_i \le C(\log x_i)^{6\mathfrak{r}}$. The weaker inequality $2\mathfrak{p}_0 > 6\mathfrak{r}$ would pay for creations but would not pay for these other events.

A predecessor in Act_{i-1} pays its event price from its own linear gain at level $i-1$. This is allowed because c_i^{ev} is nonincreasing in i. For merges, view the lineage as a tree and assign to each member its oldest creation root. At a member with q predecessors, send its $q-1$ merge units to the oldest roots of the predecessors except for the root inherited by the member. This map is injective: a root that has been used is no longer inherited and cannot be used again. The oldest root of the whole tree is never used. If the merge is at level i and its assigned root at level $n < i$, the lower flow bound gives

$$c_i^{\mathrm{ev}} \le \tfrac{1}{2} p_0(g_i) \le 2\big(p_0(g_i) - 1\big) \le 2p_0(g_n).$$

Thus its price is paid by that root's linear creation gain. This argument uses the tree structure, rather than a bound on the overlap of geometric hulls.

If $j_0(\mathfrak{l})$ is the oldest creation level, the resulting balance is

$$\begin{aligned} \mathrm{bal}(\mathfrak{l}) + c_{\mathrm{cell}}|X|_k \le\ & -\tfrac{1}{2}(1-\mathsf{w}) \sum_{(i,W)\ \mathrm{created}} c p_0(g_i)^2 (d_i'(W)+1) \\ & - \tfrac{1}{2}\sum_i |\mathrm{Act}_i| p_0(g_i) - 2p_0(g_{j_0(\mathfrak{l})}). \end{aligned} \tag{162}$$

In particular, the mark has been retained inside the original mass calculation, and

$$\mathrm{Wt}(X) := \sum_{\mathrm{top}(\mathfrak{l})=X} \mathrm{wt}(\mathfrak{l}) e^{\mathsf{w}\mathrm{Dep}(\mathfrak{l}) + c_a \mathrm{Vol}(\mathfrak{l})} \le e^{-c_{\mathrm{cell}}|X|_k}. \tag{163}$$

The unused oldest-root gain in (162) is also available when the trace is subsequently healed. No deposit already consumed by an inner operation is withheld a second time.

The global bounds use one common vacuum constant, so the label-dependent constants cannot be discarded. If $E_{(k)}$ is the vacuum constant of the first k steps, their required split is

$$E + \mathcal{E}_{k,\mathcal{S}} = \mathfrak{E}_k + \sum_{j\le k} \mathfrak{E}_j', \quad -(\tfrac{9}{2}\log x_k + C)|\mathbb{T}^{(k)}| \le \mathfrak{E}_k \le C|\mathbb{T}^{(k)}|, \quad |\mathfrak{E}_j'| \le C r_j |Z_j|_j.$$

The remaining-scale constant is $\mathfrak{E}_k = E - E_{(k)}$ and $\mathfrak{E}_K = 0$. Its logarithmic sign comes from the free-bond and gauge-fixed-site count. Four bond directions and one site gauge parameter leave three physical polarisations, giving $-\frac{9}{2}\log x$ per eliminated site for SU(2). The local Gaussian numbers, though independent of the coupling, are included once in this split. An uncontrolled constant per bare bond would not be a permitted normalisation error.

To pass from (160) to stability, group histories by the set of their live traces and use additivity of Θ_{age}. Equation (163) gives

$$e^{-E_{(k)}} \sum_h \mathfrak{t}(h) e^{\Theta^{(k)}_{\mathrm{age}}(h)} \le e^{C\mathsf{v}_k} \sum_{\mathfrak{C}} \prod_{X\in\mathfrak{C}} e^{-c_{\mathrm{cell}}|X|_k}.$$

Here $\mathfrak{C}$ ranges over compatible sets of connected unions of level-k cells. Dropping compatibility increases the sum. The graph of cells has degree at most $\Delta = 3^4 - 1$. A connected set of n cells containing a fixed cell can be encoded by a traversal of a spanning tree of length $2(n-1)$, so there are at most $\Delta^{2(n-1)}$ such sets. Choose $c_{\mathrm{cell}} = \log(2\Delta^2)$. Then

$$\sum_{X\ni\mathfrak{c}} e^{-c_{\mathrm{cell}}|X|_k} \le 1, \qquad \sum_{\mathfrak{C}} \prod_{X\in\mathfrak{C}} e^{-c_{\mathrm{cell}}|X|_k} \le \prod_X (1 + e^{-c_{\mathrm{cell}}|X|_k}) \le e^{n_k} \le e^{\mathsf{v}_k},$$

where n_k is the number of level-k large-field cells. Thus the extra cost of summing live traces is at most one per current site. The dials w, c_a change the coupling ceiling needed in (161); they do not change this extensive constant. This calculation is applied at the terminal level for the deposit mark, and at the stated stopped levels for the volume mark.

After the volume and label sums, the required upper bound and floor are

$$\sum_h e^{-E}\mathfrak{t}(h) e^{\Theta^{(K)}_{\mathrm{age}}(h)} \le e^{E_+\ell^4}, \qquad \sum_h e^{-E}\mathfrak{t}(h) \ge e^{-C_{\mathrm{low}}\ell^4}, \qquad E_+ + C_{\mathrm{low}} \le C(1 + \log x_K). \tag{164}$$

For $\mathsf{w} = 0$ the same bounds hold at each stopped level, using $E_{(k)}$, x_k and $|\mathbb{T}^{(k)}|$. Write $\widetilde{\nu}(h) = \mathfrak{t}(h)/\sum_{h'}\mathfrak{t}(h')$ for its normalised positive history masses. Dividing the two bounds gives

$$\sum_h \widetilde{\nu}(h) e^{\Theta^{(k)}_{\mathrm{age}}(h)} \le \exp[C(1 + \log x_k)|\mathbb{T}^{(k)}|]. \tag{165}$$

This extensive moment bound is not yet a volume-independent estimate for a specified cell.

The first logarithmic power in the floor is essential. Retain the never-large label and restrict each current bond to a Haar ball of radius $s \asymp g_k$. Its Haar mass is at least cs^3. The lift of this coarse field, equal to one within blocks and equal to the coarse transport across them, has all the required averages. Under the regularity estimate its background passes the tests and has action at most $Cs^2|\mathbb{T}^{(k)}|$. Consequently

$$\int \chi_k e^{-x_k A(U_k)}\, dV_k \ge \exp\left[-\left(4\log\frac{1}{cs^3} + Cx_k s^2\right)|\mathbb{T}^{(k)}|\right].$$

Since $x_k s^2 = O(1)$, this gives $C_{\mathrm{low}}(x_k) = C + 6\log x_k$. Using balls of radius ε_k would instead charge $p_0(g_k)^2$ per site and would lose the floor used in the age argument.

From the extensive mark to rarity at one cell. The extensive bound (165) is used in two separate probability arguments. First one obtains rarity of a live cell, and then decay in the age of its lineage. The first target is

$$\widetilde{\nu}(\mathfrak{c} \subset \mathcal{Z}_k) \le Ce^{-cp_0(g_k)}. \tag{166}$$

The event concerns the history law, rather than a requirement that the bare field be regular on a fixed physical box.

Write $\mathcal{E}^{(k)}_{\mathfrak{c}}(K')$ for the event that a level-k cell remains live through a terminal level $K' \ge k$. A lineage live at k either is healed later or remains live at the terminal level. In the first case its operation mass has a gain $Ce^{-cp_0(g_k)-cp_0(g_j)}$ if healing occurs at level j. Summing the masses with

the lineage trace fixed, and then over j, gives $Ce^{-cp_0(g_k)}$. This is where the prescribed-trace product bound above is used; one cannot sum an unnormalised component weight without the remaining history masses.

For the never-healed part, the terminal mark charges every cell in the lineage at each level. In particular,

$$\sum_{\mathfrak{c}} \mathbf{1}_{\mathcal{E}_{\mathfrak{c}}^{(k)}(K')} \le \frac{\Theta_{\mathrm{age}}^{(K')}}{c_a \mathsf{r}_k}, \qquad \mathsf{r}_k = M^4 \check{R}_k^5.$$

Jensen's inequality applied to the exponential moment gives a bound on the expectation of the numerator by the terminal volume times $C(1+\log x_{K'})$. The live event at level k is invariant under translations by level-k cells. The never-healed event need not be, since it also reads the coarser partitions. Thus one averages the latter event over the level-k cells and keeps a maximum only for the later-healed part. The resulting first-moment bound is

$$\widetilde{\nu}(\mathfrak{c} \subset \mathcal{Z}_k) \le Ce^{-cp_0(g_k)} + C_{g'} L^{-4(K'-k)}, \tag{167}$$

where g' is the terminal reference coupling; the constant is independent of the torus side. The volume ratio, rather than translation invariance of the never-healed event, supplies the last factor.

Continuation on the same bare lattice. Fix $0 < \mathsf{w} < 1$ and $c_a > 0$, and then a small anchor coupling $\gamma_\star$. Write $X_\star = \gamma_\star^{-2}$ and $K_\star = K(\beta, \gamma_\star)$. If the original terminal side is $2L^m$, a continuation to a first-passage cutoff $K' = K(\beta, g')$, $g \le g' \le \gamma_\star$, has terminal side $2L^{m-(K'-K)}$. Call this target admissible when that side satisfies (71) at g'. The original target is admissible. There are finitely many possible cutoffs between K and $K_\star$, so choose an admissible target $g^\dagger$ with largest cutoff and set

$$K^\dagger = K(\beta, g^\dagger), \quad p^\dagger = K^\dagger - K, \quad m^\dagger = m - p^\dagger, \quad X^\dagger = (g^\dagger)^{-2}.$$

The new bare side is still $2L^{m^\dagger + K^\dagger} = 2L^{m+K}$. Moreover, every intermediate torus satisfies (70), since the terminal volume condition implies all its predecessors.

The probabilities of old prefix events are unchanged by this continuation. Indeed a step uses only the preceding lattices and partitions, its coupling prefix and the earlier outcomes. None of these changes. Positivity and exact composition of its kernels give, history by history,

$$\sum_{h' :\, h'|_k = h} \mathsf{t}(h') = \mathsf{t}(h).$$

The total mass is the same bare Wilson partition function. After the change of physical units, the old live event therefore has the same probability in the prolonged run.

Put

$$c_{\mathrm{gap}} = \lceil C_\sharp / \lambda_\sharp \rceil + 1.$$

There are two possibilities. If $K^\dagger + c_{\mathrm{gap}} \le K_\star$, the lower drift in (69) guarantees a record low at an index $K'' \in (K^\dagger, K^\dagger + c_{\mathrm{gap}}]$. A record low is a first-passage cutoff: take the target inverse coupling $x_{K''} - B_\sharp$, or the anchor if this number is below $X_\star$. If $m^\dagger \ge m_\star(g^\dagger) + c_{\mathrm{gap}}$, where $2L^{m_\star(g')} = \ell_\star(g')$, that target would still meet the volume floor. In fact

$$m^\dagger - (K'' - K^\dagger) \ge m_\star(g^\dagger) \ge m_\star(g'').$$

This contradicts maximality. Hence in this case

$$m^\dagger < m_\star(g^\dagger) + c_{\mathrm{gap}}, \qquad N^\dagger := 16L^{4m^\dagger} \le C(L,M)\big(\log(X^\dagger + B_\sharp)\big)^{4\mathfrak{r}}. \tag{168}$$

If $K^\dagger + c_{\mathrm{gap}} > K_\star$, the upper drift bound and the terminal anchor window instead give

$$p^\dagger > \frac{g^{-2} - X_\star - B_\sharp}{B_\sharp} - c_{\mathrm{gap}}. \tag{169}$$

Thus either the new torus has polylogarithmic volume in its terminal coupling, or the continuation supplies a number of levels of order g^{-2}.

The never-healed contribution. Use the terminal exponential moment of the prolonged datum, with $\mathsf{B}(y) = C(1 + \log y)$:

$$\mathbb{E}^{\dagger} e^{\Theta_{\mathrm{age}}^{(K^{\dagger})}} \le e^{\mathsf{B}(x_{K^{\dagger}})N^{\dagger}}.$$

A cell surviving from level k to $K^{\dagger}$ lies in one terminal component's lineage. Its first nonempty level contains a creation: otherwise that level would have an earlier live predecessor. The deposit of this creation and the lower flow bound imply, on the survival event,

$$\Theta_{\mathrm{age}}^{(K^{\dagger})} \ge \mathsf{w}c\,(p_0(g_k) - 1)^2.$$

In the volume case (168), the extensive cost is at most

$$C\big(\log(X^{\dagger} + B_{\sharp})\big)^{4\mathfrak{r}+1}.$$

Choose the anchor after the dials so this is at most one quarter of the displayed deposit, uniformly for $X^{\dagger} \ge X_{\star}$. This is possible since $2\mathfrak{p}_0 > 4\mathfrak{r} + 1$. Exponential Markov then bounds survival at each specified cell by $e^{-c_{\mathrm{r}} p_0(g_k)}$, with a fixed positive c_{r} below the rate allowed by healing.

In the coupling case use the volume part of the same mark. Distinct terminal components have disjoint lineage regions, so each surviving level-k cell is counted once and

$$\sum_{\mathfrak{c}} \mathbf{1}_{\mathcal{E}_{\mathfrak{c}}^{(k)}(K^{\dagger})} \le \frac{\Theta_{\mathrm{age}}^{(K^{\dagger})}}{2c_a M^4 \check{R}_k^5}.$$

There are $n_k = N^{\dagger} L^{4(K^{\dagger}-k)}/(M\check{R}_k)^4$ level-k cells. Jensen's inequality and the moment bound therefore yield

$$\frac{1}{n_k} \sum_{\mathfrak{c}} \widetilde{\nu}^{\dagger}(\mathcal{E}_{\mathfrak{c}}^{(k)}(K^{\dagger})) \le \frac{\mathsf{B}(x_{K^{\dagger}}) L^{-4(K^{\dagger}-k)}}{2c_a \check{R}_k}. \tag{170}$$

The torus volume has cancelled. This is an average over cells; no translation invariance of the survival event is used.

To compare its depth with $p_0(g_k)$, put $u = K - k$ and $\bar{x}_u = g^{-2} + B_{\sharp}(u + 1)$. Then

$$x_k \le \bar{x}_u, \qquad K^{\dagger} - k = p^{\dagger} + u, \qquad x_{K^{\dagger}} \le g^{-2} + B_{\sharp}.$$

With $\mathfrak{P}(y) = A_0(\log y)^{\mathfrak{p}_0}$, choose $u_R(g)$ large enough that, for every $u \ge 0$,

$$4(u_R(g) + u) \log L \ge c_{\mathrm{r}} \mathfrak{P}(\bar{x}_u) + \log^{+} \frac{\mathsf{B}(g^{-2} + B_{\sharp})}{2c_a}.$$

This requires only $u_R(g) \le C(\log g^{-2})^{\mathfrak{p}_0}$. Indeed $\bar{x}_u \le (g^{-2} + B_{\sharp})(1 + u)$, and the part of $C(\log(1 + u))^{\mathfrak{p}_0}$ exceeding $4u \log L$ has a finite supremum independent of g. By (169), decreasing the original coupling ceiling makes $p^{\dagger} \ge u_R(g)$. Equation (170) is then at most $e^{-c_{\mathrm{r}} p_0(g_k)}$.

Finally the live event at level k is translation covariant, although its never-healed part need not be. Average its decomposition into healed and never-healed fates over all level-k cells. The healed contribution has the already obtained bound at every cell; the preceding average bounds the other contribution. Prefix consistency transports the result to the original run and proves (166), uniformly over the volume floor. The order of constants is the construction and the desired rate, then the dials, then the anchor, then the original small-coupling ceiling. The background estimates throughout this argument use local determining sets; demanding bare regularity throughout the torus would not give this volume-uniform statement.

For decay in age, let $\mathrm{Live}_{k,\ge\alpha}(\mathfrak{c})$ mean that the specified cell lies in a live component at level k whose unhealed lineage has age at least α. The estimate needed for two points is the weighted tail bound for the age of a live component,

$$\sum_{\alpha \ge 0} L^{b\alpha}\, \widetilde{\nu}(\mathrm{Live}_{k,\ge\alpha}(\mathfrak{c})) \le C_b e^{-c p_0(g_k)}. \tag{171}$$

Its constants and coupling ceiling are chosen after a fixed rate $b \geq 8$.

The lineage and its survival witnesses. The age needed in the two-point estimate is the age of an unhealed lineage. If j_0 is its first nonempty level, that age is $k - j_0$. For comparison, let Z_t^{lab} be the large-field region retained at scale t in the label of the current history. The label age of a live component C is

$$\mathrm{age}^{\mathrm{lab}}(C) = k - \min\{t : C \cap Z_t^{\mathrm{lab}} \neq \varnothing\},$$

where regions at different scales are viewed as point sets in the same torus. Its seed level is the earliest step at which a created region is geometrically contained in C, whether or not that region was subsequently healed; the seed age is k minus this level. Every old stored region in the label has an unhealed predecessor chain to the current component. Thus

$$\mathrm{age}^{\mathrm{lab}}(C) \leq k - j_0 \leq \mathrm{age}^{\mathrm{seed}}(C).$$

The second inequality can be strict: a region can have been healed and its former position covered by a new live component. Such a region contributes to the seed age but supplies no surviving predecessor and no deposit to the new lineage. The argument below therefore uses survival, rather than a sum over all old creations near the current cell.

For a level-n cell z, write $\mathcal{E}_z^{(n)}(k)$ for the event that it is live at every level $n, \dots, k$. The local probability estimate is made on tiles of side $l_\square \asymp M\check{R}_k$, in level-$k$ units. The side is a power of L and spans at least sixteen level-k cells. The faces are bond mirrors, and a tiling is translated by whole level-k cells. A cell of any finer level is called deep when its containing level-k cell lies in the central three quarters of the tile.

For an anchor z deep in one tile and distinct other tiles $\square_1, \dots, \square_r$, the bounds to establish are

$$\begin{gathered}\widetilde{\nu}(\mathcal{E}_z^{(n)}(k)) \leq l_\square^{-4} q_a^{c(k-n+1)}, \\ \sum_{c_i \text{ deep in } \square_i} \widetilde{\nu}\left(\mathcal{E}_z^{(n)}(k) \cap \bigcap_{i=1}^{r} \{c_i \subset \mathcal{Z}_k\}\right) \leq l_\square^{-4} q_a^{c(k-n+1)} \epsilon^r, \qquad q_a = e^{-c_a}, \quad c = 1/16.\end{gathered} \tag{172}$$

The constant c is fixed before c_a; it comes from repeating a pattern in one tile out of 2^4 and taking the chessboard root. The first inequality is also required on tori too small for this tiling. That case needs a separate verification; the large-torus reflection argument does not cover it by itself.

The extensive moment bound has the correct surplus for a repeated pattern. For this bound set $\mathsf{w} = 0$ and stop the run at level k. For an event F of that stopped run,

$$\widetilde{\nu}(F) \leq \exp\left[C(1 + \log x_k)|\mathbb{T}^{(k)}| - \inf_F \Theta_{\mathrm{age}}^{(k)}\right].$$

A cell surviving from n to k is charged in the lineage volume at every one of those levels. Multiple surviving cells are charged once for each distinct cell of each level, with weight $2c_a M^4 \check{R}_j^5$. The difficulty is to apply reflection positivity to a local event determined by those histories.

The histories use the bare field and auxiliary draws from the positive kernels. Realise each possible draw up to level k by an independent uniform variable, attached to the location of that kernel. Tile data consist of the bare field restricted to the tile and these attached variables. Reflection positivity applies to functions of the bare field. Accordingly the witness construction must eliminate the dependence on unspecified exterior draws and then integrate the interior auxiliary variables in a way compatible with products over distinct tiles.

Agreement inside a tile and threshold windows. Compare two runs with identical tile data. The local agreement statement is to be proved by induction on the level and, within a level, on the completed stages of the operation. It is needed only under conditions imposed on the first run inside the tile; the second run has no restriction outside it. Two exceptional behaviours are removed: a tested scaled fluctuation can lie very near its threshold, or a long cluster of trigger cells can reach through the layers reserved for the comparison.

Let $\delta_{\mathrm{loc}} > 0$ be the decay rate for the difference of backgrounds determined by identical data in a region. At a test of step j, with fluctuation scaled by g_i, $i \le j$, the exceptional window has width

$$\frac{2\varepsilon_j^{\mathrm{dec}}}{g_i}, \qquad \varepsilon_j^{\mathrm{dec}} = Ce^{-\delta_{\mathrm{loc}} D_j}\varepsilon_j, \tag{173}$$
$$\delta_{\mathrm{loc}} D_j = C + 4\log l_\square + (4\log L + \log 2)(k-j) + O(N_j \log L + \log p_0(g_j)).$$

Here $N_j = \check{R}_j$ is the waiting depth. The last terms allow for the finite losses of the waiting window and tests. Their bounds have to be established before D_j is fixed. The displayed distance gives the factor $l_\square^{-4}(2L^4)^{-(k-j)}$ in the window width. There are at most $Cl_\square^4 L^{4(k-j)}$ possible level-j sites in the tile. After the remaining scale factors and the conditional-density bound for the tested coordinate have been included, their product is summable as $\sum_{j\le k} 2^{-(k-j)}$. A width estimate alone does not give the probability of a window: that requires the uniform density estimate in the same scaled coordinate.

The conditional estimate at a sharp test. The density bound can be reduced to a one-coordinate estimate, provided that the actual test order and the analytic margins are retained. Identify $\mathfrak{su}(2)$ with $\mathbb{R}^3$, and let a be the scaled coordinate at one test. Write $\mathfrak{t}$ for its finite set of positive thresholds. Choose $W > 0$ so that the thresholds are at least $2W$ and are pairwise $2W$ apart. The two shells at $t \in \mathfrak{t}$ are $t \le |a| \le t+W$ and $t-W \le |a| \le t$. Suppose that $\mathcal{F}$ is real-valued, $\psi \ge 0$ is nondecreasing with the distance from t on each radial segment of either shell, and that $|\partial_r \mathcal{F}| \le 2/W$ on the shells where $\psi \ne 0$. Then, for $0 \le u \le W$,

$$\int_{\mathrm{dist}(|a|,\mathfrak{t})\le u} e^{\mathcal{F}(a)}\psi(a)\,da \le C\frac{u}{W}\int_{\mathbb{R}^3} e^{\mathcal{F}(a)}\psi(a)\,da. \tag{174}$$

Indeed, on a radial segment of length W, the exponential varies by at most e^2, and the polar Jacobian r^2 varies by at most a fixed factor since $t \ge 2W$. The average of ψ on the part nearest the threshold is at most its average on the whole segment. Integrate over directions and sum the disjoint shells. No regularity of ψ is used.

At the stage of a test σ in the positive real run, disintegrate its law with respect to all other coordinates s. Its contribution has the form

$$Z^{-1}e^{\mathcal{F}_s(a)}\chi_s^{\mathrm{st}}(a)\,\Pi(ds)\,da,$$

where χ_s^{st} is the product of the sharp tests already present. To exclude windows at later tests, multiply this density by the conditional probability of their avoidance and of the specified live-history event E. Thus the factor in (174) is

$$\psi_s(a) = Z^{-1}\chi_s^{\mathrm{st}}(a)\mathbb{P}(E \mid s, a).$$

The event concerns the histories still live at this stage; components already integrated out impose no further condition.

The conditional argument requires an interval-reading property for the later transition kernels as well as for the labels and rules: after its test, the coordinate must enter each of them only through its recorded threshold interval. The rule that labels record intervals does not by itself establish this property for a kernel that also reads a continuous stored field. It must be checked for the actual conditional law of the run. Under this property, the remaining sharp-test checks are as follows. Other sharp functions either test $|a|$ at a threshold separated by $2W$, or test $|e^{ig_i a}v - 1|$ at such a threshold. In the latter case the centre displacement must be at most cg_iW, and the error of the small exponential chart must have the same bound. These conditions make the functions constant on each shell. For a split test, they follow from the interior placement of its test cubes and the local background comparison; the test geometry is that of Bałaban[B88, equation (3.19)] and [B89I, equations (1.56)–(1.58)]. A completion test on a bond previously controlled by a passed cube test uses the stored averaged coordinate; both endpoints are in the old small-field region, so the

same interior comparison applies. For a completion test in the remaining strip, the stage data are tested first, and the broader approximate fluctuation test equals one on these shells. These are the actual support and centre conditions used by the test construction; they are needed in addition to separation of the thresholds.

There is an exception when the same bond also belongs to a collar test. The narrow approximate collar test is performed before the completion test. In the scaled coordinate its radius is $2\delta_i'/g_i$, whereas a completion shell has $|a| \geq c\delta_i/g_i$. The two centres differ by $o(\delta_i)$ in the unscaled coordinate, and $\delta_i'/\delta_i \to 0$. Consequently the earlier passed collar indicator is zero on the completion shells. The product ψ_s, rather than its future-probability factor separately, therefore vanishes there. At a collar coordinate itself, the broader test has radius $2\delta_i'/g_i$, separated from the threshold δ_i'/g_i; the local background displacement is smaller than the shell width.

If a later window tests the same level and bond at the same threshold, its avoidance indicator is nondecreasing with distance from that threshold. All other later decisions are constant on the shell. Averaging over the later coins with the interval-dependent transition laws preserves this monotonicity. This proves the required property of ψ_s, including the repeated-bond cases.

The slope estimate must use the same scaled coordinate. The quadratic gradient is an exponentially decaying kernel applied to fields bounded by $Cp_0(g_j)$, so it is bounded by $Cp_0(g_j)$. For the interaction, Cauchy's estimate divides the local supremum by its available analytic margin. Ordinary localization domains through a bond have a bounded exponentially weighted sum. Preparatory domains have instead the bound $C(\log x_j)^{4\mathfrak{r}}$, because the small stored boxes have zero cost in their length. At a collar coordinate the analytic margin is at least a fixed multiple of $\delta_i'/g_i \geq 1$. Hence their contribution is at most $C(\log x_j)^{4\mathfrak{r}}$, which is bounded by $Cp_0(g_j)$ since $\mathfrak{p}_0 > 4\mathfrak{r}$. Using only a margin of order $1/p_0$ for these domains would give an extra logarithmic factor. The slope bound follows under these actual localization and margin estimates.

Choose $W_\sigma = c/p_0(g_i)$. The flow comparison within the waiting window bounds $p_0(g_j)$ by a fixed multiple of $p_0(g_i)$, so the slope condition in (174) holds after fixing c. Integrating in s gives a loss at most $\eta_\sigma = Cu_\sigma/W_\sigma$ when the window of width u_σ at σ is removed. Remove the windows in decreasing order of their stages. At each removal the remaining event has the form just considered, and therefore

$$\mathbb{P}(E \text{ with all windows avoided}) \geq \prod_\sigma (1-\eta_\sigma)\mathbb{P}(E) \geq e^{-2\sum_\sigma \eta_\sigma}\mathbb{P}(E) \quad (\eta_\sigma \leq \tfrac{1}{2}).$$

The site count and the choice of D_j in (173) make the sum over one tile bounded. This supplies the stated conditional window estimate once the interval-reading property of the conditional law and the uniform local background, slope and analytic-margin bounds have been verified for the run.

The comparison uses a separation of CD_j level-j lattice units. In level-j cells this is $\lceil CD_j/(M\check{R}_j)\rceil$ layers. Their physical width in current, level-k units is multiplied by $L^{-(k-j)}$. These losses, together with the widths needed to exclude long trigger clusters, must have a summable total fitting inside a fixed fraction of the tile. The fine-level comparisons therefore use narrower and narrower layers in current units, despite the increasing number of tested fine sites.

Away from the exceptional windows, a background error bounded by $\varepsilon_j^{\mathrm{dec}}$ cannot change the outcome of the corresponding test. The local agreement argument must then identify the current and stored block fields, the label regions, and every live component meeting the smaller tile. Such a component is to be contained in the original tile; its label, inner history and stored block fields are common to both runs.

The assertion does not identify all stored fluctuation fields. On bonds examined by a passed fluctuation test, those fields are formed using an averaged background and can still depend on exterior data. The later operation uses them in one test; on the relevant bonds that test must pass in both runs. This is part of the agreement verification, rather than a consequence of equality of the stored block fields. One also needs the determining set to start from the local region $\Omega_0 = \Omega_1^\sim$.

Requiring regularity at every bare plaquette of the torus would let a distant rough plaquette change every background and would destroy the claimed agreement.

Trigger patterns and bare-field witnesses. A trigger at level J is a cell in Ω_J^c, or a cell carrying a tested bond where the scaled fluctuation exceeds a lower threshold. If a long trigger cluster occurs, select a pattern of

$$m = 1 + 2(k - J), \qquad c_{\mathrm{adj}}(J) = \left\lceil \frac{CD_J}{M\check{R}_J} \right\rceil + C, \qquad m_0(J) = 6(c_{\mathrm{adj}}(J) + 1)m \tag{175}$$

level-J cells. Any two pattern cells are separated by at least $2c_{\mathrm{adj}}(J) + 3$ cells, and each lies within $m_0(J)/2$ cells of the first. The step $c_{\mathrm{adj}}(J)$ is the largest advance allowed by the decoupling comparison. The factor six leaves room to select the m separated cells inside a cluster of diameter exceeding $m_0(J)$. Each selected cell either lies within two cells of a large-field cell or carries a lower-threshold fluctuation.

The same layers used in the comparison must accommodate these patterns. In current units their required bound has the form

$$\sum_{J \le k} M\check{R}_J m_0(J) L^{-(k-J)} \le l_\square/32.$$

The tile side is chosen with this fixed reserve. Establishing the agreement statement with these actual widths is the geometric and analytic verification needed here.

For a survival event deep inside the tile, the comparison has two outcomes. Either the good tile data determine survival independently of the exterior run, or the first failure of agreement is accompanied by a window or one of the patterns above. This is the proposed reduction to local witnesses. After the auxiliary draws have been treated, it must produce finite families of functions $0 \le f_{i,\iota} \le 1$ of the bare field in tile i, with a uniformly bounded cost per tile. For survival events F_i read in distinct tiles, the needed domination and its chessboard consequence are

$$\widetilde{\nu}\left(\bigcap_{i=0}^{r} F_i\right) \le C^{r+1} \sum_{\iota_0,\dots,\iota_r} \left\langle \prod_{i=0}^{r} f_{i,\iota_i} \right\rangle \le \prod_{i=0}^{r} \left(C \sum_{\iota_i} \|f_{i,\iota_i}\|_{\mathrm{ch}} \right). \tag{176}$$

The second inequality applies reflection positivity to these bare-field functions. Its use is legitimate only after the first inequality and the uniform witness construction have been established. The names of history events alone supply neither. In particular, independence of the original auxiliary variables does not by itself prove the first inequality after the tests and conditional kernels have been applied.

There are three estimates for a repeated witness pattern. A trigger fluctuation with no large-field cell within two cells is bounded using its conditional Gaussian law in its box. A cell whose component stays live until k is paid by the volume mark of the run stopped at k. A cell whose component is healed before k is paid by that component's lineage mass. The last class must remain in the witness bound even though it no longer appears in the terminal label. Conditional-density estimates handle the exceptional windows. These estimates and local agreement are the unfinished verifications behind (176).

The chessboard root and the sum over patterns. The root of a repeated witness probability changes the extensive factor into

$$\exp[C(1 + \log x_k) l_\square^4]$$

per tile. A pattern repeated in one of the 2^4 classes of tiles retains q_a^c per charged level, with $c = 1/16$. The surplus from one occupied level is much larger than this cost, since

$$l_\square^4 \log x_k \ll \mathsf{r}_k := M^4 \check{R}_k^5.$$

Here $l_\square \asymp M\check{R}_k$ and $\check{R}_k \gg \log x_k$. A fixed part of $q_a^{\mathsf{r}_k}$ pays the extensive factor, the number of current cells in one tile and the factor ϵ^{-1}, leaving the factor $l_\square^{-4}$ in (172).

The remaining pattern count is performed level by level. At level J, the first cell has at most $Cl_\square^4 L^{4(k-J)}$ positions. Each of the other $m-1$ cells is confined within $m_0(J)/2$ cells of the first. Dropping the separation condition for this upper bound gives

$$\#\{\text{patterns of level } J\} \le Cl_\square^4 L^{4(k-J)}(m_0(J)+1)^{4(m-1)}, \qquad m_0(J) \le C(x_k+k-J)^3. \tag{177}$$

Apart from the first-cell positions, its logarithm is bounded by $Cm\log(x_k+k-J)$. The mark gives a factor $q_a^{\mathsf{r}_J}$ for each of the $m = 1+2(k-J)$ cells before the fixed shares are spent. The lower flow bound, and the use of the running minimum in $\check{R}_J$, give fixed constants $c_1 > 0$ and C for which

$$\mathsf{r}_J \ge M^4\big(\log(x_k + c_1(k-J) - C)\big)^{5\mathfrak{r}}.$$

Thus a fixed fraction of the cell weights dominates that logarithmic entropy uniformly in $k-J$, after the coupling ceiling has been chosen. The comparison is uniform in the old level because the inverse coupling grows at least linearly with $k-J$.

A pattern at J together with survival from n up to $J-1$ retains the exponent

$$(J-n)^+ + m = (J-n)^+ + 1 + 2(k-J) \ge (k-n+1) + (k-J).$$

The first part gives the required survival factor $q_a^{c(k-n+1)}$. The second part is compared with the positions of the first pattern cell:

$$L^{4(k-J)} q_a^{c(k-J)} = (L^4 q_a^c)^{k-J} \le 2^{-(k-J)}, \qquad L^4 q_a^c \le \tfrac{1}{2}.$$

Summing over J is therefore geometric. The same surplus can leave a factor ϵ for every additional tile with a deep live cell. Once the witness reduction is verified, these counting steps yield (172); none of their constants depend on the number of tiles of the torus.

From survival to the weighted age series. Fix a component containing the specified current cell and having lineage age a. At its first level $n = k-a$, choose an anchor in its lineage region. It remains live through all levels $n,\dots,k$ and belongs to the current connected component. A tile contains at most $Cl_\square^4 L^{4a}$ anchors, each with survival bound $l_\square^{-4} q_a^{c(a+1)}$. Thus nearby anchors cost at most $CL^{4a} q_a^{c(a+1)}$.

For far anchors, a connected live component crosses a chain of current occupied tiles. Use the 2^4 tilings whose face planes are displaced by $\pm\lfloor n_\square/4\rfloor$ cells from a corner of the level-k cell containing the anchor, where $n_\square \ge 16$ is the number of current cells along a tile side. The anchor is deep in all these tilings, and every current cell is deep in a tile of at least one of them. From a chain of length i, one tiling contains at least $i/16$ deep cells. A tile meets at most 16 tiles of the original board, so these include cells in at least $i/16^2$ distinct tiles. For $i \ge i_0 := 2\cdot 16^2$, at least i/i_0 tiles differ from the anchor tile.

The second inequality in (172) bounds those tiles by ϵ^{i/i_0}. If C_{ent} is the entropy per link of a chain, choose $\epsilon = (2C_{\mathrm{ent}})^{-i_0}$. The extra cost, including the choice of tiling and chains, is then at most $16C_{\mathrm{ent}}^i \epsilon^{i/i_0} = 16\cdot 2^{-i}$. It is summable against the polynomial number of tiles in distance shells. Hence $CL^{4a} q_a^{c(a+1)}$ bounds all anchors, including the distant ones.

Choose the fixed rate c_a so that $L^4 q_a^c \le (2L^b)^{-2}$. This gives a pure age bound of order $(2L^b)^{-2a}$. Its geometric mean with the live-cell rarity bound (166) gives

$$\widetilde{\nu}(\mathrm{Live}_{k,=a}(\mathfrak{c})) \le Ce^{-cp_0(g_k)}(2L^b)^{-a},$$

with the constant in the first exponential decreased if necessary. The conversion from exact age to tail age can be written as

$$\begin{aligned}\sum_{\alpha\ge 0} L^{b\alpha}\widetilde{\nu}(\mathrm{Live}_{k,\ge\alpha}(\mathfrak{c})) &\le Ce^{-cp_0(g_k)}\sum_{a\ge 0}(2L^b)^{-a}\sum_{\alpha=0}^{a} L^{b\alpha}\\ &\le C_b e^{-cp_0(g_k)}\sum_{a\ge 0}2^{-a}.\end{aligned}$$

This proves the counting deduction (171) from the local witness and rarity estimates. The constants and the ceiling depend on the requested fixed $b \ge 8$. One finite c_a is not asserted to serve all rates simultaneously.

There is also a shorter estimate on the reference torus used in section 5. Split the lineage according to whether it is later healed. For the healed part sum its operation masses only over traces of age at least a. For the never-healed part use the terminal volume mark: each level with a lineage member contributes at least $\bar{c}_a = 2c_aM^4$, so the charges add without a sum over anchor positions. The resulting two-term bound is

$$\begin{aligned}\widetilde{\nu}(\mathrm{Live}_{k,\ge a}(\mathfrak{c})) &\le Ce^{-\bar{c}_a a}e^{-cp_0(g_k)}\\ &\quad + C\exp\Big[C(1+\log x_{K'})|\mathbb{T}^{(K')}| - 2p_0(g_k) - \bar{c}_a(a+K'-k+1)\Big],\end{aligned} \tag{178}$$

where K' is the last level of an admissible prolongation, including the original run as one choice. The first term has no volume factor. The second retains a terminal volume cost and consequently is not a volume-independent substitute for (172).

On the reference torus the terminal size is polylogarithmic. Its extensive exponent has degree $4\mathfrak{r}+1$ in the logarithm, whereas p_0 has the higher degree fixed earlier. With the same terminal comparisons as in the rarity argument, this cost is absorbed by $2p_0(g_k)$. Choosing $\bar{c}_a \ge b\log L + \log 2$ then gives (171) on that reference torus. The reflection comparison in section 5 transfers its two-point consequence. The general history bound at all volumes still uses the local witnesses above. Both methods split off the later-healed lineages; a mark counting only components surviving to one fixed terminal level would omit them.

The history estimates and the coupling bounds (69) are to be proved in one induction on common analyticity domains. Choose the exponents first, then L, M_1, the activity decay rate κ, and M, in the order of section 2. Fix the interaction, threshold and history reserves before the coupling ceiling. The coupling calculation checks the next inverse coupling before the next large-field step.

4. Sources and uniform bounds on correlations

The observable is followed through the integration by changing the inverse coupling. We keep the choices of the real construction fixed, carry the source through the local expansion, and separate its scalar energies from the normalized generating function. Locality removes the ultraviolet divergence from separated mixed derivatives; chessboard comparison removes the volume dependence. A graded version of this comparison gives factorial growth in the number of insertions.

4.1. The exact sourced identity.

For a plaquette weight v, write

$$A(v,U) = \sum_p v(x(p))a_p(U).$$

Fix real functions f_i and put

$$w = c + \sum_{i=1}^{n} z_i f_i, \qquad \|w\|_\flat = |c| + \sum_i |z_i|\|f_i\|_\sharp, \qquad \|f\|_\sharp = \max\{\|f\|_\infty, CM\,\mathrm{Lip}\,f\}.$$

The source radius $\rho > 0$ is fixed before g is made small. Inserting $e^{A(w,U)}$ changes $e^{-A(\beta,U)}$ to $e^{-A(\beta-w,U)}$. There is no power of the lattice spacing in this identity. For $c = 0$, its normalized integral is

$$\Gamma(z) = \frac{Z(w)}{Z(0)} = \left\langle e^{\sum_i z_i \mathcal{O}_a(f_i)} \right\rangle .$$

At fixed cutoff the variables are bounded and Γ is entire. Its logarithm is defined near zero, since $\Gamma(0) = 1$, and

$$C_n(f_1, \dots, f_n) = \partial_1 \cdots \partial_n \log \Gamma(0).$$

For $n \geq 2$ this is unchanged by subtracting any constants from the individual observables.

Proposition 4.1 (Frozen source transport)**.** Fix a real run to level k. Keep its tests, backgrounds, localizations, gauge fixings, covariances, and reference probability laws fixed when the source is introduced. There are source-independent kernels K_h such that

$$\tau_h^w(dV) = \int e^{A(w,U)} e^{-A(\beta,U)} K_h(U, dV)\, dU, \qquad \sum_h \tau_h^w(\mathcal{V}_k) = Z(w).$$

The unsourced measures are positive. The sourced measures are entire complex measures, and their total-mass identity is valid for complex sources. ■

Proof. Every operation of the fixed run is linear in the density. The averaging constraint has total integral one in the new block variable. A field test and its complementary alternative sum to one. Gauge fixing has total integral one over the gauge group, for the gauge-invariant density to which it is applied. A change of variables includes its Jacobian. Finally, replacing a component measure by its mass times a fixed probability measure preserves the mass of that component. Composing these operations gives K_h, and summing their alternatives gives the identity on total integrals. This proof is an identity of linear functionals and does not use positivity of the input density.

At finite cutoff the integration variables range over compact group spaces and $A(w, U)$ is a finite linear combination of bounded functions. The power series of its exponential is locally uniformly integrable in the source parameters. It can therefore be integrated term by term against each fixed kernel, which proves the asserted holomorphy. □

Source derivatives use the total-mass identity of proposition 4.1. Observables inside a replaced box are transported as in (77).

4.2. **The running source and its local variation.** Let φ_j be the smooth cutoff of the level-j action region Λ_j, fixed by the action sequence, with $0 \leq \varphi_j \leq 1$, equal to one in its interior and zero outside. The unsourced inverse-coupling array is

$$x_k(p) = \beta - \sum_{j \leq k} b_j \varphi_j(p).$$

Except at the final two layers of a prepared sequence, the inter-region margins exceed the smoothing length. Thus $\varphi_{j'}(p) > 0$ for $j' > j$ implies $\varphi_j(p) = 1$. If m is the last positive profile, the array is $(1 - \varphi_m(p))x_{m-1} + \varphi_m(p)x_m$; if none is positive, it is β.

At the last stage of an operation of level $t \geq 2$, the two final profiles can overlap in the thin shells. Put $a = \varphi_{t-1}(p)$ and $b = \varphi_t(p)$. Their contribution is the convex combination

$$\begin{aligned}(1-a)(1-b)x_{t-2} + a(1-b)x_{t-1} + abx_t \\ + (1-a)b(x_t + x_{t-2} - x_{t-1}).\end{aligned}$$

The additional vertex is $x_t + b_{t-1}$. Its positivity uses the one-step case of (69), namely $b_{t-1} \geq \lambda_\sharp - C_\sharp$. For a first-passage prefix define

$$x_k^{\min} = \min\big(\{x_j : 0 \leq j \leq k\} \cup \{x_t + b_{t-1} : 2 \leq t \leq k\}\big). \tag{179}$$

The second set is empty when $k < 2$. Since $x_j > g^{-2}$ along this prefix, $x_k^{\min} \geq g^{-2} - C_\sharp$. Consequently the same positive lower bound controls every label and stage profile once the coupling ceiling is small. Convexity alone does not give the lower bound for the additional vertex.

On the common regular region a constant source has running shift

$$\zeta_k(\zeta) = \zeta + \sum_{j \leq k} [b_j(\zeta) - b_j(0)].$$

For a varying source the same marginal coefficients are evaluated at local values. Let $h_y^{(j)}$ be the nonnegative tents, with $\sum_y h_y^{(j)}(p) = 1$ and physical range at most CML^{j-K}. Choose a bare plaquette p_y for each tent centre, independently of the orientation of the plaquette p at which the array is evaluated. Then

$$\zeta_k(p; w) = w(x(p)) + \sum_{j \leq k} \varphi_j(p) \sum_y h_y^{(j)}(p) [b_j(w(x(p_y))) - b_j(0)].$$

The sourced Wilson array is $x_k(p) - \zeta_k(p; w)$.

The differentiated marginal estimate required from the induction is

$$|b_{k+1}(\zeta) - b_{k+1}(0)| \leq a_{[K-k-1]} |\zeta|, \qquad \sum_{s<K} a_{[s]} \leq C \sum_{k<K} \left[g_k^3 (\log x_k)^{\mathfrak{p}_0} + \rho^{-1} g_k^{\kappa_0 - 4} \right] \leq \frac{1}{3}. \tag{180}$$

Its two terms have different origins. The small-field part is analytic in the earlier inverse-coupling shifts on discs of radii comparable to x_j. Cauchy's inequality divides its nonconstant bound $C g_k (\log x_k)^{\mathfrak{p}_0}$ by a radius of order x_k. The remainder left by a processed component is only known on a fixed source disc; its size is $C g_k^{\kappa_0 - 4}$, and differentiation costs ρ^{-1}. Neither term acquires a factor equal to the number of earlier data variables: the Schwarz estimate is applied in the maximum norm of the joint polydisc.

Here is the elementary analytic estimate used in this statement. Suppose the data coordinates have radii r_l, and a holomorphic map G is bounded by A on the corresponding polydisc. For a displacement v put $m = \max_l |v_l|/r_l$. The function $t \mapsto G(tv) - G(0)$ is bounded by $2A$ on $|t| < m^{-1}$ and vanishes at zero. Schwarz's lemma gives

$$|G(v) - G(0)| \leq 2Am \qquad (m < 1).$$

This estimate follows from one complex variable and has no factor equal to the number of coordinates. The same argument applies at a nonzero centre if the indicated polydisc about that centre remains in the analytic class. In the source induction the earlier shifts have sizes $O(|\zeta|)$; their radii are $c x_j$, and the prefix couplings are bounded below by a constant multiple of the current x_k. Thus $m \leq C|\zeta|/x_k$. Multiplying by $A = C g_k (\log x_k)^{\mathfrak{p}_0}$ gives the first term in (180). The independent remainder is handled on its fixed disc, not on the growing polydisc.

The lower flow bound $x_k \geq g^{-2} + c(K-k) - C$ makes these sensitivities summable. For example, comparison with an integral bounds the first sum by $Cg(1 + |\log g|)^{\mathfrak{p}_0}$. The remainder sum tends to zero when $\kappa_0 > 6$. Making g small after ρ gives

$$|\zeta_k(p; w)| \leq \frac{4}{3} \sup |w|, \qquad \frac{2}{3} \leq \zeta_k'(0) \leq \frac{4}{3}.$$

The latter derivative is real, since the run is real at real sources.

In particular, at a real source of the class every Wilson coefficient is at least $x_k^{\min} - \frac{4}{3} \sup |w| > 0$, once the coupling ceiling has been chosen after the source radius. This minimum includes the prepared-stage vertices. Reality of the other factors is preserved by the real changes of variables and integrations: the elimination determinants have their positive real values, the Haar density is positive in the small chart, and the remaining factors are exponentials of real functions.

4.2.1. *The source data and their comparison with constants.* We describe the induction at a first-passage datum, with the choices of the real run frozen at its own weights x_j. Put $r = 2\rho$. General sources are constructed on a class of radius r, while constant sources are constructed on the doubled disc $D(0, 2r)$. The same two radii are used at every level. The inputs from sections 2 and 3 are the uniform one-step bounds on the common complex data class, their locality at each carrier, and the component mass with its stored-volume reserve. The argument below specifies how these estimates propagate the sources; in particular, it does not obtain a new source disc by shrinking the previous one.

At depth s use the norms

$$\|f\|_{\sharp,s} = \max\{\|f\|_\infty, CML^{-s} \operatorname{Lip} f\}, \qquad \|w\|_{\flat,s} = |c| + \sum_i |z_i| \|f_i\|_{\sharp,s}.$$

The source class used at level m is $\mathcal{W}_{r,m} = \{\|w\|_{\flat,K-m} < r\}$. It decreases as m increases. For a region Y define

$$\mathsf{l}_j(Y; w) = CML^{j-K} \operatorname{Lip}_Y w, \qquad \mathsf{m}_j(Y; w, \zeta) = \max\{\sup_Y |w - \zeta|, \mathsf{l}_j(Y; w)\}.$$

The Lipschitz seminorm here is taken over the bare plaquettes in the enlarged region, with their physical distances. The quantities are monotone in Y and j. On a fixed region, $\mathsf{l}_i = L^{-(j-i)}\mathsf{l}_j$ for $i \le j$. The tree defining $d_j(X)$ joins the plaquettes of $X^{\sim 1}$ within distance $(d_j(X) + C)ML^{j-K}$. With the constant in l_j fixed sufficiently large, this gives

$$\begin{aligned} \mathsf{m}_j(X^{\sim 1}; w, w(p_z)) &\le e^{d_j(X)} \mathsf{l}_j(X^{\sim 1}; w), \\ \mathsf{l}_j(Y; w) &\le L^{-(m-j)} \|w\|_{\flat,K-m} \quad (j \le m). \end{aligned} \tag{181}$$

Here $z \in X$ and its chosen plaquette p_z lies in the indicated enlargement. This is the factor that makes the spatial variation of the source summable over the older levels.

Use the pointed form of the regular activities from section 2. At each site z of level j, retain the entire constant-source family $E^{(j),w(p_z)}(X, \mathbf{p}, z)$, its infinite-lattice companion, and its own coefficient $b_j(w(p_z))$. Its counterterm is attached to the same base point through $h_z^{(j)}$. The defect is

$$D^{(j),w}(X, \mathbf{p}, z) = E^{(j),w}(X, \mathbf{p}, z) - E^{(j),w(p_z)}(X, \mathbf{p}, z). \tag{182}$$

The Euclidean conditions on a pointed family and its coefficient are linear. Families at different sites may therefore carry different constant sources while each retains the pointwise coupling cancellation. Keeping only the full sum of the families would lose this cancellation at the individual base points.

Suppose that, for every constant $\zeta \in D(0, r)$, the regular activity has the slope

$$|E^{(j),w} - E^{(j),\zeta}|(X, \mathbf{p}, z) \le \frac{\mathsf{m}_j(X^{\sim 1}; w, \zeta)}{r} e^{-(\kappa+1)d_j(X)}. \tag{183}$$

Then (181), with $\zeta = w(p_z)$, shows that the defect of a source in $\mathcal{W}_{r,m}$ satisfies

$$|D^{(j),w}(X, \mathbf{p}, z)| \le L^{-(m-j)} e^{-\kappa d_j(X)}.$$

Defects have no further coupling subtraction. Their geometric factor is consequently retained from the time they are made; $\sum_{j \le m} L^{-(m-j)}$ is bounded independently of m. The separate remainder sum is paid by $g_j^{\kappa_0}$.

The other entries of the source data are the bare weight $u = w$, the remainder families, the boundary terms and the excess exponents in the stored component operations. On a complex data class of level k the bounds have the form

$$\begin{aligned} &\sup |u| + \sum_{j \le k} \sup_z |\widehat{b}_j(z) - b_j(0)| \le 3r, \qquad \|G^{(j)}\| \le E_0 + 1, \qquad \|D^{(j)}\| \le L^{-(k+1-j)}, \\ &\|H^{(j)}\| \le C g_j^{\kappa_0}, \qquad |B^{(j)}(X)| \le C \bar{B}_j(X), \qquad |\Psi_\omega| \le 2c_\Phi \mathsf{n}(\omega). \end{aligned} \tag{184}$$

The norms include domain decay and the supremum over base points. The coefficient $\widehat{b}_j(z)$ is attached to site z; for the actual source it equals $b_j(w(p_z))$. Here G denotes a constant-source pointed family, H a remainder, and $\bar{B}_j$ is the fixed boundary majorant of the real run. The count $\mathsf{n}(\omega)$ adds the sites stored at their own levels, including the inner component histories once each. Bounds on the oscillations of excess factors are also retained, with the carrier and hull-length majorants of section 3. The Gaussian structures, the reference measures and all the numbers of the operations are those of the real run. The shift enters the interaction and the Wilson ledger, not those structures.

The summable coefficient estimate (180) bounds the first quantity in (184) for the actual source data by $\frac{4}{3}\sup|w|$. The equality of the bare weights is plaquette by plaquette:

$$A(x_k(\cdot)-\zeta_k(\cdot;w),U)+\sum_{j\le k}A\bigg(\varphi_j\sum_z h_z^{(j)}b_j(w(p_z)),U\bigg)=A(\beta-w,U).$$

Thus the data with variable sources represent the original sourced measure, with the same normalization conventions as the unsourced run.

Localized lines of data. Fix a region W for the next partition, a source w in its level class, and an arbitrary constant $\zeta\in D(0,r)$. Put $\mathsf{m}=\mathsf{m}_{k+1}(W;w,\zeta)$. For $|t|\mathsf{m}<r$, interpolate the ordinary entries inside W between the data of ζ and those of w, and leave the entries outside W at ζ. Interpolate a whole pointed family and its coefficient at each site. An indexed ordinary term is changed only when its required enlargement lies in W; a boundary term made by a sum over sites is treated summand by summand. Denote these localized data by $\mathfrak{L}_t^W$.

The regular, remainder and boundary slopes towards the same constant keep these entries inside their complex bounds. For example, the frozen constant-source family has size at most E_0, and its interpolation costs at most $|t|\mathsf{m}/r\le 1$. The defects vanish at the constant end of the line, so at every older level their sizes are bounded by

$$|t|r^{-1}\mathsf{I}_j(W;w)\le L^{-(k+1-j)}.$$

The coefficient functions are controlled on $D(0,2r)$. A Cauchy estimate on this doubled disc gives twice their radial slope on $D(0,r)$; together with $\sum_j a_{[K-j]}\le\frac{1}{3}$, this places the interpolated weight and coefficients within the bound $3r$. All these estimates use r/m, independently of the distance of w from the boundary of its source class.

The exponents of the stored operations require a different construction. If W cuts a component, direct interpolation of its factors would select only some carriers. A bound on the total exponent and bounds on the oscillations of individual factors do not bound this partial sum: adding constants to those factors and subtracting their total from the core changes neither of the two bounds. We instead construct the operation exponents on each line by the same recursion that constructs them in the RG step.

This recursion uses only older lines. A region for the partition at level $k+2$ is also a region for the partition at level $k+1$; the next source class is contained in the preceding class; and $\mathsf{m}_{k+1}\le\mathsf{m}_{k+2}$. Hence the next line's parameter disc is contained in the older disc on which its input operation exponents have already been constructed. Apply the step to those exponents and to the prescribed ordinary data. For the next comparison, reset the ordinary entries to their affine patchwork. Only the operation exponents are retained from this output construction; equality of the entire interpolated systems at intermediate t is not required.

The exponents so produced are holomorphic in t. At $t=0$ they are those of the constant source. When $\mathsf{m}<r$, at $t=1$ locality identifies an excess factor with its value for w whenever its carrier enlargement lies in W, and identifies the core when the twice enlarged component lies in W. These are exactly the endpoint identities needed for a subsequent local comparison. The uniform step

bounds give

$$|\Psi_\omega(t)| \le 2c_\Phi \mathsf{n}(\omega), \qquad \sum_{f:R_f=R} \operatorname{osc} \psi_f(t) \le c_\times \bar{M}_i(R),$$

where $\bar{M}_i$ is the fixed carrier majorant. No partial exponent is estimated by the modulus of its full sum.

The regular output and its slope. The small-field part of the step produces a pointed family $E^{(k+1)}[\mathfrak{L}]$ before its constant-source extension is chosen. Its Gaussian normalizer $N^{\circ(k+1)}$ is common to every system in the complex class. Here the circle retains the local values at the trivial pair. The family normalized to vanish there is

$$N^{(k+1)}(X,\mathbf{p},z) = N^{\circ(k+1)}(X,\mathbf{p},z) - N^{\circ(k+1)}(X,\mathbf{1},z).$$

These local numbers are real and independent of the source and of the running coupling; their field Hessian and marginal coefficient are zero. In the unnormalized family they are counted once among the vacuum values. The separate numerical Gaussian factor counts, at each free bond, the three-dimensional scaling and the unit Gaussian integral. It does not contain the determinant of the fluctuation form: that determinant remains in the localized normalizer. This is the same division used in section 2. The uniform step estimate has the quantitative form

$$\left|E^{(k+1)}[\mathfrak{L}](X,\mathbf{p},z) - N^{\circ(k+1)}(X,\mathbf{p},z)\right| \le C\vartheta_k e^{-\kappa' d_{k+1}(X)} \le \tfrac{1}{2} e^{-\kappa' d_{k+1}(X)}, \qquad \kappa' \ge \kappa + 1, \quad (185)$$

where one can take

$$\vartheta_k = \frac{\delta_k}{\min_\iota \alpha_{\iota,k}} + M^4 g_k p_0(g_k)^3 + x_k^{-1/4}.$$

The first term is of order $(\log x_k)^{\mathfrak{p}_0 - q}$ and tends to zero. The remaining terms also tend to zero. Since the real run uses its own Gaussian weights, there is no discrepancy between the frozen weight and the real weight in this calculation. All constants are chosen before decreasing the coupling ceiling.

For the activity on X, choose $W = X^{\sim 1}$. If $\mathsf{m} = 0$, locality identifies its output with that of the constant source. Suppose next that $0 < \mathsf{m} < r$ and set $T = r/\mathsf{m} > 1$. At a fixed point of the field domain the function

$$\phi(t) = \big(E^{(k+1)}[\mathfrak{L}_t^W] - N^{\circ(k+1)}\big)(X,\mathbf{p},z)$$

is holomorphic for $|t| < T$ and has modulus at most $\frac{1}{2} e^{-\kappa' d_{k+1}(X)}$ there. Its values at zero and one are the constant-source and variable-source outputs. Schwarz's lemma applied to $\phi(t) - \phi(0)$ yields

$$|\phi(1) - \phi(0)| \le \frac{\mathsf{m}}{r} e^{-\kappa' d_{k+1}(X)}.$$

If $\mathsf{m} \ge r$, apply the two independent supremum bounds to the actual source outputs: their difference is at most $e^{-\kappa' d_{k+1}(X)}$, which is no larger than the same right side multiplied by m/r. This case needs no evaluation of the localized line at $t = 1$. This recovers (183) at level $k+1$ with constant one. The subtraction of the common normalizer is essential: the bound on the full activity is E_0, and its direct use would give slope $2E_0$ rather than the constant needed by the defect estimate.

Choose the extension at site z to be the output for the constant $w(p_z)$, with coefficient $b_{k+1}(w(p_z))$. The difference from the actual output has the required defect bound by (181). On a localized line the temporary extension has defect at most two in the same norm, which is the bound admitted during the large-field operation. The exact measure identity of the step then gives the sourced format at the new level. The constant-source extension is formed after the small-field output and before this last operation; its definition therefore uses no unknown future remainder.

Remainders, boundary terms and stored operations. The uniform output bound for the new remainder is

$$|R^{(k+1),w}(X)| \le \left(\tfrac{1}{2} g_{k+1}^{\kappa_0} + e^{-p_0(g_{k+1})}\right) e^{-\kappa d_{k+1}(X)} \le g_{k+1}^{\kappa_0} e^{-\kappa d_{k+1}(X)}.$$

The last inequality holds below one fixed coupling ceiling. Apply the same line argument to this remainder and to each new boundary term. Their slopes are twice their supremum majorants times m/r. Carried terms retain their previous slopes. When boundary terms are sums of pointed activities, the spare unit of domain decay pays the number of base points in X.

For a processed component the new part of its excess exponent has the expression in section 4.3 below. Its bound is c_{new} times the number of newly displayed sites. The older exponents use the additive count $\mathsf{n}(\omega')$, so

$$|\Psi_\omega| \le c_{\mathrm{new}} \sum_j \#_j(Z_j(\omega) \cap X) + c_\Phi \sum_{\omega'} \mathsf{n}(\omega') \le \max\{c_{\mathrm{new}}, c_\Phi\}\mathsf{n}(\omega).$$

Thus the nesting takes a maximum of constants. On a localized line the same calculation returns $\max\{c_{\mathrm{new}}, 2c_\Phi\} = 2c_\Phi$ when $c_\Phi = 2c_{\mathrm{new}}$. Applying Schwarz on the twice enlarged component gives slope $2c_{\mathrm{new}} = c_\Phi$ for its new part. The inner parts retain their slopes towards the same arbitrary constant ζ, since m is monotone in the region. No change of base constant is made at an inner component.

The oscillation norm of a family of factors is a sum of suprema, so it is useful to spell out the analytic step. For $0 < \mathsf{m} < r$, choose finitely many factors, two allowed field arguments for each, and phases that align their differences between $t = 1$ and $t = 0$. The phased sum of the corresponding differences at t is one holomorphic function with the carrier majorant as its bound. Schwarz's lemma gives twice that majorant times m/r. Taking the chosen field arguments arbitrarily close to their suprema, then increasing the finite set of factors, proves the same slope in the sum-of-oscillations norm. This uses holomorphy in t; stored near variables may remain merely Borel. For $\mathsf{m} \ge r$ the two independent supremum bounds give the same slope with the factor m/r; for $\mathsf{m} = 0$ the equality follows from locality.

At level zero the sole new datum is $u = w$ and the operation excesses vanish. At level $k+1$, first construct the constant-source increment from the coefficients and remainders through level k. The coefficient estimate (180) keeps its total sensitivity below $\frac{1}{3}$. The constant-source whole-lattice family and its coefficient then supply the extension just used. The sourced step returns the new regular, remainder, boundary and operation data with the same bounds, while locality identifies the variable-source output with the frozen-source output wherever the source is locally constant. This completes the source-data induction under the uniform complex one-step estimates. It uses the same radius at every level and introduces no factor equal to the number of previous levels, sites, or nested components.

4.3. **Complex sources and the positive component majorant.** On a regular field, the tent decomposition of the action gives

$$|A(h_y^{(k)}\zeta_k, \mathbf{U})| \le C\alpha_{0,k}^2 \sup|\zeta_k| \le C\rho g_k^2 (\log x_k)^{2q}.$$

This is a small analytic local perturbation. The domination on real fields uses the complete profile minimum in (179). Put

$$\varsigma = \frac{4\sup|w|}{3x_k^{\min}} < 1.$$

For a nonnegative plaquette weight χ and a real U,

$$\begin{aligned} e^{-(1+\varsigma)A(x_k(\cdot)\chi,U)} &\le \left| e^{-A((x_k(\cdot)-\zeta_k(\cdot;w))\chi,U)} \right| \\ &\le e^{-(1-\varsigma)A(x_k(\cdot)\chi,U)} \le 1. \end{aligned} \tag{186}$$

Indeed, $a_p(U) \ge 0$ and $|\operatorname{Re}\zeta_k(p;w)| \le \frac{4}{3}\sup|w| \le \varsigma x_k(p)$. Also $\varsigma \le C\rho g^2$ on the fixed source class. The Wilson factor therefore retains its failed-test penalty. The Gaussian and gauge-fixing penalties retain their original form because their defining choices have not changed.

Stored variables require an integrated estimate. Let ω be the local history of a processed component and Ψ_ω its source excess exponent. A layer stored at level j is regular at that level: the fourth power of its fine spacing bounds the plaquette action and cancels the reciprocal fourth power in the plaquette count per stored site. The Wilson shift, the changes of interactions, and the excesses of swallowed components consequently have the bound

$$|\Psi_\omega| \le C(1+\rho)\#(\omega).$$

The count includes each stored site at its own level. Counts add when components are swallowed. One fixed constant can thus be used throughout the nesting of histories; it is not summed once more over the levels.

For a processed component the excess has three terms:

$$\Psi_\omega = A(\hat{\zeta}\phi_{\mathrm{c}}\mathbf{1}_X, U''(\xi)) + [V^w - V^0](X^\sim) + \sum_{\omega'} \Psi_{\omega'}.$$

Here ϕ_{c} is the old-core cutoff, $U''(\xi)$ is the real background reconstructed from stored variables, V is the interaction in the component operation, and ω' ranges over swallowed islands and drop-outs. On the layer stored at level j, regularity gives

$$|\partial U'' - \mathbf{1}| \le C\check{R}_j^2 \varepsilon_j \eta_j^2 \le C\eta_j^2.$$

Hence $a_p(U'') \le C\eta_j^4$, and multiplication by the $C\eta_j^{-4}$ plaquettes per stored site proves the Wilson part of the extensive bound. On the original bare layer compactness gives $a_p \le 2$. The interaction difference is bounded by the source-data slope, and the last sum is treated by the additive site count. The new inductive constant is the maximum of the old constant and the constant for the new layer.

A drop-out is represented differently. Its carrying term retains the Wilson weight over X; the remaining operation consists of the frozen choices. Its excess exponent contains the swallowed-component excesses, while the carrying Wilson factor records the source of the drop-out itself. Allowed evaluations of the background are fibred over the stored variables: they must be layerwise regular and give the recorded test values, except for the test whose failure caused the drop-out. Replacing this set by independent evaluation sets would allow a flat field and could erase the Wilson penalty. The mass majorant uses precisely the compatible evaluations.

The reserve in the weighted history and live-age estimates is required in the form

$$\mathsf{M}(i,X) = \frac{1}{|W_4|}\sum_{\omega}\int e^{c_r \mathsf{v}(\omega)} \sup_{\mathbf{p}} |e^{\Phi_\omega(\xi;\mathbf{p})}|\, d\mu_\omega(\xi) \le e^{-(2-c')p_0(g_i)}, \qquad \mathsf{v}(\omega) = \sum_{\text{stored sites at } j} \log x_j.$$

The allowed evaluations $\mathbf{p}$ respect the stored background data. The supremum is inside both the positive integral and the history sum. After choosing g so that $c_r \log x_j \ge 2C(1+\rho)$, the modulus of e^{Ψ_ω} is absorbed in the reserve. In particular,

$$|\mathbf{T}_i'(X;w)F| \le \|F\|_\infty \mathsf{M}(i,X).$$

This estimate is valid although the sourced integrand is complex.

For a whole label, factorization uses pointwise component majorants depending on disjoint groups of stored variables, as in (138). Bounds on a conditional integral with an arbitrary fixed exterior cannot simply be multiplied. The disjoint pointwise majorants are what permit the positive integrations to factor. The remaining normalization and missing-vacuum costs are $O(1+\log x_j)$ per displayed site. The creation deposits and the raised reserves pay these costs while retaining the stated component gains.

We spell out the resulting label estimate. After replacing the integrands of processed-component operations by their absolute majorants at the reserved weights, its form is

$$\int \mathbf{T}'_k(\mathcal{S})[\chi_{k,\mathcal{S}} e^{-A(x_k(\cdot),U_k(\mathbf{V}))}]\, dV_k$$
$$\le \exp\left[C\sum_{j\le k}(1+\log x_j)\#_j(Z_j)\right]\prod_{(t,\mathfrak{c})} e^{-cp_0(g_t)}\prod_{(i,X)}\mathsf{M}(i,X).$$

The first product runs over creation cells outside the displayed processed components. The second runs only over outermost processed components. Creations swallowed by one such component have already paid for its mass and cannot be counted a second time. If a label still displays part of a drop-out's interior, use a reserve constant larger than the one spent on the source excess. The extra reserve pays for that displayed interior. This constant is chosen after ρ and before the upper bound on g.

The site count and logarithm in the source charge cost at most $CM^4\check{R}_j^5$ per level-j cell. The original creation deposit is $cp_0(g_j)^2$. The lineage credit estimate in section 3 pays the cumulative charge until the next component operation. At its parameter choices the weaker surviving exponent $cp_0(g_j)$ still pays $CM^4\check{R}_j^5\log x_j$ with room left. The weighted history sum thus retains, for every final component C, a factor

$$\exp[-cp_0(g_k)-\kappa_1|C|_k].$$

Its exponential cell decay pays the connected-set entropy. These are uses of the weighted history estimate; an unweighted component mass bound alone would not give them.

4.4. **Local source energies and the logarithm.** The scalar energy is fixed by the values of the constructed activities at the trivial pair. To show both its locality and its normalization, keep the pointed regular family and write

$$\Delta E^{(j)}(X,z;w)=E^{(j),w}(X,\mathbf{1},z)-E^{(j)}(X,\mathbf{1},z),\qquad \Delta R^{(j)}(X;w)=R^{(j),w}(X,\mathbf{1})-R^{(j)}(X,\mathbf{1}).$$

Set

$$e_j(X;w)=\sum_{z\in X}\Delta E^{(j)}(X,z;w)+\Delta R^{(j)}(X;w),\qquad e_0=0. \tag{187}$$

The Gaussian normalizer is the same in the two regular families and cancels in their difference. These values are therefore values of the interaction output, with the same subtraction as in (185). In particular, the coupling-independent local numbers retained by $N^{\circ(j)}$ cancel as well; they are not added a second time to the source energy. Their unsourced sum remains part of the vacuum energy.

The slopes towards the constant zero give

$$|\Delta E^{(j)}(X,z;w)|\le r^{-1}\|w\|_{\flat,K-j}e^{-(\kappa+1)d_j(X)},\qquad |\Delta R^{(j)}(X;w)|\le 2r^{-1}\|w\|_{\flat,K-j}g_j^{\kappa_0}e^{-\kappa d_j(X)}.$$

The number of base points is at most $CM^4(d_j(X)+1)$. Using $(d+1)e^{-d}\le 1$, and decreasing the decay rate by a fixed fraction when enlarging a domain, gives

$$|e_j(X;w)|\le CM^4r^{-1}\|w\|_{\flat,K-j}e^{-\kappa_e d_j(X)},\qquad \kappa_e>\kappa_{\mathcal{D}}. \tag{188}$$

Here $\kappa_{\mathcal{D}}$ is the entropy rate of localization domains through a fixed cube. Each energy is holomorphic in its level source class, real at real sources, zero at $w=0$, and depends only on w in $X^{\sim 1}$.

The scalar share of a label. Let $\mathcal{Z}_j(\mathcal{S})$ be the pairs (z,X) of regular terms retained by a label, and let $\mathcal{R}_j(\mathcal{S})$ be its retained remainder domains. Define

$$T_j(\mathcal{S};w)=\sum_{(z,X)\in\mathcal{Z}_j(\mathcal{S})}\Delta E^{(j)}(X,z;w)+\sum_{X\in\mathcal{R}_j(\mathcal{S})}\Delta R^{(j)}(X;w).$$

Denote by $\mathcal{E}_k(\mathcal{S};w)$ the scalar that appears with a minus sign in the exponent of its term. The numerical scaling factors, gauge normalizers and Gaussian constants are source independent. Subtracting the two scalar constants thus gives the exact identities

$$\begin{aligned}\mathcal{E}_k(\mathcal{S};w) &= \mathcal{E}_k(\mathcal{S};0) - \sum_{j\le k} T_j(\mathcal{S};w),\\ T_j(\mathcal{S}_\emptyset;w) &= \sum_X e_j(X;w),\end{aligned} \tag{189}$$

where $\mathcal{S}_\emptyset$ is the all-small-field label. Boundary terms carry no additional scalar subtraction.

A term missing from the all-small-field share contains a cube of the enlarged large-field region Z_j^+. Pin it at such a cube and sum its domain decay there. The tree bound and (188) give

$$|T_j(\mathcal{S}_\emptyset;w) - T_j(\mathcal{S};w)| \le C\#_j(Z_j^+). \tag{190}$$

The count is at the level of the missing term, including its base-point count. No sum over bare sites of a fixed physical region is used in this estimate.

For a run stopped at k, put

$$E_{(k)} = -\mathcal{E}_k(\mathcal{S}_\emptyset;0), \qquad \mathcal{E}_{(k)}(w) = \sum_{j\le k}\sum_X e_j(X;w), \qquad \widetilde{Z}_k(w) = e^{-E_{(k)}-\mathcal{E}_{(k)}(w)}Z(w).$$

At $k=K$ these are E, $\mathcal{E}$ and $\widetilde{Z}$. The uniform estimates to obtain from the weighted history sum are

$$|\widetilde{Z}_k(w)| \le e^{C_r(1+\log x_k)\mathsf{V}_k}, \qquad \widetilde{Z}_k(0) \ge e^{-C(1+\log x_k)\mathsf{V}_k}, \qquad \mathsf{V}_k = \#\mathbb{T}^{(k)}. \tag{191}$$

The permitted sources belong to $\mathcal{W}_{r,k}$. At first passage $x_K \in (g^{-2}, g^{-2}+B_\sharp]$, so the constants in the final-level estimate are independent of K, β and the torus side. Write $C_g = C(1+\log g^{-2})$ after fixing the source radius.

Upper stability. Take moduli inside every stored-variable integral, using the positive component majorants of section 4.3. The source changes the non-Wilson part of the label exponent by at most a constant depending on r per determining bond. Its stored operation excesses are absorbed by raising the fixed volume reserves. The weighted label estimate of section 3 consequently has the sourced form

$$\sum_{\mathcal{S}} |e^{\mathcal{E}_k(\mathcal{S};w)}\tau_{\mathcal{S}}^w|(\mathcal{V}_k)\exp\left[\sum_{j\le k} C(1+\log x_{j-1})\#_j(Z_j^+)\right] \le e^{c(r)\mathsf{V}_k}. \tag{192}$$

The only uncharged extensive count is the determining set of the last lattice, whose size is at most $C\mathsf{V}_k$. All older displayed bonds lie in large-field regions, where the creation payments and component masses pay their charge. The constant $c(r)$ is independent of the coupling. The uniform product and history estimates needed for this summation are the same ones used in the unsourced construction, with larger fixed reserves.

The remaining normalization is explicit. By (189),

$$\begin{aligned}&-E_{(k)} - \mathcal{E}_{(k)}(w) - \mathcal{E}_k(\mathcal{S};w)\\ &\quad = -[E_{(k)} + \mathcal{E}_k(\mathcal{S};0)] - \sum_{j\le k}[T_j(\mathcal{S}_\emptyset;w) - T_j(\mathcal{S};w)].\end{aligned}$$

The first bracket compares the label's numerical vacuum constant with that of the all-small-field label. Scaling Jacobians and gauge normalizers are omitted in its large-field region; their logarithms cost $C(1+\log x_{j-1})$ per omitted site. The source shares have the smaller cost (190). Their sum is bounded above by the weight already present in (192), together with $C(1+\log x_k)\mathsf{V}_k$ at the last lattice. Sum the exact sourced identity over labels. This proves the upper bound in (191).

The lower bound at zero source. Every unsourced term is positive, so retain the all-small-field label alone. Restrict its last block field to $|V(b)-\mathbf{1}| < cg_k$ at each bond, with c fixed small. Background regularity places this set in the kept small-field domain and bounds its Wilson action by $C\mathsf{V}_k$ after

multiplication by $x_k = g_k^{-2}$. The remaining normalized exponent is also bounded by $C\mathsf{V}_k$. A Haar ball in SU(2) of radius cg_k has volume at least $c' g_k^3$, and there are at most $4\mathsf{V}_k$ independent bond variables. Its volume therefore costs at most $C(1+\log x_k)\mathsf{V}_k$. This proves the lower bound in (191). Balls of the larger threshold radius $g_k p_0(g_k)$ would allow action $Cp_0(g_k)^2\mathsf{V}_k$ and would give the weaker floor with that logarithmic power.

At a fixed depth s, stop at $k = K - s$. All source and energy identities use the same local data through this level. The bound $x_{K-s} \le g^{-2} + B_\sharp(s+1)$ replaces C_g by $C(g,s)$; the number of sites is V_{K-s}. This gives the depth version without changing a constant according to the total cutoff.

Proposition 4.2 (Separated derivatives of the scalar energy)**.** Under these local source-energy estimates, for $n \ge 2$ and supports at mutual distance at least $d > 0$,

$$|\partial_1 \cdots \partial_n \mathcal{E}(0)| \le C\ell^4(1+d^{-4})(1+\log^+ d^{-1})(n/\rho)^n \prod_i \|f_i\|_\sharp.$$

If one support lies in a ball of radius R, the factor depending on ℓ can be replaced by a constant depending on R. ■

Proof. Locality makes a mixed derivative vanish unless the enlarged domain meets every support. In particular it must join two of them, so $d_j(X) \ge dL^{K-j}/M - C$. On the polydisc with radii $\rho/(n\|f_i\|_\sharp)$, Cauchy's inequality bounds that derivative by $(n/\rho)^n \prod_i \|f_i\|_\sharp$ times the activity bound. Summing domains through a fixed cube first uses up part of their exponential decay. The remaining estimate is

$$(n/\rho)^n \prod_i \|f_i\|_\sharp \cdot \sum_{j \le K} C\ell^4 (L^{K-j}/M)^4 e^{-c[dL^{K-j}/M - C]_+}.$$

For $L^{K-j}/M \le C\max(1, d^{-1})$ there are $O(1+\log^+ d^{-1})$ scales, and the summand is bounded by $C\ell^4 \max(1, d^{-4})$. At the finer scales the exponential beats the fourth power, and their geometric-scale sum has the same bound. This proves the first assertion. A nonzero derivative must also meet the selected support. Anchoring the domain there replaces the full-volume cube count by $(CRL^{K-j}/M + C)^4$ and proves the local version. For a single derivative there is no second support to force a long domain; the argument therefore gives no cutoff-independent uncentered mean. □

Lemma 4.3 (A local logarithm estimate)**.** Suppose h is holomorphic on $|z|_c = \sum_i c_i|z_i| < 1$, with $c_i > 0$, $h(0) = 1$, and $|h| \le Q$. Then h has a logarithm, zero at zero, on $|z|_c < (Q+1)^{-1}$, and

$$|\partial_1 \cdots \partial_n \log h(0)| \le \log 2\,[2(Q+1)n]^n \prod_i c_i.$$

■

Proof. The function $(h-1)/(Q+1)$ is bounded by one and vanishes at zero. Restricting it to each complex line through zero and applying Schwarz's lemma gives $|h(z) - 1| \le (Q+1)|z|_c$. Thus $|h - 1| < 1$ on the smaller domain, where the power series for $\log(1 + (h-1))$ defines the desired branch. On its half-radius domain this series has absolute sum at most $\sum_{m \ge 1} 2^{-m}/m = \log 2$. The polydisc of radii $[2(Q+1)nc_i]^{-1}$ lies in that domain. Cauchy's formula on a slightly smaller closed polydisc, followed by a limit of its radii, proves the estimate. □

Apply the lemma to $h = e^{-\mathcal{E}}\Gamma$, with $c_i = \rho^{-1}\|f_i\|_\sharp$ after a harmless enlargement of Q. Near zero,

$$\log \Gamma = \mathcal{E} + \log h.$$

The energy estimate and the lemma give a cumulant bound independent of the cutoff. With global stability alone Q is exponential in ℓ^4, and the resulting bound is exponential in the volume. The next comparison removes this volume dependence.

4.5. **Chessboard comparison and one common normalizer.** Partition a torus into cells of side $\mathfrak{l}$, with faces at bond mirrors and $2\mathfrak{l} \le \ell$. For F localized in one cell let $F^{[\square]}$ be its mirror copies, conjugated in odd cells, and put

$$\|F\|_{\mathrm{ch},\ell} = \left\langle \prod_{\square} F^{[\square]} \right\rangle^{(\mathfrak{l}/\ell)^4}.$$

The repeated expectation is nonnegative. The reflected Cauchy–Schwarz inequalities give the chessboard estimate [FILS78, Theorems 4.1–4.3]:

$$\left| \left\langle \prod_{\square} F_{\square} \right\rangle \right| \le \prod_{\square} \|F_{\square}\|_{\mathrm{ch},\ell}.$$

The cyclic and reflected Cauchy–Schwarz properties are supplied by the lattice reflection argument. For an exponential source the repeated expression is a quotient of partition functions, and the root replaces an extensive volume factor by its cost per cell.

We give the Gibbs comparison at fixed spacing. For a bounded real H in one cell, reflection invariance of the Gibbs state and its conditional equation imply

$$\nu(H) \le \log \|e^H\|_{\mathrm{ch},\ell} + C\beta \mathfrak{l}^4 a_K^{-3} \ell^{-1}. \tag{193}$$

Indeed, sum the reflected copies of H over the cells of a large cube. Their expectations are all $\nu(H)$. Jensen's inequality bounds the expectation of this sum by the logarithm of the expectation of its exponential. In the conditional Gibbs integral, replacing the exterior configuration by periodic boundary conditions changes only plaquettes meeting the boundary. There are $O((\ell/a_K)^3)$ of these, and the Wilson action changes by at most $C\beta$ for each one. Divide by the number $(\ell/\mathfrak{l})^4$ of cells to obtain (193). The limit $\ell \to \infty$ is taken at this fixed K, β.

Here is the linearization needed for bounded cell functions. For real G and $t > 0$ small, set $H = \log(1 + tG)$. If $b_\ell = \|1 + tG\|_{\mathrm{ch},\ell}$, then $b_\ell \ge \exp[\nu(H) - o(1)]$. The seminorm triangle inequality gives $b_\ell \le 1 + t\|G\|_{\mathrm{ch},\ell}$. Consequently, first along any chosen volume sequence and then as $t \downarrow 0$,

$$\nu(G) \le \liminf_{\ell} \|G\|_{\mathrm{ch},\ell}.$$

For complex F, choose a phase so that $\nu(\mathrm{Re}(e^{-i\alpha}F)) = |\nu(F)|$ and use the seminorm bound on the real part. For a product of finitely many cell functions, put them in one coarse cell and apply this comparison there. The fine-board chessboard inequality then gives

$$\left| \nu\left(\prod_{\square} F_{\square} \right) \right| \le \liminf_{r \to \infty} \prod_{\square} \|F_{\square}\|_{\mathrm{ch},\ell_r}. \tag{194}$$

The assertion is along every admissible sequence $\ell_r \to \infty$. This is needed when a holomorphic normalizer is selected along one such sequence. The right side can also be bounded by the product of the individual limsups. The surface error has been removed before any cutoff limit is taken.

If the product vanishes identically its comparison is immediate. Otherwise choose a common board shift and a power L^{k_0} so that all occupied fine cells lie in one cell B of side $l' = L^{k_0}\mathfrak{l}$. As L is odd, the coarse face planes are also fine face planes. Put

$$F = \prod_{\square} F_{\square}, \qquad G = \mathrm{Re}(e^{-i\alpha}F), \qquad \Phi = \log(1 + tG), \qquad 0 < t \le \tfrac{1}{2}\|F\|_{\infty}^{-1}.$$

Then $\nu(\Phi) \geq t|\nu(F)| - t^2\|F\|_\infty^2$. There are $N' = (\ell/l')^4$ coarse cells. Expanding the repeated exponential before estimating it gives

$$\left\langle \prod_{\widehat{\square}} (1+tG)^{\widehat{[\square]}} \right\rangle = \sum_Y t^{|Y|} \left\langle \prod_{\widehat{\square}\in Y} G^{\widehat{[\square]}} \right\rangle$$
$$\leq \sum_Y t^{|Y|} \left(\prod_{\square} \|F_\square\|_{\mathrm{ch},\ell} \right)^{|Y|} = \left(1 + t \prod_{\square} \|F_\square\|_{\mathrm{ch},\ell} \right)^{N'}.$$

Indeed, each reflected G is one half the sum of a product of copies of the $F_\square$ and its complex conjugate, with phases of modulus one. After expanding these two choices in each occupied coarse cell, every term has one factor in each of distinct fine cells. The fine-board chessboard estimate bounds its absolute expectation by the indicated product. The factors $2^{-|Y|}$ cancel the number of conjugation choices. No convexity inequality has been applied to the original complex product.

Apply the real Gibbs comparison in the coarse cells and divide its logarithm by N'. It follows that

$$t|\nu(F)| - t^2\|F\|_\infty^2 \leq \log\left(1 + t \prod_{\square} \|F_\square\|_{\mathrm{ch},\ell} \right) + C\beta (l')^4 a_K^{-3} \ell^{-1}.$$

At this fixed cutoff take the lower limit along any chosen $\ell_r \to \infty$, divide by t, and let $t \downarrow 0$. This proves the product domination with its lower limit. Both $\|F\|_\infty$ and l' can depend on this finite collection of functions; their contributions disappear in the stated order of limits. In particular the proof applies to a normalizer chosen on a subsequence of tori and does not impose a cutoff-uniform bound on the unnormalized products.

The same fine-board inequality gives the rule used for columns of insertions: if F is a product of fine-cell functions inside B, then

$$\|F\|_{\mathrm{ch},\ell;B} \leq \prod_{\square} \|F_\square\|_{\mathrm{ch},\ell}.$$

Indeed its reflected coarse array is such a product over fine cells, and taking the coarse root of the fine-board estimate gives this inequality. The rule keeps the actual complex fine-cell functions, rather than replacing them by pointwise moduli.

Now take a source supported in m unit balls a distance at least three from the faces of one cell. Fix one reference torus $T_{\ell'}$ with $\ell' \geq 2\mathsf{l}$. For its mirror-periodic continuation W, the finite-torus transfer estimate used in the source comparison is

$$0 \leq Z_\ell(W) \leq Z_{\ell'}(W)^{(\ell/\ell')^4}, \qquad \ell \geq \ell' \geq 2\mathsf{l}. \tag{195}$$

It applies when $W \circ \theta = \overline{W}$ at each face and W vanishes on plaquettes having a bond crossing a face. Taking the chessboard root transfers the numerator to the reference torus. Upper stability there and the floor on T_ℓ leave a vacuum-density difference. Local vacuum energies agree on nonwrapping domains, and the decaying wrapping tails bound this difference independently of the cutoff.

In detail, with $N_\ell = (\ell/\mathsf{l})^4$, the exponential seminorm and transfer inequality give

$$\|e^{\mathcal{O}_a(w)}\|_{\mathrm{ch},\ell} = \left[\frac{Z_\ell(W)}{Z_\ell(0)} \right]^{1/N_\ell}$$
$$\leq \exp\left\{ (\mathsf{l}/\ell')^4 \operatorname{Re} \mathcal{E}_{\ell'}(W) + \mathsf{l}^4 \left[C_g + \frac{E_{K,\ell',\beta}}{\ell'^4} - \frac{E_{K,\ell,\beta}}{\ell^4} \right] \right\}.$$

Upper source stability is used only in the numerator on $T_{\ell'}$; the unsourced floor is used in the denominator on T_ℓ. The vacuum densities may each diverge with the cutoff. Their difference is what the wrapping-domain estimate bounds. Estimating them separately would lose the comparison.

To define the normalizer, first divide the localized scalar energies according to their enlarged domains. A domain belongs to the first class if $X^{\sim 1}$ is contained in one board cell; all other domains

belong to the second class. Terms on which the source is zero vanish, since $e_j(X;0)=0$. Thus a term in the second class can contribute only if its enlarged domain meets both a source ball and the boundary of a cell. This division is made for the scalar energies themselves, before their absolute values are taken.

Write $N'=(\ell'/\mathsf{l})^4$. On the reference torus, the within-cell part of $\mathcal{E}_{\ell'}(W)$ is a sum over the N' cells and the levels j. For an even cell its summand is holomorphic in the original variables $z_1,\dots,z_n$. For an odd cell it is holomorphic in $\bar z_1,\dots,\bar z_n$. Conjugate the value of every odd-cell summand and then divide the sum by N'. The result is $\mathcal{E}^\circ(z)$. For example, if the odd-cell summand is

$$\phi(\bar z)=\sum_\alpha c_\alpha \bar z^\alpha, \qquad \overline{\phi(\bar z)}=\sum_\alpha \bar c_\alpha z^\alpha.$$

The outer conjugation changes no real part and restores holomorphy jointly in every source variable. Also $\mathcal{E}^\circ(0)=0$. Locality is used here in its full form: when $X^{\sim 1}$ is in one cell, the energy depends on the source in that cell alone. No values of the mirror source on neighbouring cells enter its summand.

This definition gives the exact real-part cancellation

$$\frac{1}{N'}\operatorname{Re}\mathcal{E}_{\ell'}(W)-\operatorname{Re}\mathcal{E}^\circ(z)=\frac{1}{N'}\operatorname{Re}\sum_j \sum_{X:\ X^{\sim 1}\text{ meets a cell face}} e_j(X;W).$$

The individual within-cell sums can grow with K. In particular, their first source derivatives contain the cutoff-dependent mean. They disappear from this difference identically, so they never have to be estimated by a cutoff-independent sum of moduli.

We estimate the right side by charging a nonzero face domain to a source ball that it meets. If there are several choices, one can charge it to all of them; this only enlarges the upper bound. Put $t_j=L^{K-j}/M$. A ball of fixed radius contains at most $(Ct_j+C)^4$ cubes of the level-j partition that can serve as anchors for such a domain. Its distance from the faces is at least three, so a face domain through one of these anchors has

$$d_j(X)\ge ct_j-C.$$

Use part of $e^{-\kappa_e d_j(X)}$ to sum domains through their anchor, and retain the rest for this joining distance. The tree bound then gives the contribution

$$C(Ct_j+C)^4 e^{-c[ct_j-C]_+}$$

for one ball at this level. The t_j form a geometric sequence. The polynomial factor is dominated by the exponential at its large values; only a bounded number of the remaining values lie below any fixed threshold. Thus the full cost per ball is

$$C_{\text{face}}\le C\sum_{i\ge 0}(CL^i/M+C)^4 e^{-c[cL^i/M-C]_+}<\infty.$$

The count over reflected cells is N' times this bound for each of the m source balls. Division by N' leaves at most mC_{face}. The estimate is independent of the cutoff and of the number of cells on the reference torus.

Subtract $\operatorname{Re}\mathcal{E}^\circ$ in the preceding seminorm comparison. The face estimate and the bounded difference of vacuum densities give

$$|e^{-\mathcal{E}^\circ(z)}\Gamma_{\mathfrak{s}}(z)|\le \mathsf{Q}, \qquad \mathsf{Q}=e^{mC_{\text{face}}+C_g\mathsf{l}^4}, \qquad \|w\|_\flat<\rho. \tag{196}$$

The constant C_g absorbs the bounded vacuum-density difference. Here $\mathfrak{s}$ is any torus of side at least ℓ' or any periodic Gibbs state at the same first-passage datum. For a Gibbs state, apply (194) to the exponential source and then the same finite-torus estimate. The normalizer was constructed on $T_{\ell'}$ and is unchanged when the volume or the state changes.

There are two distinct uses of complex variables in this argument. The mirror continuation W contains both z and $\bar z$; $\mathcal{E}_{\ell'}(W)$ itself need not be holomorphic in z. Its real part enters the bound and

is compared with the real part of the holomorphic function $\mathcal{E}^\circ$. In contrast, the original generating function $\Gamma_{\mathfrak{s}}(z)$ is holomorphic in z at fixed cutoff. Its insertion is a bounded function of finitely many group variables, even in an infinite-volume Gibbs state. Hence the normalized generating function in (196) is holomorphic on the original joint source domain. The modulus comparison has not changed the variables in which its derivatives are taken.

The whole finite source list is used when $\mathcal{E}^\circ(z_1, \dots, z_n)$ is formed. A within-cell energy may depend on several of these variables and retains that joint dependence. In particular the construction does not replace it by a sum of one-variable normalizers. Restricting some variables to zero gives the normalizer for that restricted source in the same analytic function. Applying the local logarithm estimate to two states therefore produces

$$\partial_1 \cdots \partial_n \log \Gamma_{\mathfrak{s}}(0) = \partial_1 \cdots \partial_n \mathcal{E}^\circ(0) + \partial_1 \cdots \partial_n \log(e^{-\mathcal{E}^\circ} \Gamma_{\mathfrak{s}})(0).$$

The first term is exactly the same for the two states. Its mixed derivative, including any cutoff-dependent part, cancels on subtraction. Agreement only on the coordinate axes would not imply this cancellation of mixed derivatives.

Proposition 4.4 (Differences of means and cumulants)**.** For tests supported in a fixed unit ball and any two allowed states or tori at one first-passage datum,

$$|C_n[\mathfrak{s}](F_1, \dots, F_n) - C_n[\mathfrak{s}'](F_1, \dots, F_n)| \le C_n(g) \prod_i \|F_i\|_\sharp, \qquad n \ge 1.$$

No separation or sign condition is required. In particular, for f in a ball of radius R and two periodic Gibbs states,

$$|\nu(\mathcal{O}_a(f)) - \nu'(\mathcal{O}_a(f))| \le C_1(R) \|f\|_\sharp.$$

■

Proof. Apply lemma 4.3 to both sides of (196), on the same source domain. Each cumulant is the corresponding derivative of $\mathcal{E}^\circ$ plus a uniformly bounded derivative of a local logarithm. Subtraction cancels the derivative of the normalizer exactly. This proof includes $n = 1$, although either individual mean can diverge with K. A fixed tent partition into unit balls proves the radius-R version. □

4.6. **A uniform bound at each fixed order.** The per-ball bound used for common extraction and the time-translation semigroup requires independence from the positions of the support balls. Their diameter as one configuration can be arbitrarily large. There is consequently no single bounded torus containing all possible configurations to which the global logarithm estimate can be applied. The chessboard estimate treats the occupied cells separately.

Proposition 4.5 (The per-ball correlation bound)**.** For $n \ge 2$, supports at mutual distance at least $d > 0$, and each support contained in a radius-R ball, one has

$$|C_n(f_1, \dots, f_n)| \le C_n^\infty(d; R) \prod_i \|f_i\|_\sharp.$$

The constants are independent of the cutoff and the positions. The torus version requires $\ell \ge Cn(R+1)$; in periodic Gibbs states there is no volume restriction. They may be chosen nonincreasing in d. ■

Proof. Take l to be the least power of L larger than $Cn(R+1)$, with C fixed sufficiently large, and require $2\mathsf{l} \le \ell$. Vary the offset of the board through one cell. For each coordinate, a given ball excludes an interval of offsets of length at most $C(R+1)$ if it is to stay a distance at least $R+2$ from every cell face. The union for n balls excludes at most $Cn(R+1)$ in that coordinate. The choice of l therefore leaves a permissible offset in all four coordinates. On the lattice one counts offsets in steps of a_K; the extra endpoint cost is at most Cna_K and is absorbed by the same choice. This is one shift of the whole board.

Group the sources in each occupied cell and denote their sum by $w_\square$. Several of the original balls may lie in one cell; we keep their source variables together there. There are $m \le n$ occupied cells. The torus chessboard bound and source stability give

$$|\Gamma(z)| \le \prod_{\square \text{ occupied}} \exp\left[(\mathsf{l}/\ell)^4 \operatorname{Re}\mathcal{E}(w_\square^{\mathrm{ch}}) + C_g \mathsf{l}^4\right]. \tag{197}$$

The norm restriction is still the original joint source restriction; each cell source uses a subset of its variables. Its reflected array has norm at most twice that restriction.

We next separate the divergent one-copy energies from the bounded interaction between copies. For pieces v_ι in balls of radius R at distance at least $D > 0$, locality gives

$$\left|\mathcal{E}\left(\sum_\iota v_\iota\right) - \sum_\iota \mathcal{E}(v_\iota)\right| \le C(D,R)\#\{\text{balls}\}.$$

For a domain meeting no more than one piece the summand is exactly zero. A remaining domain must join two pieces and has length at least $cDL^{K-j}/M - C$ at level j. If it meets r pieces, the difference of its joint and individual energies is bounded by $(r+1)Ce^{-\kappa_e d_j(X)}$ and can be charged to those r pieces. For one charged ball the sum is bounded by

$$C\sum_{j\le K}(CRL^{K-j}/M + C)^4 e^{-c[DL^{K-j}/M - C]_+},$$

which is finite uniformly in the cutoff. This explains both the bounded defect and its linear dependence on the number of pieces. In the reflected array, different occupied copies lie on opposite sides of a face with the fixed buffer $R+2$, so this defect has a bound depending on R only.

Use the cell averages of the one-copy energies as a holomorphic normalizer $\mathcal{E}_{\underline{f}}$. As before, odd-cell outer conjugation restores holomorphy without changing the real parts. Subtracting it in (197) cancels every one-copy energy exactly; the copy-interaction defect remains bounded. We obtain

$$|e^{-\mathcal{E}_{\underline{f}}(z)}\Gamma(z)| \le Q(n,R)^m \le Q(n,R)^n, \qquad Q(n,R) = e^{C(R)+C_g\mathsf{l}^4}.$$

The scalar normalizer has the ball form of the mixed-derivative estimate in proposition 4.2. In the reflected copies, a domain producing all the source derivatives must still meet every original support after folding back into the cell. Reflection folding does not increase distances, so the original separation d is a lower bound on the joining distance used in that estimate.

Apply lemma 4.3 to this normalized function and add the scalar mixed derivative. This proves the claimed fixed-order bound. Replacing d by a smaller separation in all the preceding estimates gives nonincreasing constants.

For a periodic Gibbs state use (194) and construct the same normalizer on a single auxiliary reference torus. Choose it large enough to contain the finite configuration with room to spare. The finite-torus transfer inequality brings all repeated numerators to that reference torus, and the bounded vacuum-density difference adds a fixed cell cost. The source-energy and copy-defect estimates are local. Thus the resulting constant depends on n, d, R, g but not on the auxiliary side or the diameter of the configuration. This proves the state version. □

This bound is sufficient for extraction and for translated self-pairings. Its growth in n is much larger than factorial: $\mathsf{l} \asymp n(R+1)$, while the logarithm bound involves $(2Q(n,R)^n + 2)^n$. The graded estimate below uses unit cells for all orders and records one factor for each active source, which avoids this loss.

4.7. **Graded source bounds and factorial growth.** Use unit cells. Let each v_i be supported in a ball of radius η, with $\eta \ge 4a_K$, and at distance at least η from their faces. Assume that the

different supports have mutual distance at least $d > 0$. The source induction requires a holomorphic function $\mathcal{E}''$, zero at zero, for which

$$h(z) = e^{-\mathcal{E}''(z)} \left\langle e^{\sum_i z_i \mathcal{O}_a(v_i)} \right\rangle$$

has the graded bounds

$$|h(z)| \le Q(\eta)^{|B|} \quad \text{when } z_i = 0 \ (i \notin B), \qquad Q(\eta) \le C(g)\eta^{-p_1}, \qquad \sum_i |z_i| \|v_i\|_\sharp < \rho. \tag{198}$$

The exponent p_1 is independent of n. The same energy locality estimate controls the mixed derivatives of $\mathcal{E}''$.

The source induction. The construction in section 4.2.1 propagates the source estimates together on $\|w\|_\flat < 2\rho$, with the real choices fixed, once the uniform complex one-step estimates hold. The inductive data include the local analytic activities, their scalar energies, the stability bound and floor, and the differentiated coefficient estimate (180). For the two-point comparison we additionally require (203) for the full renormalized first derivative, including its residual quadratic component, on these same complex source domains. Spatial source differences are measured relative to a local constant, and the stored excesses use the raised history reserves. These are parts of one local induction, using the same domains and coefficient sequence as the unsourced construction.

For the volume comparison use (195) on its compatible cells and tori: the reflected source satisfies $W \circ \theta = \overline{W}$ at each face, and vanishes on plaquettes with a bond crossing a face. The graded estimate (198) then records the cost of each active source with one holomorphic normalizer. Its local mixed-derivative bounds, the exponent p_1, and the constants in the induction are uniform in the number of insertions, cutoff, torus side and positions, within the displayed source-norm and support restrictions.

The cancellation underlying (198) can be organized on the fixed source domain, rather than separately for its coordinate faces. Put $c_i = \|v_i\|_\sharp$ and write $w_B = \sum_{i \in B} z_i v_i$. The common restriction $\sum_i |z_i| c_i < \rho$ gives

$$\|w_B\|_\flat \le \sum_{i \in B} |z_i| c_i < \rho, \qquad \|(w_B|_\square)^{\mathrm{ch}}\|_\flat < 2\rho \quad \text{for every occupied cell } \square.$$

For the last inequality the copies of that cell source are grouped into their two parity arrays; their number does not enter a supremum norm. Every restriction to an active set B therefore stays in the same source class used to construct the scalar energies.

Group the active pieces in their occupied unit cells. In each cell keep its entire source, including all its active variables, when forming the within-cell part of the normalizer. A localized energy can meet several pieces in the same cell. Its joint dependence is cancelled with that same energy in $\mathcal{E}''$; decomposing the normalizer further into one term per source would leave these mixed contributions uncancelled. As with $\mathcal{E}^\circ$, outer conjugation of the odd-cell terms makes this a single holomorphic function of all variables. On the face $z_i = 0$ for $i \notin B$, every term whose enlarged domain misses all active supports is zero. Thus the restriction $\mathcal{E}''|_B$ makes the cancellation for w_B without constructing a new normalizer.

Only the domains meeting a face remain in the normalized modulus bound. Charge one to an active piece whose support it meets. For a piece of radius η at distance at least η from the faces, a domain contributing at level j has

$$d_j(X) \ge c\eta t_j - C, \qquad t_j = L^{K-j}/M.$$

There are at most $(C\eta t_j + C)^4$ possible anchor cubes near that piece. After the tree sum through the anchor, the cost per piece is bounded by

$$C \sum_{j \le K} (C\eta t_j + C)^4 e^{-c[c\eta t_j - C]_+}.$$

At the fine scales $\eta t_j \geq 1$ this is a convergent geometric-scale sum of exponentially decaying quantities. At the coarser scales the anchor count is bounded and each scale costs at most a constant. There are $O(1+\log(M/\eta))$ such scales, independently of K. Consequently

$$C_{\mathrm{face}}(\eta) \leq C(1+\log(M/\eta)).$$

If a domain can be charged to several active pieces, the sum of its charges overcounts it. This yields an upper bound proportional to $|B|$, without a factor for inactive variables. The same counting applies to the difference between a joint local energy and its cell contributions: a domain meeting r contributing cells costs at most $(r+1)Ce^{-\kappa_e d_j(X)}$, which is bounded by twice the sum of its r charges.

The stability cost of a unit cell is C_g. There are at most $|B|$ occupied cells. Within-cell energies have already cancelled, and the face costs just found give

$$\log|e^{-\mathcal{E}''(z_B)}\Gamma(z_B)| \leq |B|[C_g + 2C_{\mathrm{face}}(\eta)].$$

This is (198), with $Q(\eta) = \exp[C_g + 2C_{\mathrm{face}}(\eta)]$. Since M is fixed before the small-coupling ceiling, the logarithmic face cost gives $Q(\eta) \leq C(g)\eta^{-p_1}$ with an exponent independent of g, n and the positions. In the case $B = \varnothing$ the source, scalar energy and normalized logarithm all vanish, and the normalized generating function equals one.

This calculation requires the uniform local source-energy estimates for the whole reflected array and their compatibility under restriction to every active list. Their propagation uses the same complex data construction as the regular, boundary and stored-excess terms, including its recursively produced operation exponents. The uniform one-step domain and majorant estimates for that construction remain the analytic inputs. The scalar terms cancel as in the construction of (196), before the face costs are summed.

Proposition 4.6 (From graded estimates to cumulant growth)**.** Once the graded source estimates have been established, there are fixed exponents p_2, q_0 such that, for $n \geq 2$, tests $f_i \in C_c^1(\mathbb{R}^4)$ supported in radius-R balls, $R > 0$, with mutual support distance at least $d > 0$, satisfy

$$|C_n(f_1,\dots,f_n)| \leq A(d;R)B(R)^n(n!)^{p_2}\prod_i \|f_i\|_\sharp, \qquad A(d;1) \leq C\max\{1, d^{-q_0}\}. \tag{199}$$

The estimate holds at every cutoff, on all tori above a fixed volume floor and in every periodic Gibbs state. Its constants are independent of the positions and the diameter of the configuration. ■

Proof. Write $c_i = \|v_i\|_\sharp$. For a nonempty block B, set all other sources equal to zero and apply Cauchy's formula with radii $\rho/(|B|c_i)$. The graded estimate gives

$$|\partial_B h(0)| \leq (|B|Q/\rho)^{|B|}\prod_{i\in B} c_i.$$

Since $h(0) = 1$, its logarithmic derivative is the finite algebraic identity

$$\partial_1\cdots\partial_n \log h(0) = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}\partial_B h(0).$$

There is no use here of a zero-free domain whose radius is uniform in n. For completeness, its combinatorial sum can be evaluated by an ordinary generating function. If $a_m = m^m$, then $a_m/m! \leq e^m$, and the exponential formula for partitions gives

$$\begin{aligned}\sum_\pi (|\pi|-1)!\prod_{B\in\pi}|B|^{|B|} &= n![t^n]\left[-\log\left(1-\sum_{m\geq 1}\frac{m^m}{m!}t^m\right)\right]\\ &\leq n![t^n]\left[-\log\left(1-\frac{et}{1-et}\right)\right]\\ &= (n-1)!e^n(2^n-1).\end{aligned}$$

All series in this coefficient comparison have nonnegative coefficients. It follows that

$$|\partial_1 \cdots \partial_n \log h(0)| \le (n-1)!2^n (eQ/\rho)^n \prod_i c_i. \tag{200}$$

The power of Q is exactly n, because each source index appears in one block of the partition.

Suppose first that $a_K \le c/n$, with the absolute constant chosen small. Use a tent partition at a mesh η comparable to n^{-1} and not smaller than $4a_K$. A function in a radius-R ball has at most $(CnR + C)^4$ pieces. Each piece has sharp norm at most $CMn\|f_i\|_\sharp$. For each resulting n-tuple, choose one common lattice shift of the board which keeps all its pieces away from unit-cell faces. To see the choice, vary the board offset over one unit cell. A fixed piece rules out a face strip of width $C\eta$ in each coordinate. The union for n pieces has relative measure at most $Cn\eta$. The proportionality constant in η can be made small enough that a permitted offset remains. Counting offsets on the lattice instead of in the continuous cell gives the same estimate with an additional Cna_K, still small. This shifts the board, or equivalently all entries by the same lattice vector; it does not translate the entries independently.

There are at most $(CnR + C)^{4n}$ tuples of pieces. In (200), the norm cost is $(CMn)^n$ and the graded cost is $Q(\eta)^n \le C(g)^n n^{p_1 n}$. The elementary inequality $n^n \le e^n n!$ absorbs all these costs in a geometric factor times a fixed power of $n!$.

The remaining term is $\partial_1 \cdots \partial_n \mathcal{E}''$. Its locality estimate, as in proposition 4.2, has a fixed separation power, and its factor n^n is another fixed factorial cost. The logarithm and scalar-energy contributions together give (199). In particular the separation exponent does not acquire a factor n at the level of connected functions.

If $a_K > c/n$, no cutting is needed. The elementary bound $0 \le a_p \le 2$ gives

$$|\mathcal{O}_a(f_i)| \le C(R/a_K + 1)^4 \|f_i\|_\infty \le C(nR + 1)^4 \|f_i\|_\infty.$$

For bounded variables $|X_i| \le b_i$, the same partition formula, with moments bounded by $\prod_{i \in B} b_i$, gives $|\kappa(X_1, \dots, X_n)| \le (n-1)!2^n \prod_i b_i$. This has the required factorial form, uniformly in the bare coupling.

Finally, for a periodic Gibbs state construct the normalizer on one auxiliary torus containing the whole configuration with room to spare. The within-cell cancellation and face costs are local, so their constants do not depend on that side or on the diameter of the configuration. Apply (194) on the selected volume sequence. The same graded bound and preceding algebra then prove the state estimate. □

5. The positive two-point function

The calculation has three parts. First an exact source identity is read at two observation levels. The local calculation is then compared with a free curvature covariance on one reference torus. Finally, the common source normalizer transfers the estimate to periodic Gibbs states. The running coupling fixes the logarithmic scale, while the live-age estimate controls contributions from stored finer fields.

5.1. **The test pair and the observation levels.** Fix $\vartheta > 0$. Let f, h be nonzero, nonnegative Lipschitz functions supported in balls of radius d, with the distance between their supports in $[d, 6d]$, and suppose

$$\int f \ge \vartheta \|f\|_\infty d^4, \qquad \operatorname{Lip} f \le \frac{\|f\|_\infty}{\vartheta d},$$

and the same inequalities for h. We use

$$I(f,h) = \iint f(x)h(y)|x-y|^{-8}\,dx\,dy.$$

Every pair of points in the supports is at distance at most $10d$, and all are at distance at least d. The mass lower bounds and the volume upper bounds therefore give

$$c\vartheta^2\|f\|_\infty\|h\|_\infty \le I(f,h) \le C\|f\|_\infty\|h\|_\infty.$$

The conclusion required later is

$$c(g,\vartheta)\frac{I(f,h)}{(\log(1/d))^2} \le C_2(f,h) \le C(g,\vartheta)\frac{I(f,h)}{(\log(1/d))^2}, \qquad 0 < d \le d_0(g,\vartheta), \tag{201}$$

with $c(g,\vartheta) > 0$ and g small for the fixed shape.

Choose the smallest $s = s(d)$ for which

$$C_W(s)L^{-s} \le d, \quad C_W(s) = C_{W,0}p_0(\bar{x}_s^{-1/2})^2, \quad \bar{x}_s = g^{-2} + B_\sharp(s+1).$$

Require $K - s \ge n_0 \ge 1$ and set

$$k_0 = K - s, \qquad \hat{d} = L^s d, \qquad x = x_{k_0} = g(d)^{-2}.$$

Minimality and monotonicity of C_W give $C_W(s) \le \hat{d} < LC_W(s)$. Thus the separation in block units tends to infinity, while $\log C_W(s) = o(s)$. In particular,

$$\frac{s\log L}{\log(1/d)} \longrightarrow 1.$$

The flow estimate and the terminal coupling window imply, uniformly in first-passage data,

$$x_{K-s} = x_K + c_0 s + o(s), \qquad \frac{g(d)^{-2}}{\log(1/d)} \longrightarrow \frac{11}{3\pi^2}.$$

The source response obeys

$$\zeta'_{k_0} = 1 + \sum_{j\le k_0} b'_j(0), \qquad |\zeta'_{k_0} - 1| \le C\left[g(d)p_0(g(d)) + \rho^{-1}g(d)^{\kappa_0 - 6}\right] = o(1).$$

These bounds have no volume dependence.

The second observation level is

$$k_1 = k_0 + \mathsf{n}, \qquad \mathsf{n} = \lceil 2\log_L \bar{x}_s \rceil.$$

First fix g, and then take d small so that $\mathsf{n} \le s$. This ensures $k_1 \le K$. The lower and upper flow envelopes give $\bar{x}_s \le Cx$ and $x_j \asymp x$ for $k_0 \le j \le k_1$, since $\mathsf{n} = O(\log x) = o(x)$.

At physical depth s the relevant source norm is

$$\|f\|_{\sharp,s} = \max\{\|f\|_\infty, CML^{-s}\operatorname{Lip} f\} = \|f\|_\infty$$

when $\vartheta C_W(s) \ge CM$. At level $k \ge k_0$ this norm is at most $L^{k-k_0}\|f\|_\infty$. Each source derivative at k_1 can consequently cost L^{n}. This loss determines how many extra levels are needed.

These choices separate three requirements. The scale k_0 leaves $\hat{d} \to \infty$ lattice units between the supports, enough for exponential localization errors to beat powers of the running coupling. The difference $k_1 - k_0$ is only logarithmic in x, so the coupling does not change by a fixed fraction on this segment. Nevertheless its volume rescaling is large: $L^{\mathsf{n}} \asymp \bar{x}_s^2$. Four powers of that length will be gained by translated pairs, whereas only two are lost in source differentiation. Making d small at fixed g ensures all three requirements and $k_1 \le K$ simultaneously. They impose a depth floor, not an additional limit $g \to 0$ during the calculation.

5.2. **One exact identity at both levels.** Sum histories with the same standing part before writing their effective exponent. The positive unsourced weight gives a probability law $\mathsf{P}^{(k)}$ on the standing history, the block field, and stored variables. Its standing-history marginal is the image of the unsourced history law. Let $\Phi^{(k)}(z)$ be the sourced exponent minus its unsourced value. Then $\Phi^{(k)}(0) = 0$ and frozen source transport gives

$$\Gamma(z) = \mathbb{E}_{\mathsf{P}^{(k)}} e^{\Phi^{(k)}(z)}.$$

Write $\Phi_i^{(k)} = \partial_i \Phi^{(k)}(0)$ and $\Phi_{12}^{(k)} = \partial_1 \partial_2 \Phi^{(k)}(0)$. The exact two-point formula is

$$C_2(f_1, f_2) = \mathrm{Cov}_{\mathsf{P}^{(k)}}(\Phi_1^{(k)}, \Phi_2^{(k)}) + \mathbb{E}_{\mathsf{P}^{(k)}} \Phi_{12}^{(k)}. \tag{202}$$

Indeed, $\partial_i \Gamma(0) = \mathbb{E}\Phi_i$ and $\partial_{12}\Gamma(0) = \mathbb{E}(\Phi_1\Phi_2 + \Phi_{12})$; subtracting $\partial_1\Gamma(0)\partial_2\Gamma(0)$ proves the identity. It applies at every level under consideration. The mixed derivative is present even when the original source is linear.

There are two joint measures in this argument, with different variables. We describe their common marginal. Fix a class $[h]$ of histories with one standing part, and let ξ denote the fields still stored at level k. Its unsourced effective measure has the form

$$\tau_{[h]}(dV) = \int e^{\mathcal{A}_{[h]}(V,\xi)}\, d\lambda_{[h]}(V, \xi),$$

where $\lambda_{[h]}$ is the positive measure formed from the kept gauge-fixing factors, characteristic functions, averaging constraints and Haar measures. Summing the displayed integrand over classes and dividing by $Z(0)$ defines $\mathsf{P}^{(k)}$. In particular,

$$\mathsf{P}^{(k)}([h], dV, d\xi) = Z(0)^{-1} e^{\mathcal{A}_{[h]}(V,\xi)} d\lambda_{[h]}(V, \xi).$$

The exponent belongs to the sum over histories in the class. Healed histories have already been summed into it; assigning this same exponent to each separate history would count them again.

Frozen kernels also give the measure on the bare configuration, an individual history and its final block field. Project this measure to $([h], V)$, and test it on gauge-invariant functions of V. Its value is $Z(0)^{-1}\tau_{[h]}(dV)$, the same as the marginal of $\mathsf{P}^{(k)}$. Projecting further to $[h]$ gives the image of the original history probability $\widetilde{\nu}$. Thus an event of the standing regions or their lineage ages has the same probability in all these descriptions. A history-independent gauge-invariant function of V, such as X_i below, has the same marginal as well. The stored-field source excesses are evaluated in $\mathsf{P}^{(k)}$, where their variables are present.

As in (77), the bare-field coupling gives $V = \mathcal{M}^k U$ only away from processed components. The two-point calculation uses the sourced exponents in (202) at k_0 and k_1, with their respective laws $\mathsf{P}^{(k)}$.

At k_0 define the frozen-tent interpolation by

$$\zeta'_{k_0} \tilde{f}_i(p) = f_i(x(p)) + \sum_{j \le k_0} b'_j(0) \sum_y h_y^{(j)}(p) f_i(x(p_y)).$$

It is defined globally, independently of the histories. Where the action profiles all equal one, this is exactly the derivative of the running source. Since every tent samples points at physical distance at most CML^{-s} from p, subtraction of $\zeta'_{k_0} f_i(x(p))$ gives

$$|\tilde{f}_i(p) - f_i(x(p))| \le \frac{CM}{\vartheta \hat{d}} \sum_{j \le k_0} |b'_j(0)| \|f_i\|_\infty.$$

The supports enlarge by at most CML^{-s}.

Localize the background around the support and put a regularity window χ_{k_0} on each localization cube. The resulting main insertion is

$$X_i(V) = \sum_p \tilde{f}_i(p) \chi_{k_0}(\square(p)) a_p(U_{k_0, \square(p)}(V)).$$

This is one function of V, the same in every history. Differentiating the effective exponent decomposes $\Phi_i^{(k_0)}$ into a constant, $\zeta'_{k_0} X_i$, irrelevant activities, contributions of live components, and the error in localization. The window gives

$$|X_i| \le C x^{-1} p_0(g(d))^2 \hat{d}^4 \|f_i\|_\infty.$$

For the full irrelevant part, the two-point comparison requires the differentiated regular-family bound specified in the local source induction:

$$C x^{-1} (\log x)^{3\mathfrak{p}_0 - q} \hat{d}^4 \|f_i\|_\infty. \tag{203}$$

The factor x^{-1} is essential here. Its source is the dependence of the regular family on the coupling array, weighted by $CM^4 \alpha_{0,j}^2$, and the source-slope induction. A Cauchy estimate on a fixed disc of radius ρ would give no such factor. The slope induction in the local source induction has size $x_j^{-1} (\log x_j)^{\mathfrak{p}_0 + q}$; the spatially varying part has an additional $L^{-(k_0 - j)}$. Its short-scale recurrence closes because $\mathsf{n}^2 (\log x)^{\mathfrak{p}_0 - q} = o(1)$.

The covariance of an irrelevant term with a main term has, relative to $x^{-2} \prod_i \|f_i\|_\infty$, a coefficient bounded by

$$C p_0(g(d))^2 \hat{d}^8 (\log x)^{3\mathfrak{p}_0 - q} = O((\log x)^{21\mathfrak{p}_0 - q}) = o(1).$$

This explains the choice $q > 21\mathfrak{p}_0$. The covariance of two irrelevant terms is smaller by the same estimates. The constants in the first derivatives do not contribute to a covariance.

The cubic part of the required estimate has a simple power count. Let α be a field-analyticity radius comparable to α_{0,k_0} and α_{1,k_0}, with comparison constants fixed by the analytic class. The source perturbation of the Wilson action has size $O(\alpha^2)$ per localization cube on this domain. For a differentiated activity that retains this weight, its cubic Taylor remainder on the window has size

$$\alpha^2 (\varepsilon_{k_0}/\alpha)^3 = O\big(x^{-1} (\log x)^{3\mathfrak{p}_0 - q}\big).$$

This calculation does not cover the whole insertion. After its constant and its multiple of the plaquette action have been subtracted, a quadratic component remains. The expansion (53) and the following pointed sum retain both the quadratic Hölder error and the cubic remainder, as well as the tail of the omitted domains. Applying that cancellation to the differentiated families requires their source-response bound on the same complex domains. Until this estimate and its applicability are established, (203) is an additional input to the two-point comparison. A Cauchy estimate on a fixed bare-source disc does not supply its factor x^{-1}.

5.3. **Live components and mixed derivatives.** For disjoint supports the action terms which depend on the source at one sampled plaquette have zero mixed derivative: their derivative contains $f_1(x(p_y)) f_2(x(p_y)) = 0$. A regular local activity has a mixed derivative only if its enlarged domain meets both supports. It then has tree length at least $c\hat{d}/M$. Domain summation and Cauchy's estimate bound these terms by

$$C \hat{d}^4 e^{-c\kappa \hat{d}/M} \prod_i \|f_i\|_\infty.$$

Since $\hat{d} \asymp p_0(g(d))^2$, this is smaller than every fixed power of $g(d)$.

For live components only their means are small. A component of lineage age α may contain fields from $\alpha + 1$ finer levels, with $O(L^{4(\alpha+1)})$ plaquettes per level-k cell. One first derivative can therefore grow like $L^{4\alpha}$; two derivatives or a product of two first derivatives can grow like $L^{8\alpha}$. Use the weighted history and live-age estimates with a rate at least eight, increasing it to absorb any polynomial in α:

$$\sum_{\alpha \ge 0} L^{8\alpha} \widetilde{\nu}(\mathrm{Live}_{k, \ge \alpha}(\mathfrak{c})) \le C e^{-c p_0(g_k)}.$$

The amplitude and probability use the same unhealed lineage age. A region healed and covered again has not survived through the intervening levels.

At k_0 at most $C\hat{d}^4$ cells are in a fixed multiple of the support neighbourhood. Let α_{c} be the largest live lineage age touching this neighbourhood. The union bound, followed by the nonnegative age sum, gives

$$\mathbb{E}_{\mathsf{P}^{(k_0)}}[\mathbf{1}_{\mathrm{contact}} L^{8\alpha_{\mathrm{c}}}] \le C\hat{d}^4 e^{-cp_0(g(d))}.$$

It pays the crude amplitude factors $(\log x)^C L^{C\check{R}_{k_0}}$ and any fixed power of x, because the exponent defining p_0 exceeds that defining $\check{R}$. This controls both the live part of $\mathbb{E}\Phi_{12}^{(k_0)}$ and the covariances containing live first derivatives. At k_1 the same reasoning is applied with its own law $\mathsf{P}^{(k_1)}$. There are only $O(1)$ touching cells there. The extra factor $L^{\mathsf{n}} \le L\bar{x}_s^2$ per source derivative is a power of x and is again paid by the age estimate. Each argument concerns one selected observation level, so there is no union over the full run.

We give the separate bounds entering this conclusion. Let $\mathcal{L}_i$ be the live contribution to $\Phi_i^{(k_0)}$ and $\mathcal{L}_{12}$ its mixed derivative. On a contact class of largest lineage age a, the local history and source estimates to establish are

$$|\mathcal{L}_{12}| \le (\log x)^C [e^{-c\hat{d}} + L^{C\check{R}_{k_0}} L^{8a}] \|f_1\|_\infty \|f_2\|_\infty,$$
$$|\mathcal{L}_i| \le (\log x)^C [e^{-c\hat{d}} + L^{C\check{R}_{k_0}} (a+2)^C L^{4a}] \|f_i\|_\infty.$$

The exponent $4a$ counts the stored plaquette sum for one derivative; $8a$ counts two such sums. If a component holds an island, the stored core begins at the first level of the unhealed lineage. Its label age can restart, but that only decreases the exponent relative to a. The local terms outside contact have the displayed exponentially small localization contribution.

Choose the age rate in (171) strictly larger than eight. Then $(a+2)^C L^{8a} \le C_b L^{ba}$, and the same contact count gives the weighted estimates with that polynomial inserted. Equivalently the extra geometric factor in the pointwise age tail absorbs the polynomial. Thus

$$|\mathbb{E}\mathcal{L}_{12}| \le (\log x)^C [e^{-c\hat{d}} + C\hat{d}^4 L^{C\check{R}_{k_0}} e^{-cp_0(g(d))}] \prod_i \|f_i\|_\infty,$$
$$\mathbb{E}|\mathcal{L}_i| \le (\log x)^C [e^{-c\hat{d}} + C\hat{d}^4 L^{C\check{R}_{k_0}} e^{-cp_0(g(d))}] \|f_i\|_\infty,$$
$$\mathbb{E}|\mathcal{L}_i|^2 \le (\log x)^C [e^{-c\hat{d}} + C\hat{d}^4 L^{C\check{R}_{k_0}} e^{-cp_0(g(d))}] \|f_i\|_\infty^2.$$

The generic constants and the coefficient of $\check{R}_{k_0}$ have been enlarged in the last line. Multiplication by any fixed power of x still leaves a quantity tending to zero. Indeed $\hat{d}$ is a fixed power of $\log x$ of exponent $2\mathfrak{p}_0$, while $p_0(g(d))$ has exponent $\mathfrak{p}_0$ and $\check{R}_{k_0}$ has the smaller exponent $\mathfrak{r}$.

There are two different covariance uses. For the two live first derivatives, Cauchy–Schwarz bounds the product moment by the square root of their second moments, and bounds their means by their first moments. Hence

$$|\operatorname{Cov}(\mathcal{L}_1, \mathcal{L}_2)| = o(x^{-2}) \prod_i \|f_i\|_\infty.$$

For a live first derivative and a bounded regular factor $B_i(V)$, use instead

$$|\operatorname{Cov}(\mathcal{L}_1, B_2)| \le 2\|B_2\|_\infty \mathbb{E}|\mathcal{L}_1|.$$

The main and irrelevant regular factors have supremum bounded by $Cx^{-1}(\log x)^C \hat{d}^4 \|f_2\|_\infty$. The first-moment bound therefore gives the same $o(x^{-2})$ result. Using a pointwise supremum for the live factor here would lose the age gain. Together with the mixed-derivative estimate, these are the two uses of the age theorem at k_0: one for the mean mixed derivative, one for the first-derivative covariances. The third use is at k_1 for its rough first-derivative covariance, under its own standing-history law and its own contact age. No comparison of unhealed ages between the two observation levels is required.

At the finer level (202) thus locates the two-point function in the covariance of the main insertions. At the coarser level the sourced slope bounds and the preceding live estimates give instead

$$|\operatorname{Cov}_{\mathsf{P}^{(k_1)}}(\Phi_1^{(k_1)},\Phi_2^{(k_1)})| \le o(1)x^{-2}\prod_i \|f_i\|_\infty.$$

Its mixed derivative is a scalar source-energy derivative plus a field-dependent term. The latter has an initial bound $CL^{2\mathsf{n}}$ per unit volume at level k_1.

Here the common translated-array device gains a volume factor. Translate the pair over a grid of spacing comparable to d and give the two members of each copy the same independent sign. Averaging in the signs removes all mixed-copy derivatives. The array has the same supremum and Lipschitz bounds as one copy. There are $\asymp L^{4\mathsf{n}}\hat{d}^{-4}$ copies per unit volume at k_1. Division by this number leaves the field-dependent error

$$CL^{-2\mathsf{n}}\hat{d}^4 \le C\bar{x}_s^{-4}\hat{d}^4 = o(x^{-2}).$$

The surviving scalar number is the translated average of the mixed derivatives of the local source energies at the trivial block field. This reduces the nonlinear calculation to a scalar recursion; it does not by itself evaluate that recursion.

More explicitly, for a translated-pair grid $\mathfrak{P}$ at level k_0, the scalar is

$$c_{12} = \frac{1}{|\mathfrak{P}|}\sum_{v\in\mathfrak{P}}\partial_1\partial_2\sum_{j\le k_1}\sum_X e_j(X; z_1\tau_v f_1 + z_2\tau_v f_2)\Big|_{z=0}.$$

To verify the division used above, let $F_i^\epsilon = \sum_{v\in\mathfrak{P}}\epsilon_v\tau_v f_i$. The mixed derivative of any holomorphic sourced expression is bilinear in the two source directions at zero. Its average at $(F_1^\epsilon, F_2^\epsilon)$ is therefore the sum of the mixed derivatives at the coincident translated pairs, since $\mathbb{E}\epsilon_v\epsilon_{v'} = \mathbf{1}_{v=v'}$. Common translation invariance makes the bare two-point function of each pair equal to the original value. The field-dependent part of the effective expression has a bound per coarse unit volume, so division by the number of pairs per such volume gives $L^{-2\mathsf{n}}\hat{d}^4$. This averaging is done in the original source variables. It neither replaces the reference law nor conditions the bare observable on the block field.

5.4. **Free curvature and the reference-torus comparison.** The free comparison has three parts. First identify the curvature covariance independently of the gauge used for the potential. Then prove decay for one constrained Gaussian step, with constants independent of the number of fine sites in a block. Finally sum the steps below the observation scale. The second part is the regularity argument: an operator bound in a Hilbert norm alone does not control the curvature at one bare plaquette.

Averages and conditional Gaussian projections. Work first with one colour and inverse coupling one. Let $\mathcal{E}^2_{\mathrm{ex}}$ be the exact two-cochains of the bare torus, with the counting inner product, and let P_{ex} be its orthogonal projection. A standard Gaussian F in this space has covariance P_{ex}. Indeed, in an orthonormal basis of $\mathcal{E}^2_{\mathrm{ex}}$ its coefficients are independent standard normal variables. The discrete Hodge identity gives

$$P_{\mathrm{ex}} = d\mathcal{L}_1^+ d^*, \qquad \mathcal{L}_1 = d^*d + dd^*,$$

where the superscript $+$ denotes the inverse on the orthogonal complement of the kernel. The harmonic one-cochains on the torus are constant; they do not contribute to F.

Denote the linearized contour average by Q, the straight average by Q^{st}, and the block mean of site functions by Q'. The difference of the first two is a coarse gradient. More explicitly, if $\bar{A}(y,x)$ is the integral along the contour from the centre y to x, then

$$\varphi_A(y) = L^{-4}\sum_{x\in B(y)}\bar{A}(y,x), \qquad QA = Q^{\mathrm{st}}A - d\varphi_A.$$

For $A = d\lambda$ this reads

$$Q^{\mathrm{st}}d\lambda = dQ'\lambda, \qquad Qd\lambda = d(\lambda|_{\mathrm{coarse}}).$$

Stokes' formula on each translated coarse square therefore gives

$$dQ = dQ^{\mathrm{st}} = Q^{\mathrm{pl}}d.$$

The centred blocks ensure that both potential averages send a constant one-cochain to L times that cochain. These identities persist under composition through j levels.

The straight average is onto. Assign $B(c)$ to every fine bond crossing the face of a coarse bond c, and zero to the other bonds. There are L^3 crossing bonds, each appearing in L of the straight paths, so the average is $L^{-4}L^3L\,B(c) = B(c)$. It follows by the coboundary identity that the plaquette average maps exact bare curvatures onto the exact coarse curvatures.

We also need the norm of the plaquette average to be uniform in j. For a coarse plaquette P, write

$$(Q_j^{\mathrm{pl}}F)(P) = L^{-4j}\sum_p m_P(p)F(p).$$

Here $m_P(p) \le L^{2j}$ and $\sum_p m_P(p) = L^{6j}$. Cauchy–Schwarz bounds this square by the sum of $|F(p)|^2$ over its carrier. Each bare plaquette belongs to at most four such carriers, whence $\|Q_j^{\mathrm{pl}}\| \le 2$.

Put

$$\mathcal{Y}_j = \mathcal{E}^2_{\mathrm{ex}} \ominus \ker(Q_j^{\mathrm{pl}}|_{\mathcal{E}^2_{\mathrm{ex}}}), \qquad \mathsf{S}_j = Q_j^{\mathrm{pl}}P_{\mathrm{ex}}(Q_j^{\mathrm{pl}})^*, \qquad \mathfrak{E}_j = P_{\mathrm{ex}}(Q_j^{\mathrm{pl}})^*\mathsf{S}_j^+.$$

Thus $\mathfrak{E}_j h$ is the exact bare curvature of least norm whose plaquette average is h, and

$$\mathcal{H}_j = \mathfrak{E}_j d_{(j)}, \qquad \mathsf{M}_j = \mathcal{H}_j^*\mathcal{H}_j = d^*_{(j)}\mathsf{S}_j^+ d_{(j)} \ge \tfrac{1}{4}d^*_{(j)}d_{(j)}.$$

The lower bound follows from $\|\mathsf{S}_j\| \le 4$ on its range. In particular $\mathcal{H}_j$ vanishes on closed one-cochains.

These definitions agree with minimizing the potential action under either potential average. To see this, suppose $Q_j^{\mathrm{pl}}dA = d_{(j)}B$. The difference $Q_j^{\mathrm{st}}A - B$ is closed. Write it as $d\lambda + \kappa$, where κ is constant. Subtract from A a gradient of a blockwise constant extension of λ, and subtract $L^{-j}\kappa$. This leaves dA unchanged and makes $Q_j^{\mathrm{st}}A = B$. For the contour average, use an extension with prescribed values at the coarse sites instead. Consequently

$$\{dA : Q_jA = B\} = \{dA : Q_j^{\mathrm{st}}A = B\} = \{F' \in \mathcal{E}^2_{\mathrm{ex}} : Q_j^{\mathrm{pl}}F' = d_{(j)}B\}.$$

This identifies $\mathcal{H}_jB$ with the curvature of the linear minimizer in either convention.

Conditioning the potential modulo closed forms is therefore equivalent, for curvature, to conditioning $Q_j^{\mathrm{pl}}F$. The latter determines $P_{\mathcal{Y}_j}F$. In an orthonormal basis adapted to $\mathcal{Y}_j \oplus \mathcal{Y}_j^{\perp}$ the two Gaussian components are independent, so the conditional mean is $P_{\mathcal{Y}_j}F$. The spaces decrease with j. At depth s, with $k = K - s$, this proves

$$C^{\mathrm{bg}}_{K,s} = P_{\mathcal{Y}_k}, \qquad C_K^{\mathrm{free}} - C^{\mathrm{bg}}_{K,s} = \sum_{j=1}^{k}\mathcal{C}_{(j)}, \qquad \mathcal{C}_{(j)} = P_{\mathcal{Y}_{j-1}} - P_{\mathcal{Y}_j}.$$

The slices are independent Gaussian components. Coarse translations permute the averaging blocks and commute with these projections; general bare translations need not do so.

A uniform bound for the linear minimizer. We give the regularity argument for the centred averages. It is the flat-field minimizer method of Bałaban[B84], with the averaging operators retained explicitly. Work in level-j units. The fine mesh is $\eta = \eta_j$, coarse blocks have side one, and fine-lattice norms use quadrature measure η^4. Differences contain η^{-1}, and $\mathcal{L} = \nabla^*\nabla = d^*d + dd^*$ on one-forms. Potential angles are rescaled by η^{-1}. If H^{st} is the resulting minimum-action response, then

$$\mathcal{H}_j(p,b) = \eta_j^2(d\mathsf{H}^{\mathrm{st}}\delta_b)(p).$$

The factor η_j^2 is an area factor, rather than a decay estimate. The remaining bound must be uniform in η.

Start with the scalar inverse

$$G' = (\mathcal{L} + Q'^*Q')^{-1}, \qquad \Sigma = Q'G'^2Q'^*, \qquad R = I - G'Q'^*\Sigma^{-1}Q'G'.$$

The block Poincaré inequality is uniform in the mesh:

$$\|\lambda - Q'^* Q' \lambda\|^2 \le \tfrac{1}{4} \|d\lambda\|^2.$$

Indeed, the first nonzero Neumann eigenvalue of a unit block is $4\eta^{-2} \sin^2(\pi\eta/2) \ge 4$. It follows that $\mathcal{L} + Q'^* Q' \ge c(\mathcal{L} + 1)$.

The coarse matrix Σ has a uniform lower bound as well. Choose on each block a smooth bump e_y of block mean one, zero outside that block, with $\|\mathcal{L} e_y\| \le C$. Products of the centred functions $\cos^2 \pi(x_\mu - y_\mu)$, normalized to mean one, have these properties. Their vanishing to second order at block faces controls the differences across the faces too. For $\varphi_\omega = \sum_y \omega(y) e_y$,

$$\|\omega\|^2 = \langle Q' \varphi_\omega, \omega \rangle = \langle (\mathcal{L} + Q'^* Q') \varphi_\omega, G' Q'^* \omega \rangle \le C \|\omega\| \, \|G' Q'^* \omega\|.$$

Thus $\Sigma \ge cI$. The case $j = 0$ needs no bump construction, since $\mathcal{H}_0 = d$.

The operator $I - R$ is the orthogonal projection onto $\mathrm{Ran}(G' Q'^*)$, and R projects onto $\mathcal{L} \ker Q'$. For $\lambda_0 \in \ker Q'$ the two spaces are orthogonal, since

$$\langle G' Q'^* \omega, \mathcal{L} \lambda_0 \rangle = \langle \omega, Q' G' (\mathcal{L} + Q'^* Q') \lambda_0 \rangle = 0;$$

their dimensions are complementary. This also identifies the condition $R d^* a = 0$ as fixing the gauge modes of zero block mean.

Define

$$\mathsf{K}_v = d^* d + d R d^* + Q^{\mathrm{st}*} Q^{\mathrm{st}}, \qquad \mathsf{G}^{\mathrm{st}} = \mathsf{K}_v^{-1}.$$

We prove

$$\langle a, \mathsf{K}_v a \rangle = \|da\|^2 + \|R d^* a\|^2 + \|Q^{\mathrm{st}} a\|^2 \ge c(\|\nabla a\|^2 + \|a\|^2).$$

The gauge part not seen by the curl needs particular care. Let Υa be the average of a over the fine bonds crossing the face between two neighbouring blocks. The divergence theorem, integration along lines, and the block Poincaré inequality give respectively

$$Q' d^* = d_c^* \Upsilon, \qquad \|(Q^{\mathrm{st}} - \Upsilon) a\| \le C \|\nabla a\|, \qquad \|a\|^2 \le C(\|\nabla a\|^2 + \|Q^{\mathrm{st}} a\|^2).$$

For the middle estimate a value in the straight average is compared with the crossing bond on the same line. Their distance is at most one; Cauchy–Schwarz bounds the difference by the differences along that line. The last estimate first subtracts the mean on each block and then recovers that mean from its straight average.

Choose $\lambda_0 \in \ker Q'$ with $\mathcal{L} \lambda_0 = R d^* a$, and put $a' = a - d\lambda_0$. The block inequality gives

$$\|d\lambda_0\| + \|\nabla d \lambda_0\| \le C \|R d^* a\|.$$

Now $u = d^* a'$ belongs to $\mathrm{Ran}(G' Q'^*)$ and has mean zero. Set $\psi = \mathcal{L}^+ u$ and $a'' = a' - d\psi$. Then $d^* a'' = 0$ and $\|\nabla a''\| = \|da\|$. Write $\psi_c = Q' \psi$. Since $Q^{\mathrm{st}} d \lambda_0 = 0$,

$$Q^{\mathrm{st}} a = Q^{\mathrm{st}} a'' + d_c \psi_c.$$

The flux of a'' is divergence free. Hence

$$\langle Q^{\mathrm{st}} a'', d_c \psi_c \rangle = \langle (Q^{\mathrm{st}} - \Upsilon) a'', d_c \psi_c \rangle, \qquad \|d_c \psi_c\| \le C(\|da\| + \|Q^{\mathrm{st}} a\|).$$

It remains to recover $d\psi$ from these coarse differences. Write $u = G' Q'^* \omega$. The lower bound for Σ implies $\|\omega\| \le C\|u\|$, and

$$\mathcal{L} u = Q'^* \omega_0, \qquad \omega_0 = \omega - Q' u = Q' \mathcal{L} u = d_c^* \Upsilon d u.$$

Consequently

$$\|u\|^2 = \langle \mathcal{L} u, \psi \rangle = \langle \Upsilon d u, d_c \psi_c \rangle.$$

The scalar equation bounds $\|\mathcal{L} u\|$ and $\|du\|$ by $C\|u\|$. The flux estimate then bounds $\|\Upsilon du\|$ by the same quantity, giving $\|u\| \le C \|d_c \psi_c\|$. Also

$$\begin{aligned} \|d\psi\|^2 &= \langle \psi - Q'^* Q' \psi, u \rangle + \langle d_c \psi_c, \Upsilon d \psi \rangle \\ &\le \tfrac{1}{2} \|d\psi\| \, \|u\| + \|d_c \psi_c\| (\|d\psi\| + C\|u\|). \end{aligned}$$

Since $\|\nabla d\psi\| = \|u\|$, this proves $\|d\psi\| + \|\nabla d\psi\| \le C\|d_c\psi_c\|$. The vector block inequality now bounds a''; adding the bounds for all three pieces proves the asserted gap for K_v.

The constrained minimizer has the formula

$$\mathsf{H}^{\mathrm{st}} = \mathsf{G}^{\mathrm{st}} Q^{\mathrm{st}*} (Q^{\mathrm{st}} \mathsf{G}^{\mathrm{st}} Q^{\mathrm{st}*})^{-1}.$$

To check the gauge condition, test $\mathsf{K}_v f = Q^{\mathrm{st}*} v$ against $d\lambda_0$ with $Q'\lambda_0 = 0$. The curl and average terms vanish and the remaining identity gives $Rd^* f = 0$. On the fibre $Q^{\mathrm{st}} a = B$, every potential can be moved to this gauge without changing its curl. Minimizing the form of K_v therefore minimizes $\|da\|^2 + \|B\|^2$ on that fibre. In particular

$$(Q^{\mathrm{st}} \mathsf{G}^{\mathrm{st}} Q^{\mathrm{st}*})^{-1} = I + \mathsf{M}_j.$$

For a uniform upper bound on this inverse, use smooth vector bumps concentrated near the common face in each coarse-bond tube. Their matrix of straight averages has three nonnegative entries in the bond direction, of sum one, with its central entry at least $3/4$. It is thus a uniformly invertible perturbation of its diagonal. Combining the bumps gives a_B with $Q^{\mathrm{st}} a_B = B$ and $\|\nabla a_B\| \le C\|B\|$. The variational formula yields $Q^{\mathrm{st}} \mathsf{G}^{\mathrm{st}} Q^{\mathrm{st}*} \ge cI$.

We next retain exponential localization through these inverses. Conjugate by e^{φ}, where φ has a small fixed Lipschitz constant. For the local gradients and averaging maps, the conjugation error is $O(\|\nabla\varphi\|_\infty)$ between the scales defined by $(\mathcal{L}+1)^{\pm 1/2}$. The lower bounds just proved permit a Neumann series successively for

$$\mathcal{L} + 1, \quad \mathcal{L} + Q'^* Q', \quad \Sigma, \quad \mathsf{K}_v, \quad Q^{\mathrm{st}} \mathsf{G}^{\mathrm{st}} Q^{\mathrm{st}*}.$$

Their constants and an allowed decay rate $\theta_\sharp > 0$ are independent of j, the fine mesh and the torus. For $\varphi(z) = \delta|z - x_b|$ at a smaller rate this gives

$$\|e^{\varphi} v\|_{\ell^2(\mathrm{coarse})} + \|e^{\varphi} f\|_{L^2(\eta^4)} \le C, \quad v = (Q^{\mathrm{st}} \mathsf{G}^{\mathrm{st}} Q^{\mathrm{st}*})^{-1} \delta_b, \quad f = \mathsf{H}^{\mathrm{st}} \delta_b.$$

The improvement to a pointwise bound uses the equation, including the gauge projection. Set $\Xi = Q' G' d^*$. Then

$$d^* d + dRd^* = \mathcal{L} - \Xi^* \Sigma^{-1} \Xi,$$

so that

$$(\mathcal{L}+1) f = h_1 + f, \qquad h_1 = Q^{\mathrm{st}*} v - Q^{\mathrm{st}*} Q^{\mathrm{st}} f + \Xi^* \Sigma^{-1} \Xi f.$$

Every term in h_1 is the image of coarse-lattice data. For the last term the rows of Ξ are du_y, with $u_y = G' Q'^* \delta_y$. These rows are controlled first. The scalar equation is

$$(\mathcal{L}+1) u_y = \mathbf{1}_{B(y)} + u_y - Q'^* Q' u_y.$$

The non-u_y terms on the right are obtained from uniformly bounded, exponentially localized coarse data. The scalar resolvent and its first differences then give

$$|u_y(z)| + |\nabla u_y(z)| \le C e^{-c|z-y|}.$$

Here are the two norm improvements used for this assertion and then for f. If g is the kernel of $(\mathcal{L}+1)^{-1}$, its weighted kernel norms satisfy

$$\|e^{\theta|\cdot|} \mathsf{g}\|_p \le C_p \quad (p < 2), \qquad \|e^{\theta|\cdot|} \nabla \mathsf{g}\|_p \le C_p \quad (p < 4/3).$$

All norms include the quadrature factor η^4. The heat-kernel representation proves their uniformity. The one-dimensional heat kernel is a limit of convolutions of three-point log-concave kernels; Cauchy–Binet preserves log-concavity under convolution. An exponential tilt preserves it too. A summable log-concave sequence is unimodal, so the sum of its absolute first differences is twice its maximum. Together with the heat-kernel supremum bound, this controls the first-difference kernel. Shifting the Fourier contour introduces only $e^{Ct\theta^2}$, absorbed by the resolvent factor e^{-t} for small θ. Interpolation and integration in t give exactly $p < 2$ for the resolvent and $p < 4/3$ for its gradient. Periodization retains the bounds after part of the exponential rate is spent.

Weighted Young inequalities now use

$$(p,q,r)=(8/5,2,8),\qquad (p,q,r)=(8/7,8,\infty).$$

The first application puts the solution in weighted L^8. In the scalar equation the remaining forcing is the sum of this solution and bounded coarse images, so it too lies in weighted L^8. The second application bounds the solution and its first differences. Thus it proves the estimate for u_y. Its gradient bound controls the rows of Ξ; their exponential decay and the coarse inverse bound for Σ^{-1} give the same bounded coarse-image forcing in the equation for f. Applying the two improvements there yields $|f|+|\nabla f|\le Ce^{-\delta|z-x_b|}$. This verifies the forcing estimate needed by the second Young inequality, rather than applying it to an arbitrary L^2 source. Returning to bare plaquette units proves

$$|\mathcal{H}_j(p,b)|\le C\eta_j^2 e^{-\delta_0|x(p)-x_b|/a_{K-j}}. \tag{204}$$

The rate δ_0 can be fixed independently of j and the torus. We will use a smaller rate depending on the fixed block factor L for the constrained slices.

The constrained covariance and the choice of gauge. At step j, write $d'=d_{(j-1)}$. In each block fix the lexicographic tree from its centre, and let

$$\Omega^1_{\mathrm{tr}}=\{B : B=0 \text{ on tree bonds},\ Q^{\mathrm{st}}B=0\}.$$

On this space

$$\|d'B\|^2\ge c_L\|B\|^2,$$

where $c_L>0$ is a negative power of L. Inside a block a bond value is recovered by summing curvature along the strips between the bond and the tree. Starting with the fourth direction, which vanishes on the tree, one recovers the third component along e_4, the second along e_4,e_3, and the first along e_4,e_3,e_2. Each formula uses $O(L)$ plaquettes, and a plaquette is used in at most CL^3 such formulas. On a common face, differences between parallel crossing bonds are given by curvature and the recovered in-block bonds. The equation $Q^{\mathrm{st}}B=0$ determines their mean. Cauchy–Schwarz in these formulas gives the stated inequality.

The map d' is a bijection from Ω^1_{tr} onto the exact curvatures with zero plaquette average. For surjectivity, remove the closed defect of the straight average as above and then set the tree values to zero by a gradient whose block mean is zero. This preserves the average constraint. The Poincaré inequality gives injectivity. For each coarse bond the average constraint can be solved for its single pivot bond. Thus there is a local coordinate map C_{tr} from the remaining bonds onto Ω^1_{tr}, with range $O(L)$ and $\|\mathsf{C}_{\mathrm{tr}}B'\|\ge\|B'\|$.

Let ι be the inclusion of this tree-gauge space. Conditioning a Gaussian on a linear constraint restricts its quadratic form to the kernel of that constraint. The Schur complement therefore gives

$$C_j^{\mathrm{tr}}=\iota(\iota^*\mathsf{M}_{j-1}\iota)^{-1}\iota^*=\mathsf{C}_{\mathrm{tr}}(\mathsf{C}_{\mathrm{tr}}^*\mathsf{M}_{j-1}\mathsf{C}_{\mathrm{tr}})^{-1}\mathsf{C}_{\mathrm{tr}}^*,\qquad \mathcal{C}_{(j)}=\mathcal{H}_{j-1}C_j^{\mathrm{tr}}\mathcal{H}_{j-1}^*.$$

The inverse is uniformly positive at fixed L, since its form is at least $c_L/4$.

The soft-gauge potential covariance need not equal C_j^{tr}. The identity needed here is

$$d'C_j^{\mathrm{soft}}d'^*=d'C_j^{\mathrm{tr}}d'^*.$$

Indeed, both constraints give the same set of curvatures, and on this set the quadratic action is $\langle d'B,\mathsf{S}_{j-1}^{+}d'B\rangle$. The fibre of the contour-average constraint over one curvature is an orbit of the block gauge group. The soft gauge-fixing factor has the same normalized integral on every such orbit: its quadratic part is the square of the contour potential, which changes by translation along the orbit. Integrating the gauge variables leaves the curvature Gaussian unchanged. The hard tree gauge gives that same Gaussian through the preceding bijection. Since $\mathcal{H}_{j-1}$ vanishes on closed forms, this is precisely the equality required in the slice formula.

Decay of a slice and the sum over fine scales. Put $a' = a_{K-j+1}$. From (204),

$$|\mathsf{M}_{j-1}(b,b')| \le \sum_p |\mathcal{H}_{j-1}(p,b)\mathcal{H}_{j-1}(p,b')| \le Ce^{-c|x_b - x_{b'}|/a'}.$$

The factors η_{j-1}^2 in the two kernels give η_{j-1}^4, exactly canceled by the number $O(\eta_{j-1}^{-4})$ of bare plaquettes in one unit volume at this level. The finite range of C_{tr} preserves exponential decay of its compressed form.

For completeness, a positive matrix $A \ge \gamma I$ with $|A(b,b')| \le C_A e^{-2\alpha d(b,b')}$ has an exponentially decaying inverse. Conjugating by $e^{\tau d(\cdot,b_0)}$ changes it by a matrix whose row and column sums are at most

$$C_A \tau \sup_b \sum_{b'} d(b,b') e^{-\alpha d(b,b')}.$$

Choose τ so this is at most $\gamma/2$. A Neumann series then bounds the conjugated inverse by $2/\gamma$, and its b_0-column decays as $e^{-\tau d(b,b_0)}$. Apply this to the compressed tree-gauge form. Composing its inverse with the two minimizer kernels, and spending half the rate on the triangle inequality and half on the two bond sums, gives

$$|\mathcal{C}_{(j)}(p,q)| \le C\eta_{j-1}^4 e^{-\delta|x(p)-x(q)|/a_{K-j+1}}.$$

Here $C, \delta > 0$ may depend on L, but not on j, K or the torus. The constrained inverse can have a slower rate than the unconstrained minimizer.

Let $u = x(p) - x(q)$ be the shortest torus representative and $t = |u|/a_s \ge 1$. With $k = K - s$ and $i = k - j + 1$,

$$a' = a_s L^{-i}, \qquad \eta_{j-1}^4 = L^{-4k} L^{4i}.$$

Consequently

$$|C_K^{\mathrm{free}} - C_{K,s}^{\mathrm{bg}}|(p,q) \le CL^{-4k} \sum_{i \ge 1} L^{4i} e^{-\delta t L^i} \le C(a_K/a_s)^4 e^{-ct}.$$

The sum converges because the exponential in L^i dominates L^{4i}. Beyond a fixed lower bound for t, its consecutive terms decrease geometrically. Factoring out $\sigma_{pq} = a_K^4 \pi^{-2} |u|^{-4}$ gives

$$|C_K^{\mathrm{free}} - C_{K,s}^{\mathrm{bg}}|(p,q) \le Ct^4 e^{-ct} \sigma_{pq}, \qquad t \ge cC_W(s).$$

The reference room is chosen beyond the point at which this last factor decreases.

The continuum tensor and the lattice error. The continuum Green function is $\Phi(u) = (4\pi^2|u|^2)^{-1}$. Applying the two curls to $\delta_{\mu\nu}\Phi$ gives

$$\mathbb{K}(u) = \pi^{-2}|u|^{-4}\Lambda^2(I - 2\hat{u}\hat{u}^{\mathsf{T}}), \qquad \hat{u} = u/|u|.$$

The reflection $I - 2\hat{u}\hat{u}^{\mathsf{T}}$ is an orthogonal involution. Its exterior square is therefore an orthogonal involution on the six-dimensional two-form space, and

$$\operatorname{tr} \mathbb{K}(u)^2 = 6\pi^{-4}|u|^{-8}.$$

Here is the lattice comparison in the form required below. On the unit lattice let p_t be the product heat kernel and let $p_t^{(n)}$ be its periodization on a torus of side n. The mean-zero Green function is

$$\Phi_n = \int_0^\infty (p_t^{(n)} - n^{-4})\, dt.$$

The curvature covariance consists of second differences of Φ_n. In one dimension the Fourier symbol is $\omega_1(\theta) = 2 - 2\cos\theta$, and $\omega_1(\theta) - \theta^2 = O(\theta^4)$. Fourier integration by parts bounds the second differences of the four-dimensional heat kernel, for $t \ge 1$, by

$$Ct^{-3}\langle |z|/\sqrt{t}\rangle^{-N},$$

whereas replacing them by the corresponding continuum derivatives costs at most $Ct^{-7/2}\langle |z|/\sqrt{t}\rangle^{-N}$ for $t \ge 1$. The one-sided difference symbol has a first-order relative error; this is the origin of the

half-power loss in t. For $t \le 1$ the random-walk series gives arbitrary polynomial decay away from the origin.

Split the Green-function integral at n^2. The infinite-lattice heat-kernel error integrates to $C|z|^{-5}$. For the nonzero periodic images for $1 \le t \le n^2$, the bound

$$\sum_{w \in n\mathbb{Z}^4 \setminus \{0\}} |\nabla^2 p_t(z+w)| \le C t^{-3} (\sqrt{t}/n)^6$$

integrates to Cn^{-4}. On $0 < t \le 1$ the random-walk estimate gives an image contribution $O(n^{-N})$ for any fixed N. Above time n^2, the nonzero torus modes have a spectral gap of order n^{-2}, and their second differences are bounded by $Cn^{-6}e^{-ct/n^2}$. Their integral, and the continuum heat tail, are again $O(n^{-4})$. Thus, away from the diagonal,

$$|\nabla^2 \Phi_n(z) - \partial^2 \Phi(z)| \le C(|z|^{-5} + n^{-4}).$$

Moving from the base points to the plaquette barycentres changes the tensor by another $C|z|^{-5}$. With $n = \ell/a_K$ and $2a_K \le |u| \le \ell/4$, rescaling therefore yields

$$|C_K^{\mathrm{free}}(p,q) - a_K^4 \mathbb{K}(u)_{pq}| \le C \left[\frac{a_K}{|u|} + \left(\frac{|u|}{\ell} \right)^4 \right] \sigma_{pq}.$$

For its application to the two supports, the actual distances satisfy $d/2 \le |u| \le C_\vartheta d$, with ϑ the fixed shape parameter. The witness choice gives $d \ge C_W(s) a_s$. Fixing the floor on C_W so that $C_W \ge 4$, and taking d small at fixed g, ϑ, gives

$$2a_K \le \tfrac{1}{2} C_W(s) a_s \le |u| \le C_\vartheta d \le \ell_\star/4.$$

Thus the distances lie in the interval on which the lattice comparison was proved, and $t = |u|/a_s \ge C_W(s)/2$. Enlarge the fixed floor once more so that $t^4 e^{-ct}$ is decreasing on this range. Combining the lattice and fine-slice estimates yields the relative error

$$C \left[\frac{a_K}{d} + \left(\frac{d}{\ell_\star} \right)^4 + C_W(s)^4 e^{-c C_W(s)/2} \right].$$

The constants may depend on the fixed shape. They are independent of the cutoff and of the positions of the plaquettes within the two supports.

The scalar source recursion must still be compared with this free calculation. At one step its two quantities are the covariance of sums of background plaquette energies and the mean of a bilinear form in background curvatures. They must be estimated after removal of the analytic interaction and comparison with the flat background, with errors summable through the steps. Both quantities are required: they generate the cross terms when the total covariance is squared. We isolate their quantitative consequence.

The free Gaussian algebra explains exactly which two terms the nonlinear one-step estimates must reproduce. Let M be a fixed background curvature and ξ a centred Gaussian fluctuation of covariance A. For $Q_p = |M_p + \xi_p|^2/2$, Wick's formula gives

$$\mathbb{E}_\xi Q_p = \tfrac{1}{2}|M_p|^2 + \tfrac{1}{2} \operatorname{tr} A_{pp},$$
$$\operatorname{Cov}_\xi(Q_p, Q_q) = M_p^{\mathsf{T}} A_{pq} M_q + \tfrac{1}{2} \operatorname{tr}(A_{pq} A_{qp}).$$

The first covariance term is the bilinear-curvature quantity; the second is the fluctuation half-square covariance. If the background is itself an independent centred Gaussian of covariance B, the total covariance formula adds the covariance of the conditional means. The result is

$$\tfrac{1}{2} \operatorname{tr}(B_{pq} B_{qp}) + \operatorname{tr}(A_{pq} B_{qp}) + \tfrac{1}{2} \operatorname{tr}(A_{pq} A_{qp}) = \tfrac{1}{2} \operatorname{tr}((A+B)_{pq} (A+B)_{qp}).$$

The middle term cannot be discarded when slices are summed. This calculation evaluates the free recursion; it supplies no bound for replacing a nonlinear fluctuation integral by it. Such a replacement

requires both one-step estimates and their accumulated errors. The resulting quantitative target is the following comparison.

The small parameters available in this last comparison can be made explicit. In the fluctuation integral at a level j, the localized interaction splits into the new interaction and the variation of the earlier activities. Their bounds are

$$|\mathsf{W}_j^{\mathrm{int}}(Y;B)| \le C\delta_j\Big(1+M^{-4}\sum_{b\subset Y}|B(b)|^2\Big)e^{-c\kappa d_j(Y)},$$
$$|\mathsf{W}_j^{\mathrm{old}}(Y;B)| \le C\mathsf{s}_j N_{\omega,j}e^{-c\kappa d_j(Y)}.$$

Here the Gaussian controls the quadratic growth in B, while the ratio $\mathsf{s}_j=\delta_j/\alpha_{0,j}$ controls the old terms. The quantity $N_{\omega,j}$ is their weighted regular and remainder norm from section 2. The tests add a tail of size $e^{-cp_0(g_j)^2}$. Since $x_j\asymp x$ on the segment and $N_{\omega,j}$ stays bounded, the total of these local perturbation amplitudes satisfies

$$\sum_{j=k_0}^{k_1-1}\big(\delta_j+\mathsf{s}_jN_{\omega,j}+e^{-cp_0(g_j)^2}\big)\le C(\log x)\Big[x^{-1/2}(\log x)^{\mathfrak{p}_0}+(\log x)^{\mathfrak{p}_0-q}+e^{-cp_0(x^{-1/2})^2}\Big]=o(1).$$

Thus the short segment leaves room for summable perturbations. The Cauchy–Taylor calculation must give the corresponding estimates with the two energy insertions and with the curvature bilinear insertion, with the coefficients dictated by the source recursion and errors measured against the required two-point scale x^{-2}. At real data the subsequent normalized integrations preserve a supremum bound. To use this in the comparison, the inserted recursion must carry each defect through those integrations with its correct coefficient. The local amplitude bound alone does not give this relative inserted estimate; this is the calculation needed for the comparison below.

The relative comparison on the reference torus. The comparison can be organized by carrying bounded functions through the actual unsourced run. This treats the two insertions together and retains the terms between different Gaussian slices. Let λ_{k_0} be the block-field marginal of $\mathsf{P}^{(k_0)}$, tested on gauge-invariant functions, and write $\phi_1=\tilde f$, $\phi_2=\tilde h$ and $\mathsf{O}_a=X_a$. In the rest of this calculation put

$$\mathsf{g}=x^{-1/2},\qquad p_0=p_0(\mathsf{g}),\qquad \mathsf{r}=\hat d,\qquad \eta_j=L^{-j}.$$

Thus g is the observation-scale coupling; the reference coupling g remains fixed. In particular $\mathsf{r}\asymp p_0^2$, $\mathsf{r}\ge C\check R_{k_0}$ and $\sup|\mathsf{O}_a|\le Cx^{-1}\mathsf{r}^4p_0^2\|\phi_a\|_\infty$. The comparison will be uniform in the first-passage data, the position of the pair and the fixed shape class.

The positive coupling of the run. Freeze every test, profile, reference law and gauge choice at its unsourced value. The operations in (76) then carry $G\rho_{k_0}$ as well as ρ_{k_0} for every bounded gauge-invariant Borel function G. A test selects one share of a decomposition of unity; integration, a coordinate change and a reference draw are positive kernels; merging labels adds their measures. A gauge-fixing move can be realized by drawing a gauge transformation from its normalized orbit law. This realization also preserves positivity and mass and leaves gauge-invariant insertions unchanged.

Retain the intermediate stored variables. For a term h of level j, let $\hat\tau_h$ be its positive unsourced measure and $\lambda_h[G]$ the measure carrying G. Positivity gives

$$|\lambda_h[G]|\le\|G\|_\infty\hat\tau_h,\qquad G^{[j]}:=\frac{d\lambda_h[G]}{d\hat\tau_h},\qquad |G^{[j]}|\le\|G\|_\infty.$$

If $\mathfrak{Q}_j$ is the positive map of a step, define

$$\mathbb{E}_jF=\frac{d\mathfrak{Q}_j(F\hat\tau^{(j-1)})}{d\hat\tau^{(j)}}.$$

The input here may be a disjoint union of terms that are subsequently merged. In either case $\mathbb{E}_j$ is positive, $\mathbb{E}_j 1 = 1$, and it contracts the supremum norm. The chain rule for Radon–Nikodym derivatives gives $G^{[j]} = \mathbb{E}_j G^{[j-1]}$.

At level k_1 condition these measures on the terminal class and its block field W. Classes retain the integral representation and the exponent used below. If the partition into classes is refined to record cleanness, that representation must continue to hold on every resulting class; measurability of cleanness alone does not imply this. The restriction of the positive bracket below supplies the required representation. This gives a probability kernel Π and a law $\mathbb{Q}$ with

$$\int G \, d\lambda_{k_0} = \int (\Pi G)(\mathsf{h}, W) \, d\mathbb{Q}(\mathsf{h}, W). \tag{205}$$

This is a coupling defined by the run. Where a large-field operation has drawn new variables, W need not equal $\mathcal{M}_{k_0 \to k_1} V$. No identification with conditioning on that deterministic average is used below.

Clean classes and the local law to verify. Around the pair choose, at each level $j-1$, a read set $\mathcal{S}_j$ and a larger box $\mathcal{B}_j$ whose distance from the read set is at least D spacings. Include the localization cubes of the bare plaquette functions in the read set at the first step. The straightening of the constraint and the elimination of pivot bonds have finite range, so a local insertion reads only $y|_{\mathcal{S}_j}$ after that change of variables. The number of free bonds in the box satisfies

$$n_j := \#\mathcal{B}_j \le C\big(\mathsf{r} L^{-(j-1-k_0)} + \mathsf{D}\big)^4. \tag{206}$$

Take D of order r. For d sufficiently small at fixed g, all the boxes and their required collars embed in the reference torus: their largest physical radius is at most $C(\mathsf{r} + \mathsf{D} + \check{R}_{k_0}) a_{s-\mathsf{n}}$, which tends to zero with d. All subsequent local Gaussian comparisons use this nonwrapping range. Declaring every history bad when an exceptional-mass upper bound exceeds one gives only a vacuous bound; it supplies no Gaussian identity on a wrapping box. The exact measure decomposition is kept throughout.

A clean history has no live region meeting these boxes, no creation or large-field operation there along the segment, and no such contact in the preceding window of $C\check{R}_{k_0}$ levels. The cell bounds in section 3 give the estimate

$$\epsilon_{\mathrm{uc}} \le C L^{4\mathsf{n}} e^{C\check{R}_{k_0}} (1+\mathsf{n})(\mathsf{r} + \check{R}_{k_0})^4 e^{-cp_0}. \tag{207}$$

Indeed, count cells at their own levels before applying their large-field probabilities. The $C\check{R}_{k_0}$ earlier levels cost $e^{C\check{R}_{k_0}}$. Extending this exclusion through all earlier levels would produce a divergent cell count. Regions healed farther below the segment must remain summed in the local remainders.

Restricting the positive bracket. The clean restriction can be made before the healed components are exponentiated. Use a fixed buffered neighbourhood of the pair containing every read box. At the levels of the clean window, except its lowest level, exclude each first-class component whose collar meets that neighbourhood. Also require the retained live regions to avoid it. This is a slightly smaller clean set than the one obtained by inspecting the live regions alone. Its excluded components have a live predecessor within the buffer enlarged by $C\check{R}_t$; the preceding cell count therefore still gives (207). No condition is imposed on components healed at the lowest level of the window or earlier.

Fix a level t, an output standing part o, and all its retained variables. Write $\mathfrak{e}$ for the common outer datum. The standing-part identity before exponentiation has the form

$$\sum_{(h')^\flat = o} m_{h'} = (\varsigma_o)_* \big(F_o(\mathfrak{e}) K_t(\mathfrak{e}) \mu_o(d\mathfrak{e})\big), \tag{208}$$

where $F_o > 0$, μ_o is positive, and

$$K_t(\mathfrak{e}) = \sum_{\mathfrak{X}\in\mathfrak{F}(o)} \int \prod_{X\in\mathfrak{X}} \nu_X(\mathfrak{e}|_{X^\sim}; d\xi_X)\, \exp\sum_Y V_{\mathfrak{X}}(Y;\mathfrak{e},\xi).$$

The measures ν_X, at the fixed retained datum, include the sum over the histories inside X and the normalized frame sum. This is the first, unnormalized representation of a healed component; there is no division by its mass. The map ς_o retains every variable on which a later move depends.

Two facts about this identity are used. Every string producing o occurs exactly once in the displayed family sum, and the formula holds under the retained-variable integral, before taking its image. Consequently the restriction excluding the forbidden collars is obtained by deleting those families from this same bracket. Every remaining summand has its original measure and its original $V_{\mathfrak{X}}$. Denote the restricted bracket by K_t^{cl}. Equivalently, give every forbidden component a factor z in the finite family sum. The constant coefficient of this polynomial is K_t^{cl}. Coefficient selection is made in (208), so the selected identity still holds against every bounded Borel function of the retained variables. No RG step is applied to a newly restricted exponent.

Take out the contribution of the empty family, using the same regrouping in both brackets:

$$K_t = e^{\sum_Y V_\emptyset(Y)} B_t, \qquad K_t^{\mathrm{cl}} = e^{\sum_Y V_\emptyset(Y)} B_t^{\mathrm{cl}}.$$

Here

$$B_t = \sum_{\mathfrak{X}\in\mathfrak{F}(o)} \int \prod_{X\in\mathfrak{X}} \nu_X(\mathfrak{e}|_{X^\sim}; d\xi_X)\, e^{\sum_Y W_{\mathfrak{X}}(Y;\mathfrak{e},\xi)}, \qquad W_\emptyset = 0.$$

The domains and the functions $W_{\mathfrak{X}}$ are unchanged on every retained family. Both brackets contain the positive empty term one, and all their summands are nonnegative at real data. Thus $1 \le B_t^{\mathrm{cl}} \le B_t$. In particular their logarithms are real and well defined. The possibly extensive empty-family exponential cancels in their quotient.

The abstract bracket expansion applies to both B_t and B_t^{cl} with the same majorant. Indeed, deletion is implemented by replacing the component measure by zero on the forbidden components. It changes neither compatibility nor the consistency identities of the remaining interactions, and it decreases every absolute component-mass bound. Write their localized logarithms as

$$\log B_t = \sum_Y L_t(Y), \qquad \log B_t^{\mathrm{cl}} = \sum_Y L_t^{\mathrm{cl}}(Y).$$

The local exponentiation estimate (150) in section 3.11 gives, for some fixed $c_* > 0$,

$$|L_t(Y)| + |L_t^{\mathrm{cl}}(Y)| \le C e^{-(1+c_*)p_0(g_t)} e^{-\kappa_* d_t^\natural(o;Y)}, \qquad \kappa_* > c_{\mathrm{E}}. \tag{209}$$

It also gives locality of each term. One can see the latter directly from the finite-subset formula for the exponential of $L_t(Y)$: it uses only brackets restricted to subregions of Y. Therefore

$$L_t(Y) = L_t^{\mathrm{cl}}(Y) \quad \text{unless } Y \text{ contains a deleted collared component.}$$

The logarithm branches agree since they are obtained from the same absolutely convergent connected expansion.

Let A_t be the set of level-t cubes at which a collar can witness intersection with the forbidden neighbourhood, including the fixed buffer. Put $c_t = \#A_t$. A deleted collar lies in the output small-field region and meets none of its fixed exempt hulls. Hence any cube witnessing its intersection with A_t is free. It is not necessary that the distinguished corner or the lowest cube of the component be near the pair. For every changed $L_t(Y)$ choose such a witnessing cube $q \in Y \cap A_t$. The rooted entropy bound for the hull length is

$$\sum_{Y\ni q} e^{-\kappa_* d_t^\natural(o;Y)} \le C \quad \text{for every free cube } q \text{ and } \kappa_* > c_{\mathrm{E}}.$$

It is uniform in the sizes of the fixed exempt hulls. Together with (209) it gives

$$0 \le \Delta_t(\mathfrak{e}) := \log \frac{K_t(\mathfrak{e})}{K_t^{\mathrm{cl}}(\mathfrak{e})} = \sum_Y (L_t(Y) - L_t^{\mathrm{cl}}(Y))$$
$$\le C e^{-(1+c_*)p_0(g_t)} \sum_{\substack{q \in A_t \\ q \text{ free}}} \sum_{Y \ni q} e^{-\kappa_* d_t^\natural(o;Y)} \le C c_t e^{-p_0(g_t)}. \tag{210}$$

This is a pointwise estimate at the common retained datum. The last exponent is weakened to avoid choosing a particular value of the spare constant c_*. The difference is rooted at forbidden components before summation; subtracting two extensive bounds on the full logarithms would not give this conclusion.

The clean density at every prefix. Let j_0 be the lowest level of the clean window. At j_0 the retained standing part decides whether its live region is clean; all its earlier healed histories remain summed, so its clean density is one whenever that standing part is admitted. For $t > j_0$, write $a_t = C c_t e^{-p_0(g_t)}$. At a clean output standing part, deleting the forbidden components in the one-step bracket also excludes the unclean predecessor standing parts. An offending retained region can disappear from the outer data only inside a component healed at this step. If it meets the excluded neighbourhood, that component is among the deleted ones.

Suppose inductively that, on each admitted predecessor retained space, the prefix-clean measure lies between $e^{-e_{t-1}}$ and one times the original resummed measure. Apply the original positive step kernels to these inequalities, restricting to the current clean strings. Their image from the original resummed inputs is represented by K_t^{cl}; by (210) it lies between e^{-a_t} and one times the full output term. The prefix-clean output therefore lies between $e^{-(e_{t-1}+a_t)}$ and one times that term. The argument holds before every subsequent marginalization and with merges interpreted on the disjoint union of predecessor spaces. Taking images preserves both inequalities.

More explicitly, let $\delta_t \in [0, a_t]$ be the negative logarithm of the density of the one-step-restricted output relative to the full output, after taking the common retained-variable image. If $\mathbb{E}_t^{\mathrm{ref}}$ is the backward kernel of that restricted step with its original resummed inputs, then its prefix-clean density has the exact recursion

$$D_t = \delta_t - \log \mathbb{E}_t^{\mathrm{ref}}(e^{-D_{t-1}}), \qquad 0 \le D_t \le a_t + e_{t-1}.$$

At a merge, D_{t-1} is read on the disjoint union of predecessor spaces. Thus the recursion does not identify two different conditional laws or discard any stored variable.

Radon–Nikodym derivatives consequently give, for every clean standing part h of level j, on the same retained-variable space,

$$d\hat{\tau}_h^{\mathrm{cl}} = e^{-D_h}\, d\hat{\tau}_{[h]}, \qquad 0 \le D_h \le e_j, \qquad e_j = \sum_{j_0 < t \le j} a_t \le C \sum_{j_0 < t \le j} c_t e^{-p_0(g_t)}. \tag{211}$$

One may pull this bounded Borel derivative back through a reference representation $e^{A_{[h]}}\mu_{[h]}$. Only its bounded Borel values will be used in the conditional kernels. The constants are independent of the retained variables, the torus, the cutoff and the standing part.

The cube count is

$$c_t \le C(\mathsf{r} + \check{R}_{k_0})^4 L^{4(k_1 - t)} \quad (j_0 < t \le k_1), \qquad c_t = 0 \quad (t \le j_0).$$

The flow gives $p_0(g_t) \ge c p_0$ in this finite window. Since $k_0 - j_0 \le C\check{R}_{k_0}$, geometric summation yields

$$\mathsf{n} e_{k_1} \le C L^{4\mathsf{n}} e^{C\check{R}_{k_0}} (1 + \mathsf{n})(\mathsf{r} + \check{R}_{k_0})^4 e^{-c p_0}.$$

Thus its cost is already included in (207). The normalized clean kernels have the form

$$\mathbb{E}_j F = \mathbb{E}_j^{\mathrm{ref}}(F w_j), \qquad e^{-e_{j-1}} \le w_j \le e^{e_{j-1}}, \qquad \mathbb{E}_j^{\mathrm{ref}} w_j = 1, \tag{212}$$

as follows by normalizing the two common retained-variable measures.

Normalising the clean kernels. The pointwise clean-class representation is used before taking any conditional law. Keep the same variables on both sides of (211), and at level $j-1$ write

$$d\mu^{\mathrm{cl}}_{j-1} = r_{j-1}\, d\bar{\mu}_{j-1}, \qquad r_{j-1} = e^{-D_{j-1}}, \qquad e^{-e_{j-1}} \le r_{j-1} \le 1.$$

The predecessor space is a disjoint union when several standing parts merge. Let $\mathfrak{Q}^{\mathrm{cl}}_j$ be the original positive kernel of step j, restricted to the strings that stay clean. Form the reference joint measure

$$dJ^{\mathrm{ref}}_j(y,z) = d\bar{\mu}_{j-1}(y)\mathfrak{Q}^{\mathrm{cl}}_j(y,dz).$$

It uses the original resummed input, its original counterterms and the unchanged kernels of the run. The corresponding clean joint measure is $r_{j-1}(y)dJ^{\mathrm{ref}}_j(y,z)$. Disintegrate the first joint measure over its output z, and call its backward probability kernel $Q^{\mathrm{ref}}_j(z,dy)$. The clean backward kernel is then

$$Q_j(z,dy) = \frac{r_{j-1}(y)}{Q^{\mathrm{ref}}_j r_{j-1}(z)} Q^{\mathrm{ref}}_j(z,dy).$$

Its denominator lies in $[e^{-e_{j-1}},1]$. This proves (212) with $w_j = r_{j-1}/Q^{\mathrm{ref}}_j r_{j-1}$. Positivity of the lower bound in (211) also makes the clean and resummed measures equivalent on each retained space. The backward kernels can therefore be used on common sets of full measure when they are composed. Merging predecessor terms changes the disintegration, but none of these bounds.

In particular $Q_j \ge e^{-e_{j-1}} Q^{\mathrm{ref}}_j$ as kernels of positive measures. Composing the finitely many backward kernels gives the lower bound with $P_0 = \exp(-\sum_{k_0<j\le k_1} e_{j-1})$. Keep the actual final disintegration $\mathbb{E}^W$ of the clean class over W. This positive normalized mean preserves the inequality. It is not replaced by a separately normalized reference terminal law. Consequently

$$\Pi = P_0 \Pi^{\mathrm{ref}} + (1-P_0)\Pi^{\mathrm{rest}},$$

where Π^{rest} is a probability kernel when $P_0 < 1$; when $P_0 = 1$ the two kernels agree. The reference kernel is the composition of the Q^{ref}_j, followed by that same final conditional mean. A coin with probability P_0, independent of the output class and W, realizes this decomposition. The total exceptional mass is at most the unclean mass plus $1-P_0 \le \sum_{k_0<j\le k_1} e_{j-1}$. Only bounded Borel values of D_h are used. No derivative or analyticity of this density is required.

The resulting clean measure $\mathbb{Q}_{\mathrm{cl}}$, kernel Π_{cl} and exceptional measure λ_{bad} satisfy

$$\int G\, d\lambda_{k_0} = \int \Pi_{\mathrm{cl}} G(W)\, d\mathbb{Q}_{\mathrm{cl}}(W) + \int G\, d\lambda_{\mathrm{bad}}, \qquad \lambda_{\mathrm{bad}}(1) \le C\epsilon_{\mathrm{uc}}. \tag{213}$$

The cost $\sum_{k_0<j\le k_1} e_{j-1}$ is absorbed in (207), with its unspecified positive constant c decreased if necessary. The kernel on the clean part uses $\mathbb{E}^{\mathrm{ref}}_j$ only. In the following calculation $\mathbb{E}_j$ denotes these reference kernels. Write $F^{[j]}_{\mathrm{ref}} = \mathbb{E}_j \cdots \mathbb{E}_{k_0+1} F$ for their successive transport; this differs from the original transport $F^{[j]}$ before the exceptional part has been removed.

The small-field integral on a read set. The reference kernel in (212) has the following small-field form on a clean box. Retain a bounded gauge-invariant factor F while performing the changes of variables. Criticality removes the linear term of the action; the nonlinear straightening changes only the pivots. For a small-field integral its normalized form is

$$\frac{\displaystyle\int_{\mathcal{W}_j} F\big(\mathfrak{v}(y;V_j)\big) e^{P_j(y;V_j)}\, d\gamma_{C^{(j-1)}(V_j)}(y)}{\displaystyle\int_{\mathcal{W}_j} e^{P_j(y;V_j)}\, d\gamma_{C^{(j-1)}(V_j)}(y)}, \quad \mathcal{W}_j = \prod_{b\ \text{free}} \{|y_b| < \mathsf{b}_j\}, \tag{214}$$

where $\mathsf{b}_j \asymp p_0$ and

$$\mathfrak{v}(y;V_j) = \exp\Big(i\Big[g_{j-1}\mathsf{C}y - \widetilde{\mathsf{D}}(g_{j-1}\mathsf{C}y)\Big]\Big)\, V^{\mathrm{bg}}_{j-1}(V_j).$$

The function $\widetilde{\mathsf{D}}$ begins at second order. The product runs over the free bonds of that small-field integral. The covariance is that of the actual step, before comparison with its flat value. The formula retains F in the changes of variables of Bałaban[B87, Section 2, equations (2.1)–(2.16)].

The window and the gauge fixing. The reference kernel is formed from a resummed standing part, rather than from a single inner history or from a new step applied to the exponent $A_{[h]} - D_h$. The standing-part identity keeps all variables on which a later move depends. Its strings differ in the components healed at this step. On a clean class these components lie outside the excluded neighbourhood. The box around the read set is therefore inside the ordinary small-field integration region.

The strip substitution is performed before the inner variables are conditioned. On the collar and rim it leaves the variables unchanged. In its cutoff zone the substitution switches strictly between the thresholds δ_{j-1} and $2\delta_{j-1}$. Where it has not yet reached the localized background, the approximate test at $2\delta_{j-1}$ equals one for either background. The kept test factors adjacent to the integration region equal one on the carried support and are removed there; after pivot elimination the other kept tests depend on the new field only outside that region. Each pivot value is determined from its own averaging constraint, rather than from the global averaged background. The nonlinear straightening changes only the pivot variables. The remaining restriction on each free inner bond is precisely $|y_b| < \mathsf{b}_j$. Freezing all exterior variables thus leaves the product window in (214); it does not introduce a further history-dependent inner indicator.

Condition first on the recorded frame as well. Every operation outside the box is positive and reads only its retained variables. Disintegrating these variables therefore gives a positive measure $m(d\mathsf{e})$ of exterior data. If $Z_{\mathsf{e}} > 0$ is the inner normalising integral, a normalized expectation has the form

$$\frac{\int m(d\mathsf{e}) Z_{\mathsf{e}} E_{\mathsf{e}}[F \circ \mathfrak{v}_{\mathsf{e}}]}{\int m(d\mathsf{e}) Z_{\mathsf{e}}}.$$

It is a mixture of the normalized local laws, with their partition functions included in the mixing weights. The same argument applies to merges and to the final conditioning on W. One may then sum over frames. Replacing their normalization factors by those of a single frame would change the measure.

A gauge-fixing move is realized as a probability kernel on the gauge orbit. It preserves every gauge-invariant inserted function. For such a function the invariant measure identity permits the gauge factor to be written in the small-field integral. Under the joint gauge action on the background and fluctuation coordinates, each free-bond coordinate is rotated by an adjoint action on $\mathfrak{su}(2)$, which preserves its Euclidean norm. The balls in the product window and the Haar changes of variables are therefore unchanged. Pivot coordinates remain determined by their covariant averaging constraints.

The main terms in the carried expansion, including the contracted one-line and two-line terms, are gauge-invariant functions of the current field. Apply the displayed integral formula to these terms. An arbitrary remainder is carried only by its positive normalized kernel, which contracts its supremum norm. There is no need to identify a gauge-fixing kernel with multiplication by a gauge factor on a non-invariant remainder.

Finally, use the convention which redraws every healed coupling profile and compensates its change in the Wilson counterterm at each plaquette. The total bare Wilson coefficient is unchanged. On the clean box the redrawn profiles all equal one, so the local coefficient is the running x_{j-1}. Healed components below the finite exclusion window remain summed in the original deterministic $R^{(i)}$ families. No condition excluding all earlier creations, and no conditioning on an unreconstructed deficit history, is used.

The exterior of $\mathcal{B}_j$ has to be removed from (214). Freezing it gives a mixture of laws on the product window with energies

$$H_{\mathsf{e}}(y) = \tfrac{1}{2}\langle y, Q^{\mathsf{e}}_{II} y\rangle + \langle Q^{\mathsf{e}}_{IO} y_O(\mathsf{e}), y\rangle - P_{\mathsf{e}}(y), \qquad I = \mathcal{B}_j.$$

The reference Gaussian in the box is the marginal of the full reference Gaussian. Its precision is the Schur complement $Q_{II} - Q_{IO}Q_{OO}^{-1}Q_{OI}$, rather than Q_{II}. The RG gap, exponential decay and the small Hessian of P_j make this reference energy uniformly convex after conjugation by a small exponential weight. The difference from H_{e} has gradient bounded at a bond b by $C\mathsf{b}_j e^{-c\,\mathrm{dist}(b,O)}$.

Here is the weighted comparison used for this step. Let H_0 be the reference energy on a product of convex bond windows, let H_1 be the energy with exterior datum fixed, and put $T_\varphi z = (e^{\varphi(b)} z_b)_b$. The lower gap and the small interaction Hessian must give, for every sufficiently slowly varying φ, the uniform bound

$$\langle z, T_\varphi H_0'' T_\varphi^{-1} z\rangle \geq \lambda |z|^2, \qquad \lambda > 0.$$

Write $\delta_b = \sup|\partial_b(H_1 - H_0)|$ on the window. First extend the energies off the product window. Retract each bond coordinate smoothly into a compact interior set, identically on a slightly smaller window, and compose the interaction with this retraction. Choose common smooth convex walls whose positive block-diagonal Hessians outside that smaller window dominate the extra Hessian terms of the retraction. The quadratic Gaussian part is retained. The extended energies are smooth and coercive; their force difference has at most linear growth, and the common walls cancel from it. Their block-diagonal Hessians commute with T_φ. The wall limit is taken first at a fixed smaller window and only then as that window increases to the original one.

Couple the Hamiltonian flows of H_0, H_1 with common momenta and set $v = T_\varphi(q^1 - q^0)$. After smooth convex walls have been introduced at each bond, their Hessians are bounded by a finite number Λ. For a sufficiently short time τ, the weighted coercivity and the integral form of the flow equation give

$$\mathbb{E}|v(\tau)| \leq (1 - \tfrac{1}{2}\lambda\tau^2 + C\Lambda^2\tau^4)\mathbb{E}|v(0)| + \tfrac{1}{2}\tau^2(1 + C\Lambda\tau^2)\nu_{H_1}|T_\varphi \nabla(H_1 - H_0)|.$$

Refresh the common Gaussian momentum after each interval. The two position marginals retain their Gibbs laws, since each Hamiltonian flow preserves its Gibbs-position and Gaussian-momentum product. Iteration and then $\tau \downarrow 0$ bound a Lipschitz expectation difference by ℓ_G/λ times the expected weighted force. The at-most-linear growth and the common coercive walls permit dominated convergence in this expectation as the walls increase. On the resulting hard window it is bounded by the supremum force components δ_b. Thus

$$|\nu_{H_1} G - \nu_{H_0} G| \leq \frac{\ell_G}{\lambda}\Big(\sum_b e^{2\varphi(b)}\delta_b^2\Big)^{1/2}, \quad |G(y) - G(y')| \leq \ell_G |T_\varphi(y - y')|.$$

The weighted lower bound is retained in these limits, since the walls act separately on the bond coordinates.

The product shape is also needed to differentiate a marginal density without differentiating its integration domain. If m_a is the read-set marginal and ν_a^u the conditional law of the other bonds at read-set value u, then

$$\partial_b \log(m_1/m_0)(u) = -\nu_1^u \partial_b(H_1 - H_0) - (\nu_1^u - \nu_0^u)\partial_b H_0.$$

Apply the preceding expectation estimate to the last difference, using $\varphi = -\mu\,\mathrm{dist}(\cdot, b)$. The weighted Lipschitz norm of $\partial_b H_0$ is bounded by the exponentially decaying mixed Hessian. The weighted force norm is at most $C\mathsf{b}_j e^{-c\mathsf{D}}$ when b belongs to the read set. Integrating this derivative along segments of the read-set window and using that both densities have integral one gives the density ratio in $[e^{-\Xi_j}, e^{\Xi_j}]$, with

$$\Xi_j \leq C n_j p_0^2 e^{-c\mathsf{D}}. \tag{215}$$

For the polymer part of the force, a domain reaching the exterior spends part of its exponential decay on the distance to that exterior. A domain lying inside the box instead uses the locality of the background entering its activity. In both cases the decay left over sums the differentiated domains. The same localized analytic expansion is needed for the Hessian bound in the convex comparison. A small supremum for the whole exponent would not give either spatial estimate by itself.

The exponent on the box. On the scaled fluctuation window write the exponent as the sum of the new interaction, the increments of the regular activities and the increments of the regenerated activities. Boundary activities are anchored outside the excluded neighbourhood and contribute through the exterior-force estimate. We use the redrawn-profile convention already fixed above, so that no coupling deficit remains inside a clean box.

After their critical value and Hessian have been extracted, the new action and gauge-fixing remainders start at third order in the unscaled displacement $g_{j-1}\mathsf{C}y$. Their action coefficient is $x_{j-1} = g_{j-1}^{-2}$. Thus on $|y_b| < \mathsf{b}_j \asymp p_0$ they contribute at most $C\mathsf{g}p_0^3$ per current unit volume. The Haar and change-of-variable factors have an explicit factor of g_{j-1} and obey the same coarser bound. More precisely, their odd and even parts have sizes $C\mathsf{g}p_0^3$ and $C\mathsf{g}^2p_0^4$, respectively. The odd part is harmless for the bounded-function comparison below; the finer parity estimates are not needed there.

For an earlier regular activity at level i, the displacement lies in a complex ball of radius $\alpha_{\cdot,i} \asymp g_i(\log x_i)^q$. Its mth Taylor term is therefore bounded by its activity norm times

$$\left(\frac{Cp_0}{(\log x)^q}\right)^m.$$

There is no additional positive power of g in this quotient. At a flat field the first derivative vanishes: constant gauge transformations fix the field and act by the adjoint representation on each direction, and $\mathfrak{su}(2)$ has no fixed vector. Moving the derivative to a regular nearby background costs one additional background-curvature-to-radius factor. Consequently both the first-order contribution and the quadratic contribution are bounded by $C(\log x)^{2(\mathfrak{p}_0-q)}$ per current unit volume.

This estimate must be summed after the two subtractions in (56). At age $j-1-i$ there are $L^{4(j-1-i)}$ old domains per current unit volume. The two curvature factors each supply $L^{-2(j-1-i)}$, which compensates this count exactly and leaves no decay in the age. The additional summable weight comes from subtracting the action coefficient at the pointed tents before forming the localized complex increment. The tents add to one in the interior of the box. The resulting irrelevant remainder has the weighted bound already obtained for the old interaction; its geometric age factor absorbs the polynomial change of the radii with age. This is a bound for the localized increment after cancellation of the pointed totals, rather than an assertion that each unsubtracted old term has the gain separately.

For $R^{(i)}$, the flat value is subtracted but there is no coupling counterterm. Two Cauchy derivatives cost at most Cx_i, and the two field factors contribute at most Cp_0^2. After the same cancellation of domain counts and curvature scales, the remaining sum is

$$Cp_0^2 \sum_{i<j} x_i^{1-\kappa_0/2} \le Cp_0^2 x^{2-\kappa_0/2}.$$

For the last inequality use the lower flow bound $x_i \ge x_{j-1} + c(j-1-i) - C$ and compare the sum with $\int_0^\infty (x+ct)^{1-\kappa_0/2}\,dt$. It converges precisely in the range used here, $\kappa_0 > 4$. No summable geometric age factor for these terms has been assumed.

Adding these bounds over the n_j free coordinates in the box gives

$$\theta_j := \sup|P_j| \le Cn_j\left[\mathsf{g}p_0^3 + (\log x)^{2(\mathfrak{p}_0-q)} + p_0^2 x^{2-\kappa_0/2}\right]. \tag{216}$$

The analyticity margin must be retained while obtaining the Hessian. The new interaction is localized in domains Y and is analytic on a fluctuation ball of radius $2\mathsf{b}_j$; two Cauchy derivatives give $C\mathsf{g}p_0$.

The localized old increments are analytic on radius $\alpha_{\cdot,i}/(Cg_{j-1})$, larger than the window. Two derivatives give $C(\log x)^{-2q}$ times their summable norm; the R sum is smaller at $\kappa_0 > 4$. Summing only domains containing both differentiated bonds retains exponential decay in their separation. Thus the mixed-Hessian row bound needed in the convex comparison is small, uniformly in the box size. This uses the complex increment estimate before restriction to real fields; differentiating a bound stated only at real backgrounds would not justify it.

Five comparisons for the read-set law. Let $I = \mathcal{B}_j$ be the box and let $S = \mathcal{S}_j \subset I$ be its read set. The Gaussian in the box comparison is the marginal of the full Gaussian on I. Its precision is the Schur complement already computed above. Let γ_j^0 be the flat Gaussian marginal on the free coordinates of S, with the same pivot convention, and put

$$\mathcal{W}_j^{\mathrm{read}} = \prod_{b \in S} \{|y_b| < \mathsf{b}_j\}.$$

The law γ_j^0 has one identical spatial covariance in each colour. Its upper and lower covariance bounds are uniform in j and the volume, by the compressed quadratic-form bounds.

Let Ψ be bounded and Borel on this read-set window. Extend it to the other read-set values by zero when comparing unrestricted Gaussian laws. First the density estimate (215) removes the exterior data at cost at most $(e^{\Xi_j} - 1)\|\Psi\|_\infty$. The law left on I is the box Gaussian restricted to its product window and tilted by P_j.

Second remove that tilt. For real data, $|P_j| \le \theta_j$ gives

$$\left| \frac{e^{P_j}}{\langle e^{P_j} \rangle_0} - 1 \right| \le e^{2\theta_j} - 1 \le C\theta_j,$$

where $\langle \cdot \rangle_0$ is the windowed Gaussian expectation and θ_j is small. The change of a bounded expectation is therefore at most $C\theta_j \|\Psi\|_\infty$.

Third remove the window on the entire box. Each free coordinate has bounded Gaussian variance, so a union bound gives

$$t_j \le C n_j e^{-c p_0^2}$$

for its complement. Replacing conditioning on an event of probability at least $1 - t_j$ by the unconditioned law changes a bounded expectation by at most $2t_j\|\Psi\|_\infty$.

Fourth rotate the fluctuation into the axial frame and replace its Gaussian covariance on S by the flat one. Locality and analytic dependence of the covariance, applied to the small axial potential, give the entrywise error

$$\varpi_j \le C[\mathsf{rg} p_0 + e^{-c\mathsf{D}}].$$

Here the first term is the size of the potential in that frame; the second is the error of localization. The resolvent identity is used on the box with its retained regularity margin. In dimension $m = 3\#S$, the Hilbert–Schmidt error is at most $Cm\varpi_j$. To see why it controls a bounded expectation, normalize by the flat covariance and diagonalize the resulting relative covariance. For eigenvalues close to one, comparing the finite Gaussian densities gives total variation bounded by the Euclidean norm of their deviations. If that norm is bounded below by a fixed constant, the trivial total variation bound gives the same inequality after its constant is enlarged. The uniform lower covariance bound therefore gives the cost $Cn_j\varpi_j\|\Psi\|_\infty$.

Finally restore the read-set window in the flat Gaussian. Its complement has the same Gaussian tail bound, and the cost is again at most $Ct_j\|\Psi\|_\infty$. The five comparisons prove

$$\left| \mathbb{E}_j \Psi(\widetilde{y}_j) - \gamma_j^0(\Psi \mid \mathcal{W}_j^{\mathrm{read}}) \right| \le T_j \|\Psi\|_\infty, \quad T_j \le C[(e^{\Xi_j} - 1) + \theta_j + t_j + n_j\varpi_j]. \tag{217}$$

The constants depend on the fixed geometric data and the fixed choice of radii, not on the cutoff, the class or the exterior datum. Positivity of the exterior and frame mixtures preserves all five bounds. This comparison applies to an arbitrary bounded read-set function; it uses the localized potential and the exact window as well as Gaussian covariance decay.

Lowering a plaquette and telescoping the generating function. For a bare plaquette p, let $P_p^{(j)}(A)$ be its holonomy minus one in the Landau representative of the localized background $U_{j,\square}(e^{iA})$. The input potential is given on the level-j bonds of a cube about the pair. The cube has the fixed localization margin, and the next read set includes all the bonds on which this background depends. At $j = k_0$ this includes the finitely many test cubes used in its definition.

The continuation from the trivial datum is analytic on a fixed ball $\|A\| < c$, independent of j. Its curvature bound is $C\eta_j^2\|A\|$, with the same margin in this complex ball. At zero its derivative is the linear minimizer computed above. Taylor's formula therefore gives

$$P_p^{(j)}(0) = 0, \qquad DP_p^{(j)}(0)A = i(\mathcal{H}_j A)(p), \qquad |P_p^{(j)}(A) - i(\mathcal{H}_j A)(p)| \le C\eta_j^2\|A\|^2.$$

In particular the constant is not obtained by composing a one-step regularity constant j times. It comes from the background problem at level j on this one cube. The same continuation and Cauchy estimate apply to its localized version.

We explain the one-level comparison, including the frame error. Set $a_* = C\mathsf{r}\mathsf{g}p_0$. The segment is taken so that $a_* \ll 1$. At real data put the level-j field in the axial gauge of the box and extend this gauge transformation constantly on its blocks. In this frame write the fluctuation as

$$V_{j-1} = e^{ia}V_{j-1}^{\mathrm{bg}}(V_j), \qquad a = g_{j-1}\mathsf{C}\widetilde{y}_j - \widetilde{\mathsf{D}}(g_{j-1}\mathsf{C}\widetilde{y}_j).$$

The current fields, the background input and their Landau representatives must all be taken in their regularity windows on the common interior cube of the two localizations; the smaller representative is restricted there, never extended. The background response bound gives potential size at most Ca_* there; its fine bond angles are at most $C\eta_{j-1}a_*$. These are the uniform small-representative bounds used in the following comparison.

The straightening begins at second order. Analytic dependence of the pivot elimination on the background and the Baker–Campbell–Hausdorff expansion consequently give the linear fluctuation $g_{j-1}\mathsf{C}^0\widetilde{y}_j$, with an error $O(a_*^2)$ in potential norm. Here C^0 is the flat pivot elimination. Taylor expansion of $P_p^{(j-1)}$ about the background, and replacement of its derivative there by its derivative at zero, therefore leave

$$ig_{j-1}(\mathcal{H}_{j-1}\mathsf{C}^0\widetilde{y}_j)(p)$$

as the first-order term, with error bounded by $C\eta_{j-1}^2 a_*^2$.

For the remaining background term use nesting before localization: $U_{j-1}(V_{j-1}^{\mathrm{bg}}(V_j))$ and $U_j(V_j)$ belong to the same critical orbit. Localization changes this comparison by a field whose curvature is at most $C\eta_{j-1}^2\mathsf{g}p_0e^{-c\mathsf{D}}$ on the cube of the pair. The orbit identity is modulo a spatial gauge transformation; it is not an exact identity between the chosen potentials, nor does the gauge have to be constant.

Here the small-representative bounds give the required approximation by one constant rotation. If two real bond fields W, Z on the common cube obey $Z = W^u$ and their bond angles are bounded by $C\eta a_*$, their gauge relation along a bond yields

$$|u(x + \eta e_\mu) - u(x)| \le C\eta a_*.$$

Summing along paths from a fixed root x_0 gives $|u(x) - u(x_0)| \le C\mathsf{r}a_*$. A plaquette angle is at most $C\eta_{j-1}^2\mathsf{g}p_0$, so replacing its conjugating element $u(x)$ by the common element $u(x_0)$ costs at most

$$C\eta_{j-1}^2\mathsf{g}p_0\,\mathsf{r}a_* \le C\eta_{j-1}^2 a_*^2.$$

For the localized comparison one first accounts for the small localization field and then uses this gauge calculation for the nested orbit. This last comparison concerns only the two backgrounds at the fixed output V_j; its root rotation is chosen before the current fluctuation is integrated. It may depend on V_j, but not on y_j. Absorb it into the common frame defining $\widetilde{y}_j$, once for all the

plaquettes. The result is the deterministic identity

$$P_p^{(j-1)}(A_{j-1}) = ig_{j-1}(\mathcal{H}_{j-1}\mathsf{C}^0\widetilde{y}_j)(p) + P_p^{(j)}(A_j) + R_j(p), \qquad |R_j(p)| \le \eta_{j-1}^2 d_{\mathrm{st}}, \tag{218}$$

where

$$d_{\mathrm{st}} \le C[(\mathsf{r}\mathsf{g}p_0)^2 + \mathsf{g}p_0 e^{-c\mathsf{D}}].$$

The remaining freedom is this one constant frame. Constant gauge covariance of the Landau continuation and colour isotropy of the flat Gaussian allow it to be removed by the group average in the generating function below. The error has already paid for the spatial variation of the gauge; no exact constant-gauge identification of the two localized representatives is required.

For $a = 1, 2$ choose complex variables with radii

$$|\lambda_a| \le \Lambda_a := \frac{cx}{\mathsf{r}^4 p_0^2 \|\phi_a\|_\infty}. \tag{219}$$

The constant c is small and fixed. For independent flat windowed fluctuations $\xi_{j'}$, set

$$m_{\le j}(p;\xi) = i \sum_{k_0 < j' \le j} g_{j'-1}(\mathcal{H}_{j'-1}\mathsf{C}^0\xi_{j'})(p),$$

$$Z_j(P) = \int \prod_{k_0 < j' \le j} d\gamma^0_{j'}(\xi_{j'} \mid \mathcal{W}^{\mathrm{read}}_{j'}) \exp\left\{ \frac{1}{2} \sum_{a=1}^{2} \lambda_a \sum_p \phi_a(p) |m_{\le j}(p;\xi) + P(p)|^2 \right\}.$$

Define $Y_j(V_j)$ by putting $P = (P_p^{(j)}(A_j))_p$ and averaging over the common constant gauge rotation. These are bounded, gauge-invariant Borel functions. They remain functions of the same kind even after the original profiles occupy less than one spacing: only the plaquette functions of the smooth minimizer are carried. The normalization (219) gives $|Y_j| \le C$ and

$$|Z_j(P') - Z_j(P)| \le \frac{C}{\eta_{k_0}^2 \mathsf{g}p_0} \max_p |P'(p) - P(p)|.$$

On the clean part $Y_{k_0} = \exp(\lambda_1 \mathsf{O}_1 + \lambda_2 \mathsf{O}_2)$. The definition of Z_j also gives the exact one-step integration identity obtained by adding its next independent flat fluctuation. Use (218) first and then (217). This yields

$$\|\mathbb{E}_j Y_{j-1} - Y_j\|_\infty \le C \left[L^{-2(j-1-k_0)} \frac{d_{\mathrm{st}}}{\mathsf{g}p_0} + T_j \right].$$

The outer positive normalized maps contract every accumulated error. Consequently, for $F = e^{\lambda_1 \mathsf{O}_1 + \lambda_2 \mathsf{O}_2}$,

$$\|F_{\mathrm{ref}}^{[k_1]} - Y_{k_1}\|_\infty \le C \sum_{j=k_0+1}^{k_1} \left[L^{-2(j-1-k_0)} \frac{d_{\mathrm{st}}}{\mathsf{g}p_0} + T_j \right]. \tag{220}$$

There is no volume factor $L^{4\mathsf{n}}$ in this telescoping sum. The terminal background is smaller by $L^{-2\mathsf{n}}$ in plaquette units. Symmetry of the centered Gaussian removes its linear contribution to Z_{k_1}, so replacing it by zero costs $CL^{-4\mathsf{n}}$. Removing the Gaussian windows costs $Ct^{1/2}$, where $t \le C \sum_j n_j e^{-cp_0^2}$; Cauchy–Schwarz applies because the generating function with twice the small parameter still has a bounded Gaussian integral.

The resulting full Gaussian covariance, for one colour, is

$$\mathcal{C}^x = \sum_{j=k_0+1}^{k_1} x_{j-1}^{-1} \mathcal{C}_{(j)}.$$

For $\Phi_\lambda = \lambda_1\phi_1 + \lambda_2\phi_2$, its generating function is

$$Z^0(\lambda) = \det(1 - \Phi_\lambda \mathcal{C}^x)^{-3/2}. \tag{221}$$

The power 3/2 counts the three colours of SU(2). Positivity and $\mathcal{C}^x(p,p) \le Cx^{-1}\eta_{k_0}^4$ give

$$\|(\mathcal{C}^x)^{1/2}\Phi_\lambda(\mathcal{C}^x)^{1/2}\|_1 \le \sum_a |\lambda_a| \sum_p |\phi_a(p)|\mathcal{C}^x(p,p) \le Cc p_0^{-2}.$$

Thus the determinant logarithm is its convergent trace series, and $c_1 \le |Z^0| \le C_1$ on the polydisc. Conditional on the clean part, (220) proves a bound independent of the terminal class and W. Integrating it in (213) therefore gives

$$\left| \frac{\int e^{\lambda_1 \mathsf{O}_1 + \lambda_2 \mathsf{O}_2} d\lambda_{k_0}}{Z^0(\lambda)} - 1 \right| \le C\delta, \quad \delta := \epsilon_{\mathrm{uc}} + L^{-4\mathsf{n}} + t^{1/2} + \sum_j \left[L^{-2(j-1-k_0)} \frac{d_{\mathrm{st}}}{\mathsf{g} p_0} + T_j \right]. \tag{222}$$

In particular this argument makes no use of the conditional law on a fibre of $\mathcal{M}_{k_0 \to k_1}$. The exceptional set of terminal data can be chosen independently of λ: first take a countable dense set in the polydisc, and then use continuity of the bounded generating functions.

The relative error and the mixed Gaussian slices. For small δ, take the logarithm of the ratio in (222) and apply the mixed Cauchy estimate. It gives

$$\left| \mathrm{Cov}_{\lambda_{k_0}}(\mathsf{O}_1, \mathsf{O}_2) - \frac{3}{2} \sum_{p,q} \phi_1(p)\phi_2(q)\mathcal{C}^x(p,q)^2 \right| \le C\delta \Lambda_1^{-1}\Lambda_2^{-1} \le C\delta \mathsf{r}^8 p_0^4 x^{-2} \|\phi_1\|_\infty \|\phi_2\|_\infty. \tag{223}$$

The Gaussian coefficient follows by differentiating $\frac{3}{2}\sum_{m\ge 1} m^{-1} \operatorname{tr}(\Phi_\lambda \mathcal{C}^x)^m$. In particular

$$\mathcal{C}^x(p,q)^2 = \sum_{j,j'} x_{j-1}^{-1} x_{j'-1}^{-1} \mathcal{C}_{(j)}(p,q)\mathcal{C}_{(j')}(p,q).$$

The mixed slices $j \ne j'$ are retained. They are exactly the terms which, in the free conditional-covariance calculation above, come from the bilinear curvature contribution and the covariance of the conditional means.

The bound on δ must beat the mixed Cauchy loss $\mathsf{r}^8 p_0^4 = O((\log x)^{20\mathfrak{p}_0})$. The local estimates above give its contributions, before that loss, as

$$\begin{aligned}
&C[\mathsf{r}^2 \mathsf{g} p_0 + e^{-c\mathsf{D}}],\\
&C\mathsf{n}\mathsf{r}^4[\mathsf{g} p_0^3 + p_0^2 x^{2-\kappa_0/2}],\\
&C\mathsf{n}\mathsf{r}^4 (\log x)^{2(\mathfrak{p}_0 - q)},\\
&C\mathsf{n}\mathsf{r}^4[\mathsf{r}\mathsf{g} p_0 + e^{-c\mathsf{D}}],\\
&C\mathsf{n}\mathsf{r}^4 p_0^2 e^{-c\mathsf{D}},\\
&C(\mathsf{n}\mathsf{r}^4 e^{-cp_0^2})^{1/2}, \qquad CL^{-4\mathsf{n}},\\
&CL^{4\mathsf{n}} e^{C\check{R}_{k_0}} (1+\mathsf{n})(\mathsf{r} + \check{R}_{k_0})^4 e^{-cp_0}.
\end{aligned}$$

They represent, respectively, deterministic lowering, the ordinary and remainder tilts, the earlier regular activities, covariance replacement, exterior decoupling, Gaussian windows, the terminal background and exceptional histories. With the already chosen $\mathsf{n} = \lceil 2\log_L \bar{x}_s \rceil$, every contribution tends to zero after multiplication by the Cauchy loss. Powers of g, and the remainder power $x^{2-\kappa_0/2}$, beat all logarithmic factors. The terminal term is $O(x^{-8})$ up to fixed flow-comparison constants. The exceptional term is small since $\mathfrak{p}_0 > \max\{1, \mathfrak{r}\}$. The only contribution with neither a power of g nor an exponential is bounded after the loss by

$$C(\log x)^{1+30\mathfrak{p}_0 - 2q} = o(1).$$

Thus the earlier choice $q > 21\mathfrak{p}_0$ is sufficient here too.

Finally replace $\mathcal{C}^x$ by $x^{-1}C_{K,s}^{\mathrm{bg}}$. The drift costs $C(B_\sharp \mathsf{n} + C_\sharp)/x$, the omitted terminal covariance costs $C(\mathsf{r}L^{-\mathsf{n}})^4$, and localization costs $C\mathsf{r}^4 e^{-c\mathsf{D}}$. These are comparisons of the explicit Gaussian

covariance sums. The slice envelopes sum over the two separated supports with a bounded total, so these three errors carry no mixed Cauchy loss. Combining (223) with the earlier source, localization and live-component estimates yields the target

$$\left|C_2(f,h) - {\zeta'_{k_0}}^2 \frac{3}{2} x^{-2} \sum_{p,q} \tilde{f}(p)\tilde{h}(q) C^{\mathrm{bg}}_{K,s}(p,q)^2\right| \le \epsilon_{g,\vartheta}(d) x^{-2} \|f\|_\infty \|h\|_\infty, \qquad \epsilon_{g,\vartheta}(d) \longrightarrow 0.$$

Here g and ϑ are fixed, $K - s(d) \ge n_0$, and d is small enough that the prescribed segment and boxes fit in the run and in the reference torus. The clean-class density and kernel comparison above uses the original retained-variable representation of the run and its small-field product window. The quantitative comparison also uses the uniform regularity, locality, complex increment and history bounds required by the preceding renormalisation and source arguments.

Proposition 5.1 (The reference-torus two-point function)**.** The reference-torus comparison gives

$$C_2(f,h) = {\zeta'_{k_0}}^2 \frac{9}{\pi^4} g(d)^4 I(f,h)(1+\varepsilon), \qquad \sup|\varepsilon| \longrightarrow 0 \quad (d \downarrow 0)$$

over the specified data and shape class. ■

Proof. At separation of order d, the slice error tends to zero because $\hat{d} \to \infty$. The lattice error tends to zero since $a_K/d = L^{-(K-s)}/\hat{d} \le L^{-n_0}/\hat{d} \to 0$ and $d/\ell_\star(g) \to 0$. The frozen-tent interpolation changes a test by $o(\|f\|_\infty)$ and its support by $o(d)$. The shape bounds control the resulting sum error relative to $I(f,h)$. The Riemann-sum error is bounded by $Ca_K/(\vartheta^3 d)$, which tends to zero under the same scale condition. Thus

$$\sum_{p,q} \tilde{f}(p)\tilde{h}(q) C^{\mathrm{bg}}_{K,s}(p,q)^2 = \frac{6}{\pi^4} I(f,h)(1+o(1)).$$

For jointly Gaussian vectors F, G, Wick's identity gives $\mathrm{Cov}(|F|^2/2, |G|^2/2) = \frac{1}{2}\sum_{\mu,\nu}(\mathbb{E}F_\mu G_\nu)^2$. There are three colour components for SU(2). Consequently the coefficient is $(3/2)(6/\pi^4) = 9/\pi^4$, as displayed. The absolute error in the reference-torus comparison becomes a relative error because $I(f,h)$ has a positive shape-dependent lower bound. □

5.5. **Transfer from one torus to every phase.**

Proposition 5.2 (Local transfer between states)**.** At one first-passage datum, let $\mathfrak{s}, \mathfrak{s}'$ be two permitted tori or periodic Gibbs states. Suppose $a_K \le d \le c$ and the real functions f_i are supported in one ball of radius $5d$, with $\mathrm{Lip}\, f_i \le \|f_i\|_\infty/(\vartheta d)$. There are no sign or separation restrictions. Then

$$|C_n[\mathfrak{s}](f_1,\dots,f_n) - C_n[\mathfrak{s}'](f_1,\dots,f_n)| \le C_n(g,\vartheta) d^{4-n} \prod_i \|f_i\|_\infty.$$

■

Proof. Let D be the difference of cumulants. The common-normalizer estimate in proposition 4.4 gives $|\mathsf{D}(F_1,\dots,F_n)| \le C_n \prod_i \|F_i\|_\sharp$ for arrays in a unit ball. By the lattice symmetry and reflection argument, both states are invariant under the full lattice translations, so D is invariant under a common lattice translation.

Choose a lattice grid of $N \ge cd^{-4}$ common translates of the radius-$5d$ support ball, with mutually disjoint, positively separated copies in a unit ball. This is possible with grid spacing a lattice multiple in $[Cd, (C+1)d]$, since $a_K \le d$. For $n \ge 2$ take independent signs $\epsilon_v^{(i)}$, $1 \le i < n$, and set $\epsilon_v^{(n)} = \prod_{i<n} \epsilon_v^{(i)}$. Define

$$F_i = \sum_v \epsilon_v^{(i)} \tau_v f_i.$$

Disjoint supports give $\|F_i\|_\sharp \le \|f_i\|_\sharp$. Expanding D by multilinearity, the mean of its sign factor is zero unless all translation indices coincide. Indeed, each independent sign family pairs its index with the index of the final entry. Hence

$$\mathbb{E}_\epsilon \mathsf{D}(F_1, \dots, F_n) = \sum_v \mathsf{D}(\tau_v f_1, \dots, \tau_v f_n) = N \mathsf{D}(f_1, \dots, f_n).$$

For $n = 1$ the same identity follows by taking the unsigned sum and using linearity. The array estimate now gives

$$|\mathsf{D}(f_1, \dots, f_n)| \le C d^4 \prod_i \|f_i\|_\sharp \le C_n(g, \vartheta) d^{4-n} \prod_i \|f_i\|_\infty.$$

□

For $n = 2$ the transfer cost is $C_2(g, \vartheta) d^2 \|f\|_\infty \|h\|_\infty$. By proposition 5.1 the positive leading term is comparable to $(\log(1/d))^{-2} \|f\|_\infty \|h\|_\infty$. The transfer cost divided by this term tends to zero. Choosing d small after g gives a uniformly small relative error, for instance $|\varepsilon| \le 1/2$, in every periodic Gibbs state.

Together with the uniform flow normalization this proves (201) in every such state, subject to the cutoff-depth condition. At each fixed d that condition holds eventually in a sequence $K \to \infty$. The two-sided bound is a fixed closed condition on the two-point value and therefore passes to any common continuum subsequence of pure phases. Recentring an observable changes no second cumulant.

For the profiles fixed in target theorem A, the rescaling $x = dX$, $y = dY$ gives

$$I(f_d, h_d) = d^{-8} I(f_1, h_1), \qquad I(f_1, h_1) > 0.$$

They obey one shape condition independent of d. Thus, with g small for this fixed shape, (201) is exactly (7). No uniform smallness threshold for all possible bump shapes is required.

6. The continuum limit and diagonal reflections

The volume limit is taken at fixed lattice spacing. The source estimates then permit one subsequence in the lattice spacing on which all separated correlations converge. We also need the same extraction for a Gibbs state centred at the mean of a different state. This will provide the coordinates for the phase decomposition in section 8.

6.1. **Gibbs states at fixed spacing.** At fixed K, β, enumerate the unoriented bonds of the infinite lattice. The configuration space is a countable product of compact copies of $\mathrm{SU}(2)$ and is compact and metrizable. Periodically extend the measures on tori with sides tending to infinity. Their laws have weakly convergent subsequences. Denote their limit points by $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$.

For a finite set W of bonds, the conditional Gibbs kernel integrates the bonds of W with density proportional to

$$\exp\left[-\beta \sum_{p : p \cap W \neq \varnothing} a_p(U)\right]$$

against their Haar measure, keeping the exterior bonds fixed. Only finitely many exterior bonds occur. Its normalizing integral is strictly positive and continuous in those bonds. The kernel therefore maps local continuous functions to local continuous functions. For a sufficiently large torus the conditional identity against a fixed local exterior function is the infinite-lattice identity on that finite collection of bonds. It passes to a weak limit. Approximation and the monotone class argument extend it to the conditional Gibbs identity. This proves that the limit points are Gibbs states.

The periodic Wilson measures are invariant under every lattice translation and under the cubic group W_4. These identities involve local continuous functions and persist in a weak limit. In

particular $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ is nonempty and consists of translation-invariant Gibbs states. The larger class $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ consists of Gibbs states invariant under some finite-index subgroup of lattice translations. We do not identify these two classes. The pure states needed later belong to the latter class by the Gibbs ergodic decomposition.

At this fixed spacing $\mathcal{O}_a(f)$ is a bounded local continuous function for compactly supported f. Products and cumulants are finite combinations of integrals of such functions. Thus a weak torus limit preserves every fixed moment and cumulant. Every estimate uniform on large tori passes to $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$. The comparison of section 4, under the lattice symmetry and reflection argument, gives the estimates on the larger class.

6.2. **One subsequence for every separated correlation.** The estimates for tests in small separated balls imply bounds on any fixed separated support sets. It is useful to make this extension before choosing countable test families.

Lemma 6.1 (Bounds on fixed separated supports)**.** Fix a reference coupling in the range of the state-uniform source and one-point estimates. Let $A_1, \dots, A_n$ be compact, pairwise disjoint subsets of $\mathbb{R}^4$. There is a constant $C(A_1, \dots, A_n)$, independent of the first-passage datum and of $\rho, \nu \in \mathcal{G}^{\mathrm{per}}_{K,\beta}$, such that, for real $f_i \in C^1_c(\mathbb{R}^4)$ with $\operatorname{supp} f_i \subset A_i$,

$$\left| \rho \left(\prod_{i=1}^{n} [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))] \right) \right| \le C(A_1, \dots, A_n) \prod_i \|f_i\|_\sharp, \tag{224}$$

for all sufficiently small $a = L^{-K}$. The same assertion holds for cumulants of order at least two. The constant can be chosen using only the number of entries, a positive lower bound on their support separation and radii of balls containing the individual support sets. ∎

Proof. For $n = 1$, the assertion is the one-point transfer. For $n \ge 2$, choose $R > 0$ so that distinct A_i have distance at least $6R$, and put $d = 2R$. Fix a smooth partition of unity by translates whose supports lie in balls of radius R, and retain for each A_i only the finitely many pieces that meet it. Balls belonging to distinct entries have distance at least $6R - 4R = d$: each ball can extend at most $2R$ from its point of intersection with A_i. Write

$$f_i = \sum_m f_{i,m}, \qquad f_{i,m} = \chi_m f_i.$$

Every piece is real and belongs to C^1_c; its support is contained both in A_i and in its assigned ball. The product rule in the sharp norm gives

$$\sum_m \|f_{i,m}\|_\sharp \le K_i \|f_i\|_\sharp. \tag{225}$$

The constants are fixed by the partitions. A fixed translated partition at this scale gives a bound for their number and derivatives in terms of R and the radius of a ball containing A_i. Thus the K_i have the dependence claimed in the lemma.

Choose an L-adic $\varpi > 0$ large enough that all the containing balls lie in $(-\varpi + d/2, \varpi - d/2)^4$. For all sufficiently large K, one has $a \le d/2$ and $\varpi/a \in \mathbb{N}$. Every tuple obtained by selecting one piece of each entry therefore belongs to a common small-ball support class. The state-uniform cumulant estimates apply to every subset of such a tuple, and the one-point transfer bounds its first cumulants. The finite moment–cumulant formula gives

$$\left| \rho \left(\prod_i [\mathcal{O}_a(f_{i,m_i}) - \nu(\mathcal{O}_a(f_{i,m_i}))] \right) \right| \le c_n(d; R) \prod_i \|f_{i,m_i}\|_\sharp.$$

The constant is independent of the enclosing cube and the two states. Multilinearity and (225) now give

$$\sum_{m_1,\dots,m_n} c_n(d;R)\prod_i \|f_{i,m_i}\|_\sharp \le c_n(d;R)\prod_i K_i\|f_i\|_\sharp.$$

This proves the moment assertion. The same finite expansion directly proves the cumulant assertion. The lattice conditions were required only eventually for these fixed support sets. □

In particular, for tuples $\underline{f},\underline{h}$ whose corresponding entries have supports in the same sets A_i, multilinearity gives

$$\begin{aligned}|M_a^{\rho,\nu}(\underline{f}) - M_a^{\rho,\nu}(\underline{h})| \le C(A_1,\dots,A_n)\sum_i \|f_i-h_i\|_\sharp \\ \cdot\prod_{l\ne i}(\|f_l\|_\sharp + \|h_l\|_\sharp),\end{aligned} \tag{226}$$

where $M_a^{\rho,\nu}$ denotes the reference-centred moment in (224). This estimate holds for all states under consideration with one constant. It is the continuity estimate used below, including when the support of an entry is too large to lie in one small ball separated from the others.

Proposition 6.2 (Common extraction)**.** Let (K_j,β_j) be first-passage data with $K_j\to\infty$, and choose $\nu_j\in\mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$. There is one subsequence on which the cumulants converge for every order $n\ge 2$ and every separated C^1_c tuple on $\mathbb{R}^4$. The limits satisfy the same per-ball and factorial bounds.

It is enough that the bounds hold eventually for each fixed order, radius and positive separation. The jth family may be defined only for tests in $B(0,r_j)$ with $r_j\to\infty$. The subsequence can also be chosen to make any prescribed countable list of bounded scalar sequences converge.

The same assertion holds for moments in arbitrary $\rho_j\in\mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, with the observables centred at ν_j, provided the state-uniform bounds and the one-point transfer apply to both states. ■

Proof. The bounds in lemma 6.1 hold eventually for every fixed separated support list, uniformly in the states. Equation (226) gives the corresponding continuity estimate on common support sets. Their separation is fixed throughout each approximation.

Choose all finite lists of bounded open finite unions of rational boxes whose closures are pairwise disjoint. In each support list choose a countable family of real smooth tests dense in the C^1 norm for compactly supported C^1 tests in that list. There are countably many tuples over all orders and lists. The resulting family has the following support-controlled approximation property: for a separated tuple (f_i) and any open neighbourhoods O_i of its supports, one can approximate it in the sharp norm by enumerated tuples $(f_i^{(l)})$ with $\operatorname{supp} f_i^{(l)}\subset O_i$. Indeed first choose a rational support list with $\operatorname{supp} f_i$ in its interior and its closure in O_i, retaining positive separation, and then use that list's dense family.

Equation (224) bounds every enumerated coordinate. Successively extract a convergent subsequence for the first tuple, then the second, and so on. The diagonal subsequence converges on all of them.

Every enumerated tuple is in the domain of the jth family and has its uniform bound once j is sufficiently large. Removing its finite exceptional prefix changes no extraction. Thus the same diagonal procedure applies to the stated growing domains and eventual bounds. Interleave any additional countable bounded scalar sequences with the test coordinates to select their limits at the same time.

Given any separated C^1_c tuple, enlarge its supports slightly to one of the support lists just chosen, retaining a positive separation. Approximate each entry in the sharp norm by the corresponding dense tests. The telescoping estimate controls the difference uniformly in j. For any prescribed accuracy, choose the approximation first and then j large enough that its tuple values are Cauchy. The original tuple values are consequently Cauchy. The same estimate makes their limit independent

of the approximants and proves continuity and the claimed bounds. The use of C^1 tests is essential to this particular countable extraction: the space of all Lipschitz functions is not separable in the sharp norm.

For the final assertion, the one-point transfer gives

$$|\rho_j(\mathcal{O}_{a_j}(f)) - \nu_j(\mathcal{O}_{a_j}(f))| \le C_1(R)\|f\|_\sharp.$$

Extract these linear forms along the same countable family and include their diagonal selections. A moment in ρ_j, centred at ν_j, is a polynomial in these mean differences and in the higher cumulants of ρ_j. Every factor has the required uniform bound and continuity. The identical diagonal and approximation argument therefore gives convergence of all these moments on one subsequence. □

The same argument identifies the topology of the possible limit families. Fix the per-ball bounds at every order and at rational positive separations and radii. Choose the dense test collections closed under rational linear combinations within their support sets, and enumerate the resulting separated tuples as $X^1, X^2, \ldots$. If M_q is the corresponding uniform bound for the qth value, the evaluation map takes a family into the compact countable product

$$\prod_{q\ge1}[-M_q, M_q].$$

On bounded families its coordinate topology is metrized by

$$d(C, C') = \sum_{q\ge1} 2^{-q}\min\{1, |C(X^q) - C'(X^q)|\}.$$

The support-preserving telescoping estimate implies two facts. A family is determined by these coordinates, and coordinate convergence extends to convergence at every separated C^1_c tuple. To check the second fact, first approximate the tuple at a fixed positive separation, uniformly for the whole bounded class, and only then use coordinate convergence. Thus no uniform continuity estimate as $d \downarrow 0$ is needed.

The multilinearity and compatibility conditions on these coordinates are closed finite relations. A coordinate limit satisfies them on the dense collections; the same approximation argument extends them to all tuples. For a new tuple, that approximation proves convergence of the original families themselves at the tuple. Pass to the limit in its original bound, with its actual radius and separation; no continuity of the constants in those parameters is required. Consequently the set of families satisfying the prescribed bounds and fixed closed identities is a closed subset of the displayed compact product. It is compact and metrizable, not just sequentially bounded. The choice of rational support regions and the retention of separation in the approximation are what make this countable-coordinate description valid. This topology will also be used for the limiting coordinates of phases.

For the reference state, the centred limit moments are

$$S^{\mathrm{c}}(f_1, \ldots, f_n) = \sum_{\substack{\pi \text{ a partition of } \{1,\ldots,n\} \\ |B|\ge2 \text{ for every } B\in\pi}} \prod_{B\in\pi} C_{|B|}((f_i)_{i\in B}), \qquad S^{\mathrm{c}}(\varnothing) = 1, \qquad S^{\mathrm{c}}(f) = 0. \tag{227}$$

This is the finite-cutoff moment identity for $\mathcal{O}_{a_j}(f) - \nu_j(\mathcal{O}_{a_j}(f))$, followed by a limit. It requires no convergence of either uncentred mean.

Several further consequences of the extraction will be used. First, a fixed closed relation among finitely many correlation values passes to the limit. This applies to equalities, uniform intervals and finite positive quadratic forms. If a relation has varying coefficients or bounds, one must first establish its limiting closed relation. Second, moving tests can be included whenever their sharp-norm difference tends to zero and their supports retain a common positive separation margin. The telescoping estimate proves this without any further extraction. Third, every volume-first family can be approximated by a sequence of tori whose sides exceed any prescribed floors. At stage

j choose a torus approximating the first j countable test values of ν_j to within $1/j$, with side above that floor; the same continuity argument then gives the family on every separated tuple.

In making this last diagonal choice, both the cutoff K_j and bare coupling β_j are held fixed while its torus side is chosen. At fixed spacing the finitely many required observables and products are local continuous functions, so weak approximation of ν_j approximates their moments. Cumulants are finite polynomials in those moments. If moments centred at another state are also being retained, include the required mean coordinates in this finite list. Choose the side beyond the admissibility floor of that datum and every prescribed additional floor. After this choice at each j, the approximation error tends to zero on every enumerated coordinate. The common extraction estimate extends it to the original continuum family on all separated tuples. This construction asserts no interchange of limits for an entire unselected sequence; it constructs one diagonal sequence representing the already chosen volume-first limit.

For clarity, continuity under translations is quantitative. For $f \in C_c^1$ put

$$\varpi_f(\delta) = \max\left\{ \|\nabla f\|_\infty \delta, CM \sup_{|x-y|\le\delta} |\nabla f(x) - \nabla f(y)| \right\}.$$

Then $\|\tau_v f - \tau_{v'} f\|_\sharp \le \varpi_f(|v - v'|)$ and $\varpi_f(\delta) \to 0$. For finitely many moving entries the same telescoping formula gives uniform continuity on support sets with a fixed separation margin. This will allow lattice translations and moving lattice mirrors to pass to continuum translations and mirrors.

In the case of two different centring states, full lattice symmetry also determines the form of the limiting first moment. Let $\delta_j^{\mathrm{mean}} = \rho_j(a_p) - \nu_j(a_p)$, independent of the plaquette p by the lattice symmetry and reflection argument, and put $b_j^{\mathrm{mean}} = 6a_j^{-4}\delta_j^{\mathrm{mean}}$. Then

$$\rho_j(\mathcal{O}_{a_j}(f)) - \nu_j(\mathcal{O}_{a_j}(f)) = b_j^{\mathrm{mean}} \left(\frac{a_j^4}{6} \sum_p f(x(p)) \right).$$

The parenthesis tends to $\int f$: there are six unoriented plaquette directions in four dimensions. Fixing one smooth f_0 of integral one, the mean-transfer bound makes b_j^{mean} bounded, and the common extraction makes it converge. Thus the limiting first moment is $b \int f$, with $|b|$ bounded uniformly in the chosen states. The higher cumulants remain those of ρ_j; they are not averages of component cumulants.

6.3. **Inherited symmetries and coordinate reflections.** Let σ be one of the limiting moment functionals just constructed. It may be the reference-centred functional S^{c}, or the limit in a state ρ_j with subtraction taken from ν_j. For the latter we use the mean-transfer bound as well as the source bounds. Throughout this subsection the state symmetries and positivity required on the lattice are those of the lattice symmetry and reflection argument.

Consider the lattice isometry $\mathsf{r}(x) = wx + v$, where $w \in W_4$ and $v \in a_K\mathbb{Z}^4$. Pulling back bond configurations by r preserves Haar measure and the Wilson action. The plaquette observable is sampled at the plaquette centre. Thus relabelling its sum gives the exact identity

$$\mathcal{O}_a(f) \circ \mathsf{r}^* = \mathcal{O}_a(\mathsf{r}f), \qquad (\mathsf{r}f)(x) = f(\mathsf{r}^{-1}x).$$

The same identity holds for the centred representatives, since the subtracted state is invariant. Hence every finite-cutoff moment and cumulant has the lattice covariance identity.

The lattices are nested. A vector in $\mathfrak{D}_L^4$, where $\mathfrak{D}_L = \bigcup_K a_K\mathbb{Z}$, belongs to every sufficiently fine lattice. Covariance under this dense translation group and under W_4 therefore passes to the common subsequence as a closed relation. To pass from the dense group to all translations, use the translation modulus established above. If $v_m \in \mathfrak{D}_L^4$ tends to v, each entry satisfies

$$\|\tau_{v_m} f - \tau_v f\|_\sharp \le \varpi_f(|v_m - v|) \longrightarrow 0.$$

The translated tuple retains its separation and lies in support balls of uniformly bounded radius. Telescoping its entries in the per-ball estimate proves $\sigma(\tau_v X) = \lim_m \sigma(\tau_{v_m} X) = \sigma(X)$. The same argument applies to a finite collection of different moving translations as long as a positive separation margin is retained. Reality, multilinearity and permutation symmetry pass through the finite moment–cumulant identities.

We recall the coordinate reflection calculation, because it identifies the positive form whose limits we use. At fixed $K, \beta > 0$, take a bond mirror $\Sigma = \{x_4 = c\}$ with $c \in a_K(\mathbb{Z} + \frac{1}{2})$ and its partner on the torus. Write each crossing bond as

$$U_b = (W_b^-)^{-1} W_b^+,$$

with W_b^-, W_b^+ independent Haar variables. Their quotient is Haar, so this replacement leaves integrals unchanged. It splits the remaining variables into two independent, identically distributed collections ς_-, ς_+ exchanged by reflection. For a crossing plaquette with crossing bonds b_1, b_2, its positive-side path gives a matrix $\Xi_p(\varsigma_+)$, and

$$\operatorname{tr} U_p = \operatorname{tr}\left(\Xi_p(\varsigma_+)\Xi_p(\varsigma_-)^*\right).$$

In particular its trace is a sum of a matrix entry on one side times the conjugate of the corresponding entry on the other. The real part has the same property after entries and conjugate entries are both included. Expanding the exponential of the crossing Wilson weights therefore gives, for a bounded function F on the positive side,

$$Z\left\langle \overline{F \circ \theta_\Sigma^*} F \right\rangle = e^{-\beta|\mathsf{p}_\times|} \sum_{n \geq 0} \frac{(\beta/4)^n}{n!} \sum_{\vec{\gamma}} \left| \int F e^{-\beta A_+} \xi_{\vec{\gamma}} \, d\varsigma_+ \right|^2 \geq 0. \tag{228}$$

Here $\xi_{\vec{\gamma}}$ runs over products of entries of the Ξ_p and their conjugates, and A_+ is the action from plaquettes lying entirely on the positive side. Every coefficient is nonnegative. The exponential series converges absolutely on the compact integration space, so the expansion and integrations can be interchanged. A plaquette whose centre is in the open positive side has all its corners there. Thus a test supported on that side produces an observable of precisely the variables allowed in this calculation.

A site mirror has a different decomposition. No plaquette has points on both open sides: the plaquettes either lie in the mirror or belong to one closed side. Condition on all bonds lying in the mirror. Given these bonds, the remaining bond integrations on the two sides are independent and are exchanged by reflection. The product of the two reflected integrals is the squared modulus of one side's integral, multiplied by the positive mirror weight. Integration over the mirror bonds proves positivity for functions of one closed side. This proves the finite-volume coordinate site and bond assertions without identifying the two factorizations.

For the continuum passage use the bond mirror $\Sigma_j = \{x_4 = a_j/2\}$, whose partner escapes compact sets in the volume limit. Its action on a test is $\theta_{\Sigma_j} f = \tau_{a_j e_4} \theta f$, with $\theta(x', x_4) = (x', -x_4)$. If all positive-time tests stay a fixed distance from zero, their reflected tuple remains separated. The translation modulus and per-ball estimates replace θ_{Σ_j} by θ at an error tending to zero. Consequently each finite reflection-positive quadratic form has a nonnegative limit. In the periodic Gibbs-state route the site mirror at zero is available directly from the lattice symmetry and reflection argument. The passage from finite tori to the full periodic Gibbs class is explained in the lattice symmetry and reflection argument.

6.4. **The Hilbert space and the time-translation semigroup.** Let $\mathcal{D}_+$ be the complex vector space of finite formal linear combinations of separated tuples of real smooth tests in $\{x_4 > 0\}$, including the empty tuple. If $F = \sum_i c_i X_i$, write $\overline{F} = \sum_i \overline{c}_i X_i$ and $\Theta F = \theta \overline{F}$. Concatenation is used only when the combined supports are separated. In particular $\Theta F \cdot G$ is defined for all $F, G \in \mathcal{D}_+$:

the reflected and unreflected supports lie in opposite open half-spaces. The form

$$\langle F, G\rangle_\sigma = \sigma(\Theta F \cdot G)$$

is Hermitian and positive semidefinite. Denote its null space by $\mathcal{N}$ and the completion of $\mathcal{D}_+/\mathcal{N}$ by $\mathcal{H}_\sigma$. The empty tuple gives a unit vector Ω.

Before taking the quotient, define

$$T_t^\circ F = \tau_{te_4} F \quad (t \geq 0), \qquad U^\circ(v)F = \tau_{(v,0)}F \quad (v \in \mathbb{R}^3).$$

They preserve $\mathcal{D}_+$ and fix the empty tuple. Translation invariance gives

$$\langle F, T_t^\circ G\rangle_\sigma = \sigma(\Theta F \cdot \tau_{te_4} G) = \sigma(\tau_{-te_4}\Theta F \cdot G) = \langle T_t^\circ F, G\rangle_\sigma,$$
$$\langle U^\circ(v)F, U^\circ(v)G\rangle_\sigma = \langle F, G\rangle_\sigma.$$

The second identity already implies that spatial translations preserve the null space. For time translations we need a uniform bound for translated self-pairings.

Let X, Y be two fixed positive-time tuples and let d_0 be the least separation in $\theta X \cdot Y$. For $t \geq 0$, shifting Y forward does not decrease any cross separation; separations within its entries remain fixed. Each entry remains in a ball of its original radius, though the centres move. The per-ball constants are independent of these centres and may be chosen nonincreasing in the minimum separation. The moment–cumulant formula therefore gives

$$\sup_{t\geq 0} |\sigma(\theta X \cdot \tau_{te_4} Y)| < \infty.$$

For a functional centred at a different state, the singleton blocks are bounded by the one-point transfer; the same conclusion follows. There are only finitely many tuple pairs in any $F \in \mathcal{D}_+$, so

$$c_F := \sup_{t\geq 0} |\langle F, T_t^\circ F\rangle_\sigma| < \infty. \tag{229}$$

The global torus estimate, with its volume factor, would not imply this bound. This is a use of the uniform per-ball source estimate.

Proposition 6.3 (Reconstruction of the translation operators)**.** The maps T_t° and $U^\circ(v)$ induce a strongly continuous semigroup $\mathcal{T}_t$ of positive self-adjoint contractions and a strongly continuous unitary representation $\mathcal{U}(v)$ on $\mathcal{H}_\sigma$. There is a unique self-adjoint $H \geq 0$ with $\mathcal{T}_t = e^{-tH}$, and $H\Omega = 0$. The spatial generators P_1, P_2, P_3 commute strongly with H and with each other. ■

Proof. Set $a(t) = \|T_t^\circ F\|_\sigma$. Symmetry and the semigroup law give

$$a(t)^2 = \langle F, T_{2t}^\circ F\rangle_\sigma \leq \|F\|_\sigma a(2t).$$

Iterating Cauchy–Schwarz r times and using $a(2^r t)^2 = \langle F, T^\circ_{2^{r+1}t} F\rangle_\sigma \leq c_F$ yields

$$a(t) \leq \|F\|_\sigma^{1-2^{-r}} c_F^{2^{-r-1}}.$$

If $\|F\|_\sigma > 0$, take $r \to \infty$. If $\|F\|_\sigma = 0$, the first Cauchy–Schwarz inequality already gives $a(t) = 0$. Thus $a(t) \leq \|F\|_\sigma$ for every F. Time translation preserves $\mathcal{N}$, descends to a contraction, and extends uniquely to the completion. Its symmetry makes the bounded operator self-adjoint. The semigroup law also gives

$$\mathcal{T}_t = \mathcal{T}_{t/2}^* \mathcal{T}_{t/2} \geq 0.$$

The moving-test estimate gives continuity at zero of $\langle F, T_t^\circ F\rangle_\sigma$ for every $F \in \mathcal{D}_+$. Consequently

$$\|\mathcal{T}_t[F] - [F]\|^2 = \langle F, T_{2t}^\circ F\rangle_\sigma - 2\langle F, T_t^\circ F\rangle_\sigma + \langle F, F\rangle_\sigma \longrightarrow 0.$$

Density and the contraction bound extend this convergence to all vectors. The semigroup law then proves strong continuity at every $t \geq 0$. The spatial isometries have inverses $U^\circ(-v)$ and thus extend to unitaries. The identical correlation-continuity calculation gives strong continuity in v.

Let E be the spectral measure of the positive contraction $\mathcal{T}_1$, supported in $[0,1]$. Positivity and uniqueness of the positive square root give

$$\mathcal{T}_{2^{-r}} = \int_{[0,1]} x^{2^{-r}}\, dE(x).$$

Every such operator annihilates the range of $E(\{0\})$. Strong continuity as $r \to \infty$ therefore implies $E(\{0\}) = 0$. The self-adjoint spectral integral

$$H = \int_{(0,1]} -\log x\, dE(x)$$

is nonnegative. Functional calculus identifies e^{-tH} with $\mathcal{T}_t$ first at positive dyadic t and then at every t by strong continuity. Conversely $e^{-H'} = \mathcal{T}_1$ for a nonnegative self-adjoint H' fixes its spectral measure under the one-to-one map $E \mapsto e^{-E}$, so $H' = H$. In particular every $\mathcal{T}_t$, $t > 0$, has trivial kernel. Since $\mathcal{T}_t\Omega = \Omega$, its spectral measure at Ω is concentrated at zero energy, and $H\Omega = 0$.

Spatial translations commute with $\mathcal{T}_1$ on the dense quotient and hence everywhere. A bounded operator commuting with this self-adjoint operator and its adjoint commutes with its spectral projections. Thus $\mathcal{U}(v)$ commutes with every spectral projection of H. The commuting spatial group has its joint spectral measure, and the joint spectral theorem gives a single projection-valued measure $\mathcal{P}$ for (H, P_1, P_2, P_3), with

$$e^{i\tau H}\mathcal{U}(v) = \int_{[0,\infty)\times\mathbb{R}^3} e^{i(\tau E + v\cdot q)}\, d\mathcal{P}(E,q).$$

This is the asserted strong commutation and spectral half-space. □

The countable support-controlled test families used in the extraction are dense for the reflection norm as well. Approximate each tuple entry with its support in a fixed separated positive-time set. The per-ball estimate bounds the difference of every inner product, hence the reflection norm of the difference tends to zero. Finite linear combinations with rational real and imaginary coefficients form a countable dense subset of $\mathcal{H}_\sigma$. The Hilbert space is therefore separable.

Every $w \in W_3$ preserves the positive half-space, commutes with θ, and preserves σ. It induces a unitary fixing Ω and commuting with $\mathcal{T}_t$ and H. The covariance $w\tau_v = \tau_{wv}w$ gives $\mathcal{U}(w)\mathcal{U}(v)\mathcal{U}(w)^* = \mathcal{U}(wv)$. By uniqueness of the joint spectral measure the joint spectrum is invariant under $(E,q) \mapsto (E, wq)$.

6.5. **Lattice representatives of the reflection form.** For $F = \sum_i c_i X_i \in \mathcal{D}_+$ let

$$\mathbb{F}_j = \sum_i c_i \prod_{f\in X_i} [\mathcal{O}_{a_j}(f) - \nu_j(\mathcal{O}_{a_j}(f))].$$

Define $\mathbb{G}_j$ similarly. In the reference-state volume-first route the scalar product is

$$\langle F, G\rangle_\sigma = \lim_j \rho_j\left(\overline{\mathbb{F}_j \circ \theta^*}\, \mathbb{G}_j\right). \tag{230}$$

The same formula holds along approximating tori with the bond mirror θ_{Σ_j} in place of θ. The difference between these two mirrors tends to zero by the moving-test estimate above. The reference subtraction is compatible with the reflected representative because ν_j has the lattice reflection symmetry.

For a time $t \in \mathfrak{D}_L$, $t \geq 0$, the translated representative equals the lattice time translation of $\mathbb{G}_j$ at every sufficiently fine cutoff. Thus $\langle F, \mathcal{T}_t G\rangle$ is a limit of lattice matrix elements. The same holds for spatial translations in $\mathfrak{D}_L^3$. An inequality involving finitely many such matrix elements is a closed relation on a finite list of moments. It passes to the continuum on this dense family, and then to all vectors by boundedness. This is the passage used below for the lattice energy–momentum inequality.

At this stage the spectrum is only in the half-space $[0,\infty)\times\mathbb{R}^3$ and the zero-energy space can have dimension larger than one. The spectral theorem gives the strong limit

$$\mathcal{T}_t \longrightarrow P_0 = \mathcal{P}(\{0\}\times\mathbb{R}^3) \qquad (t\to\infty).$$

The diagonal mirrors, rotation argument, and phase selection supply the additional properties needed later. The construction of the semigroup itself applies to $\mathbb{R}^4$; on a fixed torus an arbitrary positive time translation need not preserve the chosen reflection slab.

6.6. **Diagonal mirrors and the Gibbs-state transfer.** At spacing a, the coordinate mirror families are $\{x_\mu = c\}$, $c\in\frac{a}{2}\mathbb{Z}$. They include site and bond mirrors. The diagonal families are $\{x_\mu \pm x_\lambda = c\}$, $c\in a\mathbb{Z}$, for $\mu<\lambda$. There are four coordinate and twelve diagonal directions. The comparison concerns the full periodic Gibbs class, including the states selected in an ergodic decomposition.

Symmetries of periodic Gibbs states. The lattice argument seeks invariance under every lattice translation and cubic isometry, and reflection positivity at the coordinate site/bond and diagonal mirrors, for the full class $\mathcal{G}^{\mathrm{per}}_{K,\beta}$. The finite-volume kernels, temporal-layer calculation and pressure comparison below are the steps of this argument. Using the full Gibbs class is essential when an ergodic component is selected later.

Use tori adapted to each mirror and temporal layers to compare their reflection forms with a periodic Gibbs state. The period lattice matters: a cubic torus need not split along a diagonal mirror. The pressure comparison below applies to the full Gibbs class, whereas the original reference state was obtained from cubic tori.

Positive crossing kernels on adapted tori. For a chosen mirror Σ, use a period lattice rectangular in an orthogonal frame containing its normal. A diagonal normal requires a frame rotated through $\pi/4$ in the corresponding coordinate plane. The periods must preserve the bond lattice and permit the two sides to be reflected into each other. These are geometric conditions on the finite-volume approximation, independent of the ultraviolet estimates.

Choose the slab height to be an even integer in the integer layer coordinate, at least two, and large enough that the fixed test support lies in the slab. Each side of a cubic approximating torus has an even number of sites. The adapted periods also satisfy the support-dependent no-short-vector condition used in (233). These parity and lower size conditions admit arbitrarily large compatible tori at each fixed spacing and coupling.

A diagonal mirror cuts a crossing plaquette into two paths of length two. Write v for the ordered product on one path and v' for the reflected product from the other side. Apart from its constant factor, the crossing Wilson weight is

$$\exp\left(\frac{\beta}{2}\operatorname{Re}\operatorname{tr}(vv'^{-1})\right).$$

Since v' is unitary,

$$\frac{1}{2}\operatorname{Re}\operatorname{tr}(vv'^{-1}) = \frac{1}{4}\sum_{i,j}\left(v_{ij}\overline{v'_{ij}} + \overline{v_{ij}}v'_{ij}\right).$$

Each summand is a function of the variables on one side times the complex conjugate of its reflected counterpart. Every power of the sum has the same property, with nonnegative coefficients. The exponential series is uniformly convergent on the compact bond space, so the crossing weight is a uniform limit of positive reflected sums.

The plaquette weights lying wholly on one side are absorbed into the functions on that side. The decoupled Haar functional is reflection positive; for bond variables fixed by the mirror the two factors are integrated against the same Haar variable. Thus the positive-kernel criterion of [FILS78, Theorems 2.1–2.2, pp. 5–6] applies. It gives finite-volume reflection positivity in the adapted frame.

The same calculation applies to coordinate mirrors, with the usual site or bond placement. It does not by itself identify which infinite-volume Gibbs states have the resulting inequalities.

Temporal layers and trace contractions. Fix a coordinate direction as time and put the finite-volume measure in temporal gauge. A layer variable consists of the spatial bonds in one time slice. For a finite block of those bonds the coupling of two neighbouring layers has kernel

$$k_\beta(u,u') = \prod_b \exp\Big[-\beta\left(1 - \tfrac{1}{2}\operatorname{tr}(u_b u_b'^{-1})\right)\Big]. \tag{231}$$

It is real, symmetric, continuous and strictly positive on the compact product of copies of SU(2). The layer space for a bond block $\mathcal{B}$ is $L^2(\mathrm{SU}(2)^{\mathcal{B}}, \prod_b du_b)$. The positivity needed for its convolution operator is spectral, not merely pointwise. It follows from the character expansion used in [L77, DESY76/54, §II.B, Proposition 1, pp. 8–9]. For one bond, with f the defining representation, write

$$e^{-\beta}\exp\left[\frac{\beta}{4}(\chi_f + \chi_{\bar f})\right] = \sum_\rho a_\rho \chi_\rho, \qquad a_\rho = e^{-\beta}\sum_{m,n\ge 0}\frac{(\beta/4)^{m+n}}{m!n!}N_\rho(m,n).$$

Here $N_\rho(m,n)$ is the multiplicity of ρ in $f^{\otimes m}\otimes \bar f^{\otimes n}$. Every term is nonnegative. Each irreducible representation of SU(2) occurs in one of these tensor powers, so $a_\rho > 0$ for $\beta > 0$. Peter–Weyl orthogonality makes convolution by this kernel a positive scalar $a_\rho/\dim\rho$ on the matrix entries of ρ. Tensoring over the bonds gives a positive operator $\mathcal{C}$ with no zero eigenvalue on any finite collection of Peter–Weyl blocks. Its inverse need not be bounded on the full layer space; only the inverse on such finite-dimensional subspaces is used below.

After the spatial plaquette weights and the gauge projection have been included, write $\mathsf{L} \ge 0$ for the torus transfer operator. For a torus with s layers, the partition function is $\operatorname{tr}\mathsf{L}^s$. If $\Xi_1,\dots,\Xi_s$ are contractions on the layer Hilbert space, the trace Hölder inequality gives

$$\left|\operatorname{tr}(\Xi_1\mathsf{L}\cdots\Xi_s\mathsf{L})\right| \le \prod_{i=1}^{s}\|\Xi_i\mathsf{L}\|_s \le \|\mathsf{L}\|_s^s = \operatorname{tr}\mathsf{L}^s. \tag{232}$$

Here $\|\cdot\|_s$ is the Schatten norm, and the finite-volume transfer representation makes the displayed traces finite. A block operator is extended by the identity on the other layer bonds before it is used in this trace. Identity operators are allowed among the Ξ_i.

The position of a local insertion matters. It is placed at the centre of the bond kernel. If

$$k_\Xi(u,u') = (\mathcal{C}^{1/2}\Xi\mathcal{C}^{1/2})(u,u')$$

is continuous and positive, its insertion is multiplication by $e^{A(u,u')} = k_\Xi(u,u')/k_\beta(u,u')$. The two half-kernels are absorbed into the layer weights on their respective sides. This is the factorisation that produces one L between consecutive operators in (232). An arbitrary bounded function of a pair of layers does not automatically arise from a contraction in this way.

The lower pressure tangent. Let A be real, continuous and local, with support in a cube of side ra. Let ν be a periodic Gibbs state, and choose a subgroup of its periods, denoted by P, over which to repeat A. Keep P fixed. On a torus with period lattice $\Gamma \subset \mathsf{P}$ and fundamental domain D, let $\mathbb{A}$ be the resulting finite array. Require

$$|v|_\infty > (r+1)a \qquad (0 \ne v \in \Gamma),$$

so the copies and their supports have the intended nonwrapping interpretation. Write $|D|$ for the number of sites, $|\mathbb{A}|$ for the number of copies of A, and $|\partial D|$ for the number of plaquettes meeting both sides of the boundary.

The DLR equation represents the restriction of ν to D as a mixture of finite-box measures with fixed exterior bonds. Replacing these exterior bonds by periodic boundary data changes only the crossing plaquettes. Since $0 \le a_p \le 2$, their Wilson energy costs at most a constant times $\beta|\partial D|$.

Copies of A meeting that boundary contribute at most their number times $|t|\|A\|_\infty$. A support cube and its translates meet at most the fixed power of r appearing in the boundary count. Jensen's inequality for the conditional measure therefore gives the lower tangent

$$\frac{1}{|D|}\log\frac{Z_D(t\mathbb{A})}{Z_D(0)} \geq t\frac{|\mathbb{A}|}{|D|}\nu(A) - Cr^5(\beta+|t|\|A\|_\infty)\frac{|\partial D|}{|D|} \qquad (t\in\mathbb{R}). \tag{233}$$

For a fixed local observable this is a surface error. The spacing, coupling and support radius are held fixed while D grows. Thus its last term tends to zero along boxes of bounded aspect ratio, in either the coordinate or the adapted frame. No estimate uniform as $a\to 0$ is needed at this step: the volume is removed first.

For an admissible centred-kernel insertion coming from a contraction, repeat it with a fixed period cell in compatible layers and apply (232). The corresponding normalised finite-volume partition function is at most one. Combining this upper bound with (233) at $t=1$ for the actual logarithmic insertion A gives, after the volume limit, $0\geq\delta\nu(A)$, where $\delta=\lim|\mathbb{A}|/|D|>0$ is the fixed copy density. Hence

$$\nu(A)\leq 0 \tag{234}$$

for each such real local logarithmic insertion. This conclusion uses the contraction representation of the entire inserted array after the residual gauge projection. It does not assert that e^{tA} comes from a contraction for every real t, or that every unprojected contraction is an admissible insertion.

Antisymmetric functions of adjacent layers. The preceding inequality has a useful two-sided consequence. Let $Y(u,u')$ be continuous, real and invariant under simultaneous gauge transformations in the two layers, with

$$Y(u',u)=-Y(u,u').$$

First suppose that Yk_β is a polynomial in matrix entries. Its integral operator K_Y has finite Peter–Weyl support. Kernel antisymmetry gives $\mathsf{K}_Y^*=-\mathsf{K}_Y$ on the real layer space. Consequently

$$\mathsf{o}=\mathcal{C}^{-1/2}\mathsf{K}_Y\mathcal{C}^{-1/2}$$

is well defined on that finite-dimensional support and is a real antisymmetric operator, extended by zero on its orthogonal complement. The operators $\Xi_\varepsilon=e^{\varepsilon\mathsf{o}}$ are orthogonal contractions for every real ε. The simultaneous gauge invariance of Y makes both $\mathcal{C}$ and K_Y intertwine the layer gauge action. The same is true of o and Ξ_ε. Thus these particular contractions are compatible with the residual gauge projection in the transfer representation.

Their inserted kernels have the uniform expansion

$$\mathcal{C}^{1/2}\Xi_\varepsilon\mathcal{C}^{1/2}=\mathcal{C}+\varepsilon\mathsf{K}_Y+O(\varepsilon^2).$$

The remainder is a continuous kernel on the same finite-dimensional blocks. Since k_β has a positive minimum on its compact domain, the new kernel is positive for both signs of sufficiently small ε. Its ratio to k_β has a real logarithm

$$A_\varepsilon=\varepsilon Y+O(\varepsilon^2), \qquad \|A_\varepsilon-\varepsilon Y\|_\infty\leq C_Y\varepsilon^2.$$

Using (234) for positive and negative ε gives

$$\varepsilon\nu(Y)\leq C_Y\varepsilon^2.$$

Divide by ε on each side of zero and let it tend to zero. It follows that $\nu(Y)=0$.

To remove the polynomial restriction, approximate Yk_β uniformly by matrix-entry polynomials, average under the simultaneous gauge action, and antisymmetrise under $u\leftrightarrow u'$. These operations preserve polynomial finite-dimensionality and do not increase the approximation error. Division by the fixed positive kernel gives functions tending uniformly to Y. Their expectations are zero by the argument just given, and hence so is $\nu(Y)$. Thus every admissible antisymmetric observable of two adjacent layers has mean zero.

Determining a stack from two layers. The adjacent-layer identity determines the full layer law through the local Gibbs equations. We adapt the two-layer determination argument of [YM, Lemma 103.2], reproducing it here in four dimensions. Fix $\beta > 0$, choose x_4 as time, and let $\mathcal{B}_3$ be the positively oriented bonds of the spatial lattice $a\mathbb{Z}^3$. Write

$$\mathcal{U} = \mathrm{SU}(2)^{\mathcal{B}_3}, \qquad \Omega_{\mathrm{lay}} = \mathcal{U}^{\mathbb{Z}}.$$

Thus an entire spatial layer is an element of $\mathcal{U}$; it has infinitely many bonds. All resampling sets below are finite. Let u_t be the spatial bonds at time ta, and let $w_t(x)$ be the vertical bond from (x, ta) to $(x, (t+1)a)$. For a fixed $t_0 \in \mathbb{Z}$ define the temporal gauge by

$$\widehat{w}_{t_0}(x) = 1, \qquad \widehat{w}_{t+1}(x) = \widehat{w}_t(x) w_t(x), \qquad \widetilde{u}_t = \widehat{w}_t \cdot u_t,$$

where $(h \cdot u)(x, y) = h(x) u(x, y) h(y)^{-1}$. The recursion determines $\widehat{w}_t$ in both directions. Every coordinate of the map $\mathtt{tg} : U \mapsto \widetilde{u}$ is continuous and depends on finitely many original bonds. All vertical bonds become 1. In particular

$$\widetilde{u}_{t_0} = u_{t_0}, \qquad \widetilde{u}_{t_0+1} = w_{t_0} \cdot u_{t_0+1}.$$

Write $U[\widetilde{u}]$ for the bond field with these spatial layers and all vertical bonds equal to 1. A gauge-invariant continuous local F satisfies

$$F(U) = \widetilde{F}(\mathtt{tg}(U)), \qquad \widetilde{F}(\widetilde{u}) = F(U[\widetilde{u}]).$$

For finite $S \subset \mathcal{B}_3$, denote the kernel (231) on this bond set by k_S. Let $V_S(u)$ be the sum of the spatial plaquette actions meeting S. Each spatial bond belongs to four spatial plaquettes, and every plaquette action belongs to $[0, 2]$. Hence

$$0 \le V_S \le 8|S|, \qquad e^{-2\beta|S|} \le k_S \le 1.$$

If $v \in \mathrm{SU}(2)^S$, let $\widetilde{u}^{(S,t;v)}$ replace the bonds S in layer t by v. The local layer resampling kernel is

$$\begin{aligned} q_{S,t}(v; \widetilde{u}) &= k_S(\widetilde{u}_{t-1}|_S, v) e^{-\beta V_S(v \vee \widetilde{u}_t|_{S^c})} k_S(v, \widetilde{u}_{t+1}|_S), \\ (\widetilde{\gamma}_{S,t}\psi)(\widetilde{u}) &= \frac{\int \psi(\widetilde{u}^{(S,t;v)}) q_{S,t}(v; \widetilde{u})\, dv}{\int q_{S,t}(v; \widetilde{u})\, dv}. \end{aligned} \tag{235}$$

Here dv is probability Haar measure. Both $q_{S,t}$ and its normalising integral lie between $e^{-12\beta|S|}$ and 1. Consequently $\widetilde{\gamma}_{S,t}$ maps continuous local functions to continuous local functions.

The image $\rho = \mathtt{tg}_* \nu$ of any Wilson Gibbs state satisfies

$$\rho(\widetilde{\gamma}_{S,t}\psi) = \rho(\psi) \quad (S \subset \mathcal{B}_3 \text{ finite},\ t \in \mathbb{Z},\ \psi \text{ continuous and local}). \tag{236}$$

Indeed, resample only the original spatial bonds corresponding to (S, t). This leaves every $\widehat{w}_s$ unchanged. The substitution $v = \widehat{w}_t \cdot U_{S,t}$ preserves Haar measure. The plaquettes meeting these bonds are precisely the spatial plaquettes meeting S and the two vertical plaquettes at each bond of S. Their weight is (235), since a vertical plaquette becomes $1 - \frac{1}{2}\operatorname{tr}(\widetilde{u}_t(b)\widetilde{u}_{t+1}(b)^{-1})$. The original DLR equation therefore gives (236). This is a local resampling identity; no conditional independence of entire infinite layers is being asserted.

The equations are covariant under every index reflection

$$(R_\ell \widetilde{u})_t = \widetilde{u}_{\ell - t}.$$

The spatial potential is unchanged and the symmetry of k_S exchanges the two neighbouring factors in (235). Thus $R_{\ell *}\rho$ obeys the same equations whenever ρ does.

Lemma 6.4. Let $\beta > 0$. Two probability laws on Ω_{lay} satisfying (236) and agreeing on two or more consecutive entire layers agree on all layers. ∎

Proof. Suppose that the two laws agree on layers $m, \dots, n$, with $m < n$. We first extend agreement to layer $n+1$. Fix a nonempty finite $S \subset \mathcal{B}_3$ and use the coordinates

$$u' = \widetilde{u}_{n+1}|_S, \qquad v = \widetilde{u}_n|_S, \qquad c = (\widetilde{u}_n|_{S^c}, \widetilde{u}_{n-1}, \dots, \widetilde{u}_m) \in Y.$$

The space Y is a compact countable product of copies of $\mathrm{SU}(2)$. Put

$$\Phi(v;c) = k_S(\widetilde{u}_{n-1}|_S, v) e^{-\beta V_S(v \vee \widetilde{u}_n|_{S^c})}, \qquad N(u',c) = \int k_S(v,u')\Phi(v;c)\,dv.$$

The condition $m < n$ ensures that $\widetilde{u}_{n-1}$ is among the coordinates of c. These functions are continuous and local, and

$$\Phi \geq e^{-10\beta|S|}, \qquad N \geq e^{-12\beta|S|}.$$

Let ρ_i' be the image of law i on (u',c) and define the finite positive measure $\sigma_i = N^{-1}\rho_i'$. Its mass is at most $e^{12\beta|S|}$. Equation (236) at (S,n) gives

$$\rho_i(\psi) = \int \sigma_i(du', dc) \int k_S(v,u')\Phi(v;c)\psi(u',v,c)\,dv \tag{237}$$

for continuous local functions of the displayed coordinates. Take

$$\psi(u',v,c) = \frac{\varphi(v)\chi(c)}{\Phi(v;c)},$$

where φ is a finite Peter–Weyl sum on $\mathrm{SU}(2)^S$ and χ is continuous and local on Y. This is a bounded continuous local function of the already equal layers $m, \dots, n$. Subtracting the two resampling identities gives

$$\int (\mathcal{C}_S\varphi)(u')\chi(c)\,d(\sigma_1 - \sigma_2)(u',c) = 0,$$

where $\mathcal{C}_S$ is convolution by k_S.

Every Peter–Weyl block of $\mathcal{C}_S$ has a strictly positive scalar eigenvalue, as proved above. Thus $\mathcal{C}_S$ maps the finite Peter–Weyl sums onto themselves: divide each of the finitely many coefficients by its eigenvalue. These sums are uniformly dense in $C(\mathrm{SU}(2)^S)$, and the continuous local functions are uniformly dense in $C(Y)$. Finite sums of their products are uniformly dense in $C(\mathrm{SU}(2)^S \times Y)$. Since $\sigma_1 - \sigma_2$ has finite total variation, the last displayed identity implies $\sigma_1 = \sigma_2$. There is no limit of inverse operators in this argument. The set S and each Peter–Weyl test are fixed when their finitely many eigenvalues are divided out; density is applied to the resulting finite measures.

Equation (237) now gives equality on all continuous local functions of (u',v,c). Given a local test on layers $m, \dots, n+1$, choose S to contain its finitely many coordinates on layer $n+1$. This proves equality on the entire block $m, \dots, n+1$. Reflecting the indices about $m+n$ and using covariance of the chain equations proves extension to layer $m-1$. Iteration in both directions gives agreement on every finite block. Continuous local functions determine a measure on the compact product Ω_{lay}, proving the result. □

Proposition 6.5. Every periodic Wilson Gibbs state on $a\mathbb{Z}^4$ is invariant under every coordinate bond reflection. It is consequently invariant under translations by $2a\mathbb{Z}^4$. ■

Proof. We first explain the residual gauge action, so that the adjacent-layer identity applies to a full two-layer marginal. The DLR equations give local gauge invariance of every Gibbs state. At one site, resampling its eight incident bonds and making the corresponding left and right Haar substitutions leaves every plaquette trace unchanged. Thus the DLR kernel gives the same expectation before and after that gauge transformation. Composition proves invariance for finitely supported gauge transformations, which suffices for any local observable.

Under an original gauge transformation $g_t(x)$, the temporal gauge satisfies

$$\widehat{w}_t^{\,g} = g_{t_0}\widehat{w}_t g_t^{-1}, \qquad \widetilde{u}_t^{\,g} = g_{t_0} \cdot \widetilde{u}_t.$$

Hence $\rho = \mathtt{tg}_*\nu$ is invariant under the simultaneous spatial gauge action on all layers. For a continuous local $H(\widetilde{u}_{t_0}, \widetilde{u}_{t_0+1})$, average over the gauge variables at the endpoints of its spatial bonds. Call the result H^{inv}. The average preserves its expectation under ρ, remains local, and commutes with exchanging the two layers. The adjacent-layer argument, applied to the real and imaginary parts of

$$H^{\mathrm{inv}}(u, u') - H^{\mathrm{inv}}(u', u),$$

therefore proves that the entire two-layer marginal of ρ is invariant under exchange. This use of the earlier insertion calculation requires only its gauge-compatible contractions.

Let $R = R_{2t_0+1}$. Both ρ and $R_*\rho$ obey the local chain equations. On layers t_0, t_0+1 their marginals are exchanged, so they agree by the preceding paragraph. Lemma 6.4 gives $R_*\rho = \rho$. In particular this equality holds for every observable on any fixed finite stack, with no restriction on its height.

For the original bond reflection $\theta = \theta_{e_4,(t_0+1/2)a}$ one has

$$\theta U[\widetilde{u}] = U[R\widetilde{u}].$$

The factorisation through temporal gauge therefore gives $\nu(\theta F) = \nu(F)$ for every gauge-invariant continuous local F. The local gauge average below extends this equality to all continuous local functions. These functions determine the bond-field measure, so $\theta_*\nu = \nu$.

The same argument applies with any coordinate chosen as time. Equivalently, apply it to the coordinate permutation of ν; covariance of the Wilson specification makes this another periodic Gibbs state, without assuming that ν is permutation invariant. The two bond reflections at $x_\mu = a/2$ and $x_\mu = 3a/2$ compose to translation by $2ae_\mu$. This proves the asserted periods.

If $\beta = 0$, the DLR equation on the finite support of a local function sets its expectation equal to its product Haar integral. It determines the unique product Haar Gibbs state, for which all these invariances hold as well. □

All steps were at fixed spacing and coupling. Periodicity entered the adjacent-layer pressure calculation; the local chain argument requires neither a torus limit representation of the state nor ergodicity. It therefore applies to every member of $\mathcal{G}^{\mathrm{per}}_{K,\beta}$, including its periodic ergodic components. The diagonal mirror inequalities are supplied by the adapted-torus pressure comparison below, after the common periods $2a\mathbb{Z}^4$ have been established.

For the reflection inequalities below, continuous local tests also suffice for bounded Borel local tests. Fix a mirror reflection θ and a finite bond set supporting F on one closed side. Approximate F by continuous functions F_j of those same bonds in L^2 for the sum of the corresponding marginal laws of ν and $\theta_*\nu$. Then both $F_j \to F$ and $\theta F_j \to \theta F$ in $L^2(\nu)$. Cauchy–Schwarz gives

$$\left|\nu(\overline{\theta F_j}F_j) - \nu(\overline{\theta F}F)\right| \le \|\theta(F_j - F)\|_2\|F_j\|_2 + \|\theta F\|_2\|F_j - F\|_2 \longrightarrow 0.$$

Thus positivity passes to F. In particular the weakly closed continuous-test inequalities used in the pressure comparison define the full local reflection-positive class.

From bond invariance to all mirror inequalities. To pass from bond reflection invariance on gauge-invariant local polynomials to all local functions, use local gauge invariance from the DLR equations. For a local F, let

$$F^{\mathrm{inv}}(U) = \int F(U^u)\,du,$$

where the Haar average is over gauge transformations at the endpoints of bonds in the support of F. This is still a local function, $\nu(F^{\mathrm{inv}}) = \nu(F)$, and the average commutes with reflection after the endpoint set is reflected. Equality on gauge-invariant functions therefore gives equality on all local functions. Compositions of neighbouring bond reflections then give translations by $2a\mathbb{Z}^4$. At this point those translations are periods of the state, whatever its originally specified periods were.

Now fix an equidistant mirror family. Repeat a real local A so that each copy lies in a slab bounded by adjacent mirrors and the period group takes each slab to the next. On a compatible adapted torus the chessboard estimate folds $\mathbb{A}$ into a reflection-symmetric array $\mathbb{A}^{\mathrm{f}}$ and gives

$$Z_D(t\mathbb{A}) \le Z_D(t\mathbb{A}^{\mathrm{f}}).$$

Write $\delta_D = |\mathbb{A}|/|D|$, and use the same number of copies in the folded array. For $t > 0$, convexity gives the explicit upper tangent

$$\frac{1}{|D|}\log\frac{Z_D(t\mathbb{A}^{\mathrm{f}})}{Z_D(0)} \le \frac{t}{|D|}\frac{d}{dt}\log Z_D(t\mathbb{A}^{\mathrm{f}}) = t\delta_D\,\nu^{\mathrm{f}}_{D,t}(A).$$

Here $\nu^{\mathrm{f}}_{D,t}$ is the finite-volume measure with the folded perturbation. The last equality reads A in its chosen representative slab: the other copies have the same expectation by the period and reflection symmetries of the folded array. Combining this bound, the folding inequality and (233) yields

$$\nu(A) \le \nu^{\mathrm{f}}_{D,t}(A) + \frac{Cr^5(\beta + t\|A\|_\infty)}{t\delta_D}\frac{|\partial D|}{|D|}.$$

The repetition group is fixed, so δ_D tends to a strictly positive density. Let the volume tend to infinity at fixed t. A subsequential limit ν^{f}_t retains the folded-array reflection inequalities and satisfies the Gibbs specification with that fixed periodic perturbation. Next let $t \downarrow 0$. On every fixed finite region the added bounded local interaction tends to zero, so the conditional kernels tend to the unperturbed Wilson kernels. Weak limit points are therefore Wilson Gibbs states. Reflection positivity and mirror invariance are closed, giving a comparison state in the class

$$\mathcal{C}_\Sigma = \{\rho \in \mathcal{G}^{\mathrm{per}}_{K,\beta} : \rho \circ \theta_\Sigma = \rho,\ \rho \text{ is reflection positive at } \Sigma\}.$$

The compatible slabs and periods are fixed during this argument. The surface error is removed before its factor $1/t$ can cause a problem. Weak limits preserve the finite-region DLR identities and the fixed periods of the folded array. The temporal reflection argument then gives their $2a\mathbb{Z}^4$ periods as well. The conclusion is

$$\nu(A) \le \sup_{\rho\in\mathcal{C}_\Sigma} \rho(A). \tag{238}$$

The comparison state may depend on A. The compatible slabs and periods are chosen before the upper-tangent calculation, and remain fixed while the volume is removed.

The slab restriction must also be removed. Translate a given local A, possibly straddling Σ, by a vector in $2a\mathbb{Z}^4$ into one side and take the slabs wide enough to contain that translate. Bond invariance makes this translation a period of both the original and the comparison states. Thus the comparison also holds for the untranslated A. Reflection positivity is a family of closed linear inequalities in the state; mirror invariance and the DLR equations are closed as well. By the bond-reflection argument, every periodic Gibbs state is already invariant under $2a\mathbb{Z}^4$. Conversely such a Gibbs state is periodic. The class can therefore be written using this single fixed period group, whose invariance is closed under weak limits on the compact bond configuration space. The finite-range DLR identities are also closed, so this Gibbs class is compact and convex. Its closed mirror-invariant, reflection-positive subclass $\mathcal{C}_\Sigma$ is consequently compact and convex. If ν lay outside it, separation would give a real local continuous A with $\nu(A) > \sup_{\rho\in\mathcal{C}_\Sigma}\rho(A)$, contrary to (238).

This pressure and separation argument treats local tests crossing the mirror as well as tests on one side. Together with proposition 6.5, it gives the lattice symmetries and mirror inequalities used in the source comparison and phase selection.

Finally, site and bond mirrors whose positions differ by $a/2$ generate translation by a in a coordinate direction. Coordinate and diagonal mirrors generate the cubic isometries. Applying the comparison to these mirror families gives the lattice symmetries claimed at the start of this argument.

6.7. **The lattice spectral inequality.** The coordinate bond mirror $\{x_4 = -a/2\}$ gives a positive contraction T_a, and spatial translations give commuting unitaries $U_j = e^{iaP_j}$. The site-mirror space at $\{x_4 = 0\}$ has a positive contraction T_a^{s}. The relation between the two reflection forms is

$$\langle [F], [G] \rangle = \langle [F]^{\mathrm{s}}, T_a^{\mathrm{s}} [G]^{\mathrm{s}} \rangle.$$

Thus $[F] \mapsto (T_a^{\mathrm{s}})^{1/2}[F]^{\mathrm{s}}$ identifies the bond space with the closure of the range of $(T_a^{\mathrm{s}})^{1/2}$. This closed subspace is $(\ker T_a^{\mathrm{s}})^{\perp}$, on which the restriction of T_a^{s} is injective. Consequently T_a is injective and $T_a = e^{-aH}$ defines a nonnegative self-adjoint operator H. The site space may have a zero-transfer subspace; none of the arguments below identifies that subspace with finite energy.

Proposition 6.6 (The lattice cone)**.** The coordinate and diagonal reflection operators satisfy

$$(1 - T_a)^2 \geq T_a(2 - U_j - U_j^*). \tag{239}$$

Equivalently, their joint spectral variables satisfy

$$\cosh(aE) + \cos(aq_j) \geq 2.$$

■

Proof. The argument compares the coordinate and diagonal reflection pictures of the same correlation. Fix j and write $U = U_j$. The displacement $ka(e_4 + e_j) + la(e_4 - e_j)$ is represented in the coordinate picture by $T_a^{k+l} U^{k-l}$. In the picture of a diagonal mirror, the first diagonal step is a positive contraction T_+ and the other is a unitary V_-. The two commute because they represent translations.

Choose an integer $R \geq 1$ and let A be local in $\{0 \leq x_4 \leq Ra,\ |x_j| \leq Ra\}$, and first translate its reflection vector by Ra in time. Its joint spectral measure for (T_a, U) is a finite positive measure on $[0,1] \times S^1$. Let ς be its image under $(t, \zeta) \mapsto t\zeta$. Expanding a polynomial Q in monomials expresses $\int |Q|^2 \, d\varsigma$ as a finite sum of correlations of the displacements just described. For the diagonal mirror $\Sigma_d = \{x_4 + x_j = Ra\}$, write θ_d for its reflection and θ_l for reflection in $\{x_4 = -a/2\}$. The outer vectors are

$$\psi'' = [\tau_{2Rae_4} A]_d, \qquad \psi' = [\theta_d \theta_l A]_d.$$

Both representatives lie on the same allowed side of Σ_d. Expanding the polynomial gives the identity

$$\int |Q|^2 \, d\varsigma = \langle \psi', Q(T_+) \overline{Q}(V_-) \psi'' \rangle,$$

where $\overline{Q}$ denotes coefficientwise conjugation of Q. The resulting bound is

$$\int_{\mathbb{D}} |Q(z)|^2 \, d\varsigma(z) \leq C_A \sup_{[0,1]} |Q| \sup_{S^1} |Q|. \tag{240}$$

Indeed the diagonal expression is a pairing with $Q(T_+)\overline{Q}(V_-)$ between those two vectors; the spectral theorem bounds its operator norm by the product of the two suprema. The number C_A is the product of the outer-vector norms and is independent of Q.

It remains to extract the support information in this inequality. For $s > 0$ put

$$f_s(z) = \frac{\cos\left(s \operatorname{Log} \frac{1+\sqrt{z}}{1-\sqrt{z}}\right)}{\cosh(\pi s/2)}, \qquad \phi(z) = \left| \operatorname{Im} \operatorname{Log} \frac{1+\sqrt{z}}{1-\sqrt{z}} \right|.$$

Changing the branch of $\sqrt{z}$ changes the logarithm's sign and does not change its cosine. Its power series at zero contains only integer powers of z, so f_s is holomorphic in the disc. The fraction inside the logarithm maps the right half of the square-root disc to the right half-plane. In particular $0 \leq \phi < \pi/2$. The elementary formulas for the cosine of a complex number give

$$|f_s(z)| \leq 1, \qquad \sup_{0 \leq z < 1} |f_s(z)| \leq \frac{1}{\cosh(\pi s/2)}, \qquad |f_s(z)| \geq \frac{\sinh(s\phi(z))}{\cosh(\pi s/2)}.$$

For $0 < \epsilon < 1$ the Taylor polynomials of $f_s((1-\epsilon)z)$ converge uniformly on the closed unit disc. Since ς is finite, they may be inserted in (240) and then replaced by their uniform limit. It follows that

$$\int |f_s((1-\epsilon)z)|^2 \, d\varsigma(z) \le \frac{C_A}{\cosh(\pi s/2)}.$$

On a compact subset where $\phi((1-\epsilon)z) > \pi/4 + \delta$, the lower bound for the integrand has order $e^{(-\pi/2+2\delta)s}$, whereas the right side has order $e^{-\pi s/2}$. Letting $s \to \infty$ shows that the measure of that compact subset is zero. Then let $\epsilon \downarrow 0$ and exhaust the part of the closed disc where $\phi > \pi/4$ by compact subsets. Hence

$$\operatorname{supp} \varsigma \subset \{z \in \overline{\mathbb{D}} : |1+z|^2 \ge 4|z|\}.$$

Here the boundary curve follows directly from the logarithm formula: $\phi = \pi/4$ is the inverse image of the rays of arguments $\pm\pi/4$; eliminating $\sqrt{z}$ gives $|1+z|^2 = 4|z|$.

For $z = t\zeta$ one has

$$|1+t\zeta|^2 - 4t = (1-t)^2 - t|1-\zeta|^2.$$

Thus every time-translated local vector used above has spectral measure supported where $(1-t)^2 \ge t|1-\zeta|^2$. For each local A, choose R large enough for its support. The spectral projection of the complementary open set annihilates $T_a^R[A]$. It commutes with T_a^R, which is injective, so it annihilates $[A]$ as well. Local vectors are dense; hence this projection vanishes. The spectral theorem gives (239). □

At a fixed spacing this implies a linear cone with a constant smaller than one. Taking aq_j in $[-\pi, \pi]$, the curved lower boundary is

$$aE \ge \operatorname{arccosh}(2 - \cos(aq_j)), \qquad E \ge k_*|q_j|, \qquad k_* = \frac{2}{\pi}\log(1+\sqrt{2}).$$

The positive constant k_* is sufficient in the spectral estimate for clustering. Complex rotation requires the constant-one inequality in the continuum, for which one retains the curved bound until after the cutoff is removed.

6.8. **The constant-one cone in the continuum.**

Proposition 6.7 (The continuum coordinate cones)**.** Take a common limit of the separated moments, also allowing evaluation in states ρ_j with subtraction at reference states ν_j. The reflection-space semigroup and spatial translations satisfy

$$H \ge |P_j| \qquad (j = 1, 2, 3). \tag{241}$$

■

Proof. The reflection-space construction is the positive-time translation construction of [OS73, §4.1, pp. 90–92], applied to the moments on separated tests. We use its bounded semigroup operators, rather than asserting convergence of the unbounded lattice generators.

By proposition 6.6, $\sinh(aE/2) \ge |\sin(aq_j/2)|$ on the lattice spectrum. For an integer $k \ge 1$, the inequalities $\sinh(ku) \ge k \sinh u$ for $u \ge 0$ and $|\sin(kv)| \le k|\sin v|$ give the same assertion with a replaced by ka. Put $t = ka$. The polynomial operator inequality also holds in the site picture: the bond/site identification proves it on $(\ker T_a^{\mathrm{s}})^\perp$, and on $\ker T_a^{\mathrm{s}}$ it reduces to $1 \ge 0$. In that site picture, evaluated at a local reflection vector represented by F, it is the finite inequality

$$u(0) - 4u(te_4) + u(2te_4) + u(te_4 + te_j) + u(te_4 - te_j) \ge 0, \tag{242}$$

where $u(v) = \rho(\overline{\theta F}\, \tau_v F)$. All the correlations here have separated reflected supports.

Fix a positive L-adic physical displacement t. At sufficiently fine cutoffs it is an integer multiple of the lattice spacing. The lattice representatives of F are exactly covariant under these translations, because the density is evaluated at plaquette barycentres. Moment convergence therefore passes every term of (242) to the continuum. This also holds for the common reference-state subtraction:

the same finite reflection inequality holds in each evaluating state ρ_j. Strong continuity extends the resulting bounded-operator inequality from L-adic t to every $t > 0$.

Let $\mathcal{P}$ be the continuum joint spectral measure of (H, P_j). The inequality says that

$$(1 - e^{-tE})^2 - e^{-tE}(2 - e^{itq} - e^{-itq}) \geq 0$$

for $\mathcal{P}$-almost every (E, q), for every positive L-adic t. Use the countability of these t to choose one common full-measure spectral set. At each of its points divide by $4e^{-tE}(t/2)^2$ and let $t \downarrow 0$. This gives $E^2 \geq q^2$, and $E \geq 0$ gives $E \geq |q|$. This proves (241) without any exchange of an unbounded generator with the cutoff limit. □

7. From cubic symmetry to rotations

The lattice group supplies quarter-turns. To obtain the target theorem we must recover every angle. Work first in the plane with coordinates (x_4, x_1). The argument combines three estimates: the constant-one continuum cone, an insertion estimate independent of the number of other fields, and a short-distance two-point bound that improves a pure power by a factor tending to zero. Diagonal positivity supplies one further sign at the end of the argument.

For a component with mean $b \int f$, apply this argument to the centred functional. Restoring the invariant mean by (286) preserves all its rotation identities. In this section write σ for a centred limiting moment functional and $\|F\|_{(c)} = \|[\tau_{-ce_4}F]\|$ whenever the translated polynomial remains supported at positive times.

7.1. **An insertion estimate uniform in the other fields.** The first step is to bound a local insertion between arbitrary reflection vectors. Let $0 < \eta \leq 1$. Place h in a ball of radius $c\eta$ near the time-zero plane, with $|x_4| \leq \eta/8$, and let F, G be finite polynomials in separated tests supported in $x_4 > \eta/4$. The required estimate is

$$|\sigma(\Theta F \cdot (h) \cdot G)| \leq C(g)\eta^{-p_*}\|h\|_\sharp \|F\|_{(\eta/4)}\|G\|_{(\eta/4)}. \tag{243}$$

The constants must be independent of the degrees of F and G.

The proof combines source normalization with repeated reflections. First subtract the scalar source energies separately in each cell of side $l \asymp \eta$. Locality then makes the interaction between cells cost only a power of l^{-1}. Next place the normalized insertion in a long reflected column. Taking a root removes the two endpoint observables, including their potentially large supremum norms. The second operation is what makes the answer independent of the number of outer fields. The quantitative work is to carry the one-cell cancellation through all scales $a_K \ll l \leq 1$ with the same source induction; a bound for unit cells alone would not suffice.

We carry out the argument for real h. Splitting a complex test into its real and imaginary parts gives the same conclusion by linearity, with a harmless fixed enlargement of the constant.

7.1.1. *Source energies on a subunit cell.* Take a chessboard with side $l = L^{-i} \asymp \eta$ and $a_K \leq l/L^2$, whose faces are bond mirrors. Suppose first that h is supported in the middle of one cell, with a fixed positive fraction of l separating its support from every face. Let h^+ and h^- be its reflected arrays on even- and odd-parity cells, respectively. The repeated complex source is

$$W_z = zh^+ + \bar{z}h^-.$$

Copies belonging to either array have disjoint supports. Their supremum and Lipschitz bounds are those of h, including at a cell face because the source vanishes in a neighbourhood of that face. Hence

$$\|W_z\|_\flat \leq 2|z|\|h\|_\sharp.$$

This bound is independent of $N = (\ell/l)^4$, the number of cells. We use two array functions on the source domain, rather than N source variables with their norms added together.

Apply upper stability and the unsourced floor on T_ℓ itself. The vacuum factor is the same in numerator and denominator, so it cancels before any estimate of vacuum densities. The quotient and its chessboard root give

$$\|e^{z\mathcal{O}_a(h)}\|_{\mathrm{ch},\ell} = \left[\frac{Z_\ell(W_z)}{Z_\ell(0)}\right]^{1/N} \le \exp\left[\frac{1}{N}\operatorname{Re}\mathcal{E}_\ell(W_z) + C_g l^4\right], \qquad |z|\|h\|_\sharp < c\rho.$$

The repeated numerator is nonnegative by reflection positivity. This application of stability requires neither a reference torus nor the finite-torus transfer estimate.

For an enlarged domain $X^{\sim 1}$ let its copy set consist of the reflected supports of h that it meets. A term with an empty copy set is zero. Split the scalar energies into those whose copy set has one element and those whose copy set has at least two elements. For a term of the first kind, locality leaves a holomorphic function of z if the copy is even, and of $\bar z$ if it is odd. Conjugate the latter term, sum the resulting holomorphic functions and divide by N. Denote the result by $E_{h;\ell}(z)$. It is zero at zero and satisfies the exact cancellation

$$\frac{1}{N}\operatorname{Re}\mathcal{E}_\ell(W_z) - \operatorname{Re}E_{h;\ell}(z) = \frac{1}{N}\operatorname{Re}\sum_j \sum_{X:\ X^{\sim 1} \text{ meets at least two copies}} e_j(X;W_z).$$

One-copy domains can cross a face, provided they reach no second support; all remain in $E_{h;\ell}$. Their cutoff-dependent scalar contributions cancel as in the construction of (196).

We now sum the domains in the remaining class. Put $t_j = L^{K-j}/M$, the inverse physical side of a localization cube. At a fine scale, $lt_j \ge 1$, a domain meeting two copies has $d_j(X) \ge clt_j - C$. Charge it to one of the copies it meets and anchor it in one of the at most $(Clt_j + C)^4$ cubes near that support. The tree sum gives, after division by N,

$$C\sum_{lt_j \ge 1}(Clt_j + C)^4 e^{-c[clt_j - C]_+} \le C.$$

This is a geometric-scale sum. Once lt_j exceeds a fixed constant its exponential decay dominates the fourth-power count; the remaining terms form a bounded finite segment of that sequence.

At a coarse scale, $lt_j < 1$, count all localization cubes on the torus before dividing by N. The local energy bound and the tree bound give a cost $C\ell^4 t_j^4$ for all domains of that level. Thus their cost per cell is $C(lt_j)^4$, and

$$\frac{1}{N}\sum_{lt_j<1} C\ell^4 t_j^4 = C\sum_{lt_j<1}(lt_j)^4 \le C.$$

This global count is stronger than anchoring separately at every copy at a scale whose cubes contain many cells. In the latter count the additive constant in $(Clt_j + C)^4$ would cost one constant at each remaining level. Counting the cubes first is the reason that the subunit-cell cost is summable even as l tends to zero.

Combining the two scale ranges with the stability term yields

$$\left\|e^{z\mathcal{O}_a(h) - E_{h;\ell}(z)}\right\|_{\mathrm{ch},\ell} \le Q(l), \qquad Q(l) \le C(g)l^{-C}, \qquad |z|\|h\|_\sharp < c\rho. \tag{244}$$

The displayed polynomial bound is sufficient for the insertion argument; the two scalar-energy sums above are bounded even before allowing this polynomial loss. Their use requires the local source-energy and stability estimates uniformly for the two reflected array functions. Establishing that shared array induction is the source construction needed here. The cancellation and counts then give the cell estimate at the actual side l.

The difference from the graded estimate for many active sources is useful. There the coarse face domains are charged separately to active pieces of radius η, which gives a logarithmic number of bounded contributions per piece. Here the same one-copy pattern fills every cell, and the global cube count divided by N gains the factor $(lt_j)^4$ at the coarse scales. The normalizers for the two arguments serve different purposes and need not agree.

The geometric part of the chessboard argument is unchanged at $l < 1$. Dilation by L^i sends this board to a unit board and the lattice to the one of cutoff $K - i \geq 2$. It preserves the seminorm, reflection inequalities and fixed-cutoff Gibbs domination. It is not used to transfer an RG stability estimate from another datum: the stability and energy calculation just made was performed directly on the original torus.

At a fixed K, the functions $E_{h;\ell}$ are bounded on compact subdiscs, uniformly in ℓ. To see the volume independence, sum the local energy bounds through each localization cube and divide their total $C\ell^4 t_j^4$ by N. This bounds the contribution of level j by $Cl^4 t_j^4$. The sum over the finitely many levels is finite at this K, though it can increase with K. The same bound holds on a slightly larger source disc than a given compact subdisc. Montel's theorem therefore gives a sequence $\ell_r \to \infty$ and a holomorphic limit E_h, with locally uniform convergence and $E_h(0) = 0$. The sequence and limit may depend on K, β, h and l. They are chosen from finite tori before a Gibbs state or an outer reflection vector is selected.

For $D_z = e^{z\mathcal{O}_a(h) - E_h(z)}$, homogeneity gives the exact relation

$$\|D_z\|_{\mathrm{ch},\ell_r} = |e^{E_{h;\ell_r}(z) - E_h(z)}| \left\| e^{z\mathcal{O}_a(h) - E_{h;\ell_r}(z)} \right\|_{\mathrm{ch},\ell_r}.$$

The first factor tends to one locally uniformly. Hence

$$q_{D_z} := \limsup_r \|D_z\|_{\mathrm{ch},\ell_r} \leq Q(l). \tag{245}$$

This is the selected sequence throughout the column argument. No comparison of $E_{h;\ell}$ with E_h is needed outside it. Gibbs product domination in section 4 holds along every chosen volume sequence, and so applies along this one for every periodic Gibbs state.

7.1.2. *The column estimate at a fixed cutoff.* We now bound insertion of D_z between arbitrary outer reflection vectors by its chessboard seminorm. The outer factors enter through endpoint supremum norms at this fixed cutoff.

Fix a periodic Gibbs state ν' at this cutoff. For functions above a mirror at time t, use the positive form

$$\langle B, A\rangle_{\nu',t} = \nu'(\overline{B \circ \theta_t^*} A).$$

Let D be supported in a cell immediately above this mirror, between times t and $t + l$. On the space of bounded local functions above the mirror put

$$\widehat{D}A = D\,\tau_{le_4} A, \qquad D^\dagger = \overline{D \circ \theta^*_{t+l/2}}.$$

Reflection followed by translation shows that the formal adjoint of $\widehat{D}$ is $\widehat{D^\dagger}$. In fact reflection of $D^\dagger$ in the lower face is the conjugate of D translated by $-l$, and the same translation carries the reflected upper observable to the required side. Translation invariance of ν' then gives

$$\langle B, \widehat{D}A\rangle_{\nu',t} = \langle \widehat{D^\dagger} B, A\rangle_{\nu',t}.$$

All expressions here are finite products of bounded local functions; boundedness on the Hilbert quotient has not been assumed.

Set $K_D = \widehat{D^\dagger}\widehat{D}$ and $m_n = \langle A, K_D^n A\rangle_{\nu',t}$. These numbers are nonnegative. The reflected Cauchy–Schwarz inequality makes them log-convex: $m_n^2 \leq m_{n-1} m_{n+1}$. For odd n use the vectors $K_D^{(n-1)/2} A$ and $K_D^{(n+1)/2} A$ in the original form. For even n, use the corresponding two vectors after applying $\widehat{D}$. Their scalar product is m_n, and their squared norms are the two neighbouring moments. It follows that

$$m_1^n \leq m_0^{n-1} m_n.$$

The formula remains valid if a moment is zero, by Cauchy–Schwarz.

The expression for m_n is a column of $2n$ successive factors, alternating D and its reflected conjugate, with A at one end and its reflected conjugate at the other. Choose a coarse chessboard

whose side $L^{k_0}l$ contains the support of A in its first cell above the mirror, and whose faces are also fine-board faces. Take n through multiples of L^{k_0}. The ends of the column then lie on coarse faces, with the two endpoint functions in the exterior coarse cells and every D factor inside the column.

For each fixed n, apply Gibbs product domination along the selected sequence $\ell_r \to \infty$. Endpoint seminorms are bounded by $\|A\|_\infty$. On the interior coarse cells, the fine-to-coarse comparison bounds the seminorm of the product of fine-cell factors by the product of their fine seminorms. Reflection and conjugation leave that seminorm unchanged. Thus

$$m_n \le \|A\|_\infty^2 q_D^{2n}, \qquad q_D = \limsup_r \|D\|_{\mathrm{ch},\ell_r}.$$

Combining this inequality with log-convexity gives

$$\|\widehat{D}A\|_{\nu',t}^{2n} \le \|A\|_{\nu',t}^{2n-2}\|A\|_\infty^2 q_D^{2n}. \tag{246}$$

Take the $2n$th root and then let $n \to \infty$ through the allowed sequence. At this fixed cutoff A is bounded, so its supremum to power $1/n$ tends to one. Consequently

$$\|\widehat{D}A\|_{\nu',t} \le q_D\|A\|_{\nu',t}, \qquad |\nu'(\overline{B\circ\theta_t^*}D\tau_{le_4}A)| \le q_D\|B\|_{\nu',t}\|A\|_{\nu',t}.$$

The limit order matters. The volume limit is taken for each fixed K, n, then the column length tends to infinity at that fixed K, and only later is K removed. For a polynomial of degree d in the lattice density, $\|A\|_\infty$ can grow like a_K^{-4d}. Its disappearance before the cutoff limit is what yields a bound independent of the outer degrees. The coarse side may depend on A, but no such dependence survives in this inequality.

7.1.3. *Differentiation, centring, and the continuum limit.* Apply the column estimate to the holomorphic family D_z from (245). Its scalar matrix elements are holomorphic and bounded by $Q(l)\|B\|_{\nu',t}\|A\|_{\nu',t}$ on $|z|\|h\|_\sharp < c\rho$. Cauchy's formula at zero therefore gives

$$\left|\nu'\left(\overline{B\circ\theta_t^*}[\mathcal{O}_a(h) - E_h'(0)]\tau_{le_4}A\right)\right| \le C\rho^{-1}Q(l)\|h\|_\sharp\|B\|_{\nu',t}\|A\|_{\nu',t}.$$

The same normalizing function works in any second periodic Gibbs state ν. Gibbs domination and the local logarithm estimate, with one source, give

$$|\nu(\mathcal{O}_a(h)) - E_h'(0)| \le C\rho^{-1}Q(l)\|h\|_\sharp.$$

Subtracting this scalar instead changes the insertion form by at most the same bound: the intervening lattice time translation is a contraction in the reflection norm. We have obtained the estimate with centring at ν, even when the expectation is in ν'.

For this lattice contraction use the norm argument of proposition 6.3 at integer time steps. Its iterated Cauchy–Schwarz bound requires only bounded translated self-pairings, which here are at most the squared supremum of the fixed bounded local function. No continuum strong-continuity assertion is needed before taking this cutoff limit.

To use this in a physical slab of width η, choose l comparable to η with room on either side of the inserted support. A bounded number of tents of mesh comparable to η puts the pieces of h in the middle of such cells. The sum of their sharp norms costs at most a fixed power of $\eta^{-1}\|h\|_\sharp$. The translations needed to fit the cells consume only part of the slab. Remaining positive time translations are contractions, so the norms of the outer vectors are bounded by their norms after the shift $-(\eta/4)e_4$. The power $Q(l) \le C(g)l^{-C}$ absorbs these fixed geometric losses.

Finally take the common continuum subsequence. At fixed outer polynomials every matrix element and every squared outer norm is a finite collection of separated moments. The state-uniform per-ball bounds and mean transfer make their small geometric shifts equicontinuous and pass the inequality to the limiting form. The constant in the result comes from the cell estimate and the source radius; it is independent of the number of fields in the outer polynomials. This is the column and limit argument for (243); the quantitative source input is the local energy and stability induction for the reflected arrays, used in the direct derivation of (244).

7.2. **From smooth insertions to plates.** The insertion estimate and the constant-one cone from section 6 turn smooth local fields into bounded planar insertion forms. We spell out the time margins, since these are needed again when several insertions are multiplied.

Write planar displacements as $v = (v^0, v^1)$, where v^0 is time, with planar time unit vector $e_0 = (1, 0)$, and put $v_\pm = v^0 \pm iv^1$. The joint spectral representation is

$$e^{-v^0 H + iv^1 P_1} = \exp\!\left(-v_+ \frac{H - P_1}{2} - v_- \frac{H + P_1}{2}\right).$$

By (241) this is a holomorphic contraction on

$$\mathbb{V} = \{v \in \mathbb{C}^2 : \operatorname{Re} v_+ > 0, \operatorname{Re} v_- > 0\}. \tag{247}$$

The insertion estimate defines a bounded operator between the time-translated reflection vectors. Keeping its scale explicit prevents using up the time margin twice.

In detail, let h be supported in a ball about zero of radius $c\eta$. At scale $\eta/2$ define $B_{h;\eta/2}$ by

$$\Big\langle [\tau_{-(\eta/8)e_4} F'], B_{h;\eta/2} [\tau_{-(\eta/8)e_4} G'] \Big\rangle = \sigma(\Theta F' \cdot (h) \cdot G'),$$

for F', G' supported in $x_4 > \eta/8$. Then $\|B_{h;\eta/2}\| \le C\eta^{-p_*}\|h\|_\sharp$. For F, G supported in $x_4 > \eta/4$, put $\Phi = [\tau_{-(\eta/4)e_4} F]$ and $\Psi = [\tau_{-(\eta/4)e_4} G]$. The exact translation identity is

$$\sigma(\Theta F \cdot (\tau_x h) \cdot G) = \Big\langle \Phi, T_4(\tfrac{\eta}{8} e_4 + x) B_{h;\eta/2} T_4(\tfrac{\eta}{8} e_4 - x) \Psi \Big\rangle, \tag{248}$$

initially for real x with $|x_4| < \eta/8$. Here $T_4(z) = e^{-z_4 H + i \sum_{j=1}^3 z_j P_j}$. Since each $|P_j| \le H$, this operator is a contraction when $\operatorname{Re} z_4 \ge \sum_j |\operatorname{Im} z_j|$. Consequently (248) continues to

$$|\operatorname{Re} x_4| + \sum_{j=1}^3 |\operatorname{Im} x_j| < \eta/8,$$

which contains polydiscs of radius $c\eta$ in all four variables. The operator at scale η would not give the identity after the translation, because the outer fields then have only half that separation available.

Lemma 7.1 (The integral norm of a plate)**.** In the reflection space just constructed, fix $\varphi \in C_c^\infty(\mathbb{R}^2)$ in the transverse coordinates (x_2, x_3). For $\alpha \in C_c^\infty(\mathbb{R}^2)$ supported in $|x_4| < c\eta$, with c fixed sufficiently small, one has

$$|\sigma(\Theta F \cdot (\alpha \otimes \varphi) \cdot G)| \le C_\varphi \eta^{-w_*} \|\alpha\|_{L^1(\mathbb{R}^2)} \|F\|_{(\eta/4)} \|G\|_{(\eta/4)}, \qquad w_* = p_* + 5. \tag{249}$$

The insertion form has a real-analytic representative in the planar position. Smooth approximate identities concentrating at a point within the allowed time strip, with uniformly bounded L^1 norms, converge to its value there. At a fixed smaller time margin, the point limit can be taken in operator norm. ∎

Proof. The insertion estimate initially controls one derivative of a test function. We remove that derivative by a local parametrix, using the holomorphic translations in (248) to pay for the derivatives that the parametrix introduces.

In four dimensions a fundamental solution of Δ^3 is a constant multiple of $|x|^2 \log |x|$. It is C^1 at zero. Let $r = c_1\eta$, where c_1 is small enough that the ball of radius $2r$ fits inside the available translation margin. After a smooth cutoff at scale r, and subtraction of the multiple of $|x|^2 \log r$, choose $n_r \in C_c^1(\mathbb{R}^4)$ such that

$$\Delta^3 n_r = \delta_0 - s_r, \qquad h = (\Delta^3 h) * n_r + h * s_r.$$

The function s_r is smooth and supported away from zero in a ball of radius $2r$. Rescaling the cutoff gives

$$\|n_r\|_\sharp \le CMr, \qquad \|s_r\|_\sharp \le CMr^{-5}.$$

Indeed $n_r(rx)$ is r^2 times a fixed C^1 function; its first derivative costs r^{-1}. Applying six derivatives produces the r^{-4} scaling of s_r, and its first derivative costs one further power. The subtraction of $|x|^2 \log r$ removes a harmless polynomial term and makes these scaling bounds uniform as $r \downarrow 0$.

Fix the outer vectors and write $L(h) = \sigma(\Theta F \cdot (h) \cdot G)$. The insertion estimate extends this form from smooth tests to compactly supported C^1 tests by C^1 approximation. Smooth approximations to n_r may be chosen with their supports in the same small ball. Their translation forms have uniform bounds on every smaller polydisc and converge there to the holomorphic form for n_r. For every real centre y whose time coordinate lies in the smaller strip, the map $z \mapsto L(\tau_{y+z} n_r)$ is holomorphic on a four-dimensional polydisc of radius comparable to η. By the insertion bound its modulus is at most

$$C\eta^{-p_*} \|n_r\|_\sharp \|F\|_{(\eta/4)} \|G\|_{(\eta/4)}.$$

The constants are independent of the spatial centre y: spatial translations in the reflection space are unitary. Cauchy's estimates therefore bound its derivatives of total order six by the same expression times $C\eta^{-6}$.

Apply L to the parametrix identity and integrate by parts six times. The derivatives of h move onto the holomorphic function of its translation centre. The first term is bounded by

$$C\eta^{-p_*}\eta^{-6} \|n_r\|_\sharp \|h\|_{L^1} \|F\|_{(\eta/4)} \|G\|_{(\eta/4)};$$

the second is bounded directly by the insertion estimate with s_r. Since $r = c_1\eta$, both bounds are at most

$$C\eta^{-p_*-5} \|h\|_{L^1(\mathbb{R}^4)} \|F\|_{(\eta/4)} \|G\|_{(\eta/4)}.$$

Taking $h = \alpha \otimes \varphi$ proves (249), with the fixed L^1 norm of the transverse label absorbed into C_φ.

The same representation expresses L as integration against a real-analytic function of the centre on every smaller strip: the sixth derivatives of the translated n_r form and the translated s_r form both have the holomorphic extensions just used. Their integral representation identifies the density of L there. After integrating in the transverse variables against φ, it gives a real-analytic planar density. A smooth approximate identity with bounded integral norm consequently converges to that density at every point in the strip. This justifies pointlike planar insertions, which we call plates. They remain smeared in x_2, x_3.

To obtain operator-norm convergence, keep a smaller strip in the same holomorphic representation. One further Cauchy estimate bounds the first planar derivative of its density $k_{F,G}$ by

$$|\nabla_y k_{F,G}(y)| \le C_\varphi \eta^{-w_*-1} \|F\|_{(\eta/4)} \|G\|_{(\eta/4)}.$$

The constant is independent of the outer vectors. Choose nonnegative profiles α_ϵ of integral one, supported in $B(0,\epsilon)$, with $\epsilon < c\eta$. Integrating $k_{F,G}(y) - k_{F,G}(0)$ against α_ϵ bounds the difference of the smeared and point forms by the right-hand side above times ϵ. The translated reflection vectors are dense. Thus the corresponding bounded operators differ in norm by at most $C_\varphi \eta^{-w_*-1}\epsilon$, with the same outer time translations in both forms. □

All four complex translation variables are used in this proof, although the final position of a plate has only two coordinates.

7.3. **A finite Fourier expansion in the rotation angle.** The next step is analytic continuation in the angle. The operator product gives a holomorphic kernel in each ordered configuration chart. Cubic symmetry identifies its quarter-turned copies. The continuation problem is to put those charts together without losing the bounds as an ordering changes.

The passage from a single bounded insertion to an ordered kernel uses the available time intervals explicitly. Fix transverse labels $\varphi_1, \dots, \varphi_n$ and real planar positions $y_1, \dots, y_n$ in strict time order. Put $v_i = y_{i+1} - y_i$, and initially suppose that every v_i lies in a fixed narrow sector about e_0. Choose $0 < \eta \le 1$ so that

$$\mathrm{Re}(v_i)_\pm > \eta \qquad (1 \le i < n).$$

There is one common scale for this finite configuration. It is chosen below the smallest intervening tube margin, not separately according to the degrees of the products on the two sides of an insertion.

For each i, approximate a planar point mass at zero by smooth profiles of integral one and use lemma 7.1. The corresponding forms, with the outer vectors translated by $-\eta e_4/4$, converge in operator norm to a bounded operator $B_{i,\eta}$ with

$$\|B_{i,\eta}\| \le C_{\varphi_i}\eta^{-w_*}. \tag{250}$$

The two time intervals of length $\eta/4$ are included in this operator. This is a definition through its bounded form; it does not assert that an unsmeared field at time zero is a bounded operator. The range of the translated reflection vectors is dense, so the form determines the operator uniquely. The integral-norm estimate and its degree independence are what permit arbitrary separated products on either side during this construction.

Between two such operators the unused displacement is $v_i - \eta e_0/2$. It remains in $\mathbb{V}$. Translation of the smooth insertion forms gives the corresponding ordered product. At fixed η, the plate approximations converge in operator norm, and the intervening translation operators are contractions. Telescoping the finite product therefore permits the point limit and gives

$$K_n(v_1,\dots,v_{n-1}) = \langle \Omega, B_{1,\eta}\mathcal{T}(v_1 - \tfrac{\eta}{2}e_0)B_{2,\eta}\cdots\mathcal{T}(v_{n-1}-\tfrac{\eta}{2}e_0)B_{n,\eta}\Omega\rangle\,, \qquad \mathcal{T}(v) = e^{-v^0H + iv^1P_1}. \tag{251}$$

The end intervals disappear against Ω, since $H\Omega = 0$. Each intermediate factor is a holomorphic contraction. Thus the right-hand side is holomorphic while all residual displacements have positive light-cone margins.

Different permitted choices of η give the same expression at real ordered configurations, since both expressions equal the same separated correlation. Their holomorphic continuations agree on overlaps. On each compact subset of $\mathbb{V}^{n-1}$ one can choose η below its smallest margin, which defines the ordered kernel throughout that tube. This is compatibility within one ordering of the insertions. The compatibility of differently ordered angular charts is the continuation step discussed below.

There are n bounded insertion operators and $n-1$ contractions in (251). Taking their norms gives

$$|K_n(v)| \le \left(\prod_{i=1}^{n} C_{\varphi_i}\right)\eta^{-w_* n}.$$

Choose η a fixed small multiple of the minimum tube margin, capped by one. Consequently

$$|K_n(v)| \le C_{\varphi_1,\dots,\varphi_n}\min\{1, \min_{i,\pm}\operatorname{Re}(v_i)_\pm\}^{-w_* n}. \tag{252}$$

The exponent is linear in the number of insertions because each insertion costs one power in (250). A bound whose exponent or constant depended on the degrees of the outer polynomials would not give this conclusion.

The precise constant in the continuum cone enters here. For a real time-axis displacement $v = (r, 0)$, an imaginary rotation has

$$R_{is}v = (r\cosh s,\ ir\sinh s), \qquad (R_{is}v)_\pm = re^{\mp s}.$$

Both light-cone coordinates stay positive for every real s, though their minimum decreases like $e^{-|s|}$. If one knew only $H \ge k_*|P_1|$ with $k_* < 1$, the contraction domain would instead require $\operatorname{Re} v^0 \ge k_*^{-1}|\operatorname{Im} v^1|$. The same rotated point leaves that domain when $\tanh|s| > k_*$. Thus a fixed weaker lattice cone is insufficient for the arbitrarily long imaginary rotations used to remove high angular frequencies. The bounded five-point limit in section 6 supplies the constant-one cone before this argument is applied.

Continuing across ordered charts. We first continue the kernel for a configuration in a narrow sector. Its finite angular equation then extends to each ordered tube. This gives analytic coefficients on every configuration of distinct planar points, including the regular polygons used below. The final identification with correlations at projection ties is made at the level of tests in the last subsection.

Lemma 7.2 (Finite angular type)**.** Let $n \geq 2$, fix real transverse labels and put $N_n = \lfloor w_* n/4 \rfloor$. For a strictly time-ordered configuration whose successive differences lie in a common sector of aperture $2\alpha < \pi/2$, the angular kernel has an entire continuation with period $\pi/2$, and

$$K_n(R_z v) = \sum_{|m| \leq N_n} c_m(v) e^{4imz}, \qquad |K_n(R_{t+is} v)| \leq C_v e^{w_* n |s|}. \tag{253}$$

The bound is uniform in t over one real period. The coefficients extend to real-analytic functions on the configurations of distinct planar points. They are multilinear in the labels, invariant under common translations and simultaneous permutations of points and labels, and satisfy

$$c_m(R_t \mathbf{y}) = e^{4imt} c_m(\mathbf{y}). \tag{254}$$

■

Proof. For a narrow configuration, let r be the smallest length of its successive differences. In a coordinate frame whose axis is closest to the rotated sector, every real difference has time component at least

$$\delta = r \cos(\alpha + \pi/4) > 0.$$

The same bound, decreased once, holds in a small product neighbourhood of the configuration. In each of the four frames the ordered kernel defines a holomorphic angular function on a vertical strip of width $\pi - 2\alpha$. Consecutive strips have centres separated by $\pi/2$, so they cover the complex plane and overlap. Their overlaps contain real intervals. On those intervals the kernels are densities of the same correlation: a change of frame is a cubic isometry and any reordering permutes the labels as well. Their holomorphic extensions therefore agree on the overlaps. They give a single entire function with period $\pi/2$.

Imaginary rotation multiplies the two light-cone coordinates by e^s and e^{-s}. The smallest margin in the chosen frame is thus at least $\delta e^{-|s|}$. Equation (252) gives the bound in (253), with a constant uniform over a real period. For

$$c_m = \frac{2}{\pi} \int_0^{\pi/2} K_n(R_t v) e^{-4imt}\, dt,$$

Cauchy's theorem on a rectangle, with its vertical edges cancelled by periodicity, gives

$$|c_m| \leq C_v e^{w_* n |s| + 4ms}.$$

Let $s \to -\infty$ for $m > w_* n/4$, and let $s \to +\infty$ for $m < -w_* n/4$. These coefficients vanish. Fourier uniqueness and the identity theorem give the finite expansion.

We now extend its coefficients beyond the narrow configurations. For $\upsilon \in \frac{\pi}{2}\mathbb{Z}$ and a permutation π, define the ordered tube

$$\mathbb{D}_\upsilon^\pi = \{\mathbf{z} : R_{-\upsilon}(z_{\pi(i+1)} - z_{\pi(i)}) \in \mathbb{V},\ 1 \leq i < n\}.$$

Read the product kernel in this frame and ordering, with labels $\varphi_{\pi(1)}, \dots, \varphi_{\pi(n)}$. Each tube is open and convex, stable under common translations, entrywise real part, and R_{is} for real s. A nonempty intersection of two tubes therefore contains a nonempty open set of real configurations. In small product neighbourhoods of those real configurations, both kernels represent the same functional on plate profiles. Continuous densities of the same functional agree there. The identity theorem on the convex intersection makes the kernels agree throughout it.

They consequently define one holomorphic kernel $\mathbf{K}_n$ on $\mathbb{D}_n = \bigcup_{\upsilon,\pi} \mathbb{D}_\upsilon^\pi$. Its real domain is

$$\mathrm{Ord}_n = \{\mathbf{y} : \text{all } y_i^0 \text{ are distinct}\} \ \cup\ \{\mathbf{y} : \text{all } y_i^1 \text{ are distinct}\}.$$

The kernel on this union is already compatible with simultaneous permutations and translations.

For fixed s, the angular equation on a narrow configuration is a holomorphic identity in the positions. More explicitly, on a tube put $f(s;\mathbf{z}) = \mathbf{K}_n(R_{is}\mathbf{z})$. The expansion just proved gives

$$\prod_{m=-N_n}^{N_n} (\partial_s + 4m) f(s;\mathbf{y}) = 0$$

on the nonempty open set of real narrow configurations in that tube. Every derivative on the left is holomorphic in $\mathbf{z}$. The identity theorem extends the equation to the entire tube, for each real s. In angular notation its equation is

$$\prod_{m=-N_n}^{N_n} (\partial_z - 4im) F(z) = 0. \tag{255}$$

The roots are distinct. Solving the equation in s gives

$$\mathbf{K}_n(R_{is}\mathbf{z}) = \sum_{|m|\le N_n} c_m(\mathbf{z}) e^{-4ms}.$$

The coefficients are fixed linear combinations of the first $2N_n + 1$ derivatives at $s = 0$, and hence are holomorphic. On tube intersections their expansions agree for every s; independence of the exponentials makes their coefficients agree as well.

To pass from imaginary to real angles, take $\mathbf{y} \in \mathrm{Ord}_n$ and choose one of its tubes. For t in an open interval containing zero, $R_t\mathbf{y}$ stays in this tube. Since R_{is} preserves the tube, the map $t + is \mapsto \mathbf{K}_n(R_{t+is}\mathbf{y})$ is holomorphic on that whole vertical strip. On the imaginary axis it equals $\sum_m c_m(\mathbf{y}) e^{4im(t+is)}$. The identity theorem and uniqueness of the expansion at $R_t\mathbf{y}$ yield (254) for those small real t.

Let Gen_n be the open dense set of configurations of distinct planar points on which no two pair differences are perpendicular. Density follows by excluding finitely many zero sets of nonzero polynomials. This set is rotation invariant and lies in Ord_n: a time tie and a spatial tie would give perpendicular differences, or the same pair would coincide. Its entire real rotation orbit therefore lies in Ord_n. The locally constant function

$$t \longmapsto e^{-4imt} c_m(R_t\mathbf{y})$$

is constant on that orbit. Thus (254) holds for all real t when $\mathbf{y} \in \mathrm{Gen}_n$. Continuity and density extend it whenever both endpoints belong to Ord_n. This also makes the angular continuations agree on every finite chain of overlapping ordered charts. Compactness of one real angular period supplies a finite chart cover of a generic orbit and positive margins on its closed subcharts.

Finally, for any configuration of distinct planar points, choose t such that $R_t\mathbf{y} \in \mathrm{Ord}_n$. Only finitely many angles modulo π are excluded by the pair differences. Define

$$\widetilde{c}_m(\mathbf{y}) = e^{-4imt} c_m(R_t\mathbf{y}).$$

The covariance just proved makes this independent of the chosen angle. An angle admissible at one point stays admissible in a neighbourhood, so the definition is real analytic there. It extends c_m and preserves its permutation, translation and rotation identities. We henceforth write c_m for this extension. The space of distinct planar points is connected, being the complement of finitely many codimension-two coincidence subspaces. Thus a coefficient vanishing on all narrow configurations vanishes identically. This proves both the claimed extension and the compatibility of its maximum frequency with the definition using narrow configurations. □

Let $M(n)$ be the largest absolute index of a nonzero coefficient, over all positions and labels, with $M(n) = 0$ if only the constant coefficient remains. The lemma gives $M(n) \le w_* n/4$. For real labels, reality gives $c_{-m} = \overline{c_m}$. Orders zero and one are constant by normalization, translation invariance and the form of the first moment, so $M(0) = M(1) = 0$. The analytic extension is essential at a

regular polygon: its chords include perpendicular differences, so its rotation orbit need not stay in Ord_n. The next argument uses the extended coefficients at precisely these configurations.

7.4. **Convexity and the polygon argument.** Imaginary rotation preserves the reflected pairing. Write $\theta(v^0, v^1) = (-v^0, v^1)$ for time reflection. Reversing the order of the reflected insertions gives the induced map $\iota(v) = -\theta(\overline{v})$ on consecutive displacements:

$$\iota(v^0, v^1) = (\overline{v^0}, -\overline{v^1}),$$

and $R_{is}\iota = \iota R_{is}$ for real s. The ordered product and its reflected product can therefore be continued through the same positive pairing. In particular, for any finite list of separated half-configurations $Z_1, \dots, Z_N$ with their transverse labels, the reflected Gram matrix $G(s)$ at imaginary angle is satisfies

$$\sum_{i,j=1}^{N} \overline{z_i} z_j G_{ij}(s) \geq 0 \qquad (z \in \mathbb{C}^N,\ s \in \mathbb{R}). \tag{256}$$

Initially this is read for half-configurations with distinct internal times, where the ordered products are defined. Approximate an arbitrary finite list of separated half-configurations by lists of this kind. The continued coefficients are continuous, so every matrix entry at a fixed imaginary angle converges. The cone of positive semidefinite finite matrices is closed; hence (256) holds also at equal internal times. This includes the polygon halves used below.

Split one configuration into groups of k and l insertions across the time mirror. Let $G_{12}(s)$ be their pairing, and let $G_{11}(s), G_{22}(s)$ be the pairings obtained by reflecting and doubling each group. Positivity of the two-by-two Gram matrix gives

$$|G_{12}(s)|^2 \leq G_{11}(s) G_{22}(s).$$

Choose the positions and labels in an open set where the highest coefficient of the $k + l$ kernel does not vanish. Its finite angular expansion then grows at rate $4M(k + l)$ in one of the two imaginary directions. The doubled kernels grow at rates at most $4M(2k)$ and $4M(2l)$. Divide the inequality by the corresponding exponentials and let $|s|$ tend to infinity in that direction. A larger rate on the left would contradict the inequality. Hence

$$2M(k + l) \leq M(2k) + M(2l). \tag{257}$$

No lower bound on a doubled kernel is used: if its leading coefficient vanishes at the chosen configuration, its smaller rate only strengthens this restriction. In particular the nonnegative integers $m_k = M(2k)$ form a convex sequence, since the choice of group sizes $k - 1, k + 1$ gives $2m_k \leq m_{k-1} + m_{k+1}$. Here $m_0 = 0$, and the operator bound already gives $m_k \leq Ck$.

There is a second use of (256). At even order $2k$, let $M = M(2k)$ and take its leading coefficient $c_* = c_{-M}$. For a fixed finite list of k-point half-configurations, multiply the Gram matrix by e^{-4Ms} and let $s \to +\infty$. Every entry has a finite expansion, so it has a limit. Positivity is closed under this finite matrix limit:

$$\sum_{i,j} \overline{z_i} z_j c_*(\theta Z_i, Z_j) \geq 0. \tag{258}$$

Thus the highest angular coefficient itself is a positive kernel across the time mirror, even though the other coefficients need not have that property.

The covariance of the coefficients describes exactly how this positivity changes when the mirror is turned. Let $\theta_t = R_t \theta R_{-t}$. For the rotated half-configurations,

$$c_*(\theta_t R_t Z_i, R_t Z_j) = e^{-4iMt} c_*(\theta Z_i, Z_j).$$

Multiplication by e^{4iMt} therefore makes the reflected kernel positive across the line turned by t. This conclusion concerns the leading coefficient with its specified scalar phase. It does not claim reflection positivity of the full functional at a new planar mirror before rotation invariance has been proved.

Take now a regular $2k$-gon, with the same real transverse label at every vertex. Rotation through π/k permutes this labelled set. Permutation compatibility leaves its coefficient unchanged, whereas coefficient covariance multiplies it by $e^{-4iM\pi/k}$. Therefore

$$(1 - e^{-4iM\pi/k})c_*(\text{regular labelled } 2k\text{-gon}) = 0.$$

If the polygon value is nonzero, it follows that

$$k \mid 2M(2k). \tag{259}$$

If divisibility fails, the polygon value is zero. In a suitable edge-bisector reflection it is a diagonal value of the phased positive kernel above. For any positive kernel Q,

$$|Q(Z, W)|^2 \le Q(Z, Z)Q(W, W).$$

Thus $Q(Z, Z) = 0$ annihilates its whole row, not merely its value at one polygon. This gives a specific zero row with the equal-label restriction inherited from the polygon. It is the starting point of the geometric propagation, rather than its conclusion.

Propagating the zero row. Suppose that $k \nmid 2M(2k)$, and put $M = M(2k)$ and $c_* = c_{-M}$. The preceding polygon identity says that c_* vanishes at every regular $2k$-gon whose vertices carry one common real transverse label. We show that it vanishes for all labels and all distinct positions, contradicting its choice. When $k = 1$ divisibility is automatic, so assume $k \ge 2$.

Call a finite labelled set Q a zero set if c_* vanishes at every $2k$-point labelled configuration containing Q, with arbitrary remaining distinct positions and real labels. A regular $2k$-gon with common label φ is a zero set. Real analyticity of the coefficients on the full distinct-point domain is used throughout; regular polygons themselves can have simultaneous ordering ties.

We first record the consequence of the positive-kernel zero row. Let $\mathfrak{l}$ be a line and Q' a set of at most k labelled points in one of its open half-planes. Suppose that $Q' \cup \theta_{\mathfrak{l}} Q'$ contains a zero set. Complete Q' by arbitrary additional points W in that half-plane to a k-point labelled configuration $Z = Q' \cup W$. Give the new points arbitrary real labels. The reflected diagonal $\theta_{\mathfrak{l}} Z \cup Z$ contains the zero set, so its c_*-value is zero. After multiplication by the fixed phase specified above for this mirror, it is a zero diagonal of a positive kernel. Its whole row vanishes. Therefore

$$c_*(\theta_{\mathfrak{l}} Q' \cup \theta_{\mathfrak{l}} W \cup T) = 0$$

for every k-point labelled configuration T in the same open half-plane, with its positions and real labels independent of W.

For each fixed choice of these remaining labels, their $2k - |Q'|$ positions vary in a nonempty open set. The space of distinct planar positions avoiding the finitely many fixed points of $\theta_{\mathfrak{l}} Q'$ is connected: one may move points successively along planar paths avoiding the other finitely many points. Real analyticity extends the displayed zero to that entire space. Thus $\theta_{\mathfrak{l}} Q'$ is a zero set. Interchanging the half-planes proves the same for Q'. The arbitrary labels of W and T have been released by the row identity; they have not been inferred from an equal-label diagonal by polarization.

Now fix a regular $2k$-gon with cyclically numbered vertices v_i, $i \in \mathbb{Z}/(2k)$, and common label φ. We prove the following reduction: if every consecutive arc of j vertices is a zero set, then every consecutive arc of $q = \lceil j/2 \rceil$ vertices is a zero set, for $2 \le j \le 2k$. Choose such a shorter arc

$$Q' = \{v_{i_0+1}, \dots, v_{i_0+q}\}.$$

Let $\mathfrak{l}$ be the perpendicular bisector of the polygon edge $[v_{i_0}, v_{i_0+1}]$. It passes through the centre, meets no vertex, and acts by

$$\theta_{\mathfrak{l}} v_{i_0+i} = v_{i_0+1-i}.$$

The k vertices $v_{i_0+1}, \dots, v_{i_0+k}$ lie in one open half-plane, and $q \le k$. The union $\theta_{\mathfrak{l}} Q' \cup Q'$ is a consecutive arc of $2q$ vertices. Since $2q \ge j$, it contains a zero arc of length j. The zero-row consequence just proved makes Q' a zero set.

Starting with the whole polygon, repeat this reduction:

$$2k \longrightarrow k \longrightarrow \lceil k/2 \rceil \longrightarrow \cdots \longrightarrow 1.$$

It works for every integer k, with no power-of-two restriction. Each single vertex with label φ is therefore a zero set. Any planar point can be a vertex of a suitably translated regular $2k$-gon, and φ was arbitrary. Every other real label was already arbitrary in the zero-set property. Hence c_* vanishes for all real labels and all distinct planar positions. Multilinearity extends the conclusion to complex labels. This contradicts its nonvanishing and proves (259).

The remaining frequency argument is arithmetic.

For completeness, suppose $m_0 = 0$, the m_k are integers, their successive increments are nondecreasing, $m_k \le Ck$, and $k \mid 2m_k$. Every increment is at most C: a larger increment would persist by monotonicity and violate the linear bound. The increments, being integers, are therefore eventually constant, say D. Write $m_k = Dk - b$ for all large k. Divisibility says that each such k divides the fixed integer $2b$, hence $b = 0$. Since earlier increments are at most D, their sum can equal Dk only if every one is D. Therefore $m_k = km_1$. Combining this with (257) gives

$$M(2k) = kM(2), \qquad M(n) \le \tfrac{n}{2} M(2). \tag{260}$$

The divisibility condition restricts the highest index at each order. It is stronger than the assertion that angular frequencies are multiples of four. Cauchy–Schwarz supplies convexity, while the polygon argument supplies this extra restriction.

7.5. **The ultraviolet estimate for two plates.** The angular base case requires only $t^8\Gamma(t) \to 0$. We first derive it from the two-point estimate of section 5, and then obtain it from a weaker bound supplied by source stability. The second derivation separates rotations from the precise nontriviality calculation.

The equal-label kernel is read from the same separated two-point distribution as the ordered kernels. It retains positivity at the diagonal mirror from section 6; this is the premise used when identifying its diagonal reflection picture below.

For equal real transverse labels the two-point kernel has the positive spectral representation

$$K_2(v) = \int_{E \ge |q|} e^{-v^0 E + i v^1 q} \, d\rho(E, q), \qquad \rho \ge 0.$$

Set $\Gamma(t) = \int e^{-tE} \, d\rho$. The ordered plate kernel is finite at positive unit separation, so $\Gamma(1) = K_2(e_0) < \infty$. Hence $e^{-E}\rho$ is finite, and $\rho(\{E \le N\}) \le e^N \Gamma(1)$; these sets exhaust its nonnegative-energy carrier. Thus the measure is sigma-finite before the Tonelli computations below. The upper two-point estimate from section 5, for nonnegative bumps of maximum one whose size and separation are comparable to d, has the form

$$0 \le C_2(f, h) \le u(d) \|f\|_\infty \|h\|_\infty, \qquad u(d) \longrightarrow 0.$$

The vanishing of u comes from the running coupling. It must be available at the fixed shape used in the decomposition of plates.

Choose a nonzero smooth bump $0 \le \beta \le 1$ supported in $[1, 2]$, and the cubic B-spline $\gamma = c_\gamma \mathbf{1}^{*4}_{[-1/2,1/2]}$, normalized by $\|\gamma\|_\infty = 1$. For $\kappa \in [1, 2]$, put

$$\alpha_{t,\kappa}(s, y) = \beta(s/t)\gamma(8y/(\kappa t)), \qquad f_{t,\kappa} = \alpha_{t,\kappa} \otimes \varphi.$$

These planar profiles have width and height comparable to t, are placed at positive time of order t, and have maximum at most one. With sufficiently small transverse cells, their reflected pairs fit in balls of radius t and have separation between t and $6t$. The comparison needed below is

$$\Gamma(Ct) \le Ct^{-4} \sup_\kappa \|[f_{t,\kappa}]\|^2. \tag{261}$$

To verify it, write $\operatorname{sinc} u = \sin u/u$, with value one at zero. The spatial Fourier transform of γ is $c_\gamma \operatorname{sinc}(q/2)^4$, while

$$\int \beta(s/t) e^{-sE}\, ds \geq cte^{-2tE} \qquad (E \geq 0).$$

The Fourier–Laplace representation of the reflected norm gives

$$\|[f_{t,\kappa}]\|^2 \geq ct^4 \int e^{-4tE} \operatorname{sinc}(\kappa t q/16)^8 \, d\rho(E,q).$$

The spline is C^2 and can be used by smooth approximation in C^1, retaining the positive-time support. The sharp-norm bounds give convergence of the reflected norms; the transforms are dominated by Ct^2e^{-tE}, so the spectral formula passes to the limit as well. No finite undamped mass of ρ is required.

The dilation average satisfies

$$J(u) := \int_1^2 \operatorname{sinc}(\kappa u)^8 \, d\kappa \geq c(1+|u|)^{-8}.$$

Indeed, J is continuous and strictly positive on each compact interval, since the dilated transforms have no common zero. For $|u| \geq 2\pi$,

$$J(u) \geq 2^{-8}|u|^{-9} \int_{|u|}^{2|u|} \sin^8 v \, dv \geq c|u|^{-8},$$

using the positive mean of $\sin^8 v$. Tonelli's theorem and the cone $|q| \leq E$ now give

$$\int_1^2 \|[f_{t,\kappa}]\|^2 \, d\kappa \geq ct^4 \int \frac{e^{-4tE}}{(1+t|q|)^8} \, d\rho(E,q) \geq ct^4 \Gamma(5t).$$

Here $(1+u)^8 \leq Ce^u$ for $u \geq 0$. This proves (261) with fixed constants.

For a fixed transverse label, a profile $f_{t,\kappa}$ decomposes into $O(t^{-2})$ transverse cells, each a difference of controlled nonnegative four-dimensional bumps. The reflection norm of each piece is bounded by $Cu(ct)^{1/2}$. The triangle inequality yields

$$\|[f_{t,\kappa}]\| \leq Ct^{-2} u(ct)^{1/2}.$$

Together with (261) this gives

$$t^8 \Gamma(t) \leq Cu(ct) \longrightarrow 0. \tag{262}$$

The exponent eight contains both the planar transform loss and the squared count of transverse cells. No estimate on a point insertion is inferred directly from a bound on an arbitrary test.

A bound sufficient for rotations without the relative comparison. The limit in (262) needs less than the precise asymptotic law of section 5. We give a second derivation from the coupling dependence of the source bounds. In particular, the relative Gaussian comparison is needed for item (v), but need not be used to prove rotations.

We first record that dependence. With L, M, ρ, the analytic reserves and the activity rates fixed, the constants in the local source-energy bound and the comparison of vacuum densities are independent of the reference coupling $0 < g' \leq \gamma$. At terminal inverse coupling y, the upper stability and lower-floor exponents can be taken to satisfy

$$E_+(2\rho; y) + C_{\text{low}}(y) + \delta_{\text{vac}} \leq C(1 + \rho + \log y). \tag{263}$$

This is the normalization used in the source argument: each omitted Gaussian factor costs $C(1 + \log x_j)$ per site, the weighted history estimate pays those costs, and the final small ball of radius $x_k^{-1/2}$ has bounded action and Haar-volume cost $C \log x_k$. The constants are chosen before the coupling ceiling. The auxiliary torus may have to grow as g' decreases. Its side does not enter the

per-cell estimate (197); division by the number of reflected cells leaves the same cell volume. Thus, for two tests in unit balls at a fixed positive separation, that proof gives

$$|C_2(f,h)| \le Ce^{C(1+\log(g'^{-2}+B_\sharp))}\|f\|_\sharp\|h\|_\sharp \tag{264}$$

in every periodic Gibbs state at first-passage data for g'. This retains the uniform source and history estimates required earlier; zero-source stability alone would not give it.

Lemma 7.3 (A subpower bound at the observation scale)**.** Fix a shape of two nonnegative bumps whose size and separation are comparable to d, where $0 < d \le d_0 < 1$. The source bounds imply, at all sufficiently fine first-passage cutoffs,

$$|C_2(f,h)| \le u_*(d)\|f\|_\infty\|h\|_\infty, \qquad u_*(d) = C\big(g^{-2} + B_\sharp(C + \log_L(1/d))\big)^C. \tag{265}$$

The cutoff threshold may depend on d, but the bound does not. In particular $d^2u_*(d) \to 0$. The same bound holds in continuum limits along these states. ■

Proof. Call k a record level if $x_k < x_i$ for all $i < k$. The first-passage level K is a record level. If i is one record level, the lower drift bound shows that another occurs within $j_0 = 1 + \lceil C_\sharp/\lambda_\sharp \rceil$ steps, unless the run has already ended. Choose b on the lower side of $\log_L(1/d)$, within $j_0 + 1$ of it, so that $K - b$ is a record level and $L^{-j_0-1} \le L^b d \le 1$. Take $d_0 \le L^{-j_0-2}$, so $b \ge 1$. It is possible once $K \ge C + j_0 + \lceil \log_L(1/d) \rceil$.

Set $g_b^{-2} = x_{K-b} - B_\sharp$. For $b > 0$ this is positive, $g_b < g$, and $(K - b, \beta)$ is a first-passage datum for g_b: all earlier x_i exceed x_{K-b}. The upper drift bound gives $x_{K-b} \le g^{-2} + B_\sharp(b + 1)$. The coupling sequence is the same sequence in either choice of units, since its defining local operations and extraction do not depend on the terminal reference coupling.

The dilation $x \mapsto L^b x$ takes a spacing-L^{-K} Gibbs state to one at spacing $L^{-(K-b)}$, with the same Wilson weight. It also gives the exact equality

$$C_2(f,h) = C_2(f^{[b]}, h^{[b]}), \qquad f^{[b]}(x) = f(L^{-b}x).$$

The sup norms do not change, and $\mathrm{Lip}(f^{[b]}) = L^{-b}\,\mathrm{Lip}\, f \le C\|f\|_\infty$, with a constant depending on the fixed shape and j_0 only. Apply (264) at g_b to this pair at fixed separation. This gives (265). For fixed d the cutoff requirement is eventually satisfied, so the estimate passes to the continuum before d decreases to zero. □

To keep the decay between transverse cells, we use a point kernel in four dimensions. The construction of that kernel is included here because a bound on a smeared plate alone does not give it.

Lemma 7.4 (The point kernel and its axis bound)**.** Before proving rotations, the two-point function has a real-analytic kernel D off the origin, invariant under cubic isometries, with

$$D(te_4 + z) = \int e^{-tE+ip\cdot z}\, d\rho_4(E,p), \qquad t > 0, \qquad \rho_4 \ge 0, \qquad |p_j| \le E. \tag{266}$$

Here $z \in \mathbb{R}^3$ is embedded as a spatial vector. Its axis value $s_a(t) = D(te_4)$ is finite, nonnegative and nonincreasing, and

$$|D(x)| \le s_a(|x|_\infty) \le s_a(|x|/2), \qquad x \ne 0. \tag{267}$$

If $C_2(\theta h, h) \le u(d)\|h\|_\infty^2$ for the fixed-shape mirror bumps of size d, then

$$s_a(t) \le Ct^{-8}u(ct). \tag{268}$$

■

Proof. Use the joint spectral measure Π of H, P_1, P_2, P_3 on the one-field subspace. For positive-time tests write $\widetilde{f}(E,p) = \int f(x) e^{-x_4 E + ip\cdot\vec{x}}\,dx$ and $m_f = \langle [f], \Pi(\cdot)[f]\rangle$. Convolution of two such tests gives $[h * f] = \widetilde{h}(H,P)[f] = \widetilde{f}(H,P)[h]$, hence

$$|\widetilde{h}|^2 m_f = |\widetilde{f}|^2 m_h.$$

Choose a nonnegative time bump translated strictly into positive time, and in each spatial variable use $b = (v * v^\vee)^{*2}$, where $v(x) = e^{-x}\mathbf{1}_{[0,1]}(x)$ and $v^\vee(x) = v(-x)$. This is a compactly supported C^1 shape and

$$\widehat{b}(q) = \frac{|1 - e^{-1+iq}|^4}{(1+q^2)^2} \geq c(1+q^2)^{-2} > 0.$$

The product test f_* therefore has a nowhere-zero transform on $E \geq 0$. Set $\rho_4 = |\widetilde{f_*}|^{-2} m_{f_*}$. Here $m_{f_*}(\mathbb{R}^4) = \|[f_*]\|^2 < \infty$. The sets where $|\widetilde{f_*}| \geq 1/N$ exhaust the spectral carrier and have ρ_4-mass at most $N^2\|[f_*]\|^2$. This makes ρ_4 sigma-finite despite division by a transform that need not be bounded away from zero globally.

The exchange identity gives $m_f = |\widetilde{f}|^2 \rho_4$ for every positive-time test. Its carrier satisfies the three coordinate cone inequalities because Π does.

Apply this identity to $h_t(x) = a(x_4/t)\prod_{j=1}^3 b(x_j/t)$, translating the fixed time shape a high enough that $(\theta h_t, h_t)$ belongs to a fixed mirror-pair shape class. The time transform is bounded below by cte^{-CtE}; the spatial transforms are bounded below by $ct(1+t^2p_j^2)^{-2}$. The cone absorbs all polynomial losses into an exponential, so

$$\|[h_t]\|^2 \geq ct^8 \int e^{-C'tE}\,d\rho_4.$$

The left side is finite by the per-ball bounds. Thus every positive time damps ρ_4 to a finite measure. The spectral integral in (266) is well-defined and holomorphic when the imaginary spatial displacement is smaller than the positive time margin. Polarization and the spectral norm formula identify it with the original two-point distribution on $x_4 > 0$. Cubic changes of frame extend it over $x \neq 0$; the extensions agree on overlaps because they represent the same distribution. This also proves real analyticity there. The mirror-pair bound in the same inequality gives (268), after changing fixed constants.

Taking absolute values in the spectral integral gives $|D(te_4 + z)| \leq s_a(t)$. A cubic isometry can put the coordinate of largest modulus in the positive time direction. Monotonicity of the Laplace transform proves (267). □

For the fixed transverse label φ, the plate kernel is

$$\Gamma(t) = \int_{\mathbb{R}^2} D(te_4 + \zeta)(\varphi \star \varphi)(\zeta)\,d\zeta, \qquad (\varphi\star\varphi)(\zeta) = \int \varphi(y)\varphi(y+\zeta)\,dy.$$

Its spectral measure is the image of $|\widehat{\varphi}(p_2,p_3)|^2\rho_4$ under $(E,p) \mapsto (E,p_1)$; in particular it is the same positive measure defining Γ above. Cauchy–Schwarz gives $|\varphi\star\varphi| \leq \|\varphi\|_2^2$. Split its compact support
area is $O(2^{2i}t^2)$, while (268) bounds the kernel by $C(2^i t)^{-8}u(c2^i t)$. Hence

$$\Gamma(t) \leq C_\varphi \left[t^{-6} \sum_{2^i t \leq \delta} 2^{-6i} u(c2^i t) + s_a(\delta) \right]. \tag{269}$$

After multiplication by t^8, the sum is at most $C\sup_{0<d\leq c\delta} d^2 u(d)$, because its remaining weights are 2^{-8i}. First choose δ small, then let $t \to 0$. Thus $d^2u(d) \to 0$ implies (262). Taking $u = u_*$ from lemma 7.3 supplies this condition from the source estimates alone. The coordinate cone and diagonal positivity are still required in the subsequent harmonic argument.

In the common angular continuation, at a fixed separation $v = re_0$, move the angle contour in (253). Positivity and the cone bound the coefficient by

$$|c_m(re_0)| \le e^{4ms}\Gamma(cre^{-|s|}).$$

The power bound eliminates $|m| > 2$, and the strict little-o in (262) eliminates $|m| = 2$. A mere $O(t^{-8})$ bound would leave precisely that endpoint unresolved.

7.6. **The last two harmonics and diagonal positivity.** The remaining possible terms are indexed by $m = 0, \pm 1$. At $v = re_0$ put $a_m = c_m(re_0)$. Imaginary rotations give the transform $a_0 + a_1e^{-4s} + a_{-1}e^{4s}$. At the diagonal angle $\pi/4$ the nonconstant terms have the opposite sign. Diagonal reflection positivity represents that rotated two-point kernel by another positive measure ρ_{d}.

Let $b_{\pm 1}$ be the Lorentz boosts of the planar energy–momentum variables with rapidities ± 1. The identity of the two transforms at the three remaining harmonics yields the required measure identity

$$(\cosh 4 - 1)\rho_{\mathrm{d}} + (b_1)_*\rho + (b_{-1})_*\rho = (\cosh 4 + 1)\rho. \tag{270}$$

For the uniqueness step, each measure has nonnegative energy and finite positive-time damping. For ρ and ρ_{d} this follows from the plate bounds; the boosted measures satisfy it because their energies are at least $e^{-1}E$. Let M be the difference of the two sides of (270). For each fixed real spatial displacement y, its transform

$$F_M(z, y) = \int e^{-zE+iyq}\, dM(E, q)$$

is holomorphic on $\operatorname{Re} z > 0$. Under the stated angular continuation, the three-harmonic identity makes it zero on the common real ordering sector $t > 0$, $|y| < ct$, for a fixed $c > 0$. For this fixed y, the identity theorem extends its vanishing from $t > |y|/c$ to the whole right half-plane. Thus, for every $t_0 > 0$, the two finite positive measures obtained by damping the sides of (270) by e^{-t_0E} have the same joint Fourier transform, obtained by setting $z = t_0 + iu$. Fourier uniqueness [G04, Theorem, p. 1] makes these damped measures equal, and hence proves (270). This uses only nonnegative energy for ρ_{d}. Positivity now gives $\rho_{\mathrm{d}} \le C\rho$, so ρ_{d} is supported in the same cone. Its transform can therefore also be evaluated at all the imaginary rotations under consideration. It follows that

$$a_0 - a_1e^{-4s} - a_{-1}e^{4s} \ge 0 \quad (s \in \mathbb{R}), \qquad a_{-1} \ge 0. \tag{271}$$

The second inequality is positivity of the leading Gram coefficient on a reflected time-axis pair with equal labels. Multiply the first inequality by e^{-4s} and let $s \to +\infty$ to get $a_{-1} \le 0$. Thus $a_{-1} = 0$. Translation and the covariance of the coefficient extend its vanishing from that axis to every pair, polarisation extends it to unequal labels, and reality gives $a_1 = 0$. Consequently the two-point argument gives $M(2) = 0$. The polygon propagation and (260) then pass this conclusion to every n. Thus the ultraviolet estimate settles the base case, while the geometric argument carries it through the hierarchy.

7.7. **From ordered plates to separated tests.** The analytic coefficients above were extended through configurations with projection ties. We must identify their sum with the original correlation on test functions there. Equality of densities only on the generic set would not exclude a distribution supported on its complement. The following local translation identity supplies the needed passage at the level of tests.

Take two clusters A, B on opposite sides of a coordinate slab of width $2\delta > 0$, smaller than their normal separation. Remove the two intervals of length δ and let a, b be the resulting reflection vectors. For a vector u tangent to the slab,

$$\sigma(A \cdot \tau_{ru}B) = \langle a, e^{-2\delta H+iru\cdot P}b\rangle = \int e^{-2\delta E+iru\cdot q}\, d\varsigma_{A,B}(E, q). \tag{272}$$

The complex measure $\varsigma_{A,B}$ has finite total variation. The coordinate cones give $|u \cdot q| \le |u|_1 E$, so the expression is holomorphic on $|\operatorname{Im} r|\,|u|_1 < 2\delta$. On every smaller strip the exponential also dominates each fixed derivative. For a different coordinate normal, use its reflection space. This formula applies while the normal gap remains open, regardless of ties in a tangential projection. Plate point limits are taken at a fixed smaller gap using the insertion estimates already proved.

Put $\widetilde{K}_n(\mathbf{y}) = \sum_m c_m(\mathbf{y})$, with the extended analytic coefficients. For fixed transverse labels and pairwise disjoint compact planar profiles, the identity to prove is

$$\sigma((\alpha_i \otimes \varphi_i)_i) = \int \prod_i \alpha_i(y_i)\, \widetilde{K}_n(\mathbf{y})\, d\mathbf{y}.$$

It holds whenever the profiles are ordered in time or in the planar spatial coordinate, by the construction of $\mathbf{K}_n$. We prove it on a neighbourhood of every configuration of distinct planar points, by induction on the number of pairs tied in their spatial coordinate y^1.

If $y_i^1 = y_j^1$ and $y_i \neq y_j$, then $y_i^0 \neq y_j^0$. Choose a time level strictly between these times, avoiding all the other times. It divides the configuration into an earlier cluster and a later one. Small product neighbourhoods retain a positive time gap. Slide the later cluster by re_1. For a sufficiently small positive r_0, the slide removes every spatial tie between the two clusters and creates no new one. Ties within each cluster are unchanged, so the number of tied pairs decreases.

The original correlation is real analytic in r by (272). The integral of $\widetilde{K}_n$ against the translated profiles is real analytic as well: the time gap prevents collisions during the slide, and a real-analytic function on the distinct-point domain has a bounded holomorphic extension on a neighbourhood of each compact set of positions used in the integral. Near r_0 the two expressions agree by the induction hypothesis, for every choice of profiles in the chosen product neighbourhoods. The identity theorem in r gives agreement at zero. To globalise, cover the compact product of the profile supports by finitely many products of discs where the local identity holds. Split each profile by a smooth partition with supports small enough that every nonzero product term lies in one member of this cover. Multilinearity proves the formula for the original profiles. It is an identity of distributions on each separated product neighbourhood; thus no additional term supported only at a simultaneous time and spatial ordering tie remains.

Once the polygon and two-point arguments have shown $M(n) = 0$, this formula and a change of planar variables give rotation invariance for plates with distinct planar supports:

$$\begin{aligned}\sigma(((\alpha_i \circ R_{-t}) \otimes \varphi_i)_i) &= \int \prod_i \alpha_i(y_i)\, \widetilde{K}_n(R_t\mathbf{y})\, d\mathbf{y} \\ &= \int \prod_i \alpha_i(y_i)\, \widetilde{K}_n(\mathbf{y})\, d\mathbf{y}.\end{aligned}$$

Consider next a configuration of distinct points in $\mathbb{R}^4$ whose planar projections are distinct. In small product neighbourhoods, the planar projections lie in disjoint discs. Fourier approximation on product boxes, multiplied by fixed product cutoffs, approximates each smooth test in C_c^∞ by finite sums $\alpha(x_4, x_1)\varphi(x_2, x_3)$ with supports in those neighbourhoods. Multilinearity, the plate identity and separated-test continuity give rotation invariance on such neighbourhoods.

It remains to allow coincident planar projections while the full points remain distinct. Suppose one pair has the same projection. Its points differ in x_2 or x_3. A hyperplane of constant value of that coordinate separates them and, after choosing the level away from all other points, separates the configuration into two clusters. Retain a positive normal gap in small neighbourhoods and slide one cluster by re_4. A small nonzero displacement removes all planar coincidences between the clusters without creating new ones. The number of coincident pairs therefore decreases.

Fix the rotation angle t and compare $\sigma(R_tX)$ with $\sigma(X)$ during this slide. The transverse separating hyperplane is preserved by R_t, and both e_4 and R_te_4 are tangent to it. Both expressions are thus real analytic in r by (272), in the corresponding coordinate picture. They agree near the

displaced configuration by induction, and hence at the original one. This again is an identity for arbitrary tests in product neighbourhoods, rather than only an identity of point values away from the tie set. Induction and a partition of unity cover every configuration outside the full coincidence diagonals.

The separated-test continuity bound finally extends the identity to the C^1 tests used earlier. Rotations in this plane, their cubic conjugates in all coordinate planes, and a coordinate reflection generate $\mathrm{O}(4)$. The moment–cumulant formula transfers the identity between moments and connected functions. Thus, once the sharp cone and degree-independent insertion bounds are available, the ordered angular kernels, their analytic coefficient extension, the polygon argument and the two-point base case give rotation invariance on every separated tuple. The uniform construction estimates used to obtain those inputs remain the analytic tasks identified earlier.

8. Selecting a phase and proving clustering

The centred limit can contain several translation-invariant phases. Even if it is Euclidean invariant and reflection positive, correlations between distant groups need not factor. Phase selection therefore requires information beyond the compactness argument of section 6. The useful estimate is a rate of approach to the translation-invariant plateau, uniform both in the lattice spacing and in the Gibbs state.

8.1. **The finite-lattice decomposition and its coordinates.**

The Gibbs ergodic decomposition. The ergodic decomposition at a fixed spacing is a property of the Wilson Gibbs specification. To apply the Gibbs decomposition theorem, regard one lattice site as carrying the four positively oriented outgoing bonds. Its state space is $E = \mathrm{SU}(2)^4$, a compact standard Borel space, and the configuration space is $E^{\mathbb{Z}^4}$. The plaquette interaction is bounded and has finite range. Its normalized finite-region kernels are translation covariant, and the reference state is translation invariant. The Gibbs decomposition theorem [G11, Theorem 14.17, p. 298] therefore decomposes it into translation-ergodic states satisfying these same Wilson DLR kernels. Selecting a component changes the state, not the interaction or the conditional law in a finite region. Every finite bond set is contained in finitely many of these site variables. Conditioning the finite-site Wilson kernel further on the unused outgoing bonds recovers the original bond kernel, so this encoding preserves the specification used throughout the manuscript.

This standard Gibbs theorem supplies the decomposition used here. Reflection positivity and the uniform estimates for its components are obtained by applying the lattice reflection and source arguments to the full Gibbs class. The uniform clustering estimate below is what permits pure states to vary with the cutoff.

In this section a pure state means a translation-ergodic Gibbs state. Let $\mathcal{I}$ be the sigma-algebra of translation-invariant events. The Gibbs ergodic decomposition gives

$$\nu = \int \rho \, D_\nu(d\rho), \qquad E_\nu[A \mid \mathcal{I}](U) = \rho_U(A)$$

for bounded local A, with D_ν concentrated on ergodic Gibbs states. The conditional states are shift invariant and belong to the full periodic Gibbs class. The source estimates therefore apply to almost every ergodic component with the same constants.

For the reference sequence ν_j define

$$X_j^{\nu_j} = \prod_{f \in X} [\mathcal{O}_{a_j}(f) - \nu_j(\mathcal{O}_{a_j}(f))]. \tag{273}$$

The subtraction is common to all states at datum j. The source normalizer gives

$$|\rho(\mathcal{O}_{a_j}(f)) - \nu_j(\mathcal{O}_{a_j}(f))| \le C_1(R)\|f\|_\sharp \tag{274}$$

for support in a ball of radius R, uniformly in j and ρ. This bounds the first coordinates even when the two uncentred means diverge separately.

Proposition 8.1 (Compact component coordinates)**.** Let S^{ref} be a common volume-first limit of the reference-centred moments. There are a compact metrisable coordinate space X, a probability measure λ of full support, and continuous coordinates $\xi \mapsto \sigma_\xi(X)$ for every separated tuple, such that

$$S^{\mathrm{ref}}(X) = \int_{\mathsf{X}} \sigma_\xi(X)\, d\lambda(\xi). \tag{275}$$

Each σ_ξ is a limit of the moments in pure states ρ_j, with the subtraction at ν_j as in (273). For any finite list of tuples,

$$\int_{\mathsf{X}} \prod_{i=1}^{q} \sigma_\xi(X^i)\, d\lambda(\xi) = \lim_{|v_2|\to\infty} \cdots \lim_{|v_q|\to\infty} S^{\mathrm{ref}}(X^1 \cdot \tau_{v_2} X^2 \cdots \tau_{v_q} X^q). \tag{276}$$

The separated correlations of each σ_ξ satisfy the uniform clustering estimate in (6). ∎

Proof. Use a countable family $(X_m)_{m\ge1}$ of real separated tuples with the support-controlled approximation property constructed above. Include the empty tuple. At datum j evaluate

$$\rho \longmapsto (\rho((X_m)_j^{\nu_j}))_{m\ge1}.$$

For each fixed coordinate, (224) gives a bound B_m for all sufficiently large j, uniformly in ρ. The finitely many earlier values are bounded too: at fixed spacing each field product is a bounded local continuous function. Enlarge B_m to include them. Thus one compact product of intervals contains all the evaluation measures.

The product $\mathsf{K} = \prod_m [-B_m, B_m]$ is compact and metrisable. For example the metric

$$d(u,v) = \sum_{m\ge1} 2^{-m} \min\{1, |u_m - v_m|\}$$

induces its product topology. Push D_{ν_j} forward to a probability measure $\widehat{D}_j$ on this product and pass to a weakly convergent subsequence with limit λ. Put $\mathsf{X} = \operatorname{supp}\lambda$. For each coordinate the defining decomposition gives

$$\int u_m\, d\widehat{D}_j(u) = \nu_j((X_m)_j^{\nu_j}).$$

Both sides have limits, proving (275) first on the countable family.

Fix $\xi \in \mathsf{X}$. Every open neighbourhood of ξ has positive λ-mass and hence positive $\widehat{D}_j$-mass for all sufficiently large j. Its preimage contains a pure state because D_{ν_j} is concentrated on such states. Choose successively neighbourhoods restricting the first N coordinates to accuracy $1/N$, and let N increase slowly with j. This gives pure ρ_j whose coordinates converge to ξ.

For an arbitrary separated tuple $X = (f_i)$, choose pairwise disjoint compact neighbourhoods A_i of its supports and approximate it using the countable family with all approximants supported in these same A_i. The estimate (226) is uniform in j and the pure evaluating state. Consequently the limit of $\rho_j(X_j^{\nu_j})$ exists, is independent of the chosen pure-state sequence converging to ξ, and will be denoted $\sigma_\xi(X)$.

Each approximating tuple is a coordinate, hence is continuous in ξ. The same estimate controls its error uniformly over all $\xi \in \mathsf{X}$, so the function $\xi \mapsto \sigma_\xi(X)$ is continuous. It is real, multilinear and symmetric, and takes the value one at the empty tuple. Coordinate functions separate X, since they are its product coordinates. Integrating the uniform approximation also extends (275) to every tuple. These statements hold at every point of the support; no tuple-dependent exceptional set is introduced.

Here is the additional step needed for products of coordinates. The lattice estimate (277), proved below using the shifted Fourier filter, is uniform in the spacing and the state when the products

have fixed physical supports. In a pure state its plateau is the product of the two expectations. Thus, for fixed tuples A, B, the difference between $\rho_j(A_j^{\nu_j}\tau_v B_j^{\nu_j})$ and $\rho_j(A_j^{\nu_j})\rho_j(B_j^{\nu_j})$ tends to zero as $|v| \to \infty$, uniformly in j and in the pure evaluating state. The reflection sizes, rather than the divergent supremum norms, give the uniform bound needed here.

Integrate this inequality against D_{ν_j}. At each fixed large displacement the reference correlation converges to $S^{\mathrm{ref}}(A \cdot \tau_v B)$, while the integral of the product of the two coordinates converges by weak convergence on K. The uniform error permits the two limits to be interchanged. This proves (276) for $q = 2$. Repeat with a product of the first $q - 1$ translated groups as A and the last group as B. Those earlier translations are kept fixed during the last distant-group limit. Their containing cube, and hence the constant, may depend on them, but remains independent of j. Induction proves the stated iterated limits without requiring a uniform bound in all displacements at once.

The same pure-state inequality passes directly to each chosen sequence ρ_j and gives (6) for σ_ξ. The constants are independent of ξ. Moment inequalities, reflection positivity and the source bounds pass to these limits as closed finite conditions. All coordinates therefore concern the same continuum family. □

The image measure is intrinsic to the chosen reference hierarchy. To see this, do not fix the subsequence used for weak compactness. Every convergent subsequence of $\widehat{D}_j$ has the mixed moments in (276): its proof uses the already fixed limit S^{ref} and the state-uniform plateau error. Thus, for every coordinate monomial $u_{m_1} \cdots u_{m_q}$, all weak limit points have the same integral. Finite linear combinations of such monomials form an algebra containing the constants and separating points of the compact product K. They are uniformly dense in $C(\mathsf{K})$ by Stone–Weierstrass. Two probability measures with these same monomial moments therefore agree on every continuous function and are equal.

It follows that $\widehat{D}_j$ converges along the entire given sequence of reference states whose correlations converge to S^{ref}. This is uniqueness of the coordinate law for that hierarchy. It is not uniqueness of the continuum hierarchy as the bare data vary. Compactness supplies a limit point; the plateau identities identify every such point and remove only the additional subsequence in the coordinate-measure construction.

The description also does not depend on the initially chosen dense family of tuples. Given two such families, use their union as a third countable family. Equicontinuity determines the missing coordinates continuously from either dense family. The two projection maps from the common support are therefore continuous and injective, as well as onto the respective supports: a compact image carrying the full image measure must contain its support, and conversely every image of a support point belongs to that support. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. Under these identifications, the measures and all tuple evaluations agree. This is the compact moment-problem meaning of the component space; it is stronger than an arbitrary barycentric representation.

For completeness the plateau identities identify these functionals with translation components of the reference hierarchy, rather than just with an arbitrary convex decomposition. In its reflection space let P_0 be the projection onto $\ker H$. The spectral theorem gives $e^{-tH} \to P_0$ strongly as $t \to \infty$. Taking a time plateau in (276) gives

$$\langle [F], P_0[G] \rangle = \int_{\mathsf{X}} \overline{\sigma_\xi(F)}\sigma_\xi(G)\, d\lambda(\xi).$$

Hence $P_0[F] \mapsto (\xi \mapsto \sigma_\xi(F))$ extends isometrically from the invariant subspace into $L^2(\lambda)$ and sends the vacuum to the constant one. Every compact separated tuple can be translated far enough in the positive time direction that all its entries have positive-time support. Translation invariance of each σ_ξ preserves its value. Thus all coordinate functions occur in the image of the positive-time polynomial vectors. They separate points of X, contain the constants, and are closed under

conjugation. Their linear span has a uniform closure closed under multiplication: for positive-time tuples X, Y, uniform clustering approximates $\sigma_\xi(X)\sigma_\xi(Y)$ by $\sigma_\xi(X\cdot\tau_{te_4}Y)$, uniformly in ξ, as $t\to\infty$. For sufficiently large t this concatenated tuple is separated and still has positive-time support. Stone–Weierstrass therefore makes that closure $C(\mathsf{X})$, which is dense in $L^2(\lambda)$. The isometry is consequently onto.

Each limiting pure-state functional clusters, so its own invariant subspace is generated by its vacuum. This is the translation-ergodic component property used in target theorem A. Selecting any point of X supplies the required component. The representing pure lattice states belong to the enlarged periodic Gibbs class; they need not themselves be limits of cubic-torus measures at fixed spacing.

8.2. **The plateau estimate and reflection sizes.** The lattice estimate measures the approach to the invariant plateau using reflected two-copy correlations. These reflection sizes stay bounded for products of centred densities at fixed physical separation, even when their pointwise supremum norms diverge as the spacing tends to zero. This makes the rate useful for the continuum limit.

Let a be fixed, let $r\ge 1$ be an integer, and let A, B be bounded bond observables supported in $[-ra, ra]^4$. For $x\in a\mathbb{Z}^4$ the required estimate is

$$\begin{aligned}\Big|\nu(\overline{A}\,\tau_x B) - \nu\Big(\overline{E_\nu[A\mid\mathcal{I}]}E_\nu[B\mid\mathcal{I}]\Big)\Big| \\ \le C\mathcal{N}_r(A)^{1/2}\mathcal{N}_r(B)^{1/2}(ra/|x|)^2, \qquad |x|\ge Cra.\end{aligned} \tag{277}$$

The constant C is absolute. To define the reflection size $\mathcal{N}_r(A)$, let θ_μ reflect x_μ at zero, and let γ range over the sixteen coordinate sign changes. Put

$$\mathcal{N}_r(A) = \max_{\mu,\gamma}\nu\Big(\overline{\theta_\mu\gamma A}\,\tau_{2rae_\mu}\gamma A\Big). \tag{278}$$

These are nonnegative reflected two-copy correlations. Their maximum is at most $\|A\|_\infty^2$, but that weaker estimate will not be used for density products in the continuum limit.

Choose a coordinate with $|x_\mu|$ maximal and use it as time. After exchanging A, B if necessary, assume $x_\mu>0$. In the site-mirror Hilbert space put

$$\Phi_A = [\tau_{rae_\mu}\theta_\mu A], \qquad \Phi_B = [\tau_{rae_\mu}B], \qquad n = x_\mu/a - 2r.$$

Then the correlation is a pairing of these vectors with $T_a^n U(x_\perp)$, and their squared norms are among the reflection sizes. Write P_0 for the projection onto $\ker(1-T_a)$. The lattice cone with constant $k_*>0$ puts zero-energy vectors at zero momentum, so $P_0U(v)=P_0$. Reflection positivity and the ergodic averages identify the zero-energy pairing with

$$\langle\Phi_A, P_0\Phi_B\rangle = \nu\Big(\overline{E_\nu[A\mid\mathcal{I}]}E_\nu[B\mid\mathcal{I}]\Big).$$

This is the plateau subtracted in (277). It must be removed before estimating the decaying spectral part.

8.3. **The spectral estimate behind the inverse-square rate.** One may first split A into its even and odd parts under θ_μ. The estimate controls each part by the reflection size of A. If $\theta_\mu A=\varepsilon_A A$, with $\varepsilon_A\in\{1,-1\}$, the definition of Φ_A gives, for an integer $m\ge 1$ and $w\in\mathbb{Z}^4$ with $w_\mu=0$,

$$\nu(\overline{A}\,\tau_{a(2rme_\mu+w)}A) = \varepsilon_A\langle\Phi_A, T_a^{2r(m-1)}U(aw)\Phi_A\rangle.$$

Thus time $2rma$ corresponds to the power t^{m-1} in the spectral measure of Φ_A, where $t=e^{-2raE}$. The filter must combine both time signs into the transform of a nonnegative function of T_a, with this shift and the reflection parity taken into account.

A filter for the shifted time powers. The required coefficients can be chosen independently of the state, spacing and support size. Use normalized Haar measure on the circle. Fix two nonnegative even smooth functions h_+, h_- of integral one, supported respectively where

$$\tfrac{1}{2} \le \cos\alpha \le \tfrac{\sqrt{3}}{2}, \qquad -\tfrac{\sqrt{3}}{2} \le \cos\alpha \le -\tfrac{1}{2}.$$

Put $W_0 = h_+ - h_-$ and $w_m = \int W_0(\alpha) e^{-im\alpha}\, d\alpha/(2\pi)$. These coefficients are real and even, $w_0 = 0$, and $\sum_m |w_m| < \infty$, by twice integrating by parts. The symbol W_0 vanishes on a fixed neighbourhood of zero.

For $0 < t < 1$, the Poisson kernel

$$P_t(\alpha) = \frac{1 - t^2}{1 - 2t\cos\alpha + t^2} = \sum_{m\in\mathbb{Z}} t^{|m|} e^{im\alpha}$$

gives the shifted transfer weight

$$\Psi(t) = 2\sum_{m\ge 1} w_m t^{m-1} = t^{-1} \int W_0(\alpha) P_t(\alpha)\, \frac{d\alpha}{2\pi}.$$

The series defines a continuous function on $[0,1]$, with the $m = 1$ term interpreted as a constant also at $t = 0$. To check its sign and size, let $D_t(\alpha) = 1 - 2t\cos\alpha + t^2$. Since both $h_\pm$ have integral one,

$$\Psi(t) = 2(1 - t^2) \iint \frac{h_+(\alpha) h_-(\beta)(\cos\alpha - \cos\beta)}{D_t(\alpha) D_t(\beta)}\, \frac{d\alpha\, d\beta}{(2\pi)^2}.$$

On these supports the cosine difference is at least one, and $D_t \le 4$. Consequently

$$\Psi(t) \ge \tfrac{1}{8}(1 - t^2), \qquad \Psi(e^{-s}) \ge \tfrac{1}{8}\tanh s \quad (s \ge 0).$$

Continuity includes both endpoints in these inequalities. In particular $\Psi \ge 0$, and $\Psi(1) = W_0(0) = 0$.

For a field of reflection parity ε_A, take $b^{\mathrm{filt}}_m = \varepsilon_A w_m$ for $m \ne 0$. Reflection invariance and $\theta_\mu A = \varepsilon_A A$ identify the positive and negative old-time correlations at the same transverse displacement. Combining them with the preceding shifted-power identity gives exactly $\Psi(t)$, since $\varepsilon_A^2 = 1$. The transverse symbol of the same sequence is $W_{\varepsilon_A} := \varepsilon_A W_0$, so it vanishes near zero, and its absolute coefficient sum has one fixed bound for both parities. Thus one sequence has both required spectral readings, including $t = 0$.

Use the full joint spectral measure of T_a^{2r} and transverse translations on Φ_A, with transfer variable $t \in [0,1]$ and $q \in \mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ measured per lattice step. The endpoint $t = 0$ is retained in the coefficient identity. The next calculation shows that this filter gives a projected weighted density bounded by $Cr^3\mathcal{N}_r(A)$. Only after that calculation can zero transfer and the invariant endpoint $t = 1$ be separated for the energy integral.

The role of the filter is best seen before the final energy integral. Keep e_μ as the original time direction, and let $\bar\mu_A$ be the projection to $q \in \mathbb{T}^3$ of the weighted spectral measure of the full bounded transfer spectrum, including $t = 0$. The filter represents its Fourier coefficients in the form

$$\widehat{\bar\mu}_A(w) = \sum_{m\ne 0} b^{\mathrm{filt}}_m\, \nu\Big(\overline{A}\, \tau_{a(2rme_\mu + w)} A\Big), \qquad \sum_{m\ne 0} |b^{\mathrm{filt}}_m| \le C, \tag{279}$$

with the appropriate parity convention. In this formula $w_\mu = 0$ and both m and w are in lattice units. In the original site-reflection space, the positive-time terms contain powers $T_a^{2r(m-1)}$, not T_a^{2rm}. Thus the coefficients must produce the specified Ψ with that shift. Reflection combines the two signs of m; neither an arbitrary summable sequence nor an arbitrary nonnegative function of T_a gives the required identity.

Absolute summability permits the series to be evaluated in each reflection picture. For $|w|_\infty < 4r$, the reflected norm bound on each correlation gives

$$|\widehat{\bar\mu}_A(w)| \le C\mathcal{N}_r(A).$$

There are at most Cr^3 such displacements. For the remaining displacements choose $\lambda \neq \mu$ with $|w_\lambda| = |w|_\infty$ and take e_λ as time, reversing its sign if necessary. The support of A still lies in a cube of radius r in this picture, and its reflection vectors have squared norms at most $\mathcal{N}_r(A)$. Their new time separation is

$$n_\lambda = |w_\lambda| - 2r \geq \tfrac{1}{2}|w|_\infty.$$

All translations in the old time direction are now spatial unitaries. Therefore the sum over m in (279) becomes the bounded spectral multiplier

$$W_{\varepsilon_A}(2rQ_\mu) = \sum_{m\neq 0} b_m^{\mathrm{filt}} e^{2irmQ_\mu}, \qquad \|W_{\varepsilon_A}(2rQ_\mu)\| \leq \sum_m |b_m^{\mathrm{filt}}|,$$

where Q_μ is the old-direction momentum measured per lattice step. This is a different spectral reading of the same scalar correlation series, not a comparison of the two Hilbert spaces by an assumed rotation.

The required symbol vanishes on a fixed neighbourhood of zero on the circle. Consequently its support after the substitution $2rq$ has $|q| \geq c/r$ in the principal momentum interval. There can be other zero intervals because of periodicity; these only remove more spectral support. On the remaining support, the cone in the new picture gives $aE_\lambda \geq c/r$. All other transverse translations are unitary and commute with this joint spectral multiplier. Hence

$$\|T_{\lambda,a}^{n_\lambda} W_{\varepsilon_A}(2rQ_\mu)\| \leq Ce^{-cn_\lambda/r}.$$

Cauchy–Schwarz for the two reflection vectors now gives

$$|\widehat{\bar{\mu}}_A(w)| \leq C\mathcal{N}_r(A) e^{-c|w|_\infty/r} \qquad (|w|_\infty \geq 4r). \tag{280}$$

The choice of λ can be any of the three directions transverse to the original time. This is where all four reflection pictures, rather than one energy–momentum inequality, enter the density estimate.

The number of lattice displacements in the shell $j \leq |w|_\infty < j+1$ is at most $C(j+1)^2$. Combining the near count with (280) yields

$$\sum_{w\in\mathbb{Z}^3} |\widehat{\bar{\mu}}_A(w)| \leq C\mathcal{N}_r(A) \left[r^3 + \sum_{j\geq 4r} (j+1)^2 e^{-cj/r} \right] \leq Cr^3\mathcal{N}_r(A).$$

Thus these coefficients have an absolutely convergent Fourier series. The continuous function represented by that series has the same Fourier coefficients as the finite measure $\bar{\mu}_A$. Equality on trigonometric polynomials, followed by uniform density, identifies the measure with that density. In particular

$$\left\| \frac{d\bar{\mu}_A}{dq} \right\|_\infty \leq Cr^3\mathcal{N}_r(A).$$

Proposition 8.2 (Decay to the invariant plateau)**.** The coefficient construction above and the reflection calculation give (277), with a constant independent of the state and lattice spacing. ■

Proof. Use the time direction and vectors Φ_A, Φ_B chosen above. First suppose A has definite reflection parity. Let ς_A be the full joint spectral measure of T_a^{2r} and transverse translations on Φ_A. Its transfer variable is $t \in [0,1]$; momentum $q \in [-\pi,\pi]^3$ is measured per lattice step. Put

$$d\mu_A(t,q) = \Psi(t)\, d\varsigma_A(t,q), \qquad \bar{\mu}_A = \text{its projection to } q.$$

The Fourier summation above gives $d\bar{\mu}_A/dq \leq Cr^3\mathcal{N}_r(A)$ on this full spectrum.

The site transfer operator can have a kernel. This contributes at $t = 0$, and the shifted $m = 1$ term in the filter identity can see it. For the decay estimate, however, $n \geq 1$, so T_a^n annihilates this subspace. The plateau is the separate endpoint $t = 1$, removed by $1 - P_0$. We may therefore restrict the remaining integral to $0 < t < 1$ and set

$$E = -\frac{1}{2ra}\log t > 0.$$

The projected weighted measure on this restriction is a positive submeasure of $\bar{\mu}_A$ and inherits its density bound. On this finite-energy subspace the lattice cone gives $aE \geq c|q|$ and

$$\Psi(t) = \Psi(e^{-2raE}) \geq c \tanh(2raE).$$

On positive energies the weight is strictly positive. For $q \neq 0$ the cone consequently bounds the integrand after removal of the weight:

$$\frac{e^{-naE}}{\Psi(e^{-2raE})} \leq C \frac{e^{-cn|q|}}{\tanh(cr|q|)}.$$

The projected weighted measure is absolutely continuous, so the point $q = 0$ contributes no weighted mass. Strict positivity of the weight at positive energy then excludes positive-energy mass over that point as well. The unweighted zero-energy atom has already been removed by $1 - P_0$. Integration gives

$$\begin{aligned} \langle \Phi_A, T_a^n (1 - P_0) \Phi_A \rangle &\leq C r^3 \mathcal{N}_r(A) \int_{\mathbb{T}^3} \frac{e^{-cn|q|}}{\tanh(cr|q|)} \, dq \\ &\leq C \mathcal{N}_r(A) ((r/n)^2 + (r/n)^3). \end{aligned} \tag{281}$$

Here is the last integration explicitly. For $u > 0$, $1/\tanh u \leq C(1 + u^{-1})$. Enlarge the momentum cube to $\mathbb{R}^3$ for the resulting nonnegative majorant and use radial coordinates. Up to absolute constants the two integrals are

$$\int_0^\infty s^2 e^{-cns} \, ds = Cn^{-3}, \qquad \frac{1}{r} \int_0^\infty s e^{-cns} \, ds = Cr^{-1} n^{-2}.$$

Multiplication by r^3 gives exactly the two powers in (281). In the range $n \geq Cr$ the cubic term is bounded by a constant times the quadratic term.

For general A, split into its even and odd reflection parts. The reflection seminorms of each part are at most those entering $\mathcal{N}_r(A)$: use the triangle inequality for the corresponding reflection norm and the coordinate sign changes in its definition. Applying the estimate to the two parts and then the triangle inequality changes only an absolute constant. The same applies to B. Since T_a and the transverse translations commute, Cauchy–Schwarz for the positive operator $T_a^n(1 - P_0)$ gives

$$|\langle \Phi_A, T_a^n U(x_\perp)(1 - P_0) \Phi_B \rangle| \leq C \mathcal{N}_r(A)^{1/2} \mathcal{N}_r(B)^{1/2} (r/n)^2.$$

The removed projection has precisely the invariant plateau identified above. When $|x| \geq Cra$, choosing a largest coordinate gives $n = x_\mu/a - 2r \asymp |x|/a$. This proves (277). □

The exponent two comes from the three transverse momentum dimensions and the first-order zero in the lower filter bound. The proof uses a positive lower cone constant at fixed spacing, and uses all three transverse directions when the filter is read with a different time axis. It makes no positive-gap assertion. The constant-one continuum cone, which was essential for rotations, is not needed in this radial integration.

8.4. **Closedness of pure states and the continuum passage.** At a fixed spacing, the uniform rate also makes the set of ergodic periodic Gibbs states weakly closed. Suppose $\rho_i \to \rho$ with each ρ_i ergodic. For a continuous local A, apply (277) to a distant translate of A in each ρ_i. Its plateau is $|\rho_i(A)|^2$. Pass to the weak limit at each fixed displacement, and then send that displacement to infinity. The plateau formula for ρ gives

$$\rho(|E_\rho[A \mid \mathcal{I}]|^2) - |\rho(A)|^2 = 0.$$

For a countable dense family of local A, this makes the invariant conditional state constant almost surely; thus ρ is ergodic. This is why support points of D_ν can be selected as pure states, rather than merely almost-everywhere representatives.

For the continuum, fix separated tuples $X = (f_1, \dots, f_n)$ and $Y = (h_1, \dots, h_m)$. The required reflection-size bound must be uniform in the selected pure state, not just finite for each of its limiting functionals.

The reflected support geometry. Choose one $R > 0$ so that distinct supports within either X or Y have distance at least $6R$, ignoring an internal separation condition for a one-entry tuple. Put $d = 2R$ and use this radius in both partitions of lemma 6.1. Thus the containing balls of pieces from distinct entries of either tuple are separated by d. For X, write

$$f_i = \sum_\alpha f_{i,\alpha}, \qquad X^\alpha = (f_{1,\alpha_1}, \dots, f_{n,\alpha_n}), \qquad L_X = \sum_{\alpha_1,\dots,\alpha_n} \prod_i \|f_{i,\alpha_i}\|_\sharp.$$

Then $L_X = \prod_i \sum_\alpha \|f_{i,\alpha}\|_\sharp \le (\prod_i K_i) \prod_i \|f_i\|_\sharp$. Make the corresponding choices for Y. Use one L-adic $\varpi > 0$ large enough that all these balls lie in $(-\varpi + d/2, \varpi - d/2)^4$. These choices depend on the original supports alone.

For every coordinate direction μ, sign change γ, and two piece tuples X^α, X^β, consider

$$\theta_\mu \gamma X^\alpha \cdot \tau_{2\varpi e_\mu} \gamma X^\beta.$$

Within each half the balls have separation d. Between the two halves, the μ-coordinates of the first half are at most $\varpi - d/2$, and those of the second are at least $\varpi + d/2$. Thus the two halves also have separation d. All balls are contained in $(-3\varpi + d/2, 3\varpi - d/2)^4$. The whole $2n$-entry tuple belongs to one small-ball support class. This checks the per-ball estimate for the reflected copies, even when a whole original support cannot fit into a small separated ball.

For all sufficiently large j, $r_j = \varpi/a_j$ is an integer and $a_j \le d/2$. Put $A_j = X_j^{\nu_j}$. Multilinearity of the centred field gives

$$A_j = \sum_\alpha (X^\alpha)_j^{\nu_j}.$$

The reference state is invariant under the lattice symmetries, so the same centring is used in the reflected copies. The state-uniform moment estimate on the preceding small-ball class gives

$$\begin{aligned} \mathcal{N}_{r_j}(A_j) &\le c_{2n}(d; R) \sum_{\alpha,\beta} \prod_i \|f_{i,\alpha_i}\|_\sharp \prod_i \|f_{i,\beta_i}\|_\sharp \\ &= c_{2n}(d; R) L_X^2 =: C_X^2. \end{aligned} \tag{282}$$

The bound holds for every evaluating $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$, with the reflection size taken in that state. The same argument gives $\mathcal{N}_{r_j}(Y_j^{\nu_j}) \le C_Y^2$. The constants can depend on the fixed reference coupling and support geometry, but not on j, the evaluating pure phase, ξ, or the chosen volume-first limit. Bounds by $\|A_j\|_\infty^2$ would instead diverge.

Continuity and clustering on arbitrary supports. For any fixed disjoint support neighbourhoods, (226) passes to every component sequence. It proves the same local continuity estimate for each σ_ξ, uniformly in ξ, including for the tuples with reflected copies just used. It also supplies the fixed-support local bounds needed by the Hilbert-space and translation construction; small-ball support restrictions are not imposed on its polynomial vectors.

Apply (277) in a pure evaluating state. Its invariant conditional expectations are constants, so the plateau is the product of its two moments. Equations (277) and (282) give, at lattice displacements v_j,

$$\left|\rho_j(X_j^{\nu_j} \tau_{v_j} Y_j^{\nu_j}) - \rho_j(X_j^{\nu_j}) \rho_j(Y_j^{\nu_j})\right| \le C C_X C_Y \varpi^2 |v_j|^{-2}, \qquad |v_j| \ge C\varpi.$$

For fixed sufficiently large v, take $v_j \in a_j \mathbb{Z}^4$ tending to v. Their tuples retain common disjoint support neighbourhoods. Translation continuity of C_c^1 functions in the sharp norm and (226) permit these moving lattice representatives to pass to the same limit as the fixed continuum tests. This yields (6) with

$$r_{X,Y} = C\varpi, \qquad C_{X,Y} = C C_X C_Y \varpi^2,$$

after enlarging the fixed constants if necessary. Both are uniform over the component space and the volume-first limits at the chosen reference coupling.

Equivalently, one may first apply the bound to the finitely many small-ball pairs and sum their cluster errors. Indeed, multilinearity gives exactly

$$\begin{aligned}&\sigma_\xi(X\cdot\tau_v Y)-\sigma_\xi(X)\sigma_\xi(Y)\\&\qquad=\sum_{\alpha,\beta}\left[\sigma_\xi(X^\alpha\cdot\tau_v Y^\beta)-\sigma_\xi(X^\alpha)\sigma_\xi(Y^\beta)\right].\end{aligned}$$

The partitions and the common containing cube are fixed once for X, Y, before varying the state or taking a large displacement. Thus the sum of the class constants is finite and has precisely the uniformity needed for the component plateau identities.

The connected two-point bound passes independently through the cumulants of ρ_j. Its positivity cannot be explained solely by variation of the component mean.

8.5. **The component mean and the invariant plateau.** The subtraction in (273) is at the reference state, not at the selected pure state. Choose a fixed smooth integral-one test f_0. Translation invariance and the first-mean transfer give a bounded real number

$$b_\xi=\sigma_\xi((f_0)),\qquad \sigma_\xi((f))=b_\xi\int f.$$

It is continuous on X, since tuple evaluations are continuous. The common reference mean is zero, so the barycentre identity gives $\int b_\xi\,d\lambda=0$. On the lattice its value is represented by

$$b_\xi=\lim_j 6a_j^{-4}(\rho_j(a_p)-\nu_j(a_p)).$$

There are six unoriented plaquette planes in four dimensions. Their expectation differences are equal by the lattice symmetries; the Riemann sum of the fixed test then gives the displayed factor. Neither uncentred expectation has to converge separately.

It is useful to distinguish two different second moments. For separated f, h, define the component's connected covariance by

$$C_{2,\xi}(f,h)=\sigma_\xi((f,h))-b_\xi^2\left(\int f\right)\left(\int h\right).$$

Since the reference first moment is zero, its ordinary and connected second moments agree. Averaging the component identity gives

$$S_2^{\mathrm{ref}}(f,h)=\int_{\mathsf{X}}C_{2,\xi}(f,h)\,d\lambda(\xi)+\kappa\left(\int f\right)\left(\int h\right),\qquad \kappa=\int_{\mathsf{X}}b_\xi^2\,d\lambda(\xi).\tag{283}$$

The first term contains connected component covariances. Using ordinary component second moments there would count their means twice. In particular the barycentre itself remains $S_2^{\mathrm{ref}}(f,h)=\int\sigma_\xi((f,h))\,d\lambda$.

The uniform component clustering bound permits the large-translation limit to pass through this integral. It gives

$$\lim_{|v|\to\infty}S_2^{\mathrm{ref}}(f,\tau_v h)=\kappa\left(\int f\right)\left(\int h\right).$$

Thus variation of the component mean can create a nonzero reference plateau even though every component clusters. Conversely the positive short-distance covariance in target theorem A is the connected covariance of each selected phase, obtained from the state-uniform two-point comparison. It is not produced solely by this variation of means. Equality of the means alone would not show that the components agree on higher tuples.

Time clustering and the vacuum. In the reflection space of one component the spectral theorem gives $e^{-tH} \to P_0$ strongly, where P_0 projects onto $\ker H$. For positive-time polynomial vectors,

$$\lim_{t\to\infty} \langle [F], e^{-tH}[G]\rangle = \langle [F], P_0[G]\rangle.$$

Time clustering identifies the left side with $\overline{\sigma_\xi(F)}\sigma_\xi(G)$. Since $\langle \Omega, [G]\rangle = \sigma_\xi(G)$, these are exactly the matrix elements of the rank-one projection $|\Omega\rangle\langle\Omega|$. The polynomial vectors are dense and both operators are bounded; consequently $P_0 = |\Omega\rangle\langle\Omega|$. Conversely, if this projection identity holds, the same strong limit gives time clustering on all such vectors. This equivalence uses no rate. The rate in (277) is needed earlier to preserve clustering when the cutoff and the evaluating pure state both vary.

There is an explicit reason that the cone and all the mirror positivities cannot replace that uniform estimate. Start with a clustering component functional σ and shift its field by a constant $c \neq 0$. On a tuple $X = (f_1, \dots, f_k)$ set

$$\sigma_c(X) = \sum_{T\subset\{1,\dots,k\}} c^{|T|} \prod_{i\in T} \int f_i \, \sigma((f_i)_{i\notin T}), \qquad \widetilde{\sigma} = \tfrac{1}{2}(\sigma + \sigma_c). \tag{284}$$

The shift commutes with translations, rotations and reflection: a test is replaced by the corresponding field insertion plus the real constant $c\int f$. Reflection positivity follows by applying the original positive form to this shifted polynomial. It holds at each of the coordinate and diagonal mirrors. Averaging the two positive forms preserves their positivity. The bounded five-point inequality used for the cone is linear in the functional for every fixed test polynomial, so it is preserved by this averaging as well.

The reconstruction used for this example applies to the mixture itself. To check this without assuming it is one chosen lattice approximation, first realize σ_c using the same approximating states with their centring constants decreased by $c\,a_j^4/6$. The added lattice constant is

$$c_j(f) = \frac{c\,a_j^4}{6} \sum_p f(x(p)) \longrightarrow c\int f.$$

The sum ranges over the six plaquette families. Multiplying out their field products gives exactly (284). Thus σ_c has the same kind of local continuity bounds, symmetry and reflection form as σ.

For each fixed polynomial the contraction inequality

$$\sigma(\theta\overline{Z}\cdot\tau_{2te_4}Z) \le \sigma(\theta\overline{Z}\cdot Z)$$

and the five-point inequality are linear inequalities in the functional. They pass from the two shifted functionals to their equal mixture, as do their invariances and reflection positivity. The local continuity bounds pass with the larger constant. Symmetry, multilinearity and normalization at the empty tuple pass too.

These properties construct the reflection Hilbert space of $\widetilde{\sigma}$, its strongly continuous contraction semigroup and spatial unitary translations by the same argument used above. The five-point inequality can then be applied to this reconstruction to obtain the coordinate cones. This uses no ergodicity of the mixture. The equivalence between time clustering and a one-dimensional vacuum space uses only the reconstruction and the strong limit $e^{-tH} \to P_0$, so it applies to $\widetilde{\sigma}$ as well. The nonzero plateau below therefore does prove a second vacuum direction for this mixture, while making no such claim for the selected clustering components.

Write $\sigma((f)) = b\int f$. At large time separation the two components give the mean products $b^2\int f\int h$ and $(b+c)^2\int f\int h$. The mean of their equal mixture is $(b+c/2)\int f$. Hence

$$\lim_{t\to\infty}\left[\widetilde{\sigma}((f,\tau_{te_4}h)) - \widetilde{\sigma}((f))\widetilde{\sigma}((h))\right] = \left[\frac{b^2+(b+c)^2}{2} - \left(b+\frac{c}{2}\right)^2\right]\left(\int f\right)\left(\int h\right) = \frac{c^2}{4}\left(\int f\right)\left(\int h\right).$$

For a positive-time test of nonzero integral this is a nonzero invariant covariance. Its reflection vector has an invariant projection not proportional to the vacuum. Thus the mixture has at least two invariant vectors despite satisfying the mirror positivities and the cone. Selecting pure approximants and carrying their clustering through a state-uniform rate is the step that excludes this phenomenon for the component in target theorem A.

The round spectral cone after rotations. Once rotation invariance of the component has been proved, its spatial rotations act on the reflection space by

$$U(R)[F] = [(R\oplus 1)F], \qquad R\in \mathrm{O}(3).$$

The reflected form is unchanged because these rotations commute with time reflection. The action is therefore isometric on the dense polynomial vectors and extends to a unitary representation. Continuity in the tests gives strong continuity. It fixes the vacuum, commutes with H, and rotates spatial translations:

$$U(R)e^{iv\cdot P}U(R)^{-1} = e^{i(Rv)\cdot P}.$$

The joint energy–momentum spectrum is consequently invariant under spatial rotations. Applying this to the already proved coordinate inequality $H \geq |P_1|$ gives $H \geq |u\cdot P|$ for every unit vector $u\in\mathbb{R}^3$. On the joint spectrum, $\sup_{|u|=1}|u\cdot p| = |p|$, and hence its support lies in $\{E\geq |p|\}$. This round cone is a consequence of the established rotation invariance. The earlier rotation argument used the constant-one inequalities in the individual coordinate planes and does not use this conclusion as an input.

9. Extension to flat Schwartz tests

We pass from separated tuples to the flat Schwartz space in (3). The state-uniform cumulant estimates control one derivative per entry and a fixed power of the support separation. Flatness supplies the derivatives needed to sum near coincidences. We keep the factorial and derivative counts, since the growth clause of target theorem A requires more than qualitative distributional extension. The construction leaves contact terms unspecified.

9.1. **Distributions away from coincidences.**

Proposition 9.1 (The local distribution)**.** Let T be a multilinear functional on smooth separated tuples. Suppose it satisfies a product sharp-norm bound whenever the entries have supports in prescribed mutually separated compact sets. Then it defines a unique distribution on the open configuration space

$$(\mathbb{R}^4)^n_{\neq} = \{(x_1,\dots,x_n) : x_i\neq x_j \text{ if } i\neq j\}.$$

On each separated product region its order is at most Cn. It agrees with T on tensor products supported in that region. ∎

Proof. Let F have compact support in $\prod_i O_i$, where the O_i are mutually separated open sets. Choose $\psi_i\in C_c^\infty(O_i)$ equal to one on the corresponding support projections. Put the supports in the interior of a cube of side P and extend F periodically, using that it is zero near the cube boundary. Its Fourier series gives

$$F = \sum_{\xi\in(2\pi/P)\mathbb{Z}^{4n}} \widehat{F}(\xi)\bigotimes_i [\psi_i(x_i)e^{i\xi_i\cdot x_i}].$$

The sharp norm of the ith factor is at most $C_i(1+|\xi_i|)$. Consequently the summand after applying T is bounded by a constant times $|\widehat{F}(\xi)|\prod_i(1+|\xi_i|)$. Integration by parts bounds the Fourier coefficient by $C_k(1+|\xi|)^{-k}\|F\|_{C^k}$. Taking $k>5n$, for example an even integer at most $6n+2$, makes the sum absolutely convergent: the n sharp derivatives consume n powers, and the frequency lattice has dimension $4n$.

Apply the convergent series as the definition of $T(F)$ on the product region. It is a distribution of the stated finite order. On a tensor, Fourier partial sums converge in the sharp norm of each factor; the multilinear bound therefore proves agreement with the original T. Finite sums of tensors are dense in the smooth topology on such a product region. Two local definitions which agree on tensors consequently agree on their overlap. A partition of unity on the configuration space patches them to a distribution, and the same density proves uniqueness. □

We now keep track of the constants needed near the diagonals. For tests supported in unit balls, (199) has the form

$$|C_n(f_1,\dots,f_n)| \le C(g)^n (n!)^{p_2} \max\{1,d^{-q_0}\}\prod_i \|f_i\|_\sharp, \qquad n\ge 2,$$

where q_0 does not depend on n. For component-centred moments, the moment–cumulant formula sums over partitions with block sizes at least two. There are at most $n!$ such partitions and at most $n/2$ blocks. Also $\prod_{B\in\pi}|B|!\le n!$. Hence

$$|S_n^0(f_1,\dots,f_n)| \le C(g)^n (n!)^{p_2+1} \max\{1,d^{-q_0 n/2}\}\prod_i \|f_i\|_\sharp.$$

Rounding the separation exponent upward makes it a nonnegative integer $a_n\le Cn$. The separation loss is therefore linear in the order and the growth remains a fixed power of $n!$.

9.2. **A quantitative extension on flat tests.**

Proposition 9.2 (Flat-test extension)**.** Suppose $n\ge 2$ and the separated functional T has the bound

$$|T(f_1,\dots,f_n)| \le c_T \max\{1,d^{-a}\}\prod_i \|f_i\|_\sharp$$

for supports in unit balls at mutual distance at least d, where $a\ge 0$ is an integer. Its configuration-space distribution has a unique continuous extension to $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$, with

$$|\widetilde{T}(\varphi)| \le C^{n+a}(n!)^C c_T\,|\varphi|_{Cn+a}. \tag{285}$$

The absolute constants can depend on the fixed sharp-norm convention. ■

Proof. Write $r(x)=\min_{i<j}|x_i-x_j|$. Use dyadic scales $\delta=1,1/2,1/4,\dots$, with the first region covering $r\ge 1$ and the subsequent regions covering $r\asymp\delta$. One can make the smooth shell cutoffs directly from a fixed monotone radial function which vanishes for radius at most one and is one for radius at least two: take products over all pairs and subtract the products at successive scales. Their sum is one off the diagonals. At the inner shell of scale δ, all pair distances are at least $c\delta$ and one pair distance is at most $C\delta$.

Refine each shell by products of cubes of side $c\delta$, with a fixed small c. The slightly enlarged cubes in any product meeting the shell still have mutual distance at least $c'\delta$. Their diameters are less than one, so the assumed tuple bound applies there. Fixed smooth cutoffs in the scaled cubes permit the Fourier construction in proposition 9.1. Choose the one-variable cutoffs of Gevrey order two, with derivative bounds $C^k(k!)^2$. Only Cn derivatives of them occur in the Fourier estimate. Products over the $O(n^2)$ pairs and the n cubes consequently cost at most $C^n(n!)^C$: differentiating $k\le Cn$ times introduces at most $(Cn^2)^k$ choices, again a fixed factorial power. No derivative of order a is applied to these cutoffs.

Here is the estimate on one scaled product cube. Rescale it to a fixed product cube and apply the Fourier series just described. The separation factor is at most $C^a\delta^{-a}$. The first derivative in each sharp norm costs at most $C\delta^{-1}$, giving $C^n\delta^{-n}$. Thus for its smooth cutoff υ the local distribution obeys

$$|T(\upsilon\varphi)| \le C^{n+a}(n!)^C c_T \delta^{-a-n} \max_{|\gamma|\le Cn} \delta^{|\gamma|} \sup_{\text{cube}} |\partial^\gamma \varphi|.$$

The frequency sum is over a fixed scaled cube. For example, using $(1-\Delta)^{3n+1}$ in the Fourier coefficient estimate costs $(Cn)^{Cn}$ in its coefficients, another fixed power of $n!$. This accounts for the dimension dependence in the displayed bound.

Flatness supplies as many powers of δ as needed. On an inner shell choose a pair at distance at most $C\delta$ and move one of its points to the other. Every derivative of φ vanishes at the resulting point of that diagonal. Taylor's formula, with s further derivatives applied to φ itself, gives

$$|\partial^\gamma \varphi(x)| \le C^{s+Cn}\delta^s |\varphi|_{Cn+s}(1+|x|^2)^{-(Cn+s)/2}, \qquad |\gamma| \le Cn,$$

after increasing the absolute derivative constant. Along the Taylor segment only one four-dimensional coordinate moves a distance $O(\delta) \le O(1)$, so the Schwartz weights are comparable with a factor C^{Cn+s}. The Taylor factorial in the denominator may be discarded. In particular this step costs C^s, rather than $s!$. For the outer shell take $\delta = 1$ and use the direct Schwartz bound, which has the same form without a small-scale gain.

We sum over cube positions using the Schwartz weight. A product grid in $4n$ dimensions with mesh $c\delta$ satisfies, for an exponent larger than $4n$ by a fixed multiple of n,

$$\sum_z (1+|x_z|^2)^{-(Cn+s)/2} \le C^{n+s}\delta^{-4n}.$$

This follows by comparison with the integral over the product cubes; the integral of the radial weight is bounded by C^n when its exponent is at least $6n$. We have used the coarser count δ^{-4n}; the smaller count obtained after separating a common translation is not needed. The complete scale bound is therefore

$$\sum_z |T(\upsilon_{\delta,z}\varphi)| \le C^{n+a+s}(n!)^C c_T \delta^{-a-5n+s}|\varphi|_{Cn+s}.$$

Choose $s = a+6n+1$. A positive power of δ remains, and the dyadic sum is bounded by an absolute constant. The derivative index $Cn+s$ is $a+C'n$ after renaming the absolute constant. The factor C^s is absorbed by C^{n+a}, proving (285). The argument has differentiated the geometric cutoffs only $O(n)$ times; the further a derivatives belong to the flat test. Thus no additional factorial in a has been concealed in the estimate.

The absolutely convergent sum defines the extension. The density statement proved below shows that it is uniquely determined by its values on the separated tensors, and hence by the original functional. □

Lemma 9.3 (Density on the flat test spaces)**.** Finite sums of smooth compactly supported separated tensors are dense in $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$ for its Schwartz topology. For tests flat outside the positive-time configuration space, the tensor entries can all be chosen with compact support in $\{x_4 > 0\}$. ■

Proof. We describe the approximations in an arbitrary fixed seminorm $|\cdot|_k$; repeating the choices for increasing k gives a sequence converging in the Schwartz topology. Let χ_R be a smooth cutoff equal to one on the ball of radius R and zero beyond radius $2R$. Its derivatives satisfy $|\partial^\alpha \chi_R| \le C_\alpha R^{-|\alpha|}$. Leibniz's rule and the extra Schwartz decay give, for every fixed $m > 0$,

$$|(1-\chi_R)\varphi|_k \le C_{k,m} R^{-m} |\varphi|_{k+m}.$$

This removes infinity first. The cutoff does not affect flatness.

Next let A_ε be a smooth cutoff which is zero if some pair distance is at most ε and one if all pair distances are at least 2ε. It can be the product of the pair cutoffs used in the dyadic decomposition.

For fixed n, k its derivatives of order at most k are bounded by $C_{n,k}\varepsilon^{-k}$. On the support of $1 - A_\varepsilon$, at least one point lies within $C\varepsilon$ of a pair diagonal. Flatness and Taylor's formula, applied to any derivative of $\chi_R\varphi$, give a factor ε^s for any chosen s. Thus, with $s > k$,

$$|(1 - A_\varepsilon)\chi_R\varphi|_k \le C_{n,k,s,R}\varepsilon^{s-k}|\chi_R\varphi|_{k+s} \longrightarrow 0.$$

The large derivatives of the cutoff have therefore been paid for by flatness. There is no claim of a uniform bound for tests with nonzero jets on a diagonal.

The remaining test has compact support a positive distance from the diagonals. Cover its support by finitely many products of mutually separated open cubes and choose a subordinate smooth partition of unity. On each product region the Fourier construction of proposition 9.1 approximates the localized test by finite sums of smooth tensors, with supports in the same separated open sets. Choose the Fourier truncation large enough for the fixed C^k accuracy on this compact support. Summing the finitely many pieces proves the first assertion.

For the positive-time assertion, after the cutoff at infinity also cut off the region where any time coordinate is at most ε. Smoothness and vanishing outside the open half-space imply that all derivatives vanish on its boundary. The same Taylor and Leibniz estimates make this cutoff error tend to zero in each fixed seminorm. The separated product cover can now be chosen inside the positive half-space, as can all Fourier tensor factors. These operations prove density in the required positive-time flat space. □

Applying the proposition with $a = a_n \le Cn$ and the moment bound above gives constants independent of n such that

$$|S_n^0(\varphi)| \le A_{\mathrm{c}}B_{\mathrm{c}}^n(n!)^p|\varphi|_{kn}.$$

The zero and first centred moments are 1 and 0, respectively. This proves the required quantitative extension for a component centred at its own mean.

9.3. **Restoring the component mean.** The selected phase retains the subtraction of the reference state. Its first moment can be $b\int f$, with $|b|$ uniformly bounded by one-point transfer. Its higher cumulants are the selected phase's cumulants. Consequently its own centred moments S_n^0 need not be the reference mixture's centred moments S_n^{c} in (227).

For $A \subset \{1, \dots, n\}$ define $\psi_A = \int \varphi\, dx_A$. Integration preserves Schwartz decay and flatness on every diagonal of the remaining variables: every derivative in those variables still vanishes on that diagonal before the integration. The separated-tuple binomial identity extends as

$$S_n(\varphi) = \sum_{A \subset \{1,\dots,n\}} b^{|A|} S^0_{n-|A|}(\psi_A), \qquad S_0^0 = 1. \tag{286}$$

Increase k so that $k > 4$. If y denotes the remaining variables, then

$$(1 + |x|^2)^{kn/2} \ge (1 + |y|^2)^{k(n-|A|)/2} \prod_{i \in A} (1 + |x_i|^2)^{k/2}.$$

Each factor on the right is bounded by the corresponding power of $1 + |x|^2$, and the exponents sum to $kn/2$. Differentiating under the integral, and using this inequality, yields

$$|\psi_A|_{k(n-|A|)} \le C_{\int}^{|A|}|\varphi|_{kn}, \qquad C_{\int} = \int_{\mathbb{R}^4} (1 + |y|^2)^{-k/2}\, dy < \infty.$$

For A the whole index set this includes the integral of φ against the constant zero-variable functional. Since $(n - |A|)! \le n!$, (286) and the binomial theorem give the growth bound with

$$B = B_{\mathrm{c}} + |b|C_{\int}$$

and an enlarged prefactor to include S_0^0. The constant mean changes only the geometric growth factor. Its uniform bound therefore permits the same four growth constants for all components under consideration.

9.4. **Passage of the Euclidean properties.** The seminorms have the useful tensor inequality

$$|\varphi \otimes \psi|_k \le |\varphi|_k |\psi|_k.$$

Indeed $1 + |x|^2 + |y|^2 \le (1 + |x|^2)(1 + |y|^2)$, and each derivative of the tensor is the product of one derivative of each factor. This inequality and the growth bound control every term of a finite reflection-positive quadratic form under test-function approximation.

Take a smooth Schwartz test flat outside the positive-time configuration space. It can first be cut off at infinity, then near the coincidence sets, and then near the time boundary, all in its Schwartz topology. The boundary cutoff is allowed because every derivative vanishes where one time coordinate is zero, by smoothness of its zero continuation. After these cutoffs the tensor approximation can be made within a finite union of separated positive-time product regions.

For two such tests, reflection of the first followed by tensor product with the second is flat on the cross diagonals as well as the original diagonals. A negative-time point and a positive-time point cannot coincide; at their common boundary all derivatives vanish. The reflection-positive quadratic form on the tensor approximants is thus defined entirely on separated tuples. It is nonnegative by the previous sections. Every one of its finitely many terms converges by the growth bound, so its limit is nonnegative.

More explicitly, for a sequence with degrees at most N the quadratic form contains only distributions S_{m+n} with $m + n \le 2N$. Their growth bounds require derivatives of order at most $2kN$. Apply lemma 9.3 to each test in a Schwartz seminorm at least this large. The tensor inequality bounds each difference of reflected products by the sum of one factor difference times bounded factor seminorms. Every term of the finite quadratic form therefore tends to its original value. Positivity on the separated tensor approximants proves positivity on the test sequence. No estimate uniform in an infinite number of degrees is needed for this passage.

The identical density argument passes symmetry and the Euclidean invariance established in section 7 to the flat test space. For each fixed Euclidean transformation the pullback is continuous in the Schwartz topology and preserves flatness. The separated identities therefore determine the identities of the extended distributions. This gives the test-space assertions of target theorem A while leaving their values on general tests at coincidences unspecified.

9.5. **Assembly of the construction.** We now put the proof outline in the order needed for target theorem A. This also specifies which limits must be shared by the different parts of the argument.

First carry out the renormalisation, history and source constructions on the same finite prefixes. The scalar coefficients extracted from the effective density determine the next inverse coupling before the next large-field operation. Their bounded one-loop partial-sum error, the small ordinary correction and the summable remainder correction give the two-sided drift. For bare inverse couplings tending to infinity, stop when the run first enters the fixed reference window. The stopping indices K_j tend to infinity, so $a_j = L^{-K_j} \to 0$. All endpoint couplings stay in one small-coupling range. This avoids an exact endpoint tuning or an assertion that the chosen sequence is unique.

At each such datum take the volume limit first. Compactness of the bond configuration space gives a reference Wilson Gibbs state ν_j. The pressure and reflection arguments are designed to give the same coordinate and diagonal reflection forms in this state and in every Gibbs state used later. The common scalar source normalizer then controls the centred correlations uniformly in these choices. Select one subsequence on a countable dense family of separated tuples; the per-ball equicontinuity bounds extend convergence to all separated tuples. Call the resulting hierarchy S^{ref}.

Next decompose each ν_j into translation-ergodic Gibbs states. Georgii's decomposition theorem applies to the same Wilson specification, as explained in section 8. Map these states into the compact space of correlation coordinates. The uniform plateau estimate identifies mixed coordinate moments with iterated limits in which groups of insertions are translated far apart. Compact moment determinacy then identifies a support point of the limiting coordinate law with a component of S^{ref}. It can be approached by pure lattice states ρ_j, evaluated with the original subtraction at

ν_j. Passing to a further subsequence preserves the already chosen reference hierarchy. Denote the component by σ.

The purpose of the state-uniform estimates is visible here. A reflection inequality or a two-point comparison known only for the reference tori would not automatically hold in ρ_j. Bounds for the whole admitted Gibbs class allow the same limits to carry reflection positivity, reality, permutation symmetry, translations and cubic symmetries to σ. The bounded five-point transfer inequalities give the constant-one coordinate cones without passing unbounded generators through the limit.

Apply the rotation argument to the component centred at its own mean. The cell normalization and long-column limit give the degree-independent insertion bound. The plate construction and ordered products give the analytic kernels and their linear angular growth. The ordered-tube differential equation gives global analytic coefficients, and slab translations identify them with the original functional at projection ties. The polygon zero-row induction gives divisibility at every order. The weaker ultraviolet estimate from source stability eliminates the high two-point harmonics; diagonal reflection positivity removes the remaining two-point harmonics. Convexity together with the polygon divisibility then removes them at every order. Support-controlled approximation by plates returns the rotation identity to ordinary separated tests.

The common first-derivative comparison bounds the component's first moment. Translation invariance identifies it as $b \int f$, with $b \in \mathbb{R}$ finite. Restoring this constant mean preserves the rotation identities and the reflection form. The graded source bound, the moment–cumulant formula and the diagonal-flat extension give (5), with one derivative order linear in n. The positive-time approximation in this section passes reflection positivity to the finite Schwartz sequences in (4). This is the route to clauses (i)–(iii) on the stated test spaces.

For clustering, apply the lattice plateau estimate to representatives of two fixed tuples in ρ_j. Their physical support cube is fixed, and their reflection sizes are uniformly bounded by the separated moment estimates. Their plateaux are products because the states are ergodic. The inverse-square error survives the cutoff limit and gives (6) for the selected component. No factorization of the reference mixture is needed.

For the last clause, apply proposition 5.1 and proposition 5.2 to the fixed bump pair. The positive leading term has size $g(d)^4 d^{-8}$, with $g(d)^4 \asymp (\log(1/d))^{-2}$. The state-transfer error is at most $Cd^2 \|f_d\|_\infty \|h_d\|_\infty$; its ratio to the leading term is $O(d^2(\log(1/d))^2)$, which tends to zero. Decreasing d_0 absorbs this error uniformly. Reference subtraction does not change a second cumulant, and the finite mean product cancels in the connected difference. This gives the lower and upper bounds in (7).

9.6. **Parameters and compatibility.** The main construction cannot be assembled from unrelated estimates that happen to have the same form. Its small-field, large-field and source representations must be the representations of one measure. The exact kernels fix the common measure identity. The analytic representations must additionally be constructed on one compatible array of domains; equality of the represented measures alone does not identify those arrays. At a finite prefix, all tests and stored-field operations use the same geometry and real choices. A source changes only the weight on which the frozen operations act. The marginal coefficient is extracted from that common density, and the history contribution is included before the next coupling is fixed. These are finite induction steps, not conditions imposed on an eventual continuum limit.

After fixing the test, remainder and analytic exponents, choose L and M_1, then the activity decay rate κ, and then the larger interaction scale M and interaction bound E_0, as in section 2. The further geometric enlargements and analytic reserves must accommodate the layered inverses, localization domains, shell preparation and reflected source arrays. The logarithmic test exponents must both pay the failed sharper tests and leave room for the generated fields; the live-age rate is chosen strictly above eight so that it absorbs two stored plaquette sums and their polynomial

factors. The ceiling for the running coupling is chosen last, after the fixed two-point shape and source radii. Sending the cutoff to infinity does not change any of these fixed choices.

Several different limit orders occur, and none can replace another. The normalizer is selected along a volume sequence at fixed cutoff. The column length then tends to infinity at that fixed cutoff, so its endpoint supremum disappears before it can grow with the number of outer insertions. The common continuum subsequence is selected only after the state-uniform estimates are available. In the four spectral pictures the shifted filter identity uses the full bounded transfer spectrum; the zero-transfer part is removed only by a later positive time step, and the invariant plateau is removed separately. Finally, the cluster limits used to identify the component law are iterated with the earlier translations fixed.

These compatibility checks explain the role of the detailed estimates throughout the outline. The analytic work still to complete includes the uniform RG and history bounds on the actual evaluation domains and the common complex one-step, generated-background and complex-increment estimates used by the source and relative two-point comparisons. The differentiated estimate (203) also remains to be established for the full insertion, including the residual quadratic component, on the required complex source domains. Their local analytic verifications include the seam right-inverse energy estimate, the scale-weighted auxiliary system bounds, the retained-form coercivity comparison, and the generated-gauge and stage estimates on their common domains. The local source propagation, energy normalization, component majorants and free curvature comparison have been developed above under those inputs. The history-window argument also requires the interval-reading property of the actual later conditional laws. Recording interval labels and separating sharp thresholds do not by themselves establish that property. The fixed-spacing reflection transfer and the angular and polygon arguments have been given explicitly. Their continuum application still depends on the uniform construction estimates supplying the sharp cone, insertion bounds and source stability. The weaker short-distance bound separates the rotation argument from the relative two-point comparison. Completing these analytic estimates and conditional-law verifications would turn this construction into a proof of the target theorem.

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